TYPICAL CHARACTERISTICS (CONT'D)

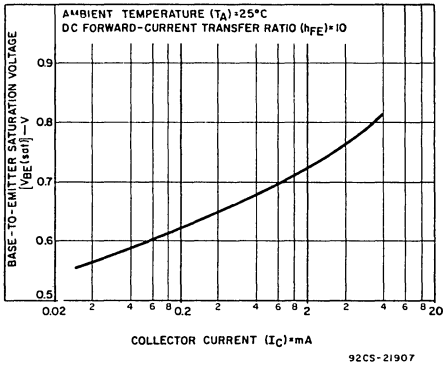


Fig.7 – Base-to-emitter saturation voltage vs. collector current for p-n-p transistor Q4.

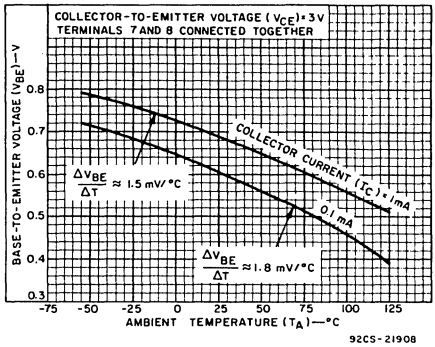


Fig.8 – Base-to-emitter voltage vs. ambient temperature for p-n-p transistor Q4.

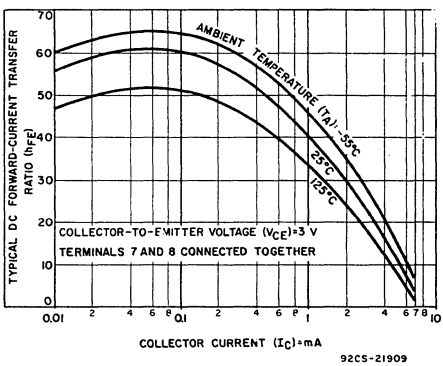


Fig.9 – DC forward-current transfer ratio vs. collector current for p-n-p transistor Q4.

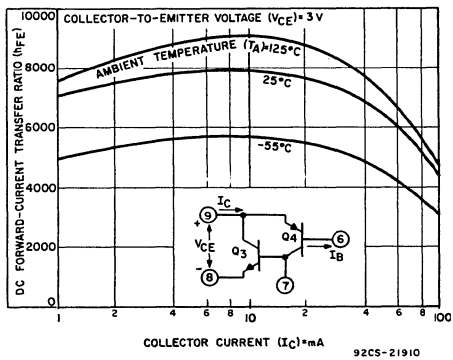


Fig.10 – DC forward-current transfer ratio vs. collector current for transistor pair Q3, Q4.

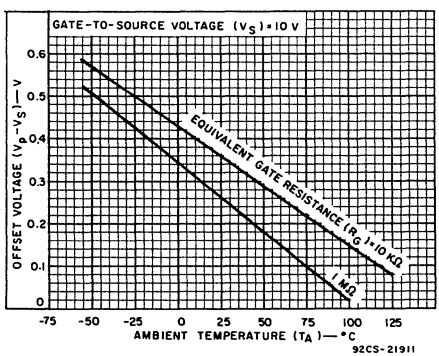


Fig.11 – Offset voltage vs. ambient temperature for Q1 (PUT).

TYPICAL CHARACTERISTICS (CONT'D)

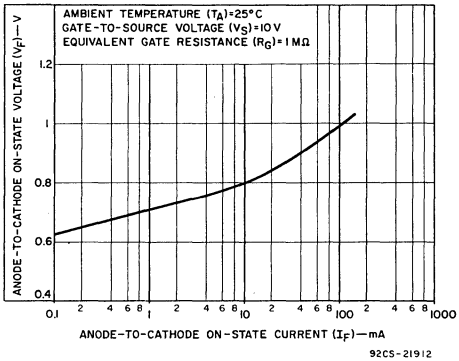


Fig.12 — Anode-to-cathode on-state voltage vs. anode-to-cathode on-state current for Q1 (PUT).

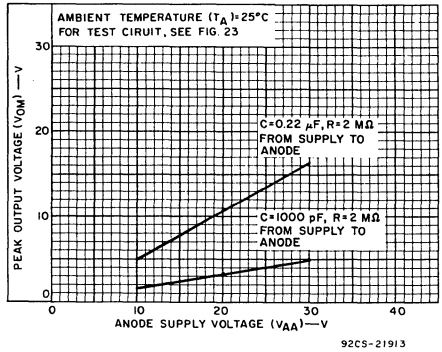


Fig.13 — Peak output voltage vs. anode supply voltage for Q1 (PUT).

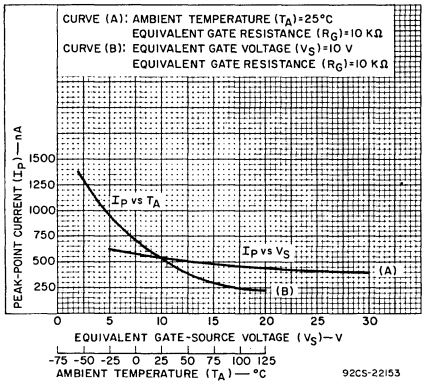


Fig.14 — Peak-point current vs. gate-source voltage and ambient temperature for Q1 (PUT).

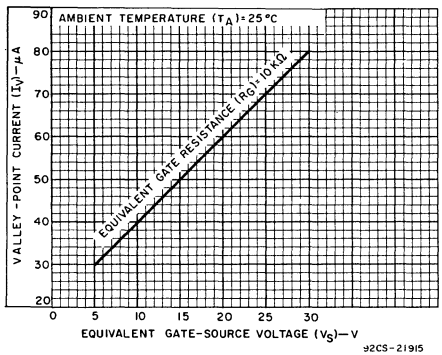


Fig.15 — Valley-point current vs. gate-source voltage for Q1 (PUT).

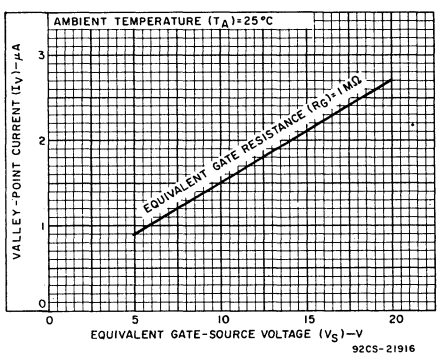


Fig.16 — Valley-point current vs. gate-source voltage for Q1 (PUT).

TYPICAL CHARACTERISTICS (CONT'D)

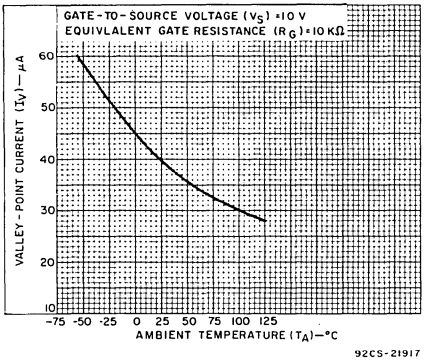


Fig.17 — Valley-point current vs. ambient temperature for Q1 (PUT).

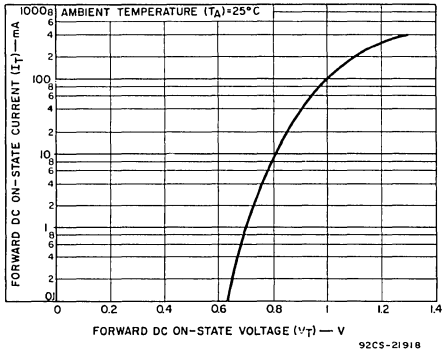


Fig.18 — Forward DC on-state current vs. on-state voltage for Q2 (SCR).

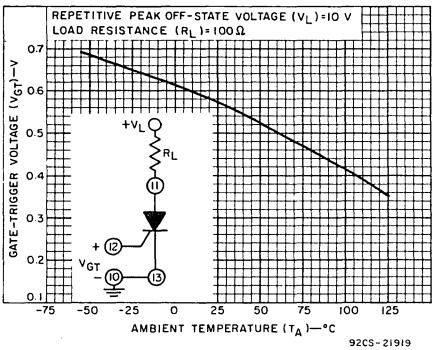


Fig.19 — Gate-trigger voltage vs. ambient temperature for Q2 (SCR).

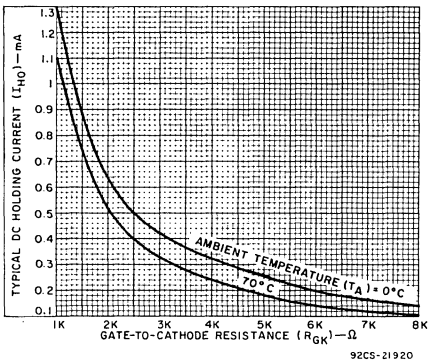


Fig.20 — Typical DC holding current vs. gate-to-cathode resistance for Q2 (SCR).

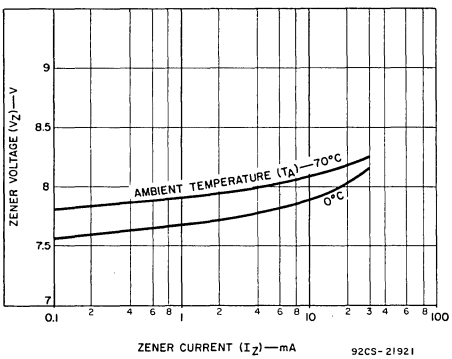


Fig.21 — Zener voltage vs. zener current for Z1.

OPERATING CONSIDERATIONS FOR CA3097E

1. Composite p-n-p/n-p-n Transistors Q3, Q4 (See Fig. 3)

To use Q3 as an individual n-p-n transistor, join terminals no. 6 and no. 9 to disable p-n-p transistor Q4.

The appropriate terminal connections are then:

Collector..... terminal 9
Base ..... terminal 7
Emitter..... terminal 8

To use Q4 as an individual p-n-p transistor, join terminals no. 7 and no. 8 to disable n-p-n transistor Q3.

The appropriate terminal connections are then:

Collector..... terminal 7
Base ..... terminal 6
Emitter..... terminal 9

To use Q3 and Q4 as a composite use terminals 6, 7, 8, and 9 as required.

2. Programmable Unijunction Transistor Q1 (PUT)

The programmable unijunction transistor is essentially an anode-gate SCR. The volt-ampere characteristic of the device is shown in Fig. 22. When an equivalent Thevenin source (V_S, R_G), as shown in Fig. 22, is applied to the gate terminal the device will be "off" if the anode-voltage is negative with respect to the gate voltage. Under this condition, any current flow is exclusively leakage current. When the anode voltage becomes more positive than the gate voltage by an increment equal to the threshold voltage (V_T = 0.4 V typ.), the device can turn "on" only if the current available at the anode terminal **is greater** than the specified peak-point current. The PUT will then switch through its negative-resistance region to the "on" state (low anode-to-gate voltage). It should be noted that I_P is not the maximum current allowed through the device, but is the current required at the peak of the V-I curve. I_P is typically a very low value of current.

After the PUT has switched to its low-impedance state, the device will remain "on" if the anode-current (I_A) exceeds the valley-point current (I_V). If $I_A < I_V$, the PUT will switch back to its high-impedance "off" state. Thus, the PUT can be made to "latch" or recover, depending on I_V. Since I_V is a function of the "on"-state gate current (which depends on R_G and V_S) a choice of R_G and/or V_S will determine the operating mode, i.e., "off" state→"on" state or "off" state → "on" state → "off" state. The value of I_V increases directly as a function of V_G and inversely with R_G. The PUT in the CA3097E has a low I_P.....I_P = 15 nA at V_S = 10 V, R_G = 1 MΩ. This low value of I_P indicates that an extremely large value of anode-supply resistor, e.g. 60 MΩ (typ.), can be used in timing circuits requiring long RC time constants. This becomes important when considering the size of the external

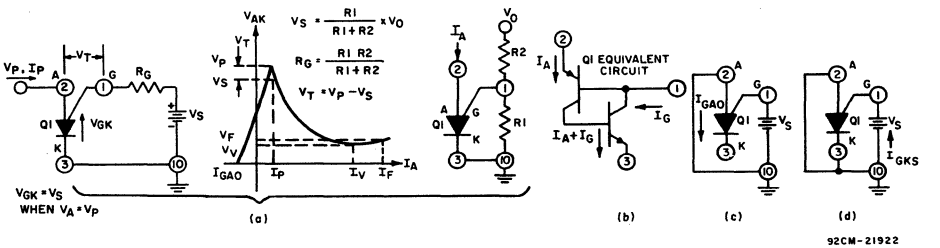


Fig. 22 — General anode characteristics for Q1 (PUT).

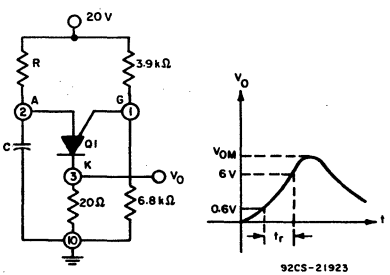


Fig. 23 — Output pulse characteristics for Q1 (PUT).

OPERATING CONSIDERATIONS (CONT'D)

timing capacitor to be used. Consequently, the use of the PUT in the CA3097E is advantageous since it has a lower I_P than most discrete PUT's.

Temperature Compensation of Switching Point

As described previously, the PUT will switch to its low-impedance state when its anode voltage is approximately a diode-drop above the gate voltage. Since the anode-to-gate threshold voltage vs. temperature characteristic is similar to that of a typical silicon-diode junction, a compensating series diode such as used in the circuit of Fig. 29 (Z1 connected as forward-biased diode) considerably reduces the effect of temperature on the switching point.

Bypassing Anode Current

If the PUT gate equivalent source is such that $I_A > I_V$, the PUT will remain "on". A method for turning the PUT off is by shunting current away from the anode until $I_A < I_V$. An example of this technique is the oscillator circuit of Fig. 29. Q3 transistor is turned "on" after the PUT fires and shunts current away from the anode, thereby forcing $I_A < I_V$. The PUT then turns "off" allowing C_T to recharge through R_T, to repeat the cycle.

Protecting The PUT Against Discharge Current Of The Capacitor

A current-limiting resistor in series with the PUT is normally required to dissipate capacitive discharge energy (see Figs. 23 and 29).

Silicon Controlled Rectifier, Q2 (SCR)

The SCR should be used with a 1 kΩ (or less) resistor connected between the cathode and gate terminals if the SCR is to be subjected to its maximum forward and reverse voltage ratings (V_{DXM} and V_{RXM}). Selecting a value for R_{GK} of 1 kΩ (or lower) increases the capability of the device to withstand greater dv/dt and increases the noise immunity of the SCR against false triggering at the gate. Practical considerations such as available current drive from the triggering devices (e.g., a PUT) will determine the lowest value of R_{GK} at which the SCR will fire with a $V_{GK} \approx 0.55$ V. With a value of 500Ω for R_{GK}, the trigger source must be capable of supplying 1.1 mA. R_{GK} should be non-inductive within the frequency band of the noise transients normally encountered in a particular application.

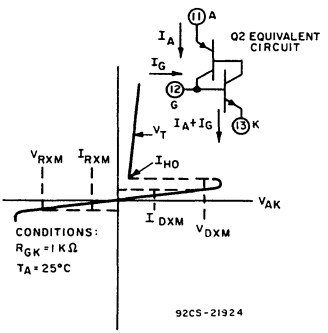


Fig. 24 – Principle voltage-current characteristics for Q2 (SCR).

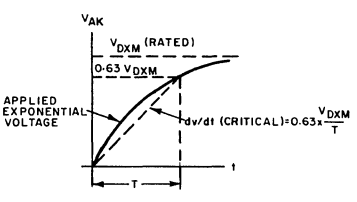
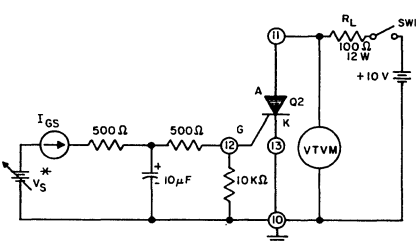


Fig. 25 – Definition of critical rate of rise of off-state voltage for Q2 (SCR).



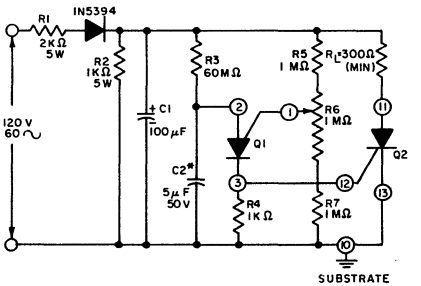
WITH SWI CLOSED, INCREASE V_S UNTIL SCR FIRES (VTVM DROPS FROM 10V TO APPROXIMATELY 1V). I_{GS} (TRIGGER) IS MEASURED JUST PRIOR TO THIS TRIGGERING POINT. NOTE THAT I_{GS} MAY DECREASE AS V_S IS INCREASED DUE TO CURRENT DRAWN OUT OF THE GATE TERMINAL OF THE SCR AS IT TURNS ON. TO UNLATCH THE SCR OPEN SWI.

* V_S SHOULD BE CAPABLE OF SUPPLYING MILLIVOLT INCREMENTS NEAR THE TRIGGER POINT

92CS-21926

Fig. 26 – Test circuit for determining I_{GS} in Q2 (SCR).

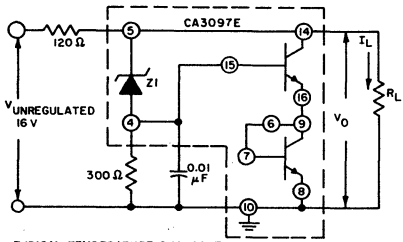
APPLICATIONS CIRCUITS



TIMING PERIOD ≈ 200 SEC. WITH 1 MΩ POT CENTERED
TIMING CYCLE BEGINS WHEN AC IS APPLIED
* SPRAGUE TYPE 4308, 5 μF AT 50 V
SPRAGUE TYPE 6308, 5 μF AT 50 V
OR EQUIVALENT

92CS-21927

Fig. 27 – AC line-operated one-shot timer.



TYPICAL TEMPERATURE CHARACTERISTIC

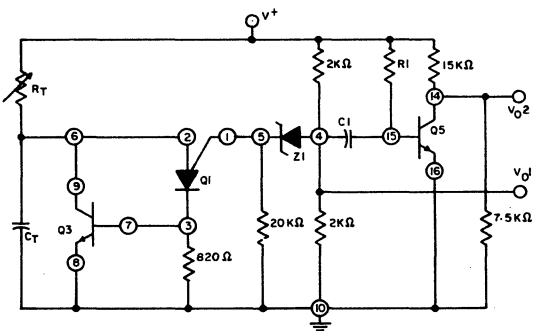
$@\ R_L = 330\,\Omega\ \frac{\Delta V_O/V_O}{\Delta T} \times 100 = \pm 0.01\ \%/°C$

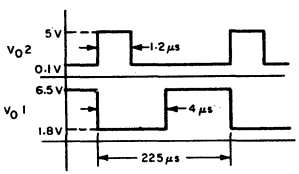
92CS-21928

TYP. LOAD REGULATION @ I_L = 0 TO 40 mA, $(\Delta V_O/V_O) \times 100$ = −3% (NO LOAD TO FULL LOAD)

TYP. LINE REGULATION @ $R_L = 330\,\Omega$, $\frac{\Delta V_O/V_O}{\Delta V_{UNREG.}} \times 100 = \pm 0.55\%/V$

Fig. 28 – Temperature-compensated shunt regulator.





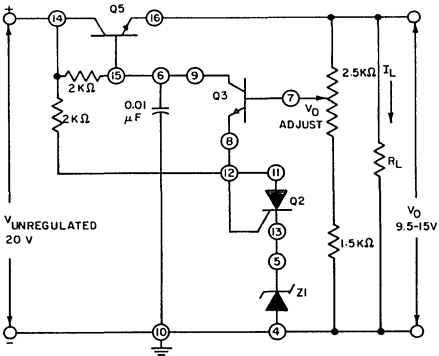
PULSE RATE ADJUSTED BY VARYING R_T OR C_T.
OUTPUT PULSE WIDTH ADJUSTED BY R_1C_1
DIFFERENTIATING TIME CONSTANT

TYPICAL OPERATION FOR:
$V^+ = 15$ V, $C_T = 0.1\ \mu F$, $R_T = 4.3K\Omega$
$C_1 = 82$ pF, $R_1 = 60K\Omega$

92CM-21929

Fig. 29 – Pulse generator.

APPLICATIONS CIRCUITS



TYPICAL LOAD REGULATION @ V_O=12 V, I_L=0 TO 40 mA

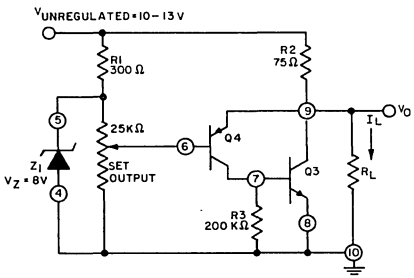
$$\frac{\Delta V_O}{V_O} \times 100 = \pm 0.4\ \%$$ (NO LOAD TO FULL LOAD)

TYPICAL LINE REGULATION @ V_O=12 V

$$\frac{\Delta V_O / V_O}{\Delta V_{UNREG.}} \times 100 = \pm 0.45\ \%/V$$

92CS-21930

Fig. 30 — Series voltage regulator.



TYPICAL LOAD REGULATION @ V_O=7 V, I_L=0 TO 40 mA

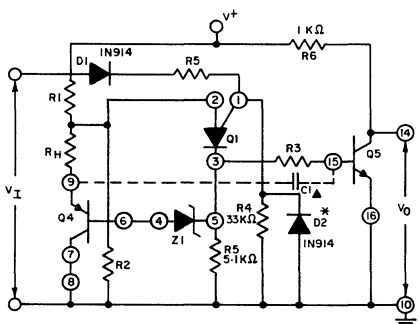
$$\frac{\Delta V_O}{V_O} \times 100 = -1.1\ \%$$

TYPICAL LINE REGULATION @ V_O=7 V, I_L=20 mA

$$\frac{\Delta V_O}{V_O} = \pm 0.85\ \%/\text{VOLT}$$
$$\Delta V_{UNREGULATED}$$

92CS-21931

Fig. 31 — 5 to 7.5 V shunt regulator.



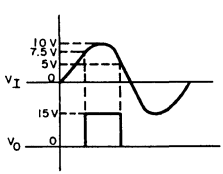
▲ OPTIONAL SPEED-UP CAPACITOR
* REQUIRED IF V_I SWINGS BELOW GROUND

TYPICAL OPERATING CONDITIONS:
FREQUENCY IN = 0-10 KHz
SUPPLY VOLTAGE (V^+) = 15 V
R1, R2, R_H = 5.1 KΩ
R3 = 6.2 KΩ, R_S = 300Ω
C_I = 820 pF
$V_{TH}U$ = 7.5 V, $V_{TH}L$ = 5 V
HYSTERESIS VOLTAGE = 2.5 V

UPPER THRESHOLD VOLTAGE $(V_{TH}U) \approx V^+ \frac{R2}{R1+R2}$

LOWER THRESHOLD VOLTAGE $(V_{TH}L) \approx \frac{(V^+)\left(\frac{R2\ R_H}{R2+R_H}\right)}{\frac{R2\ R_H}{R2+R_H} + R1}$

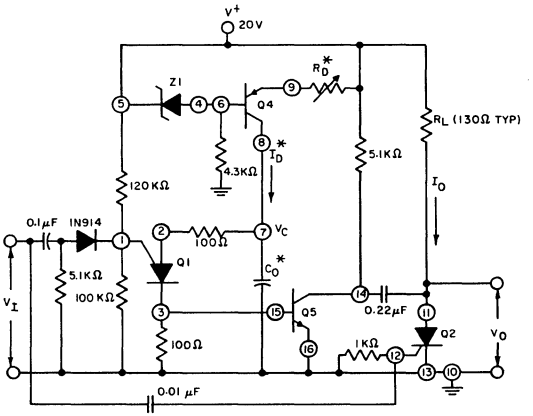
HYSTERESIS VOLTAGE = $V_{TH}U - V_{TH}L$

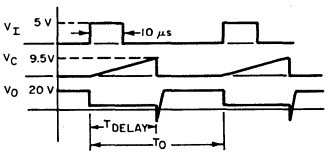


92CM-21932

Fig. 32 — Schmitt trigger.

APPLICATIONS CIRCUITS (CONT'D)



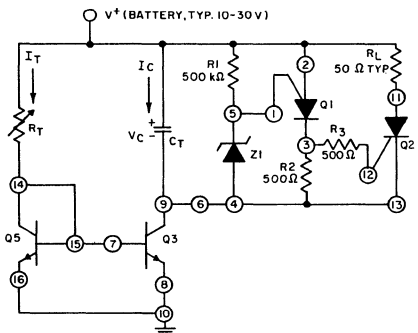


*MONOSTABLE DELAY TIME SET BY ADJUSTMENT OF I_D (VARY R_D) OR BY C_D. I_D MUST BE GREATER THAN I_V OF Q1 (PUT) FOR MONOSTABLE OPERATION.

Q2 (SCR) SWITCHING TIMES:
GATE-CONTROLLED TURN-ON TIME (t_{gt}) = 50 ns (TYP.)
CIRCUIT-COMMUTATED TURN-OFF TIME (t_q) = 10 μs (TYP)

92CM-21933

Fig. 33 – Monostable multivibrator with variable delay.



T_{OFF} = TIMING PERIOD (NO LOAD CURRENT)
PUT FIRES WHEN $V_C \approx 8$ V

$$V_C = \frac{I_C (T_{OFF})}{C_T}, \quad I_C \approx I_T \text{ (Q3, Q5 MATCHED)}$$

I_T SET BY ADJUSTING R_T, $I_T \approx \frac{V^+ - 0.7}{R_T}$

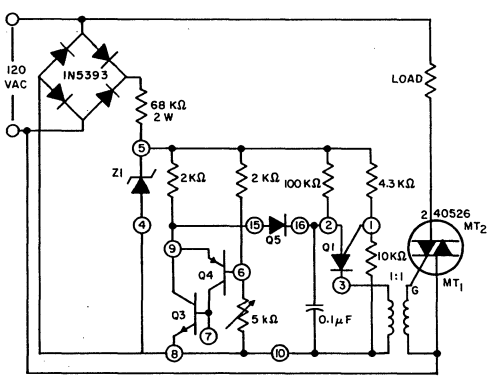
T_{ON} = CAPACITOR DISCHARGE TIME THROUGH LOAD. LOAD TURNS OFF WHEN SCR ANODE CURRENT FALLS BELOW HOLDING CURRENT (I_{HO}). TYPICAL $I_{HO} = 1.2$ mA

EXAMPLE: FOR TIMING PERIOD OF 8.3 MIN.
$C_T = 1000\ \mu F$, $I_T = 16\ \mu A$

$$R_T = \frac{V^+ - 0.7}{I_T} \text{ (FOR } V^+ = 16 \text{ V, } R_T \approx 1 M\Omega)$$

92CS-21934

Fig.34 – Low-current-drain battery-operated long interval astable timer.



NOTE: SHORT TERMINAL 15 TO 14 WHEN USING Q5 AS A DIODE

92CS-22178

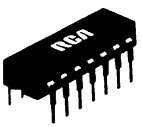
Fig.35 – Phase control circuit.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3099E



14-Lead Dual-in-Line Plastic Package H-1517

Programmable Comparator - - With Memory

Features:

- Programmable operating current
- Micro-power standby dissipation
- Directly controls current up to 150 mA
- Low input on/off current of less than 1 nA for programmable bias current of 1 μA
- Built-in hysteresis: 10 mV max.
- Programmable hysteresis: 10 mV to V^+
- Dual reference input
- High sensor range: 100 Ω to 100 MΩ
- Stable predictable switching levels
- Temperature-compensated reference voltage

Applications:

- Control of relays, heaters, LED's, lamps, photo-sensitive devices, thyristors, solenoids, etc.
- Signal reconditioning
- Phase and frequency modulators
- On/off motor switching
- Schmitt triggers, level detectors
- Time delays
- Overvoltage, overcurrent, overtemperature protection
- Battery-operated equipment
- Square and triangular-wave generators

RCA-CA3099E* Programmable Comparator is a monolithic silicon integrated circuit designed to control high-operating-current loads such as thyristors, lamps, relays, etc. The CA3099E can be operated with either a single power supply with maximum operating voltage of 16 volts, or a dual power supply with a maximum operating voltage of ± 8 volts. It can directly control currents up to 150 mA. It operates with microwatt standby power dissipation when the current to be controlled is less than 30 mA. The CA3099E contains the following six (6) major circuit-function features (Figure 1):

1. **Differential amplifiers and summer;** the circuit uses two differential amplifiers, one to compare the input voltage with the "high" reference, and the other to compare the input with the "low" reference. The resultant output of the differential amplifiers actuates a summer circuit which delivers a trigger that initiates a change in state of a flip-flop.

* Formerly Developmental No. TA6189.

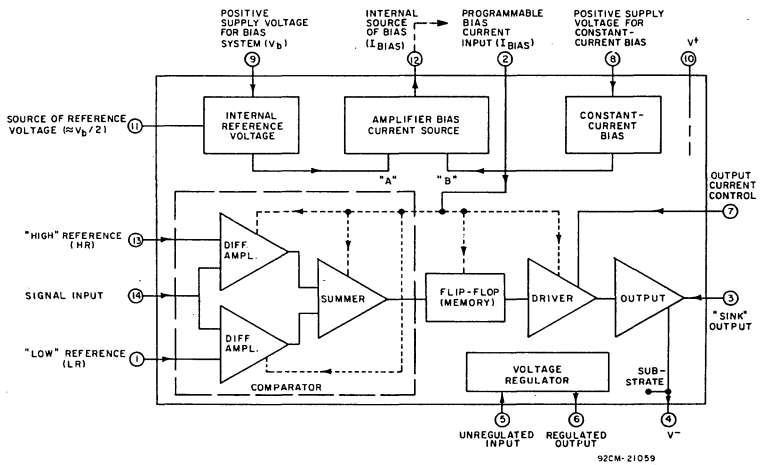


Fig. 1—Block diagram of CA3099E programmable comparator.
(See page 3 for general description of circuit operation.)

Major Circuit-Function Features (Cont'd)

2. **Flip-flop;** the flip-flop functions as a bistable "memory" element that changes state in response to each trigger command.
3. **Driver and output stages;** these stages permit the circuit to "sink" maximum peak load currents up to 150 mA at terminal 3.
4. **Programmable operating current;** the circuit incorporates a separate terminal to permit programming the desired quiescent operating current and performance parameters.
5. **Internal sources of reference voltage and programmable bias current;** an integral circuit supplies a temperature-compensated reference voltage ($V_b/2$) which is about 1/2 of the externally applied bias voltage (V_b). Additionally, integral circuitry can optionally be used to supply an uncompensated constant-current source of bias (I_{bias}).
6. **Voltage regulator;** provides optional on-chip voltage regulation when power for the CA3099E is provided by an unregulated supply.

Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

Supply Voltage Between Terminals 10 and 4, 9 and 4, 8 and 4	16	V
Output Voltage Between Terminals 7 and 4, and 3 and 4	16	V
Differential Input Voltage Between Terminals 14 and 1, and Terminals 13 and 14	10	V
Operating Voltage Range:		
Term. 14	0 V to V^+	
Term. 13	2.0 V to V^+	
Term. 1	0 V to V^+ minus 2.0 V	
Load Current (Term. 3)	150	mA
Input Current to Voltage Regulator (Term. 5)	25	mA
Programming Bias Current (Term. 2)	1	mA
Output Current Control (Term. 7)	15	mA
Power Dissipation:		
Up to T_A = 55°C	750	mW
Above T_A = 55°C. Derate Linearly at	6.67	mW/°C
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Termperature (During Soldering):		
At distance not less than 1/32 inch (0.79 mm) from seating plane for 10 s maximum	+265	°C

ELECTRICAL CHARACTERISTICS AT T_A = 25°C *(Unless otherwise indicated)*

CHARACTERISTICS	SYMBOL	TEST CONDITIONS T_A = 25°C Unless Otherwise Indicated	FIG. No.	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNIT
Reference Voltage	V_{REF}	Term. 9 = 12 V, Term.4 = Grd, Term.11 = Test	–	5.7	6	6.3	V
Reference Voltage Temperature Coefficient			–	–	100	–	μV/°C
Regulated Supply Voltage	V_{REG}	Term.5 1K to 12V, Term.4 = Grd, Term.6 10K to Grd	5	6	7.2	8	V
Regulated Supply Voltage Temperature Coefficient			5	–	2.9	–	mV/°C
Input Offset Voltage: "Low" Reference	$V_{IO\ (LR)}$	V_{LR} = Grd, V_{HR} = 3 V, I_{BIAS} = 100 μA	20, 6	−8	−3	2	mV
"High" Reference	$V_{IO\ (HR)}$	V_{HR} = Grd, V_{LR} = −3 V, I_{BIAS} = 100 μA	20, 7	−5	±1	5	
"Low" Reference Temp. Coefficient		−55°C to +125°C	20, 8	–	4.5	20	μV/°C
"High" Reference Temp. Coefficient		−55°C to +125°C	20, 9	–	±8.2	±20	
Min. Hysteresis Voltage	$V_{IO(HR-LR)}$	V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21, 10	–	3	10	mV
Min. Hysteresis Voltage Temperature Coefficient		−55°C to +125°C	11	–	6.7	20	μV/°C
Output Saturation Voltage	$V_{CE(SAT)}$	V_I = 4 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21,12,13	–	0.72	1.2	V
Total Supply Current: I_{TOTAL} "ON"	I_{TOTAL}	V_I = 4 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21,14,15	600	710	800	μA
I_{TOTAL} "OFF"		V_I = 8 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21,14,15	420	560	750	
Input Bias Current: $I_{B(p-n-p)}$	I_{IB}	V_I = 4 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21,16,17	–	33	200	nA
$I_{B(n-p-n)}$		V_I = 8 V, V_{REG} = 6 V, V^+ = 12 V, I_{BIAS} = 100 μA	21,16,17	–	20	60	
Output Leakage Current	$I_{CE(OFF)}$	Current from Term.3 when Q46 is "OFF"	–	–	–	10	μA
Internal Bias Current	I_{IBC}		18,19	120	200	280	μA
Switching Times:		I_C = 100 μA, I_{BIAS} = 100 μA, V^+ = 5 V, V_{REG} = 2.5 V					
Delay	t_d		22	–	600	–	ns
Fall	t_f		22	–	50	–	
Rise	t_r		22	–	500	–	
Storage	t_s		22	–	4.5	–	μS

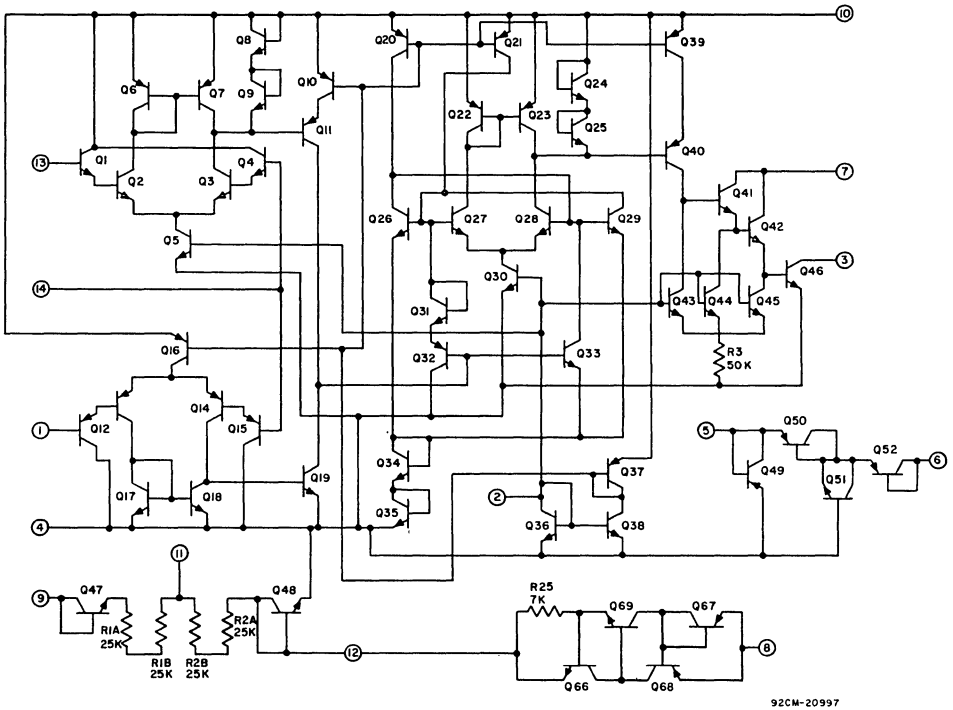


Fig.2–Schematic diagram of CA3099E.

General Description of Circuit Operation (Refer to Fig.1)

When the signal-input voltage of the CA3099E is equal to or less than the "low" reference voltage (LR), current flows from an external power supply through a load connected to terminal 3 ("sink" output). This condition is maintained until the signal-input voltage rises to or exceeds the "high" reference voltage (HR), thereby effecting a change in the state of the flip-flop (memory) such that the output stage interrupts current flow in the external load. This condition, in turn, is maintained until such time as the signal again becomes equal to or less than the "low" reference voltage (VR).

The CA3099E comparator is unique in that it contains circuit provisions to permit programmability. This feature provides flexibility to the designer to optimize quiescent power consumption, input-circuit characteristics, hysteresis, and additionally permits independent control of the comparator, namely, pulsing, strobing, keying, squelching, etc. Programmability is accomplished by means of the bias current (I_{bias}) supplied to terminal 2. As an alternative to externally supplied bias current, the CA3099E contains an internal source of regulated bias current accessible at terminal 12. This internal source of bias current is developed by two alternative methods; in the first method, bias voltage (V_b) applied at terminal 9 develops a source of temperature-compensated reference voltage ($\approx V_b/2$) at terminal 11 and additionally supplies a source of bias current at terminal 12 via line "A". Alternately, when a positive supply voltage is applied at terminal 8, a source of constant-current biasing is provided at terminal 12 via line "B".

An auxiliary means of controlling the magnitude of load-current flow at terminal 3 is provided by "sinking" current into terminal 7. The CA3099E contains an on-chip voltage regulator which may optionally be used to regulate the voltages and bias currents (exclusive of the load current at terminal 3) needed for the operation of the IC.

Fig. 2 is the schematic diagram of the CA3099E. Figs. 3 and 4 are, respectively, functional and logic diagrams of CA3099E operation.

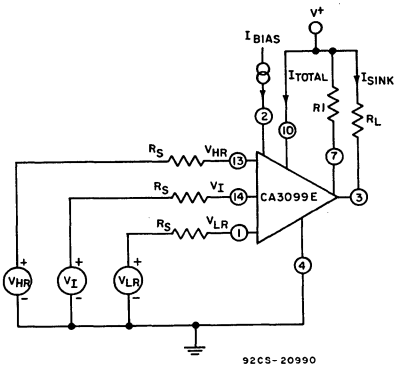


Fig. 3 – Functional diagram.

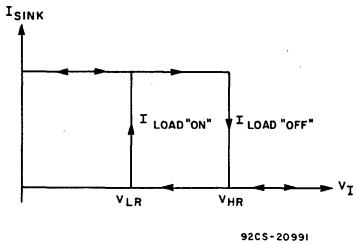


Fig. 4 – Logic diagram.

TYPICAL CHARACTERISTIC CURVES

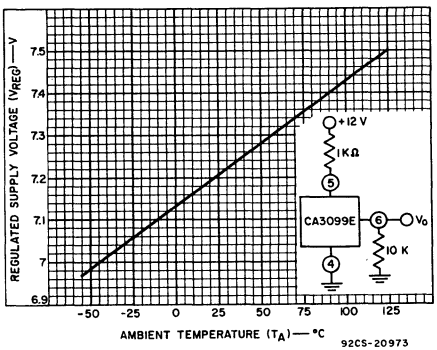


Fig. 5 – Regulated supply voltage vs. ambient temperature.

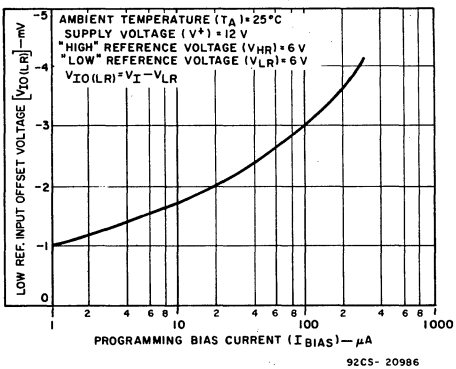


Fig. 6 – Input-offset voltage ("low" reference) vs. programming bias current.

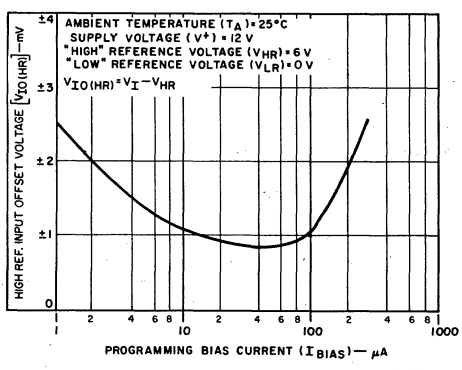


Fig. 7 – Input-offset voltage ("high" reference) vs. programming bias current.

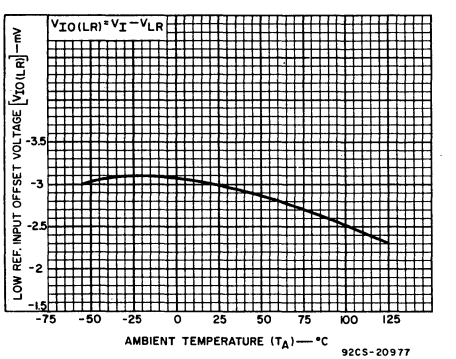


Fig. 8 – Input-offset voltage ("low" reference) vs. ambient temperature.

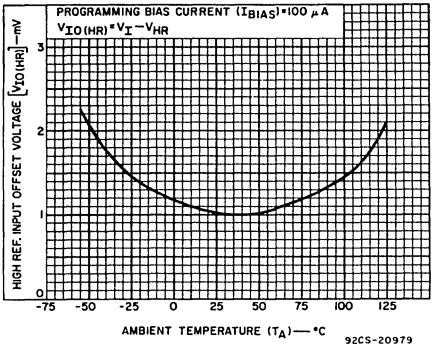


Fig. 9 – Input-offset voltage ("high" reference) vs. ambient temperature.

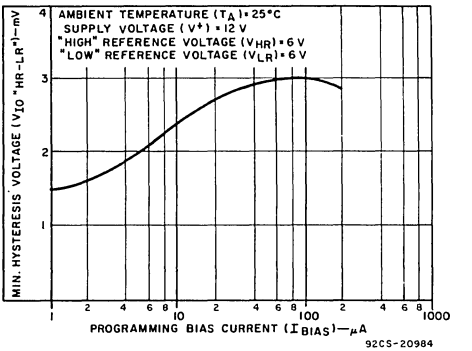


Fig. 10 – Min. hysteresis voltage vs. programming bias current.

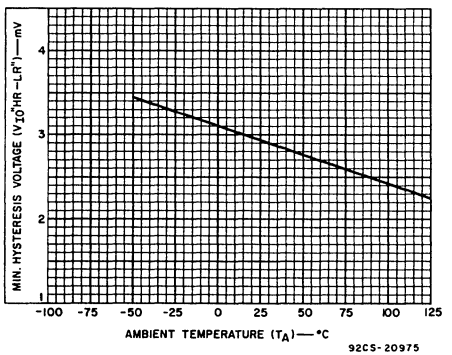


Fig. 11 – Min. hysteresis voltage vs. ambient temperature.

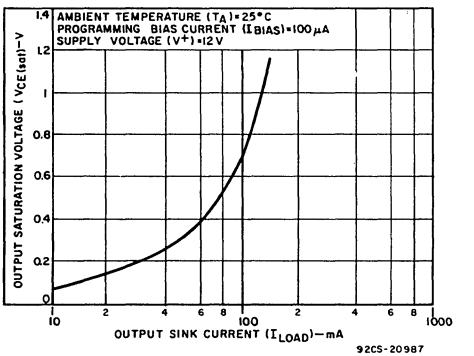


Fig. 12 – Output saturation voltage vs. output sink current.

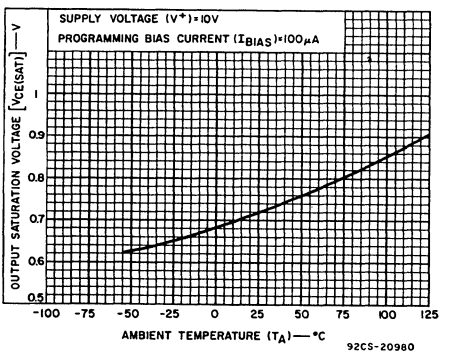


Fig. 13 – Output saturation voltage vs. ambient temperature.

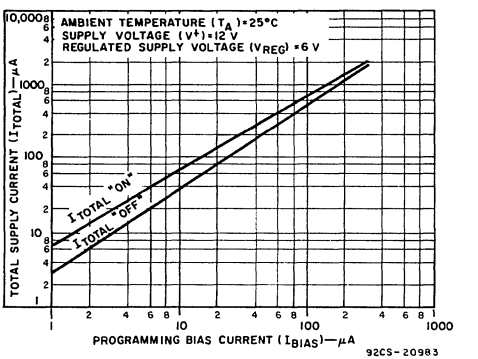


Fig. 14 – Total supply current vs. programming bias current.

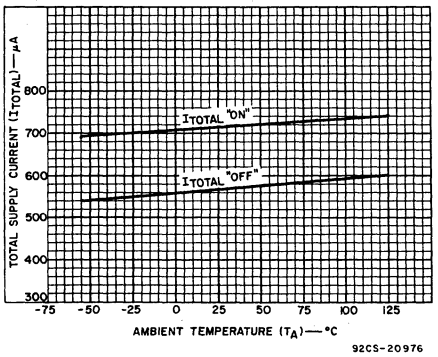


Fig. 15 – Total supply current vs. ambient temperature.

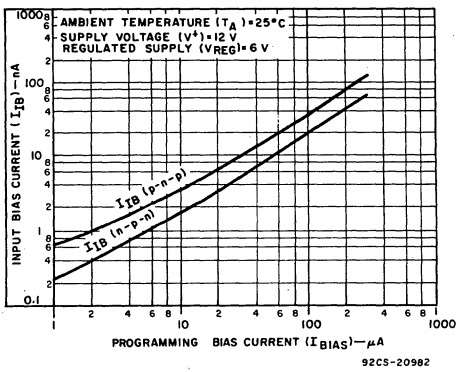


Fig. 16 – Input bias current vs. programming bias current.

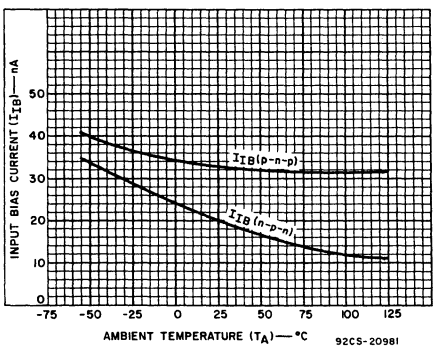


Fig. 17 – Input bias current vs. ambient temperature.

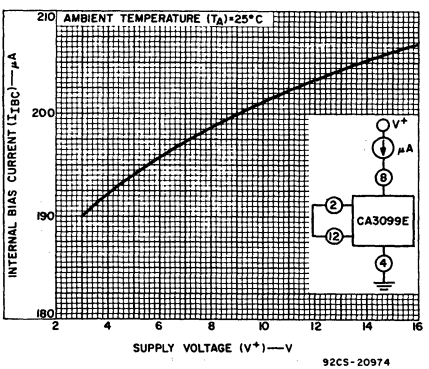


Fig. 18 – Internal bias current vs. supply voltage.

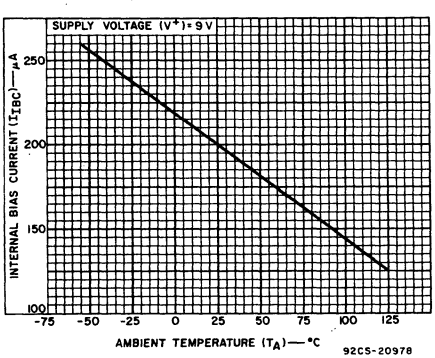


Fig. 19 – Internal bias current vs. ambient temperature.

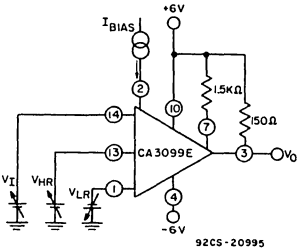


Fig. 20 – Input-offset voltage test circuit.

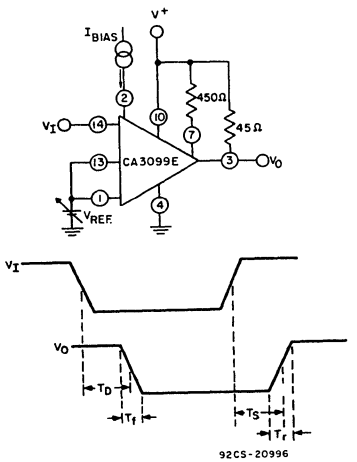


Fig. 22 – Switching time test circuit.

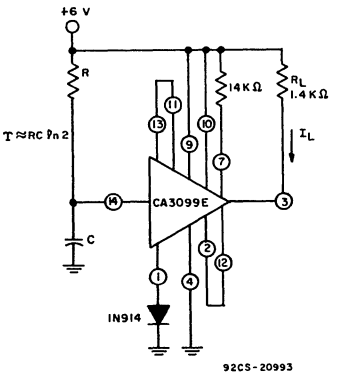


Fig. 23(b) – Time delay circuit: "sink" current interrupted after T seconds.

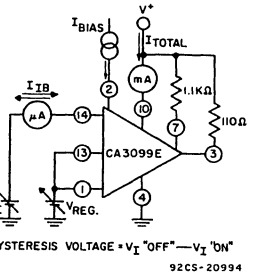


Fig. 21 – Min. hysteresis voltage, total supply current, and input bias current test circuit.

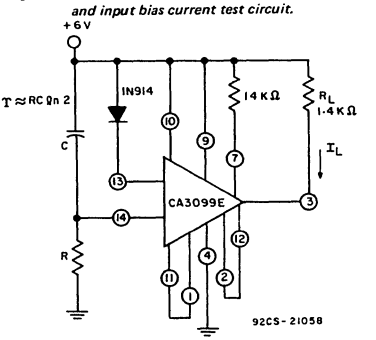
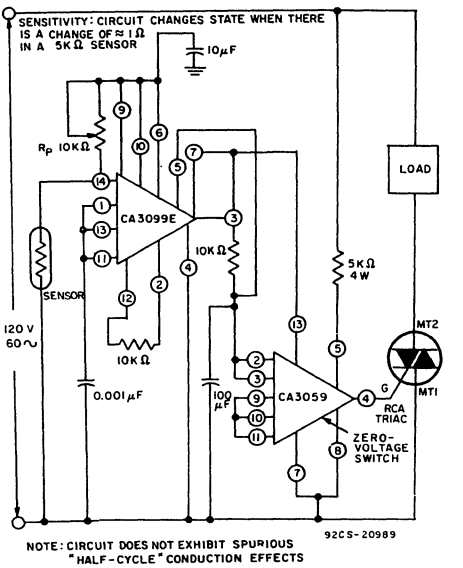


Fig. 23(a) – Time delay circuit: Terminal 3 "sinks" after T seconds.



NOTE: CIRCUIT DOES NOT EXHIBIT SPURIOUS "HALF-CYCLE" CONDUCTION EFFECTS

Fig. 24 – Sensitive temperature control.

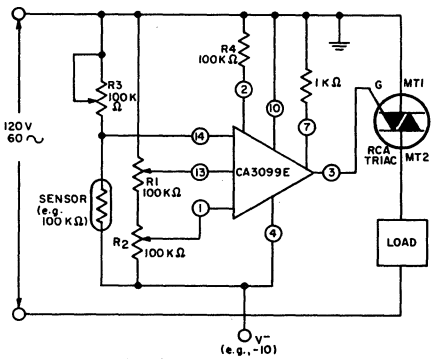


Fig. 25 – OFF/ON control of triac with programmable hysteresis.

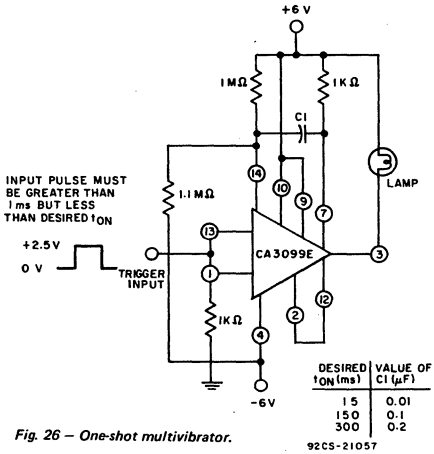


Fig. 26 – One-shot multivibrator.

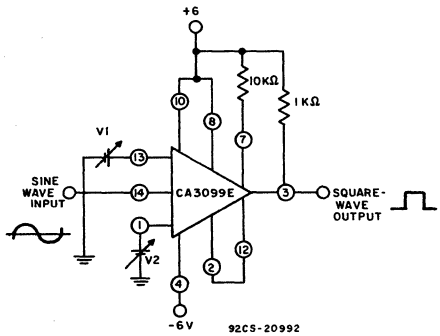


Fig. 27 – Sine-wave to square-wave converter with duty-cycle adjustment (V_1 and V_2).

RCA Solid State Division

Linear Integrated Circuits

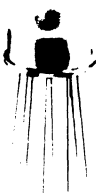
Monolithic Silicon

CA3100S CA3100T



8-LEAD TO-5 with Dual-In-Line Formed Leads ("S" Suffix)

H-1787



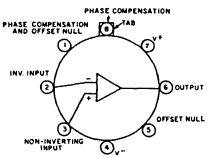
8-LEAD TO-5 ("T" Suffix)

H-1528

Wideband Operational Amplifier

Features:

- High open-loop gain at video frequencies – 42 dB typ. at 1 MHz
- High unity-gain crossover frequency (f_T) – 38 MHz typ.
- Wide power bandwidth – V_O = 18 V p-p typ. at 1.2 MHz
- High slew rate – 70 V/μs (typ.) in 20 dB amplifier
  25 V/μs (typ.) in unity-gain amplifier
- Fast settling time – 0.6 μs typ.
- High output current – ± 15 mA min.
- LM118, 748/LM101 pin compatibility
- Single capacitor compensation
- Offset null terminals



Functional Diagram 92CS-22569

RCA-CA3100S, CA3100T* is a large-signal wideband, high-speed operational amplifier which has a unity gain crossover frequency (f_T) of approximately 38 MHz and an open-loop, 3 dB corner frequency of approximately 110 kHz. It can operate at a total supply voltage of from 14 to 36 volts (±7 to ±18 volts when using split supplies) and can provide at least 18 V p-p and 30 mA p-p at the output when operating from ±15 volt supplies. The CA3100 can be compensated with a single external capacitor and has dc offset adjust terminals for those applications requiring offset null. (see Fig. 19).

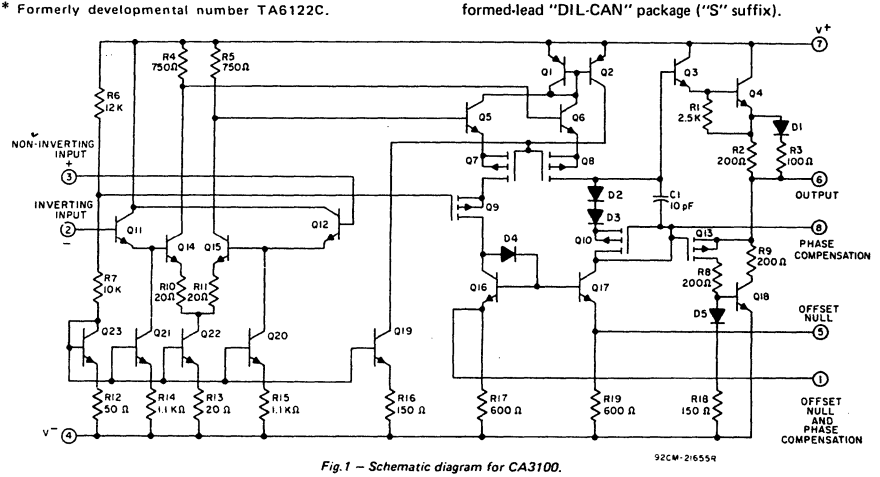
* Formerly developmental number TA6122C.

Applications:

- Video amplifiers
- Fast peak detectors
- Meter-driver amplifiers
- High-frequency feedback amplifiers
- Video pre-drivers
- Oscillators
- Multivibrators
- Voltage-controlled osc.
- Fast comparators

The CA3100 circuit contains both bipolar and P-MOS transistors on a single monolithic chip.

The CA3100 is supplied in either the standard 8-lead TO-5 package ("T" suffix), or in the 8-lead TO-5 dual-in-line formed-lead "DIL-CAN" package ("S" suffix).



92CM-21655R

Fig. 1 – Schematic diagram for CA3100.

MAXIMUM RATINGS, *Absolute-Maximum Values at* T_A = 25°C:

SUPPLY VOLTAGE (Between V^+ and V^- terminals)	36	V
DIFFERENTIAL INPUT VOLTAGE	±12	V
INPUT VOLTAGE TO GROUND*	±15	V
OFFSET TERMINAL TO V^- TERMINAL VOLTAGE	±0.5	V
OUTPUT CURRENT	50	mA●
DEVICE DISSIPATION:		
Up to T_A = 55°C	630	mW
Above T_A = 55°C Derate Linearly at	6.67	mW/°C

AMBIENT TEMPERATURE RANGE:

Operating	−55 to +125°C
Storage	−65 to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 s max.	300 °C

* If supply voltage is less than ± 15 volts, the maximum input voltage to ground is equal to the supply voltage.

● CA3100S, CA3100T does not contain circuitry to protect against short circuits in the output.

TYPICAL CHARACTERISTIC CURVES

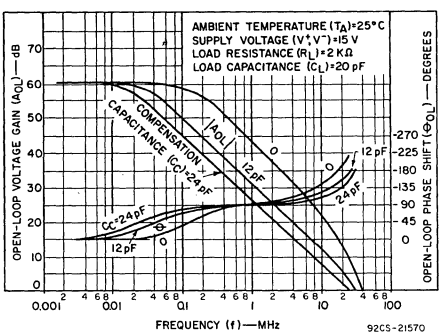


Fig.2 – Open-loop gain, open-loop phase shift vs. frequency.

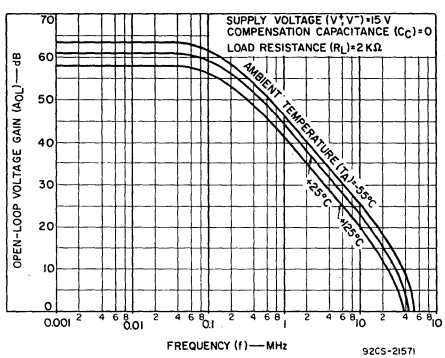


Fig.3 – Open-loop gain vs. frequency and temperature.

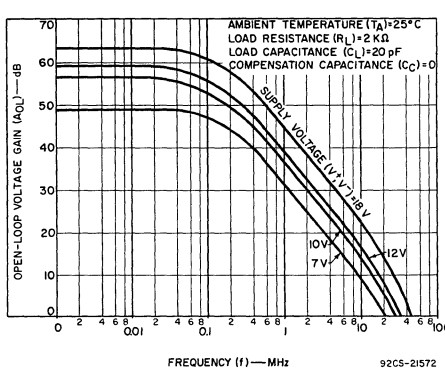


Fig.4 – Open-loop gain vs. frrquency and supply voltage.

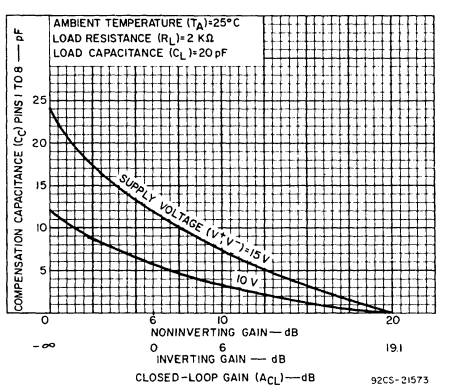


Fig.5 – Required compensation capacitance vs. closed-loop gain.

ELECTRICAL CHARACTERISTICS, *At T_A = 25°C:*

CHARACTERISTICS	SYMBOL	TEST CONDITIONS SUPPLY VOLTAGE (V^+,V^-)=15 V UNLESS OTHERWISE SPECIFIED	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
STATIC						
Input Offset Voltage	V_{IO}	V_O = 0 ± 0.1 V	—	±1	±5	mV
Input Bias Current	I_{IB}	V_O = 0 ± 1 V	—	0.7	2	μA
Input Offset Current	I_{IO}	V_O = 0 ± 1 V	—	±0.05	±0.4	μA
Low-Frequency Open-Loop Voltage Gain ◆	A_{OL}	V_O = ± 1 V Peak, F = 1 kHz	56	61	—	dB
Common-Mode Input Voltage Range	V_{ICR}	CMRR ≥ 76 dB	±12	+14 −13	—	V
Common-Mode Rejection Ratio	CMRR	V_I Common Mode = ± 12 V	76	90	—	dB
Maximum Output Voltage Positive	V_{OM}^+	Differential Input Voltage = 0 ± 0.1 V	+9	+11	—	V
Maximum Output Voltage Negative	V_{OM}^-	R_L = 2 KΩ	−9	−11	—	V
Maximum Output Current Positive	I_{OM}^+	Differential Input Voltage = 0 ± 0.1 V	+15	+30	—	mA
Maximum Output Current Negative	I_{OM}^-	R_L = 250 Ω	−15	−30	—	mA
Supply Current	I^+	V_O = 0 ± 0.1 V, R_L ≥ 10 KΩ	—	8.5	10.5	mA
Power-Supply Rejection Ratio	PSRR	ΔV^+ = ± 1 V, ΔV^- = ± 1 V	60	70	—	dB
DYNAMIC						
Unity-Gain Crossover Frequency	f_T	C_C = 0, V_O = 0.3 V (P-P)	—	38	—	MHz
1-MHz Open-Loop Voltage Gain	A_{OL}	f = 1 MHz, C_C = 0, V_O = 10 V (P-P)	36	42	—	dB
Slew Rate: 20-dB Amplifier	SR	A_V = 10, C_C = 0, V_I = 1 V (Pulse)	50	70	—	V/μs
Slew Rate: Follower Mode	SR	A_V = 1, C_C = 10 pF, V_I = 10 V (Pulse)	—	25	—	V/μs
Power Bandwidth ▲: 20-dB Amplifier	PBW	A_V = 10, C_C = 0, V_O = 18 V (P-P)	0.8	1.2	—	MHz
Power Bandwidth ▲: Follower Mode	PBW	A_V = 1, C_C = 10 pF, V_O = 18 V (P-P)	—	0.4	—	MHz
Open-Loop Differential Input Impedance	Z_I	F = 1 MHz	—	30	—	KΩ
Open-Loop Output Impedance	Z_O	F = 1 MHz	—	110	—	Ω
Wideband Noise Voltage Referred to Input	e_N(Total)	BW = 1 MHz, R_S = 1 KΩ	—	8	—	μV_{RMS}
Settling Time [To Within ± 50 mV of 9 V Output Swing]	t_s	R_L = 2 KΩ, C_L = 20 pF	—	0.6	—	μs

▲ Power Bandwidth = $\frac{\text{Slew Rate}}{\pi V_O \text{ (P-P)}}$ ◆ Low-frequency dynamic characteristic

TYPICAL CHARACTERISTIC CURVES (Cont'd)

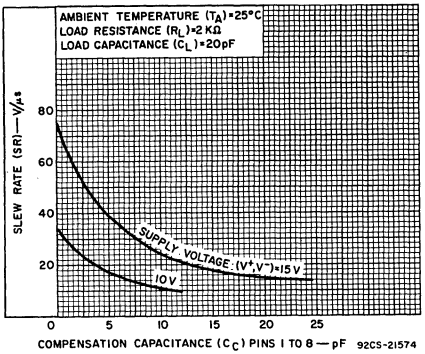


Fig.6 – Slew rate vs. compensation capacitance.

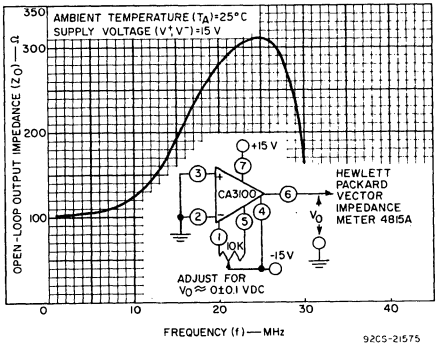


Fig.7 – Typical open-loop output impedance vs. frequency.

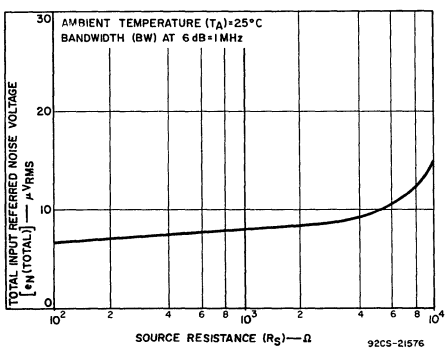


Fig.8 – Wideband input noise voltage vs. source resistance.

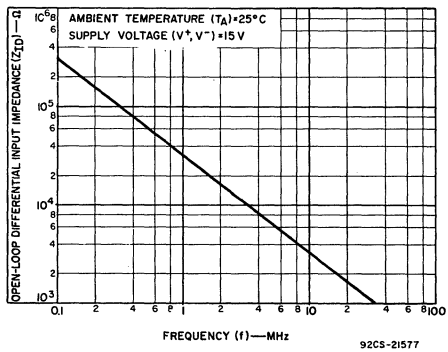


Fig.9 – Typical open-loop differential input impedance vs. frequency.

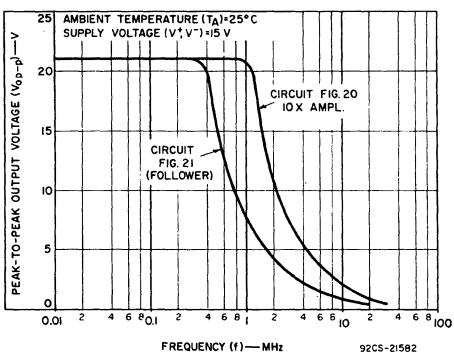


Fig.10 – Maximum output voltage swing vs. frequency.

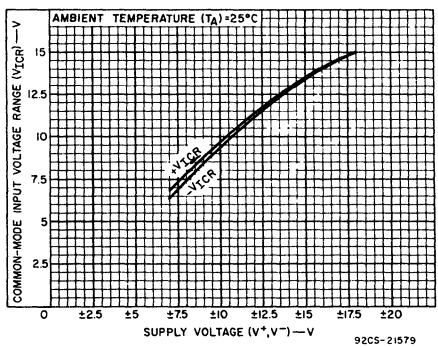


Fig.11 – Common-mode input voltage range vs. supply voltage.

TYPICAL CHARACTERISTIC CURVES (Cont'd)

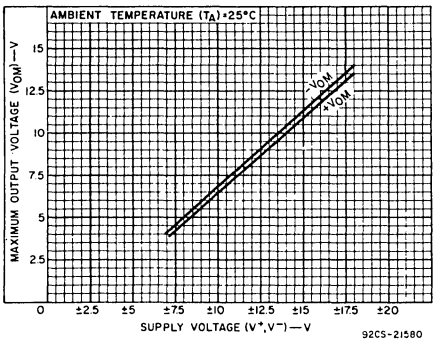


Fig.12 – Maximum output voltage vs. supply voltage.

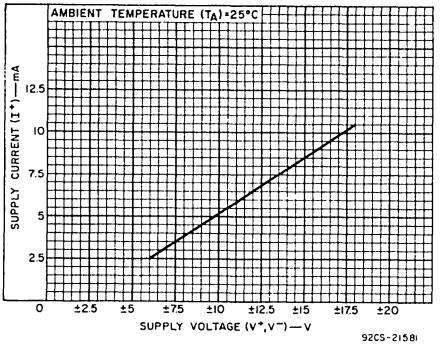


Fig.13 – Supply current vs. supply voltage.

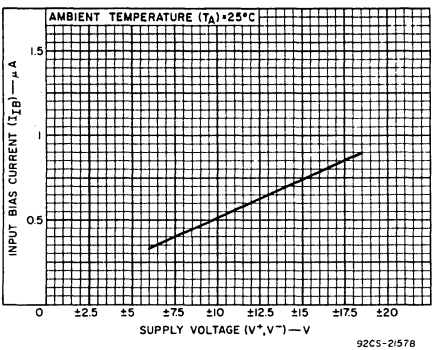


Fig.14 – Input bias current vs. supply voltage.

TYPICAL APPLICATIONS

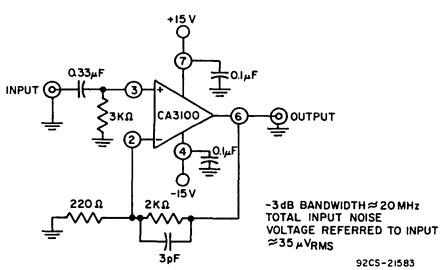


Fig.15 – 20 dB video amplifier.

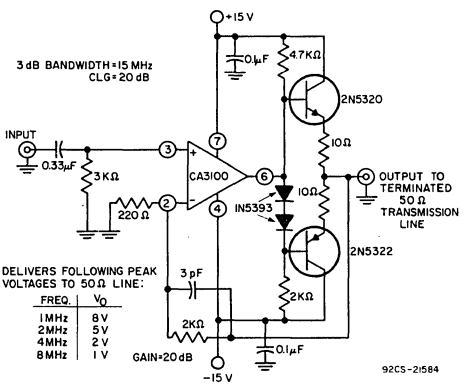


Fig.16 – 20 dB video line driver.

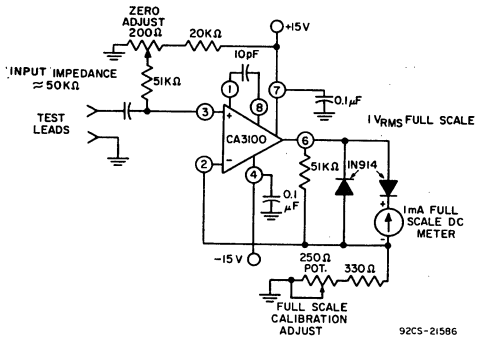


Fig.17 - 1 MHz meter-driver amplifier.

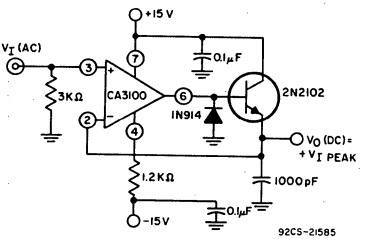


Fig.18 – Fast positive peak detector.

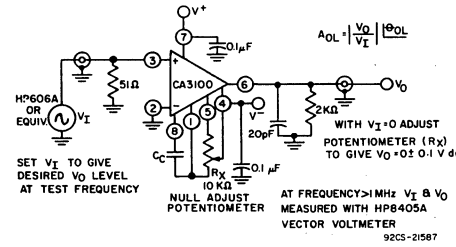


Fig.19 – Open-loop voltage gain test circuit.

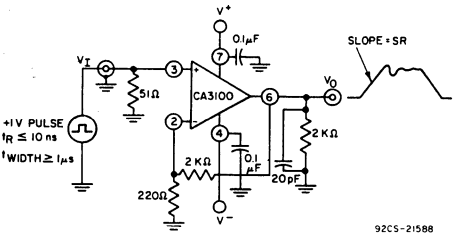


Fig.20 – Slew rate in 10X amplifier test circuit.

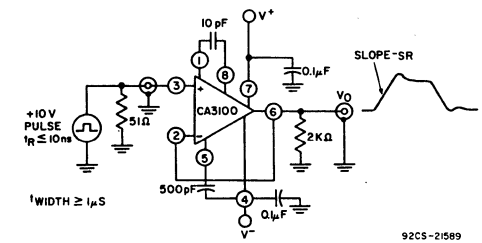


Fig.21 – Follower slew rate test circuit.

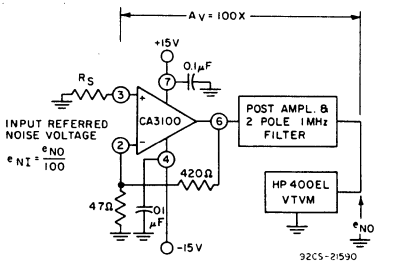


Fig.22 – Wideband input noise voltage test circuit.

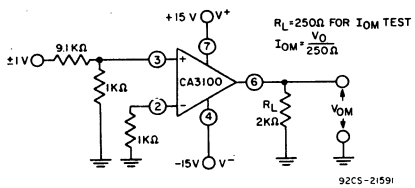


Fig.23 – Output voltage swing (V_{OM}), output current swing (I_{OM}) test circuit.

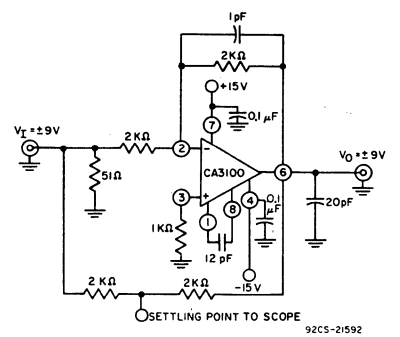


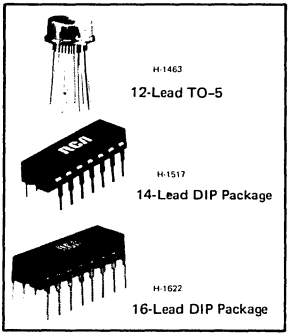
Fig.24 – Settling time test circuit.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3118AT CA3146AE CA3183AE
CA3118T CA3146E CA3183E



High-Voltage Transistor Arrays

Applications

- General use in signal processing systems in DC through VHF range
- Custom designed differential amplifiers
- Temperature compensated amplifiers
- Lamp and relay drivers (CA3183AE, E)
- Thyristor firing (CA3183AE, E)

Features

- Matched general-purpose transistors
- V_{BE} matched ±5mV max.
- Operation from DC to 120 MHz (CA3118AT, T; CA3146AE, E)
- Low-noise figure: 3.2dB typ. at 1kHz (CA3118AT, T; CA3146AE, E)
- High I_C: 75mA max. (CA3183AE, E)

RCA-CA3118AT, CA3118T, CA3146AE, CA3146E, CA-3183AE, and CA3183E* are general-purpose high-voltage silicon n-p-n transistor arrays on a common monolithic substrate.

Types CA3118AT and CA3118T consist of four transistors with two of the transistors connected in a Darlington configuration. These types are well suited for a wide variety of applications in low-power systems in the DC through VHF range. Both types are supplied in a hermetically sealed 12-lead TO-5 type package and operate over the full military temperature range. (CA3118AT and CA3118T are high-voltage versions of the popular predecessor type CA3018.

Types CA3146AE and CA3146E consist of five transistors with two of the transistors connected to form a differentially-connected pair. These types are recommended for low-power applications in the DC through VHF range.

Types CA3183AE and CA3183E consist of five high-current transistors with independent connections for each transistor. In addition two of these transistors (Q1 and Q2) are matched at low-current (i.e. 1mA) for applications where offset parameters are of special importance. A special substrate terminal is also included for greater flexibility in circuit design.

The types with an "A" suffix are premium versions of their non-"A" counterparts and feature tighter control of breakdown voltages making them more suitable for higher voltage applications.

For detailed application information, see companion Application Note, ICAN-5296 "Application of the RCA CA3018 Integrated Circuit Transistor Array."

*Formerly Developmental Types Nos.

CA3118AT	– TA6091	CA3146E	– TA6181
CA3118T	– TA6182	CA3183AE	– TA6094
CA3146AE	– TA6084	CA3183E	– TA6183

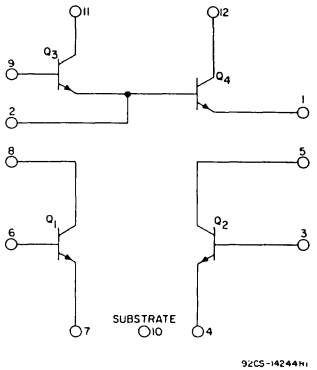
TYPE	P_T● max. mW	I_C max. mA	V_{CEO} max. V	V_{CBO} max. V	V_{CE} sat. at 10 mA typ. V	h_{FE} at 1 mA, & V_{CE}=5V typ.	V_{IO} Diff. Pair at 1 mA max. mV	I_{IO} Diff. Pair at 1 mA max. μA	T_A Range (Operating) °C
VALUES APPLY FOR EACH TRANSISTOR									
CA3118AT	300	50	40	50	0.33	95	±5	2	
CA3118T	300	50	30	40	0.33	95	±5	2	
CA3146AE	300	50	40	50	0.33	95	±5	2	−55 – +125
CA3146E	300	50	30	40	0.33	95	±5	2	
CA3183AE	500	75	40	50	0.16	75	±5	2.5	
CA3183E	500	75	30	40	0.16	75	±5	2.5	

Caution on Total Package Power Dissipation: The maximum total package dissipation rating for the CA3118 Series circuits is 450 mW at temperatures up to +85°C, then derate linearly at 5 mW°C. The maximum total package dissipation rating for the CA3146 and CA3183 Series circuits is 750 mW at temperatures up to +55°C, then derate linearly at 6.67 mW°C.

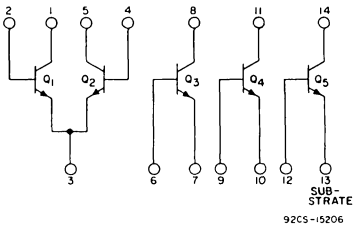
MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

Power Dissipation:		
Any one transistor —		
CA3118AT, CA3118T, CA3146AE, CA3146E	300	mW
CA3183AE, CA3183E	500	mW
Total package —		
Up to 85°C (CA3118AT, CA3118T)	450	mW
Up to 55°C (CA3146AE, CA3146E, CA3183AE, CA3183E)	750	mW
Above 85°C (CA3118AT, CA3118T)	derate linearly 5	mW/°C
Above 55°C (CA3146AE, CA3146E, CA3183AE, CA3183E)	derate linearly 6.67	mW/°C
Ambient Temperature Range:		
Operating —	−55 to +125	°C
Storage (all types)	−65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265	°C
The following ratings apply for each transistor in the device:		
Collector-to-Emitter Voltage (V_{CEO}):		
CA3118AT, CA3146AE, CA3183AE	40	V
CA3118T, CA3146E, CA3183E	30	V
Collector-to-Base Voltage (V_{CBO}):		
CA3118AT, CA3146AE, CA3183AE	50	V
CA3118T, CA3146E, CA3183E	40	V
Collector-to-Substrate Voltage (V_{CIO}): ■		
CA3118AT, CA3146AE, CA3183AE	50	V
CA3118T, CA3146E, CA3183E	40	V
Emitter-to-Base Voltage (V_{EBO}) all types	5	V
Collector Current —		
CA3118AT, CA3118T, CA3146AE, CA3146E	50	mA
CA3183AE, CA3183E	75	mA
Base Current (I_B) — CA3183AE, CA3183E	20	mA

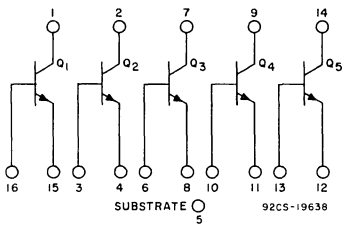
■ The collector of each transistor is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.



CA3118AT, CA3118T,



CA3146AE, CA314E



CA3183AE, CA3183E

Fig. 1 – Schematic diagrams of high-voltage arrays.

COMPARISON OF RELATED PREDECESSOR TYPE WITH TYPES IN THIS DATA BULLETIN

	DATA FILE NO.	V_{CEO} min.	V_{CBO} min.	$V_{CE\ sat.}$ typ. V	V_{BE} typ. V	I_C max. mA	C_{CB} typ. pF	C_{CI} typ. pF	C_{EB} typ. pF
				I_C=10 mA	I_C=1 mA				
CA3018	338	15	20	0.23	0.715	50	0.58	2.8	0.6
CA3018A	338	15	20	0.23	0.715	50	0.58	2.8	0.6
CA3118AT		40	50	0.33	0.730	50	0.37	2.2	0.7
CA3118T		30	40	0.33	0.730	50	0.37	2.2	0.7
				I_C=10 mA	I_C=1 mA				
CA3046	341	15	20	0.23	0.715	50	0.58	2.8	0.6
CA3146AE		40	50	0.33	0.730	50	0.37	2.2	0.7
CA3146E		30	40	0.33	0.730	50	0.37	2.2	0.7
				I_C=50 mA	I_C=10 mA				
CA3083	481	15	20	0.4	0.74	100	–	–	–
CA3183AE		40	50	1.7	0.75	75	–	–	–
CA3183E		30	40	1.7	0.75	75	–	–	–

NOTE: Related predecessor types are shown in shaded areas.

STATIC ELECTRICAL CHARACTERISTICS – CA3118 and CA3146 Series

CHARACTERISTICS	SYMBOL	TEST CONDITIONS $T_A = 25^oC$		LIMITS CA3118AT, CA3146AE Min.	CA3118AT, CA3146AE Typ.	CA3118AT, CA3146AE Max.	CA3118T, CA3146E Min.	CA3118T, CA3146E Typ.	CA3118T, CA3146E Max.	UNITS
For Each Transistor:										
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_C = 10\mu A$, $I_E = 0$		50	72	–	40	72	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	$I_C = 1mA$, $I_B = 0$		40	56	–	30	56	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	$I_{CI} = 10\mu A$, $I_B = 0$, $I_E = 0$		50	72	–	40	72	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	$I_E = 10\mu A$, $I_C = 0$		5	7	–	5	7	–	V
Collector-Cutoff Current	I_{CEO}	$V_{CE} = 10V$, $I_B = 0$		–	see curve	5	–	see curve	5	μA
Collector-Cutoff Current	I_{CBO}	$V_{CB} = 10V$, $I_E = 0$		–	0.002	100	–	0.002	100	nA
DC Forward-Current Transfer Ratio	h_{FE}	$V_{CE}=5V$	$I_C=10mA$	–	85	–	–	85	–	–
			$I_C=1mA$	30	100	–	30	100	–	
			$I_C=10\mu A$	–	90	–	–	90	–	
Base-to-Emitter Voltage	V_{BE}	$V_{CE} = 3V$, $I_C = 1mA$		0.63	0.73	0.83	0.63	0.73	0.83	V
Collector-to-Emitter Saturation Voltage	V_{CEsat}	$I_C = 10mA$, $I_B = 1mA$		–	0.33	–	–	0.33	–	V
For transistors Q3 and Q4 (Darlington Configeration):										
Collector-Cutoff Current (CA3118AT and CA3118T only)	I_{CEO}	$V_{CE} = 10V$, $I_B = 0$		–	–	5	–	–	–	μA
DC Forward-Current Transfer Ratio (CA3118AT and CA3118T only)	h_{FE}	$V_{CE} = 5V$, $I_C = 1mA$		1500	9000	–	1500	9000	–	–
Base-to-Emitter (Q3 to Q4)	V_{BE}	$V_{CE}=5V$	$I_E = 10mA$	–	1.46	–	–	1.46	–	V
			$I_E = 1mA$	–	1.32	–	–	1.32	–	
Magnitude of Base-to-Emitter Temperature Coefficient	$\left\lvert\frac{\Delta V_{BE}}{\Delta T}\right\rvert$	$V_{CE} = 5V$, $I_E = 1mA$		–	4.4	–	–	4.4	–	mV/°C
For transistors Q1 and Q2 (AS a Differential Amplifier):										
Magnitude of Input Offset Voltage $\lvert V_{BE1} - V_{BE2}\rvert$	$\lvert V_{IO}\rvert$	$V_{CE} = 5V$, $I_E = 1mA$		–	0.48	5	–	0.48	5	mV
Magnitude of h_{FE} Ratio (CA3118AT and CA3118T only)		$V_{CE} = 5V$, $I_{C1} = I_{C2} = 1mA$		0.9	1.0	1.1	0.9	1.0	1.1	–
Magnitude of Base-to-Emitter Temperature Coefficient	$\left\lvert\frac{\Delta V_{BE}}{\Delta T}\right\rvert$	$V_{CE} = 5V$, $I_E = 1mA$		–	1.9	–	–	1.9	–	mV/°C
Magnitude of V_{IO} ($V_{BE1} - V_{BE2}$) Temperature Coefficient	$\left\lvert\frac{\Delta V_{IO}}{\Delta T}\right\rvert$	$V_{CE} = 5V$, $I_{C1} = I_{C2} = 1mA$		–	1.1	–	–	1.1	–	$\mu V/^oC$
Magnitude of Input Offset Current $\lvert I_{IO_1} - I_{IO_2}\rvert$ (CA3146AE and CA3146E only)	I_{IO}	$V_{CE} = 5V$, $I_{C_1} = I_{C_2} = 1\ mA$		–	0.3	2	–	0.3	2	μA

DYNAMIC ELECTRICAL CHARACTERISTICS – CA3118 and CA3146 Series

CHARACTERISTICS	SYMBOL	TEST CONDITIONS T_A = 25°C	CA3118AT CA3146AE			CA3118T CA3146E			UNITS
			Min.	Typ.	Max.	Min.	Typ.	Max.	
Low Frequency Noise Figure	NF	f = 1kHz, V_{CE} = 5V, I_C = 100μA, Source resistance = 1 kΩ	–	3.25	–	–	3.25	–	dB
Low-Frequency, Small-Signal Equivalent-Circuit Characteristics:									
Foward-Current Transfer Ratio	h_{fe}	f = 1kHz, V_{CE} = 5V, I_C = 1mA	–	100	–	–	100	–	–
Short-Circuit Input Impedance	h_{ie}		–	2.7	–	–	3.5	–	kΩ
Open-Circuit Output Impedance	h_{oe}		–	15.6	–	–	15.6	–	μmho
Open-Circuit Reverse-Voltage Transfer Ratio	h_{re}		–	$1.8x10^{-4}$	–	–	$1.8x10^{-4}$	–	–
Admittance Characteristics:									
Foward Transfer Admittance	Y_{fe}	f = 1MHz, V_{CE} = 5V, I_C = 1mA	–	31-j1.5	–	–	31-j1.5	–	mmho
Input Admittance	Y_{ie}		–	0.35+j0.04	–	–	0.3+j0.04	–	mmho
Output Admittance	Y_{oe}		–	0.001+j0.03	–	–	0.001+j0.03	–	mmho
Reverse Transfer Admittance	Y_{re}			See curve			See curve		mmho
Gain-Bandwidth Product	f_T	V_{CE} = 5V, I_C = 3mA	300	500	–	300	500	–	MHz
Emitter-to-Base Capacitance	C_{EB}	V_{EB} = 5V, I_E = 0	–	0.70	–	–	0.70	–	pF
Collector-to-Base Capacitance	C_{CB}	V_{CB} = 5V, I_C = 0	–	0.37	–	–	0.37	–	pF
Collector-to-Substrate Capacitance	C_{CI}	V_{CI} = 5V, I_C = 0	–	2.2	–	–	2.2	–	pF

STATIC ELECTRICAL CHARACTERISTICS – CA3183 Series

CHARACTERISTICS	SYMBOL	TEST CONDITIONS T_A = 25°C	LIMITS CA3183AE			LIMITS CA3183E			UNITS
			Min.	Typ.	Max.	Min.	Typ.	Max.	
For Each Transistor:									
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C=100μA, I_E=0	50	–	–	40	–	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C=1mA, I_B=0	40	–	–	30	–	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI}=100μA, I_B=0, I_E = 0	50	–	–	40	–	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 500μA, I_C = 0	5	–	–	5	–	–	V
Collector-Cutoff Current	I_{CEO}	V_{CE} = 10V, I_B = 0	–	–	10	–	–	10	μA
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10V, I_E = 0	–	–	1	–	–	1	μA
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE}= 3 V, I_C = 10mA	40	–	–	40	–	–	–
		V_{CE} = 5V, I_C = 50mA	40	–	–	40	–	–	
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3V, I_C = 10mA	0.65	0.75	0.85	0.65	0.75	0.85	V
Collector-to-Emitter Saturation Voltage	*V_{CEsat}	I_C = 50mA, I_B = 5mA	–	1.7	3.0	–	1.7	3.0	V
For Transistors Q1 and Q2 (As a Differential Amplifier):									
Absolute Input Offset Voltage	$\lvert V_{IO}\rvert$	V_{CE} = 3V, I_C = 1mA	–	0.47	5	–	0.47	5	mV
Absolute Input Offset Current	$\lvert I_{IO}\rvert$		–	0.78	2.5	–	0.78	2.5	μA

* A maximum dissipation of 5 transistors x 150mW = 750mW is possible for a particular application.

TYPICAL STATIC CHARACTERISTICS CURVES – CA3118 and CA3146 SERIES

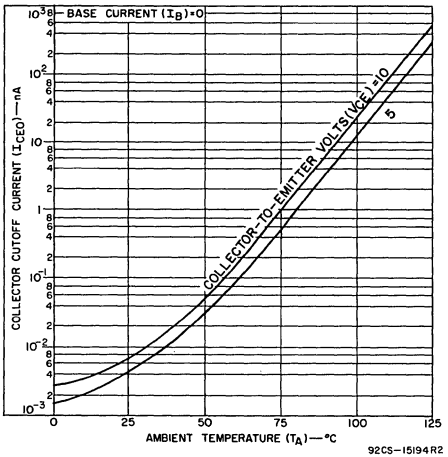


Fig. 2 – I_{CEO} vs. T_A for any transistor.

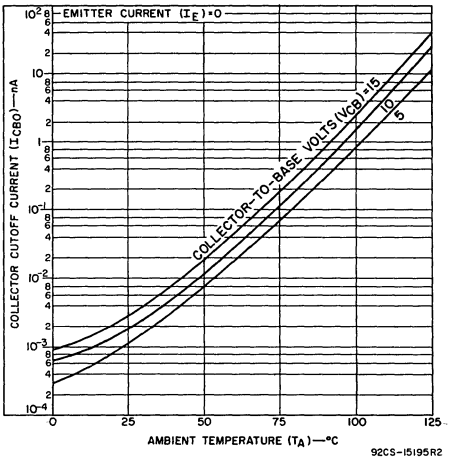


Fig. 3 – I_{CBO} vs. T_A for any transistor.

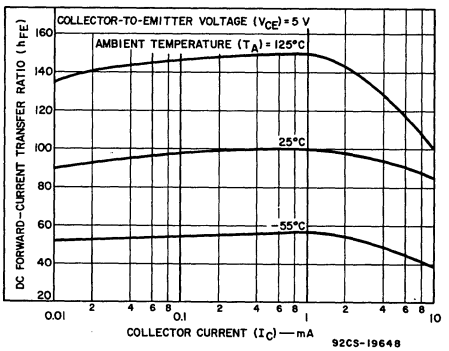


Fig. 4 – h_{FE} vs. I_C for any transistor.

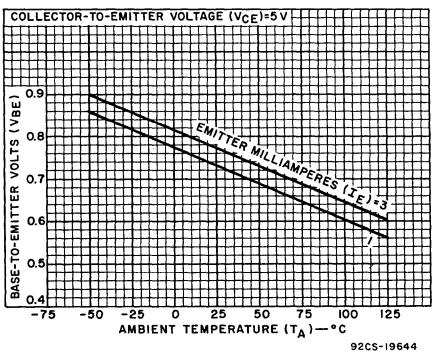


Fig. 5 – V_{BE} vs. T_A for any transistor.

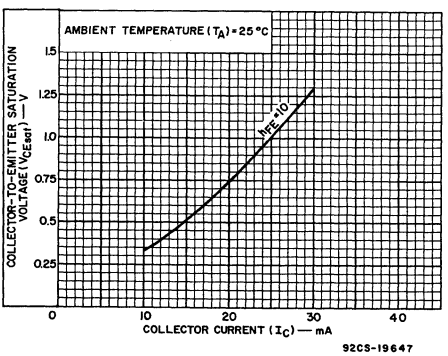


Fig. 6 – V_{CE} sat vs. I_C for any transistor.

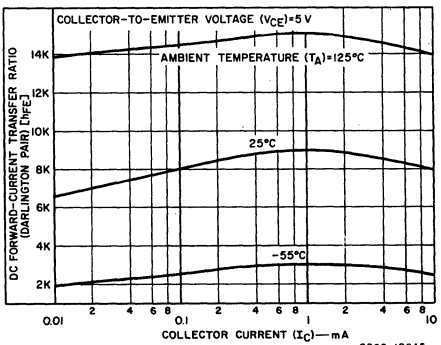


Fig. 7 – h_{FE} vs. I_C for Darlington pair (Q3 and Q4) for types CA3118AT and CA3118T.

TYPICAL STATIC CHARACTERISTICS CURVES – CA3118 and CA3146 SERIES

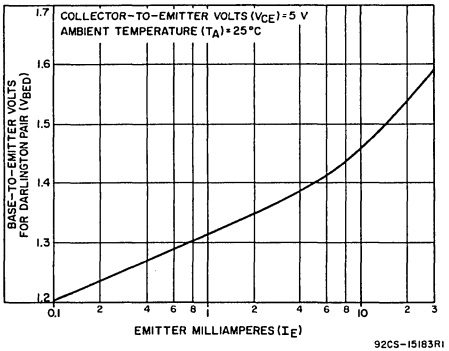


Fig. 8 – V_{BE} vs. I_E for Darlington pair (Q3 and Q4).

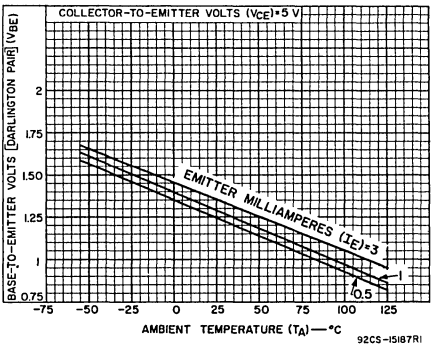


Fig. 9 – V_{BE} vs. T_A for Darlington pair (Q3 and Q4).

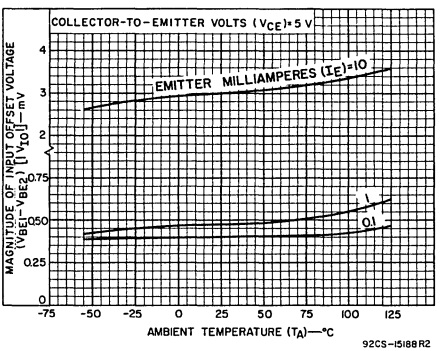


Fig. 10 – V_{IO} vs. T_A for Q1 and Q2.

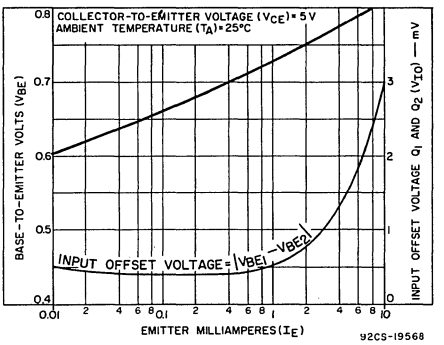


Fig. 11 – V_{BE} and V_{IO} vs. I_E for Q1 and Q2.

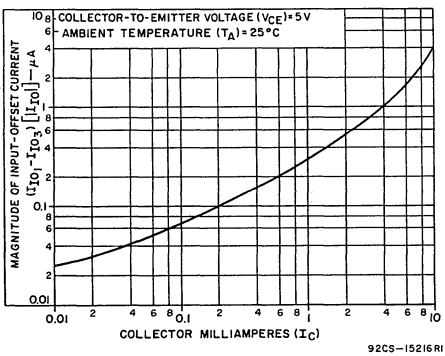


Fig. 12 – I_{IO} vs. I_C (Q1 and Q2) for types CA3146AE and CA3146E.

TYPICAL DYNAMIC CHARACTERISTICS CURVES (FOR ANY TRANSISTOR) – CA3118, CA3146 SERIES

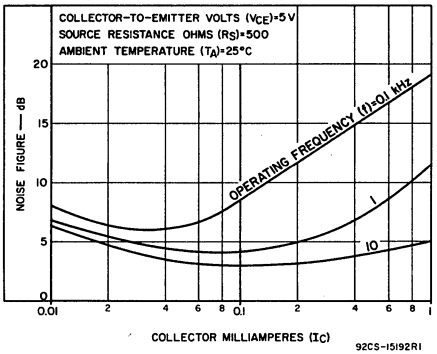


Fig. 13 – NF vs. I_C @ R_S = 500Ω.

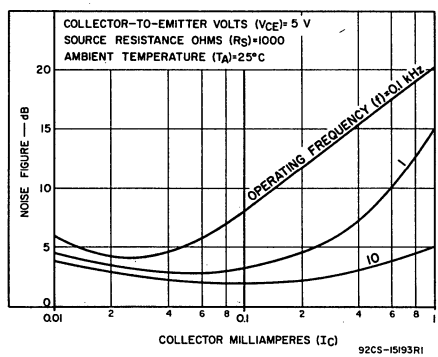


Fig. 14 – NF vs. I_C @ R_S = 1kΩ.

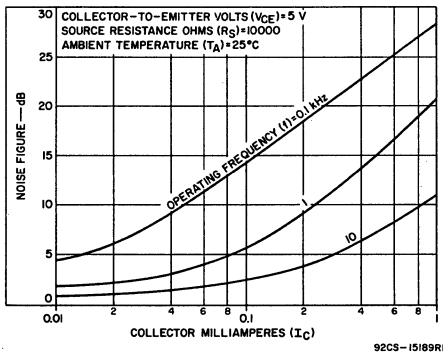


Fig. 15 – NF vs. I_C @ R_S = 10kΩ.

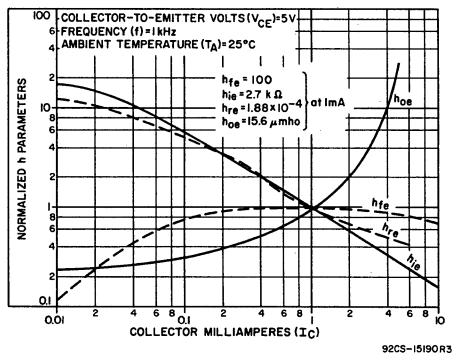


Fig. 16 – h_{fe}, h_{ie}, h_{oe}, h_{re} vs. I_C.

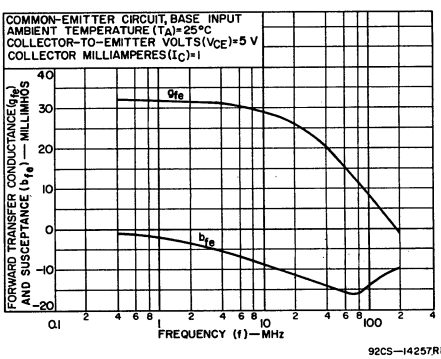


Fig. 17 – y_{fe} vs. f.

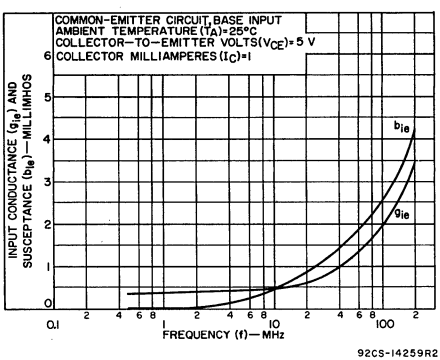


Fig. 18 – y_{ie} vs. f.

TYPICAL DYNAMIC CHARACTERISTICS CURVES (FOR ANY TRANSISTOR) – CA3118, CA3146 SERIES

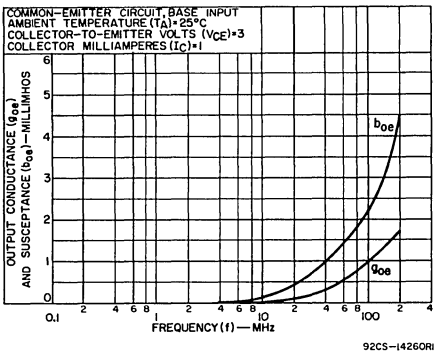


Fig. 19 – y_{oe} vs. f.

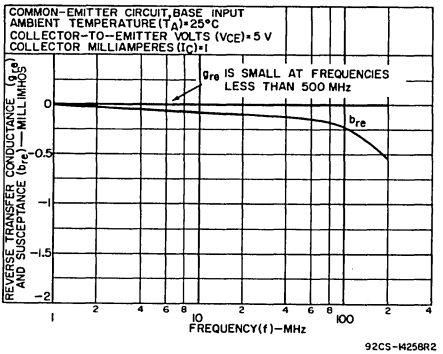


Fig. 20 – y_{re} vs. f.

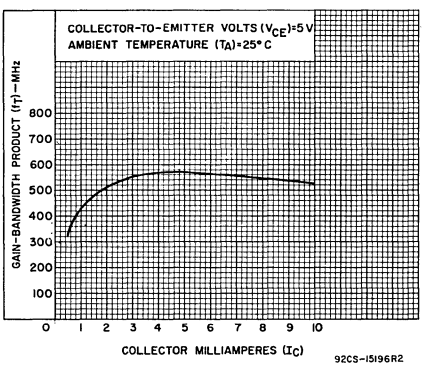


Fig. 21 – f_T vs. I_C.

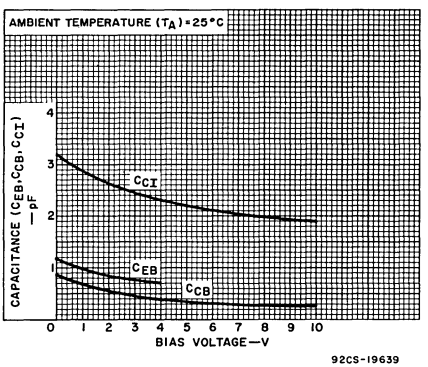


Fig. 22 – C_{EB}, C_{CB}, C_{CI} vs. bias voltage

TYPICAL STATIC CHARACTERISTICS CURVES – CA3183 SERIES

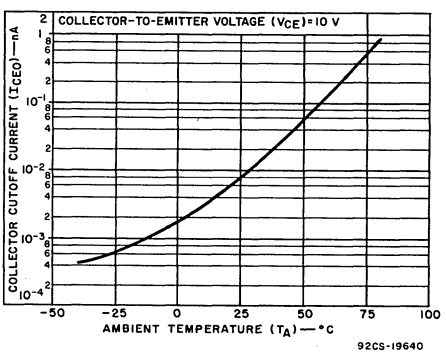


Fig. 23 – I_{CEO} vs. T_A for any transistor.

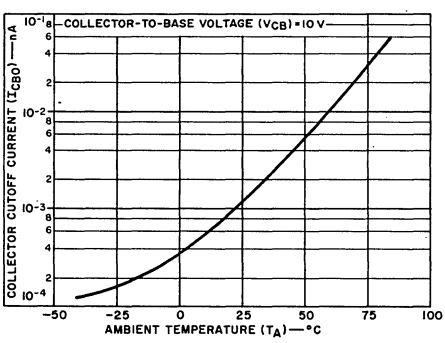


Fig. 24 – I_{CBO} vs. T_A for any transistor.

TYPICAL STATIC CHARACTERISTICS CURVES – CA3183 SERIES

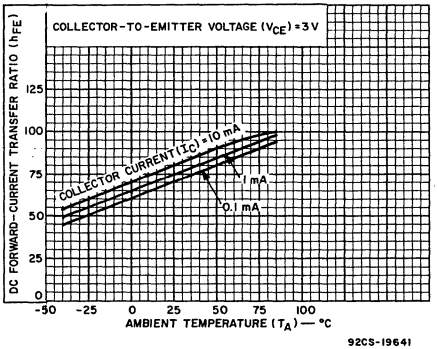


Fig. 25 – h_{FE} vs. T_A for any transistor.

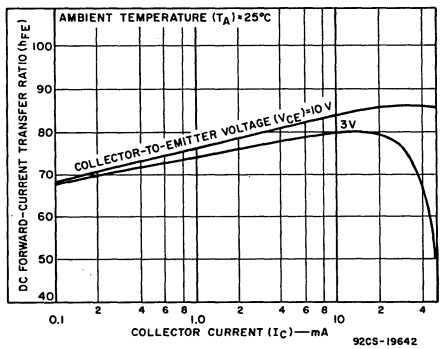


Fig. 26 – h_{FE} vs. I_C for any transistor.

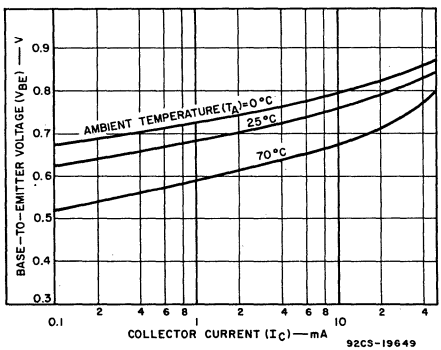


Fig. 27 – V_{BE} vs. I_C for any transistor.

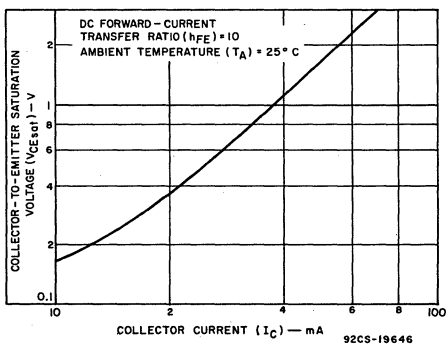


Fig. 28 – $V_{CE\ sat}$ vs. I_C for any transistor.

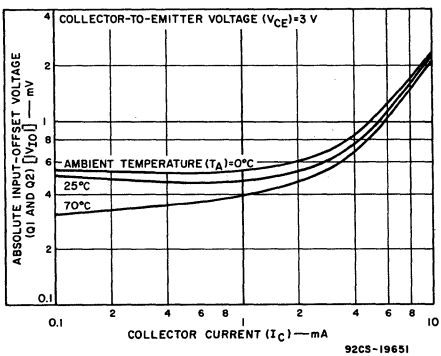


Fig. 29 – $|V_{IO}|$ vs. I_C for differential amplifier (Q1 and Q2).

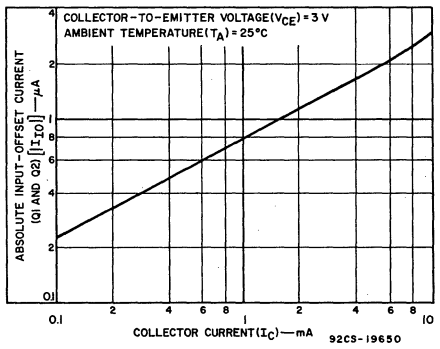


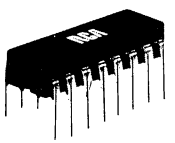
Fig. 30 – $|I_{IO}|$ vs. I_C for differential amplifier (Q1 and Q2).

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3120E



16-Lead Dual-In-Line Plastic Package

TV Signal Processor ("Jungle" Circuit)

For Color and Monochrome Receivers

Features

- Internal impulse noise processing
- Sync separator – low impedance, dual polarity
- Strobed AGC system
- IF AGC output
- Delayed outputs for forward or reverse AGC tuners
- Automatic noise threshold and AGC detector level control
- High-impedance video input
- Low-impedance video output
- Choice of external time constants for sync separator
- Negative power supply not required
- RF AGC delay externally controlled
- **Application Note ICAN-6302, "Description and Application of the RCA-CA3120E Integrated-Circuit TV-Signal Processor"**

RCA-CA3120E is a monolithic silicon integrated circuit TV signal processor for use in color or monochrome receivers. The circuit provides low-impedance video output signals, stripped synchronization signals in both polarities, and AGC output signals for IF (reverse) and tuner (forward and/or reverse).

The circuit design of the CA3120E features impulse noise inversion, delay techniques to reduce the deleterious effects of impulse noise in the receiver AGC and sync circuits. In addition, the CA3120E incorporates standard AGC strobing techniques.

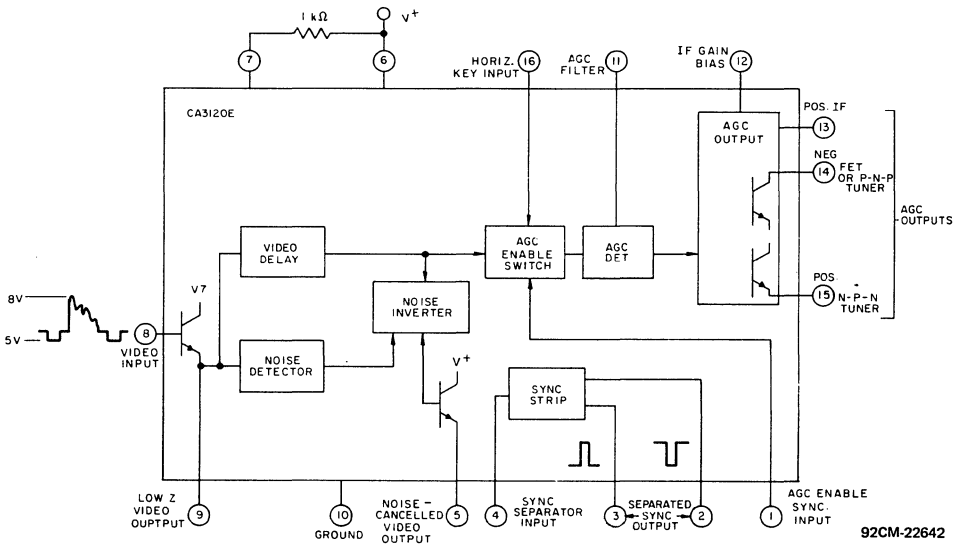


Fig. 1 – Simplified block diagram of the CA3120E.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, Supply Voltage (V^+) = 24 V and Referenced to Test Circuits and Test Conditions (Figs. 6, 7, and 8).

CHARACTERISTICS	TERMINAL MEASURED AND SYMBOL	LIMITS			UNITS
		Min.	Typ.	Max.	
Supply Current (Pulse Test)	I_{T24}	20	–	40	mA
AGC Threshold (Sync Tip Level at Video Input)	V_{TH}	4.5	–	5.5	V
Video Input Amplitude (White Positive)	V_8	–	3	–	Vp-p
Video Output Amplitude (Low Impedance)	V_9	–	3	–	Vp-p
Noise Cancelled Video Output at V_{TH} (Black Positive, Gain ≅ 2)	V_5	6.6	–	9.2	V
AGC to Noise Separation	$V_{TH\ (SEP)}$	1.1	–	2.2	V
Sync Input Current for Full Amplitude Outputs	$I_{4\ (ON)}$	–	–	70	μA
Maximum Leakage Current at Terminal 4	$I_{4\ (OFF)}$	–	–	±6	μA
Sync Outputs:					
Negative Sync Low	$V_{2(L)}$	0	–	2.6	V
Negative Sync High	$V_{2(H)}$	23.8	–	24	V
Positive Sync Low	$V_{3(L)}$	0	–	0.2	V
Positive Sync High	$V_{3(H)}$	20.1	–	24	V
AGC Filter:					
Charge Current (Pulse Test)	$I_{11(CH)}$	12	–	36	mA
Discharge Current	$I_{11(DISCH)}$	1.1	–	2.6	mA
Leakage Current	$I_{11(LEAK)}$	–	–	±6	μA
AGC Enable:					
Horizontal Keying	$V_{16\ (ON)}$	3	–	6	V
Negative Sync Input Current	$I_{1\ (ON)}$	–	1	–	mA
Maximum IF Gain-Clamp Voltage	V_{11}	4.8	–	5.7	V
Maximum IF Gain Bias	V_{12}	4.2	–	5.2	V
IF AGC Voltage:					
Low	$V_{13\ (LOW)}$	0	–	3.3	V
High	$V_{13(HIGH)}$	5.7	–	6	V
Tuner Currents:					
Reverse AGC (FET) OFF Current	$I_{14\ (OFF)}$	–	–	±6	μA
Reverse AGC (FET) ON Current	$I_{14\ (ON)}$	1.8	–	5.5	mA
Forward AGC (n-p-n) OFF Current	$I_{15\ (OFF)}$	–	–	±6	μA
Reverse AGC (n-p-n) ON Current	$I_{15\ (ON)}$	4.5	–	15	mA
Internal Noise-Lockout Time	T	1	–	63	μs

MAXIMUM RATINGS, Absolute-Maximum Values at $T_A = 25^oC$

DC SUPPLY VOLTAGE	30	V
DEVICE DISSIPATION:		
Up to $T_A = 55^oC$	750	mW
Above $T_A = 55^oC$	Derate linearly at 7.9	mW/°C

AMBIENT TEMPERATURE RANGE:	
Operating	−40 to +85 °C
Storage	−65 to +150 °C
LEAD TEMPERATURE (During soldering):	
At a distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265 °C

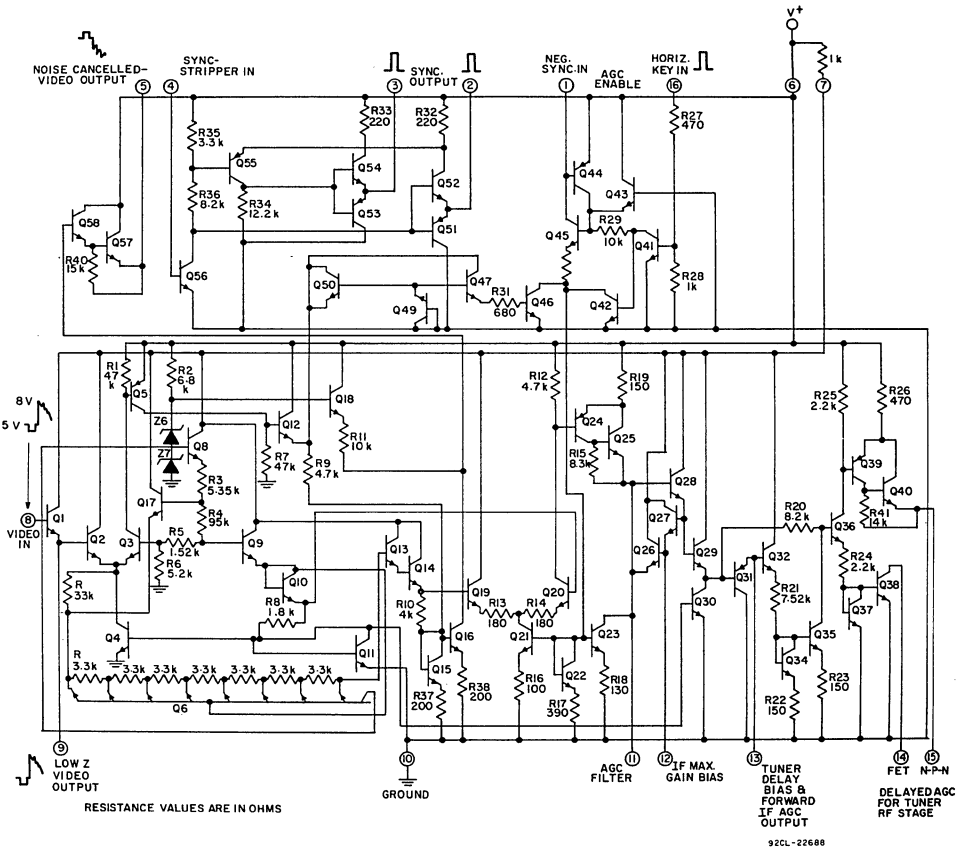


Fig. 2—Schematic diagram of the CA3120E.

CIRCUIT DESCRIPTION*

An AGC sample-and-hold system generates control voltages proportional to the video level. The sync-tip voltage is compared to an internal reference voltage during the horizontal sychronization (retrace) interval. The control voltages (AGC outputs) are supplied to the tuner's RF stage and the IF amplifier to maintain the video level at a constant value.

The composite positive and negative output sync signals are developed across a low impedance source (totem-pole circuit) at an amplitude of approximately 20 volts peak-to-peak.

Video Chain and Impulse Noise Inverter - The input video signal applied at Terminal 8 is white "positive" with a required amplitude in the range of 2 to 4 volts. The DC level of the sync peaks, AGC threshold voltage (V_{TH}) is approximately 5 volts. The level is maintained at 5 volts by the AGC loop in the circuit, comprised of the CA3120E and the TV receiver RF and IF amplifiers. A low source impedance video signal is available from the emitter of Q1 (Terminal 9 in Fig. 2). The external resistor (R_{X1} in Fig. 9) reduces the dissipation of Q1. The emitter-follower output of Q1 is direct coupled to a differential comparator stage (Q2, Q3). Unless a negative-going noise pulse is present, Q2 functions as an emitter follower and also cuts off transistors Q3, Q5, and Q17.

The output of Q2 is applied through a signal delay network, consisting of transistor Q60 and associated resistors, to the Darlington followers (Q13 and Q14). The delayed video signal at Q14 is fed via its emitter to an AGC comparator Q19 and to the junction of a noise-cancelling amplifier stage (Q16). The noise-cancelled video signal is inverted and amplified by Q16 and then connected to a Darlington emitter-follower output stage (Q57, Q58).

If impulse noise is present on the video signal, Q3 conducts and turns on transistors Q5 and Q12. Q5 inverts and "stretches" the noise pulse width. The output of Q5 is applied to an emitter follower stage (Q12). The signal from Q12, in turn, is applied to the summing junction to the noise-cancelling amplifier Q16. The noise pulse, which has now been amplified, inverted and stretched, is added to the delayed video signal from the emitter of Q14.

Because the video signal has been delayed approximately, 300 nanoseconds and the noise pulse has been widened ("stretched") approximately 500 nanoseconds, the output of the combined signal no longer contains impulse noise signals. The derived noise-gating pulse "surrounds" and effectively eliminates the effects of the impulse noise.

The noise-cancelled video signal, amplified and buffered, is available at Terminal 5 for use in the sync-separator stage. The peak-to-peak amplitude of the noise-cancelled output signal is approximately twice the amplitude of the input video signal at Terminal 8.

Sync Separator (See Figure 3) - The sync separator stage (Q56) clamps the detected sync tips to a fixed reference voltage (≅ 0.7 V) across its base-emitter junction, and amplifies a portion of the sync signal to provide dual polarity sync-signal outputs at Terminals 2 (negative) and 3 (positive). The output signals, are derived from low-impedance complementary emitter-follower stages; a base current of 100 microamperes into Terminal 4 is sufficient to generate full-amplitude sync signals.

The choice of coupling the noise-cancelled video signal from the emitter-follower (Terminal 5) to the sync separator (Terminal 4) is a user option. Fig. 4 shows three typical coupling networks.

Fig. 5 illustrates the operation of the AGC circuits. An input ramp signal, simulating the potential to which the AGC filter capacitor may be charged, is applied to Terminal 11. The forward IF AGC output voltage appears at Terminal 13. Under low-signal level conditions (represented by A to B in Fig. 5), the output level is approximately 1.4 volts less than the voltage applied to Terminal 12.

The circuit designer should select the voltage at Terminal 12 to provide the maximum IF gain required for the system. At intermediate signal level conditions (represented by B to C in Fig. 5), the IF AGC signal follows the AGC filter potential. The tuner(s) will operate at maximum gain for good signal-to-noise ratios at these equivalent input signal levels. Point C is a turnover point determined by the open-circuit potential of the tuner-delay bias potentiometer. At this potential, further change in the IF AGC output is inhibited (for good dynamic range) and the tuner AGC potentials are activated (represented by C to D).

The output at Terminal 14 with suitable level shifting is used for tuners requiring reverse AGC, such as MOSFET or electron-tube types. The output at Terminal 15 is used for tuners requiring forward AGC, such as tuners utilizing n-p-n bipolar transistors.

* For additional information refer to the "IEEE Transactions on Broadcast and TV Receivers", August 1970, pp. 185–195, Vol. BTR No. 3.

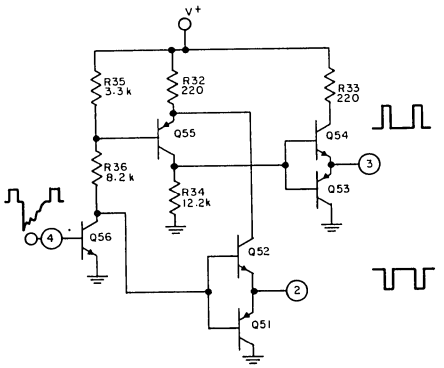


Fig. 3 — Sync separator stage.

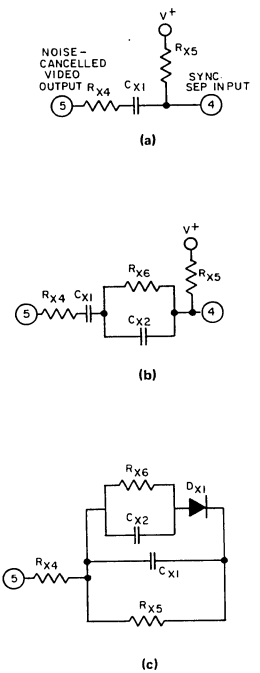


Fig. 4 — Typical coupling networks (Term. 5 to Term. 4).

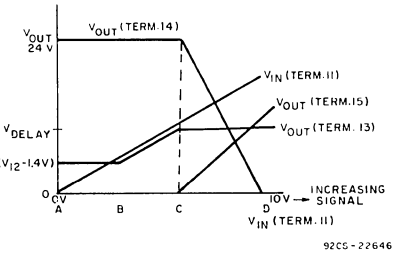


Fig. 5 — Typical operation of the AGC circuits using the CA3120E.

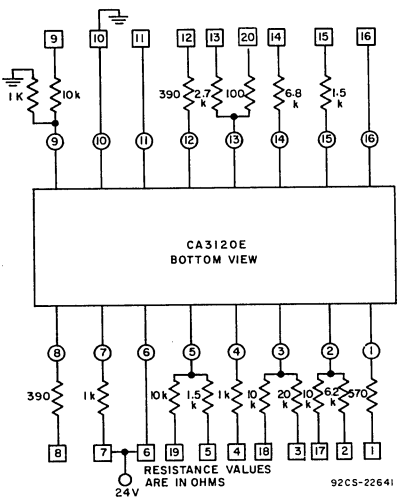


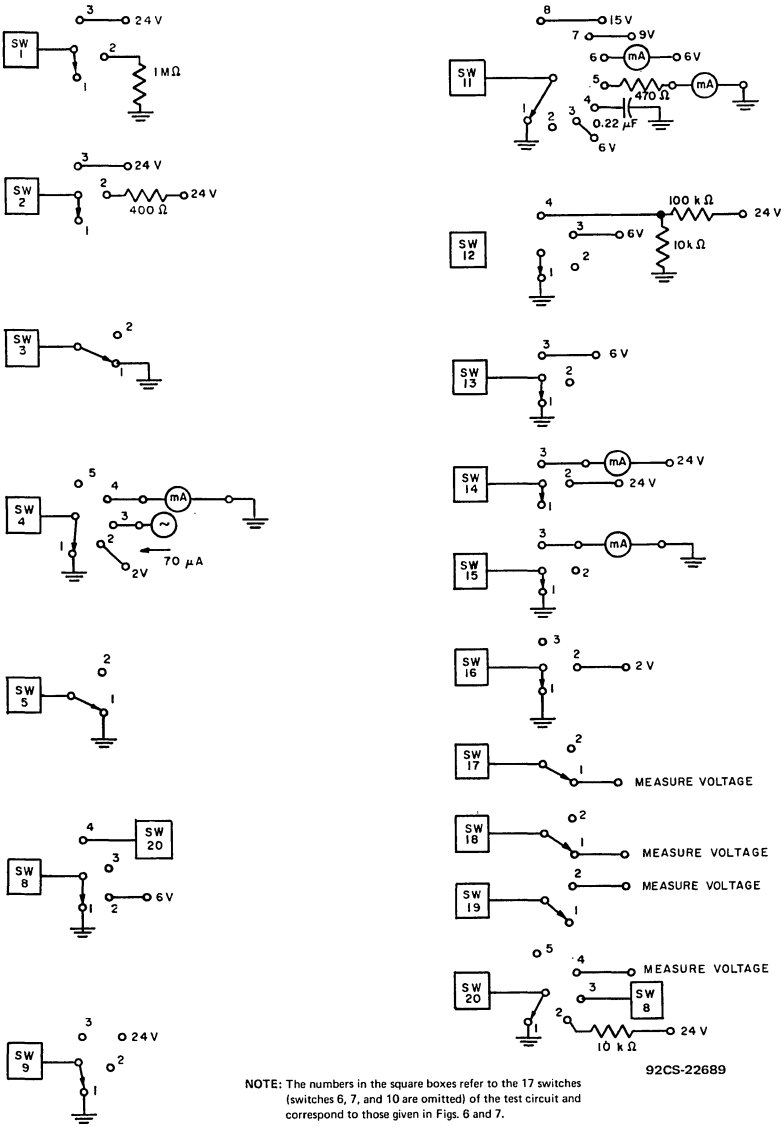
Fig. 6 — Test circuit for measuring electrical characteristics of the CA3120E. Refer to Figs. 7 and 8 for switch selector positions.

CHARACTERISTIC	TEST CONDITIONS — SWITCH NUMBERS / SWITCH POSITION																	TERMINAL MEASURED
	1	2	3	4	5	8	9	11	12	13	14	15	16	17	18	19	20	
I_{T24}	**2**	3	1	2	**1**	2	3	1	**1**	3	2	1	2	2	2	**1**	5	2 6 7 9 14
V_{TH}	2	1	2	1	1	4	3	4	4	3	1	2	2	2	2	1	3	8
V_5	**2**	1	2	1	1	4	3	4	4	3	1	2	2	2	2	2	3	19
$V_{TH(SEP)}$	3	1	2	1	1	*	3	3	4	1	1	2	1	2	2	2	1	*
$I_{4(OFF)}$	3	1	2	4	2	1	1	1	1	1	1	2	1	2	2	1	1	I_4
V_{2L}	1	2	2	3	2	1	1	1	1	1	1	2	1	1	2	1	1	V_{17}
V_{2H}	3	3	1	1	2	1	1	1	1	1	1	2	**1**	1	2	1	**1**	V_{17}
V_{3L}	3	3	1	1	2	1	1	1	1	1	1	2	1	2	1	1	1	V_{18}
V_{3H}	**3**	**3**	1	3	2	1	1	1	1	1	1	2	1	2	1	1	1	V_{18}
$I_{11(CH)}$	2	1	2	5	2	1	1	5	4	3	1	2	2	2	2	1	5	I_{11}
$I_{11(DISCH)}$	2	1	2	5	1	2	**3**	6	4	3	1	2	2	2	2	1	5	I_{11}
$I_{11(LEAK)}$	**2**	1	2	5	2	1	1	6	4	3	2	2	1	2	2	1	5	I_{11}
V_{11}	2	1	2	5	1	2	3	2	3	3	1	2	2	2	2	1	5	V_{11}
V_{12}	3	1	2	5	2	1	1	3	4	3	1	2	1	2	2	1	5	V_{12}
$V_{13(LOW)}$	3	1	2	5	2	2	3	1	1	2	1	2	1	2	2	1	2	V_{13}
$V_{13(HIGH)}$	3	1	2	5	2	2	3	7	4	3	2	1	1	2	2	1	4	V_{20}
$I_{14(OFF)}$	3	1	2	5	2	2	3	3	4	3	3	1	1	2	2	1	5	I_{14}
$I_{14(ON)}$	3	1	2	5	2	2	3	8	4	3	3	1	1	2	2	1	5	I_{14}
$I_{15(OFF)}$	3	1	2	5	2	2	3	3	4	3	2	3	1	2	2	1	5	I_{15}
$I_{15(ON)}$	3	1	2	5	2	2	3	8	4	3	2	3	1	2	2	1	5	I_{15}

CAUTION: Remove power before selecting or adjusting switches.

* Reduce voltage at Terminal 8 until V_{19} decreases. $V_{TH(SEP)} = V_{TH} - V_8$.

Fig. 7 – Test condition values for associated switches 1 through 20 (switches 6, 7, and 10 are omitted). Refer to Figs. 6 and 8 for test circuit and test-condition selector-switch arrangements.



NOTE: The numbers in the square boxes refer to the 17 switches (switches 6, 7, and 10 are omitted) of the test circuit and correspond to those given in Figs. 6 and 7.

CAUTION: Remove power before selecting or adjusting switches

Fig. 8 — Test condition selector switch arrangement for measuring the electrical characteristics of the CA3120E.

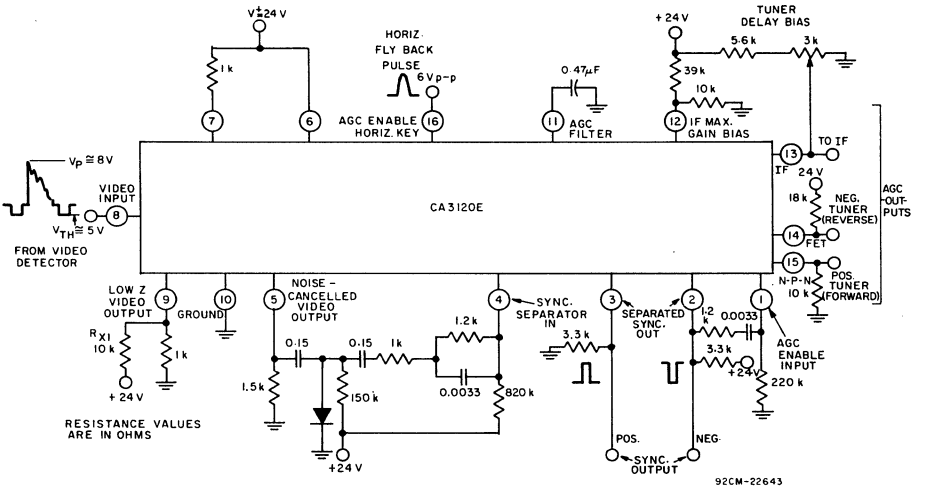


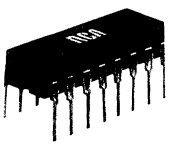
Fig. 9 – Typical application using the CA3120E.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3121E



16-Lead Dual-In-Line Plastic Package H-1622

TV Chroma Amplifier/Demodulator

Provides Complete System for Processing Chroma When Used with RCA-CA3070

Features

- Excellent linearity in dc chroma gain-controlled circuit
- Improved filtering reduces 7.2 MHz output from the color demodulators
- Current limiting for short-circuit protection
- Good tolerance to B+ supply variations
- Good temperature coefficient stability

RCA-CA3121E is a monolithic silicon integrated circuit chroma amplifier/demodulator with ACC and killer control for color-TV receivers. It is designed to function compatibly with the CA3070 in a two package chroma system. Figs. 3 and 4 show a functional block diagram and the outboard circuitry of a typical two-package chroma system incorporating the CA3121E and CA3070, respectively.

The CA3121E is supplied in a 16-lead dual-in-line plastic package.

MAXIMUM RATINGS at $T_A = 25^\circ C$

Supply Voltage	30 V
Device Dissipation:	
Up to $T_A = 55^\circ C$	750 mW
Above $T_A = 55^\circ C$	derate linearly 7.9 mW/°C
Operating Temperature Range	−40 to +85°C
Lead Temperature (During Soldering)	
At distance 1/16" ±1/32" (1.59 ±0.79 mm) from case for 10 s max.	+265°C

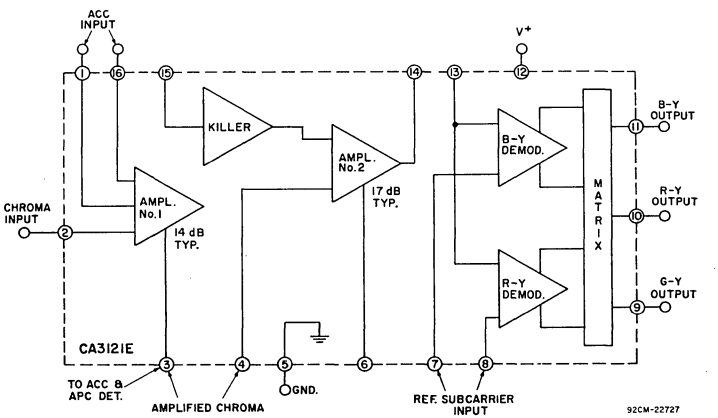


Fig. 1 — Functional block diagram of the CA3121E.

ELECTRICAL CHARACTERISTICS at T_A = 25°C and Referenced to Test Circuit (Fig. 8)

CHARACTERISTIC	TERMINAL MEASURED AND SYMBOL	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
Supply Current	I_T	–	–	40	44	mA
Input Sensitivity	V_2	Vary Eg; set V_4 for 55 mV RMS	6	10	15	mV RMS
Second-Stage Sensitivity	V_4	Vary Eg; set V_{11} for 2 V RMS	25	55	100	mV RMS
Output Voltage (Killer off)	V_{11}	Switch Position: S1=2, S2=2, S3=2 Adjust killer potentiometer until output drops	–	–	70	mV RMS
Demodulator Characteristics:						
Output Voltages	V_9, V_{10}, V_{11}		13	14.3	15.6	V
DC Output Balance (Between any 2 outputs)	–	–	–0.6	–	+0.6	V
Unbalance	V_9, V_{10}, V_{11}	Eg=0; Switch Position: S1=1, S2=1, S3=1	–	–	0.8	Vp-p
Relative Outputs— R-Y	V_{10}	Vary Eg; set V_{11} for 2 V RMS	1.4	1.52	1.68	V RMS
G-Y	V_9		0.3	0.4	0.5	V RMS
Relative Phase— R-Y	V_{10}	Vary Eg; set V_{11} for 2 V RMS; read phase of V_{10} and V_9	–101	–106	–111	degrees
G-Y	V_9	with V_{11} as reference	112	104	96	degrees
Max. Output Voltage	V_{11}	Eg = 750 mV	2.8	–	–	V RMS

CIRCUIT OPERATION

The CA3121E consists of three basic circuit sections: (1) amplifier No. 1, (2) amplifier No. 2, and (3) demodulator. Amplifier No. 1 contains the circuitry for automatic chroma control (ACC) and color-killer sensing. The output of amplifier No. 1 (Terminal 3) is coupled to the Chroma Signal Processor (CA3070 or equivalent) for ACC and automatic phase control (APC) operation and to the input of amplifier No. 2 (Terminal 4) containing the chroma gain control circuitry. The signal from the color-killer circuit in amplifier No. 1 acts upon amplifier No. 2 to greatly reduce its gain.

The output from amplifier No. 2 (Terminal 14) is applied, through a filtering network, to the demodulator input (Terminal 13). The demodulator also receives the R-Y and B-Y demodulation subcarrier signals (Terminals 7 and 8) from the oscillator output of the chroma signal processor. The R-Y and B-Y demodulators and the matrix network contained in the demodulator section of the CA3121E reconstruct the G-Y signal to achieve the R-Y, G-Y, and B-Y color difference signals. These high-level outputs signals with low impedance outputs are suitable for driving high-level R, G, B output amplifiers. Internal capacitors are included on each output to filter out unwanted harmonics. For additional operating information and signal waveforms, refer to Television Chroma System (utilizing RCA-CA3070, CA3071, CA3072), File No. 468.

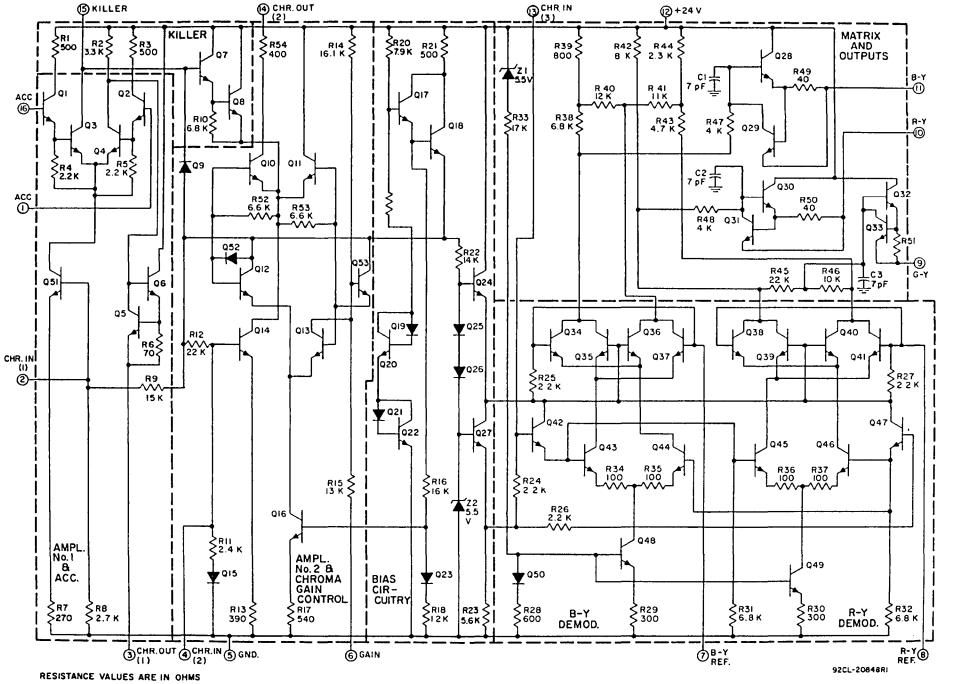


Fig. 2 – Schematic diagram of the CA3121E.

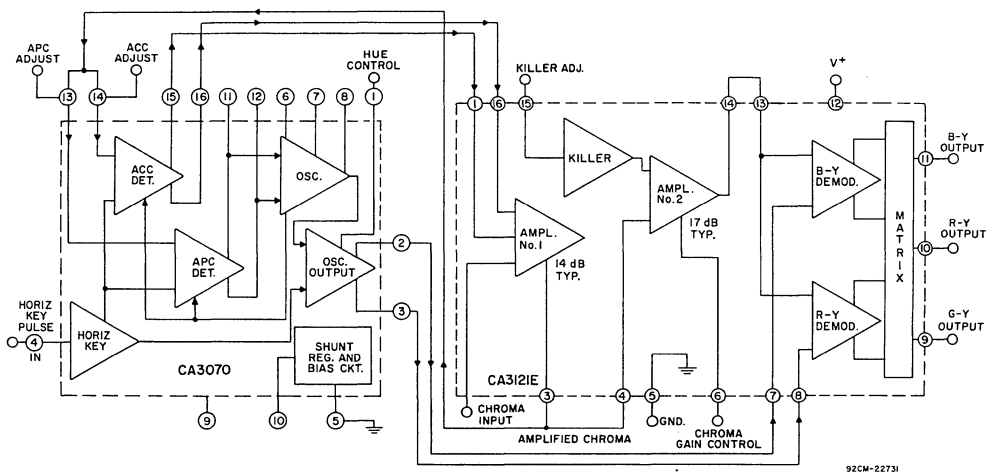


Fig. 3 – Simplified functional diagram of a two-package TV chroma system utilizing the CA3121E and CA3070.

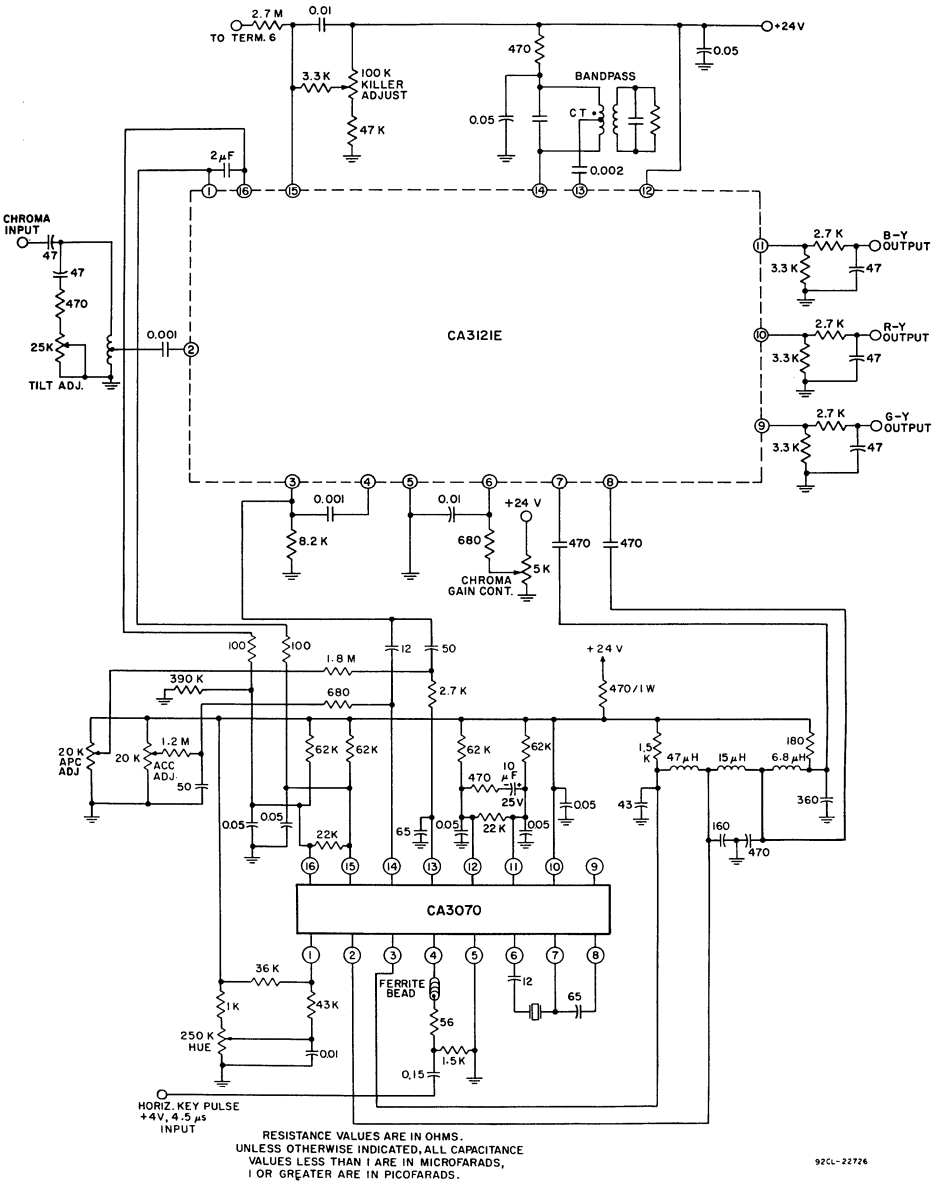


Fig. 4 — Outboard circuitry of a typical two-package chroma system for color-TV receivers utilizing the CA3121E and CA3070.

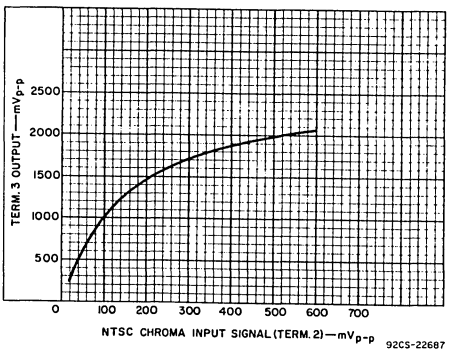


Fig. 5 – Typical ACC plot for the CA3121E when used with the CA3070.

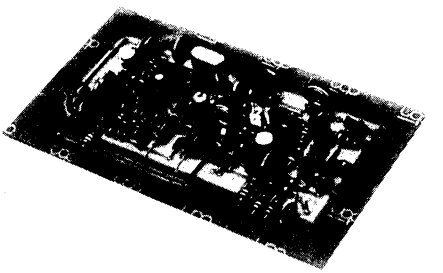


Fig.6 – Photograph of the component side of the circuit board (4 in. x 7 in.) of the two-package chroma system utilizing the CA3121E and CA3070.

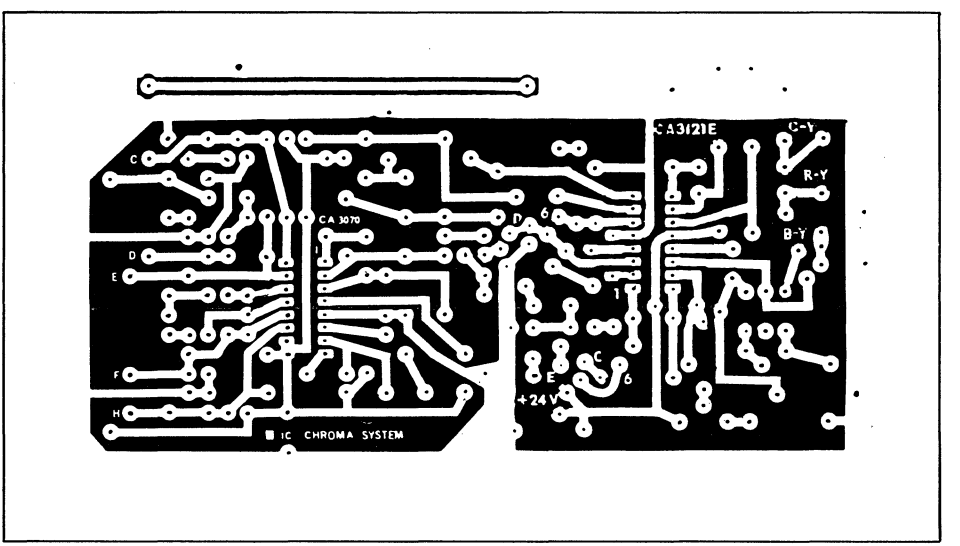
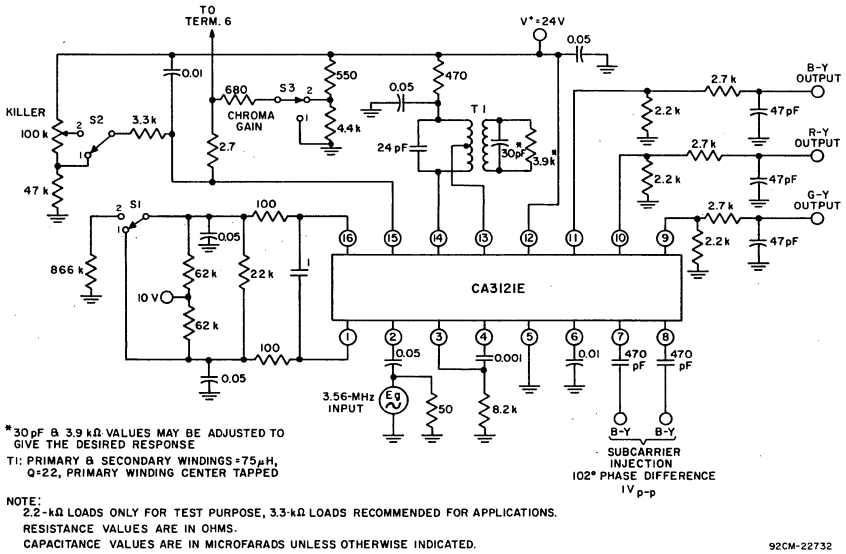


Fig.7 – Photograph of the foil side of the circuit board (4 in. x 7 in.) of the two-package chroma system utilizing the CA3121E and CA3070.



*30 pF & 3.9 kΩ VALUES MAY BE ADJUSTED TO GIVE THE DESIRED RESPONSE

T1: PRIMARY & SECONDARY WINDINGS=75 μH, Q=22, PRIMARY WINDING CENTER TAPPED

NOTE:
2.2-kΩ LOADS ONLY FOR TEST PURPOSE, 3.3-kΩ LOADS RECOMMENDED FOR APPLICATIONS.
RESISTANCE VALUES ARE IN OHMS.
CAPACITANCE VALUES ARE IN MICROFARADS UNLESS OTHERWISE INDICATED.

92CM-22732

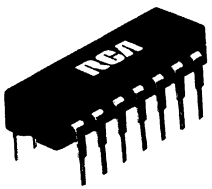
Fig. 8 – Typical characteristics test circuit for the CA3121E.

RCA
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Linear Integrated Circuits

Monolithic Silicon

CA3123E



14-Lead Dual-in-Line Plastic Package

H-1517

AM Radio Receiver Subsystem

Includes RF Amplifier, IF Amplifier, Mixer, Oscillator, AGC Detector, and Voltage Regulator

Features:

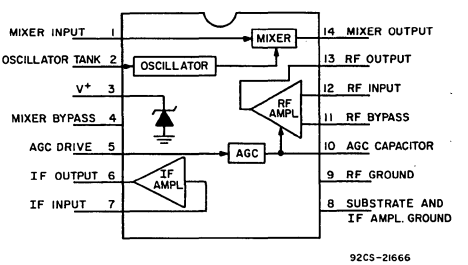
- Low-noise, low-R_b' rf stage in cascode connection — eliminates Miller-Effect regeneration and allows controlled power rise by the choice of external components
- Mixer-oscillator stage with internal feedback — eliminates need for tapped or multi-winding oscillator coils
- Cascode if amplifier with controlled output impedance and negligible Miller Effect — eliminates regeneration and selectivity skewing
- Frequency-counter AGC circuit — allows control of AGC response by selection of the coupling capacitor
- Integral regulation with built-in surge protection
- Separately accessible amplifiers

The CA3123E* is a monolithic silicon integrated circuit that provides an rf amplifier, if amplifier, mixer, oscillator, AGC detector, and voltage regulator on a single chip. It is intended for use in super-heterodyne AM radio receiver applications particularly in automobiles. The CA3123E is supplied in a 14-lead dual-in-line plastic package and operates over the temperature range of −55° to 125°C.

* Formerly RCA Dev. No. TA6155

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE:	
At Terminal No. 3 (V^+)	9 V
At Terminal No. 6 (IF Output)	40 V
At Terminal No. 13 (RF Output)	20 V
At Terminal No. 14 (Mixer Output)	20 V
DC CURRENT:	
Into Terminal No. 3 (V^+)	35 mA
DEVICE DISSIPATION:	
Up to T_A = 55°C	750 mW
Above T_A = 55°C	derate linearly 6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	−55 to +125°C
Storage	−65 to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16" ± 1/3" (1.59 mm ± 0.79 mm) from case for 10 s max.	265°C



Terminal assignment diagram.

ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
Static Characteristics In Circuit of Fig. 3						
DC Voltage:						
At Terminals 1, 4	V_1, V_4			4.7		V
At Terminals 2, 3, 14	V_2, V_3, V_{14}			6.8		V
At Terminal 5	V_5			0.25		V
At Terminal 6	V_6			12		V
At Terminal 7	V_7			0.76		V
At Terminals 8, 9	V_8, V_9			0		V
At Terminals 10, 11	V_{10}, V_{11}			0.71		V
At Terminal 12	V_{12}			0.71		V
At Terminal 13	V_{13}			4.0		V
DC Current:						
Into Terminals 1, 4, 5, 7 8, 9, 10, 11, 12	$I_1, I_4, I_5, I_7, I_8, I_9, I_{10}, I_{11}, I_{12}$			0		mA
Into Terminal 2	I_2			1.2		mA
Into Terminal 3	I_3			15		mA
Into Terminal 6	I_6			4.3		mA
Into Terminal 13	I_{13}			4.5		mA
Into Terminal 14	I_{14}			0.170		mA
Performance Characteristics In Circuit of Fig. 3						
Sensitivity		Input Signal to Dummy Antenna at f_{IN}=1 MHz, 30% AM Modulation at f_{MOD}=400 Hz, for 11 mV output at V_O	–	2.3	5	μV
Signal-to-Noise Ratio	S/N	Ratio of Output at V_O with Modulation ON and then OFF, Input Signal=100 μV, 30% AM Modulation at f_{MOD}=400 Hz	34	43	–	dB
Overload Distortion		Input Signal set at 1 MHz, 90% AM Modulation, Distortion at V_O must be $\leq$ 10%	160000	400000	–	μV

Dynamic Characteristics For Indicated Stages In Circuit of Fig. 3

Stage	Parallel Capacitance		Parallel Resistance		Transconductance
	Input pF	Output pF	Input Ω	Output Ω	μmhos
RF Amplifier	80	6	750	2×10^6 min.	140000
IF Amplifier	35	3.5	950	10^4	80000
Mixer	6	2	2000	2×10^6 min.	2500 (Mixer) 3000 (Amplifier)

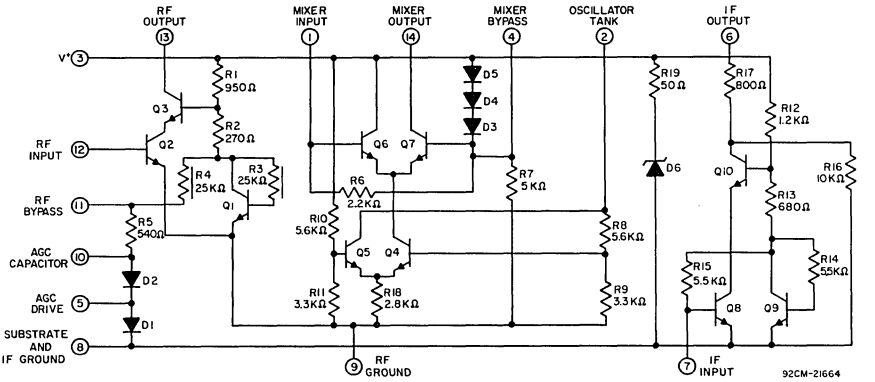
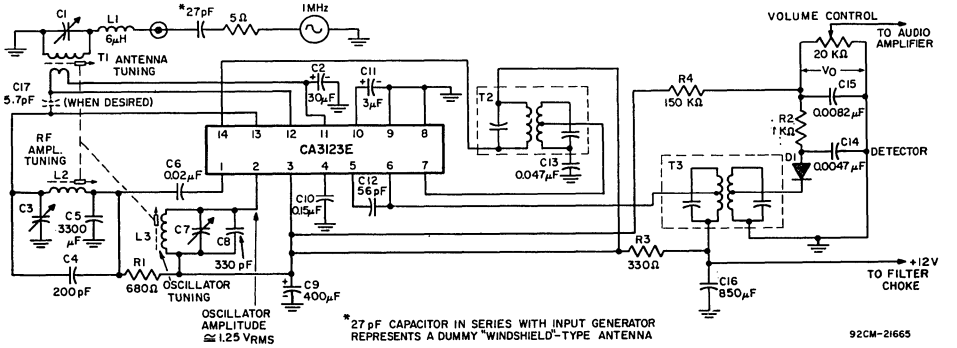


Fig. 2– Schematic diagram of CA3123E.



Transformer	Symbol	Frequency	Inductance μh (≈)	Capacitance pF (≈)	Q (≈)	Total Turns To Tap Turns Ratio	Coupling
First IF:							
Primary	T_2	262 kHz	2840	130	60	none	
Secondary			2840	130	60	or 30:1 / 31:1	critical ≈0.017 ≈1/Q
Second IF:							
Primary	T_3	262 kHz	2840	130	60	8.5:1	–
Secondary			2840	130	60	8.5:1	critical ≈0.017 ≈1/Q
Antenna:							
Primary	T_1	1 MHz	195	$(C_1)-130$	65		
Secondary		Adjusted to an impedance of 75 Ω with primary resonant at 1 MHz. Coupling should be as tight as practical. Wire should be wound around end of coil away from tuning core.					
Coils	L_1	7.9 MHz	6		50		
	L_2	1 MHz	55		50		
	L_3	1.262 MHz	41		40		

Fig. 3– Schematic diagram of AM radio receiver using CA3123E.

TYPICAL CHARACTERISTICS

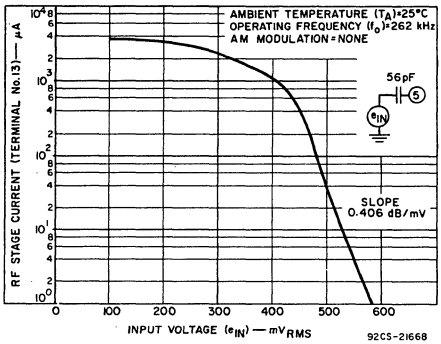


Fig.4 – Control of RF stage by signal into Terminal No.5.

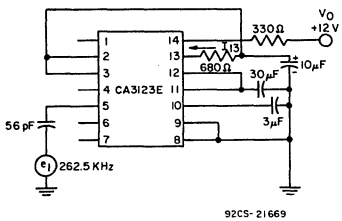


Fig.5 – Test circuit for Fig. 4.

PERFORMANCE CHARACTERISTICS IN CIRCUIT OF FIG. 3

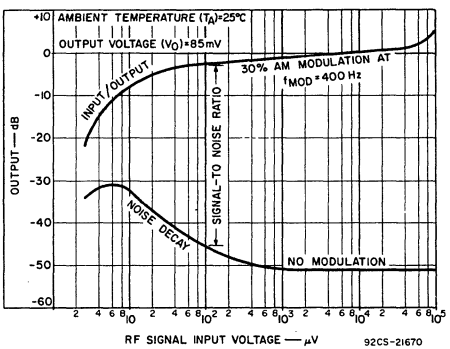


Fig.6 – Signal-to-noise performance.

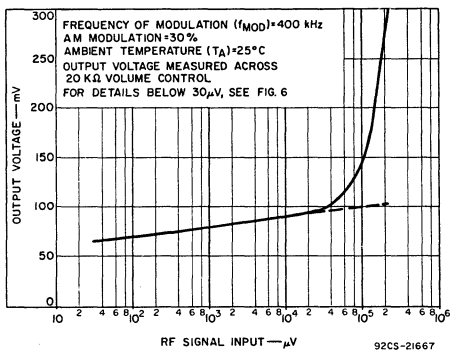


Fig.7 – AGC curve showing voltage rise (controlled by external capacitance of 5.7 pF: C_{17}, Fig.3).

Change in slope in the vicinity of 40000 μV signal input voltage is the result of the use of C_{17} (5.7 pF) in Fig.3. The dotted curve indicates expected performance if C_{17} = 0.

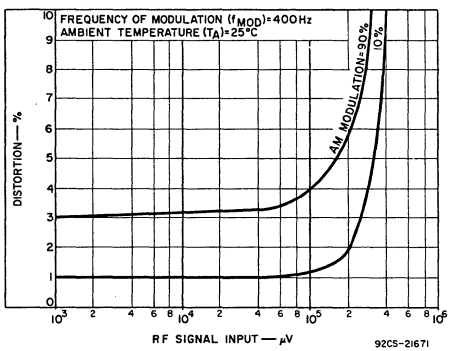


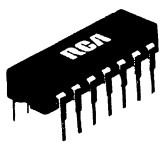
Fig.8 – Overload response.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3125E



14-Lead Dual-In-Line Plastic Package

H-1517

Television Chroma Demodulator

Features:

- Luminance input
- Blanking control input
- Three separate demodulators with independent phase control
- Low output offset voltage ..... 0.4 V

RCA-CA3125E is a monolithic silicon integrated-circuit chroma demodulator having three separate demodulators with independent phase control. It is designed to function compatibly with the CA1398E IC Chroma Processor as well as other commercially available Chroma Processors in R-G-B Systems of color-TV receivers. Fig. 2 shows a functional block diagram of a 2-package TV Chroma System incorporating the CA3125E and CA1398E. The CA3125E is supplied in a 14-lead dual-in-line plastic package.

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25°C$

SUPPLY VOLTAGE	25 V
SUPPLY CURRENT	20 mA
AMBIENT-TEMPERATURE RANGE:	
Operating	−40°C to +85°C
Storage	−65°C to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 s max.	265°C

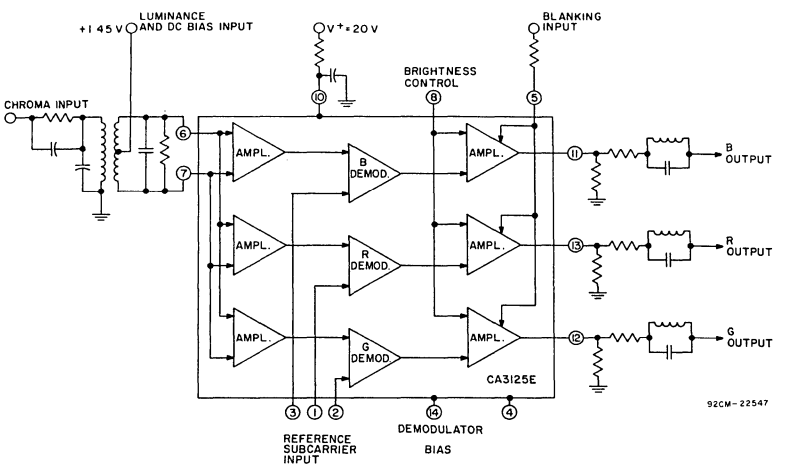


Fig. 1 – Functional block diagram of the CA3125E.

TYPICAL STATIC CHARACTERISTICS AT $T_A = 25°C$, $V^+ = +20$ VOLTS

SUPPLY CURRENT ..... 9.6 mA

BRIGHTNESS CONTROL VOLTAGE:

Measured with 8 volts at Terminals 11, 12, and 13 ..... 1.4 V

MAX. OUTPUT DIFFERENCE VOLTAGE:

Measured between any two of Terminals 11, 12, and 13 ..... ±0.4 V

MAXIMUM DC DETECTOR UNBALANCE VOLTAGE:

DC voltage shift on Terminals 11, 12, and 13 when Terminals 1, 2, and 3 are alternately biased 0.5 volt positive, then negative with reference to Terminal 14 ..... +150 mV

TYPICAL DYNAMIC CHARACTERISTICS AT $T_A = 25°C$, $V^+ = +20$ volts

BLUE CHROMA GAIN:

Peak-to-peak voltage at Terminal 11 with 1.0 volt peak-to-peak applied differentially between Terminals 6 and 7, and with a subcarrier injection voltage of 1 volt peak-to-peak 7.36 Vp-p

RED GAIN RATIO:

$$\frac{\text{Peak-to-peak voltage at Terminal 13}}{\text{Peak-to-peak voltage at Terminal 11}} \times 100 \text{ ..... } 100\%$$

GREEN GAIN RATIO:

$$\frac{\text{Peak-to-peak voltage at Terminal 12}}{\text{Peak-to-peak voltage at Terminal 11}} \times 100 \text{ ..... } 30\%$$

LUMINANCE GAIN:

Peak-to-peak voltage measured at Terminals 11, 12, and 13, with a peak-to-peak voltage of 0.1 volt applied to Terminals 6 and 7 (common mode), and with no subcarrier injection ..... 0.7 Vp-p

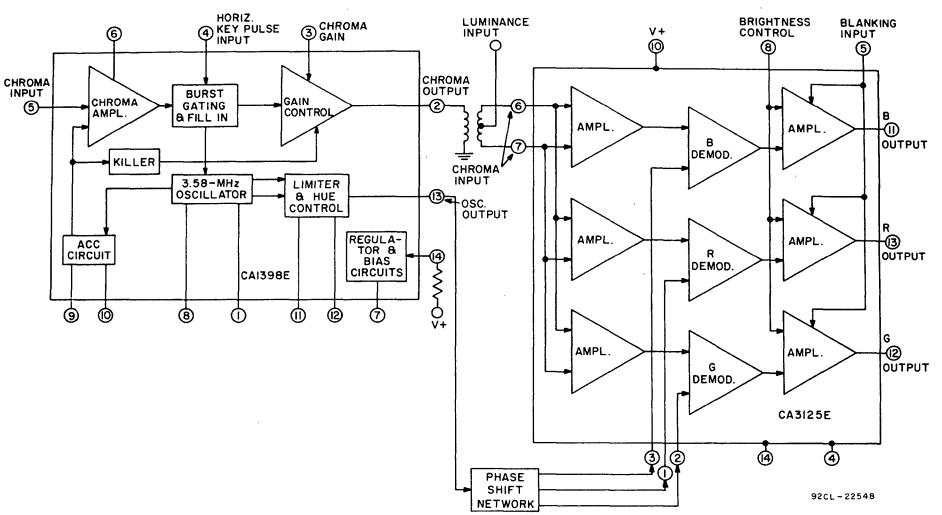


Fig. 2 – TV chroma system functional block diagram.

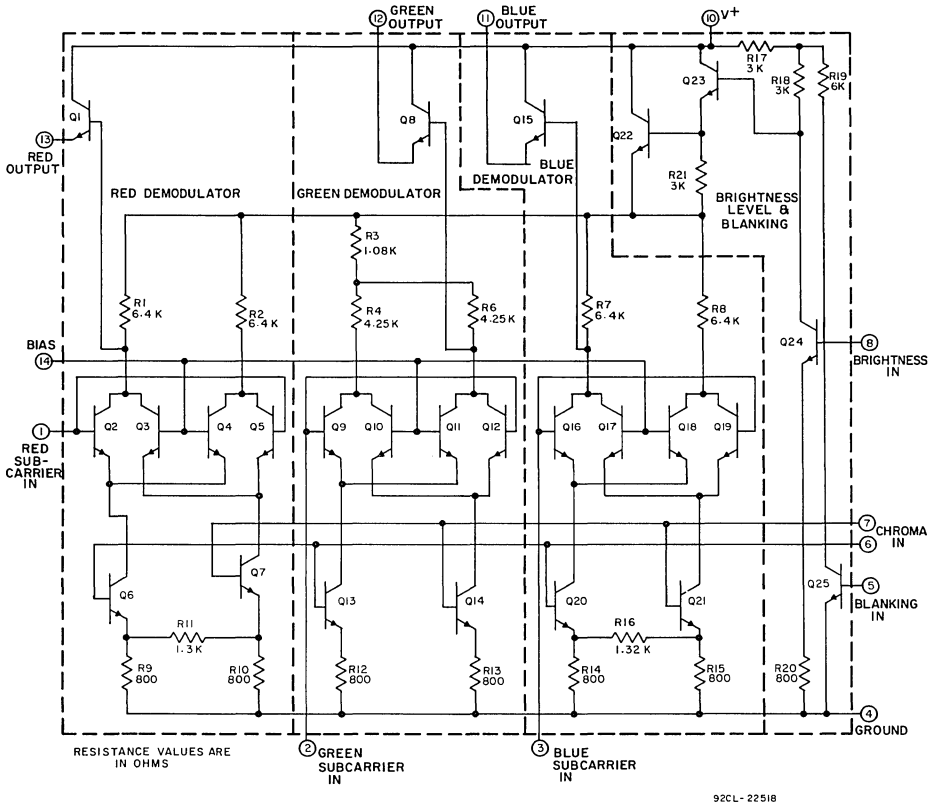


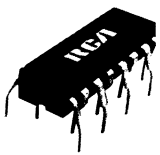
Fig. 3 – Schematic diagram of the CA3125E.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3126Q



16-lead Quad-in-line plastic package

H-1706

TV Chroma Processor

Features:

- Phase-locked subcarrier regeneration utilizes sample-and-hold techniques
- Automatic chrominance control (ACC)/killer detector employs sample-and-hold techniques
- Supplementary ACC with an overload detector to prevent oversaturation of the picture tube
- Sinusoidal subcarrier output
- Keyed chroma output
- Emitter-follower buffered outputs for low output impedance
- Linear dc saturation control
- Internal zener-regulated reference potentials
- Only the initial crystal filter tuning is required. . . no killer or ACC adjustments required at any time
- Few external components required
- Compensation for temperature and supply variations
- All terminals protected against short circuits

RCA-CA3126Q is a monolithic silicon integrated circuit designed for chroma processing applications in color TV receivers. It is compatible with the CA3067 chroma demodulator as well as other chroma demodulators.

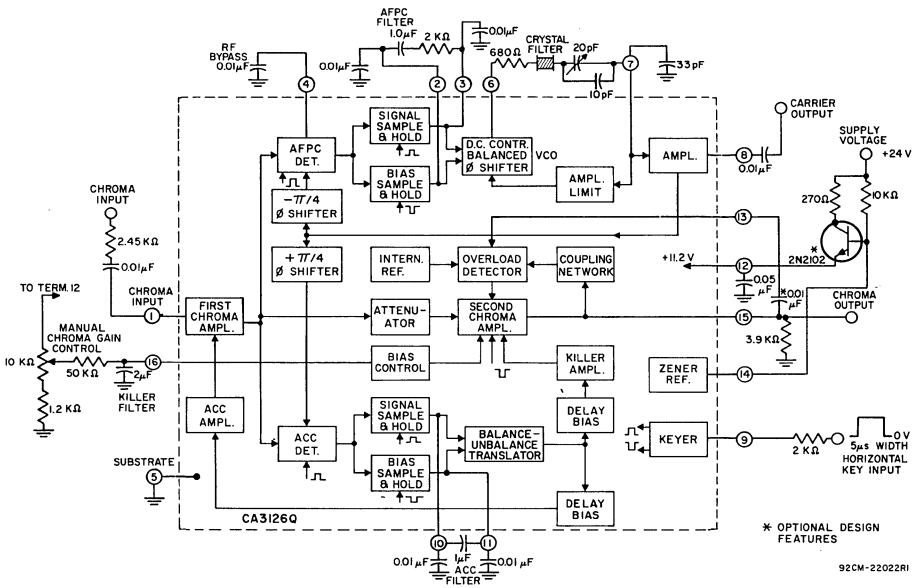


Fig. 1—Block diagram of CA3126Q TV Chroma Processor.

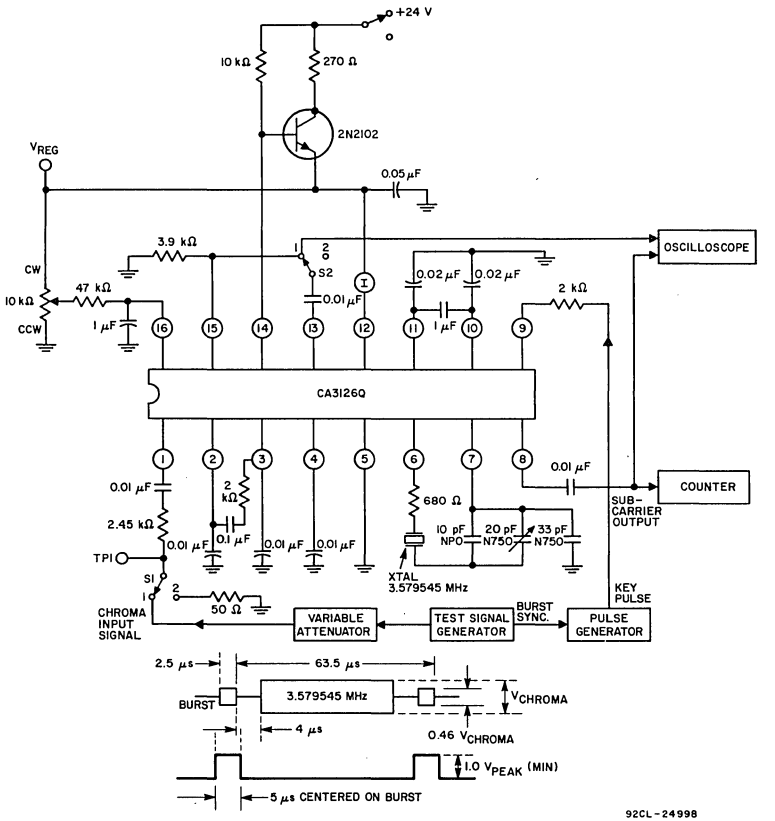


Fig. 2—Test circuit for CA3126Q.

CIRCUIT DESCRIPTION

The following paragraphs briefly describe the circuit operation of the CA3126Q (shown in Figs. 1 and 3). A detailed description of the operation of various portions of the CA3126Q is given in ICAN-6247, "Application of the CA3126Q Chroma-Processing IC Using Sample-and-Hold Techniques".

The chroma input is applied to Terminal 1 through the desired band-shaping network. A 2,450-ohm resistor should be placed in series with Terminal 1 to minimize oscillator pickup in the first chroma amplifier. This amplifier supplies signals to the second chroma amplifier and to the ACC and AFPC detectors. The first chroma amplifier is gain-controlled by the ACC amplifier.

A horizontal keying pulse is applied to Terminal 9. This pulse must be present to ensure proper operation of the oscillator circuit. The subcarrier burst is sampled during the keying interval in the AFPC detector. The error voltage, produced at Terminal 2 and proportional to the burst phase, is compared to the quiescent bias voltage at Terminal 3 by the sample-and-hold circuitry. This "compared" voltage controls the phase-shifting network in the phase-locked loop. The operation of the AFPC loop is independent of any external adjustments or voltages except for an initial capacitor adjustment to set the free-running frequency.

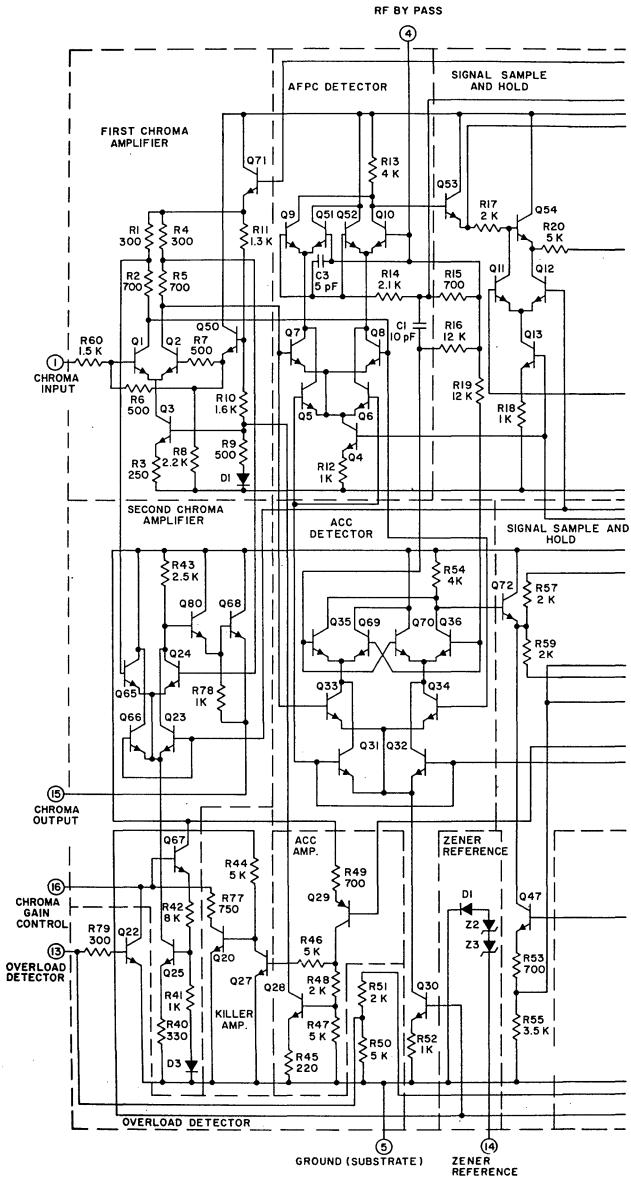


Fig. 3—Schematic diagram of the CA3126Q.

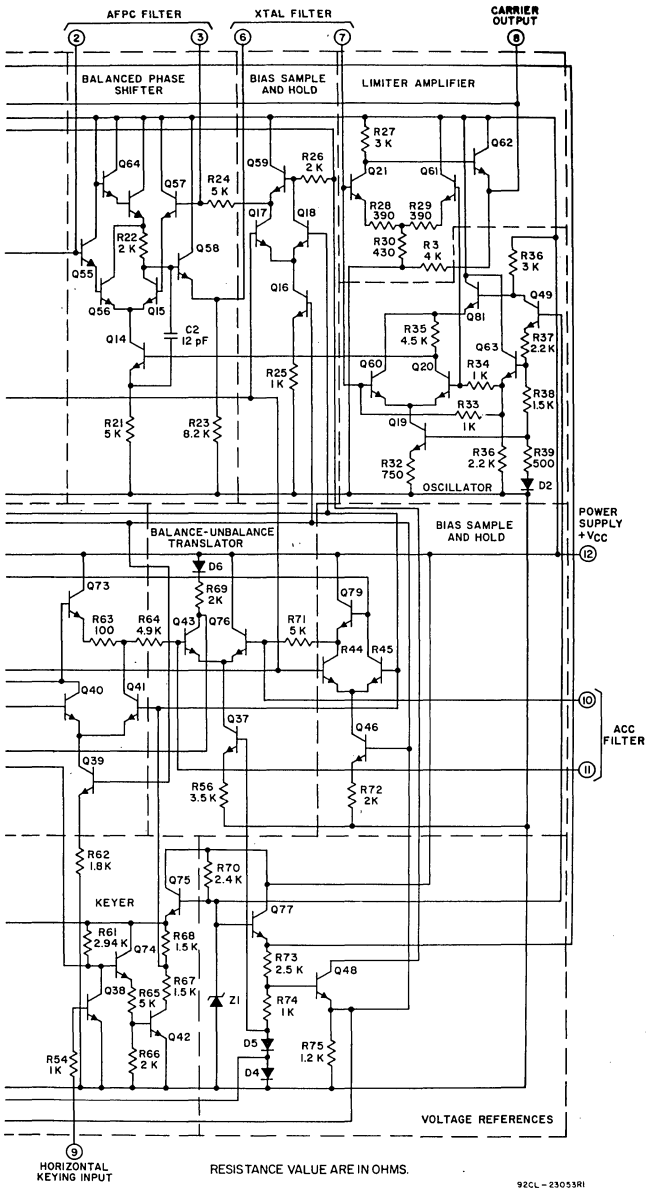


Fig. 3—Schematic diagram of the CA3126Q (cont'd).

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25^oC$

DEVICE DISSIPATION:
- Up to $T_A = 55^oC$ ..... 750 mW
- Above $T_A = 55^oC$ ..... derate linearly 7.9 mW/^oC

DC SUPPLY VOLTAGE (Across Terms. 5 and 12)[▲] ..... 13.2 V

DC CURRENT:
- Into Term. 12 ..... 38 mA
- Into Term. 14 ..... 20 mA

DC VOLTAGE (Terminal 9):
- Negative Rating ..... −5 V
- Positive Rating ..... 3 V

AMBIENT TEMPERATURE RANGE:
- Operating ..... −40 to +85^oC
- Storage ..... −65 to +150^oC

LEAD TEMPERATURE (During Soldering):
- At a distance not less than 1/32 in. (0.79 mm) from case for 10 seconds max. ..... +265^oC

[▲]This rating does not apply when using the internal zener reference in conjunction with an external pass transistor.

ELECTRICAL CHARACTERISTICS

Test Conditions: $T_A = 25^oC$, chroma control at maximum position for all characteristics tests except for chroma output test. For this test, control should be set at minimum position. Electrical characteristics referenced to test circuit, Fig.2.

CHARACTERISTIC	TERMINAL, MEASUREMENT, AND SYMBOL	SWITCH POS.		CHROMA INPUT	LIMITS			UNITS
		S1	S2	TP1	Min.	Typ.	Max.	
Static Characteristics								
Voltage Regulator	V_{12}	2	2	0	10.1	11.2	12.1	V
Supply Current	I_{12}	2	2	0	16	25	38	mA
Dynamic Characteristics (See Note 1)								
Pull-in Range*	V_8	*	2	0.5 V_{p-p}	±250	—	—	Hz
Oscillator Output	V_8	2	2	0	0.6	1.0	—	V_{p-p}
100% Chroma Output	V_{15}	1	2	0.5 V_{p-p}	1.4	2.7	—	V_{p-p}
Overload Detector	V_{15}	1	1	0.5 V_{p-p}	0.4	—	0.7	V_{p-p}
Minimum Chroma Output	V_{15}	1	2	0.5 V_{p-p}	—	—	20	mV_{p-p}
200% Chroma Output	V_{15}	1	2	1 V_{p-p}	70	100	140	% of 100% reading
20% Chroma Output	V_{15}	1	2	0.1 V_{p-p}	40	—	105	
Kill Level	V_{TP1}	1	2	vary	5	—	60	mV_{p-p}

Note 1: Except for pull-in range testing, tune oscillator trimmer capacitor for free-running frequency of 3.579545 MHz ± 10 Hz.
*Set Switch 1 to Position 2, detune oscillator ± 250 Hz, set Switch 1 to Position 1, and check for oscillator pull-in.

The regenerated oscillator signal at Terminal 8 is applied internally to the AFPC and ACC detectors through +45- and −45-degree phase-shifter networks to establish the proper phase relationship for these detectors. The ACC detector, which also samples the burst during the keying interval, produces a correction voltage proportional to the burst amplitude. The correction voltage is compared to the quiescent bias level using sample-and-hold circuitry similar to that used in the AFPC portion of the circuit. The "compared" voltage is applied internally to the ACC amplifier and killer amplifier. Because the amplifier gains and killer threshold are determined by the ratios of the internal resistors, these functions are independent of external voltages or controls.

The attenuated chroma signal is fed to the second chroma amplifier, where the burst is removed by keyer action. The killer amplifier, the chroma gain control, and the overload detector control the action of the second chroma amplifier, whose gain is proportional to the dc voltage at Terminal 16. The overload detector (Terminal 13) receives a sample of the chroma output (Terminal 15) and detects the peak of the signal. The detected voltage is stored in an external capacitor connected to Terminal 16. This stored voltage on Terminal 16 affects the gain of the second chroma in the same manner as the chroma gain control.

APPLICATIONS INFORMATION

General Considerations

The block diagram shown in Fig. 1 is typical of the type of circuit used in the practical application of the CA3126Q. Several items are critical for proper operation of the circuit.

1. A series resistor of approximately 2,450 ohms (or high source impedance) must be used at the chroma input, Terminal 1. This high impedance minimizes pickup of unbalanced currents, particularly of the subcarrier oscillator signal.
2. When the overload detector is used, a large resistor (nominally 47,000 ohms) must be placed in series with Terminal 16 to set the required RC time constant. The same RC network series serves to set the killer time constant.
3. The setting of the free-running oscillator frequency requires the presence of the keying pulse. The free-running frequency will be erroneous if Terminal 1 is dc shorted during the setting operation because of the dc offset voltage introduced to the AFPC detector.
4. Care must be taken in PC board designs to provide reasonable isolation between the oscillator portion of the circuit (Terminals 6, 7, and 8) and the chroma input (Terminal 1).

Overload Detector

The overload detector accomplishes two purposes:

1. It prevents oversaturation due to low burst-to-chroma ratios.
2. It prevents overload conditions due to noise.

Both of these conditions are discussed in more detail in ICAN-6247. The extent to which the overload detector is used depends upon the individual receiver design goals. If greater than 0.5-volt peak-to-peak output is desired, the chroma output at Terminal 15 can be tapped to yield any desired degree of overload detector action.

Chroma Gain Control

The chroma gain control operates by varying the base bias on current source transistor Q25. To ensure proper temperature tracking of the chroma gain control, it is essential that the control be operated from a supply source derived from the reference voltage at Terminal 12. Because the control operates from a current source, chroma gain is much more predictable and far less temperature sensitive than controls that steer current by means of a differential amplifier. The typical chroma gain characteristic for the CA3126Q is shown in Fig. 4.

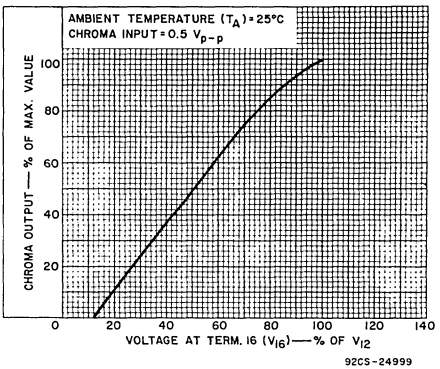


Fig. 4—Chroma gain control.

Subcarrier Regenerator Oscillator

The oscillator filter consists of a 3.579545-MHz crystal, a 680-ohm resistor, and a 10-pF capacitor connected in series across Terminals 6 and 7. A 33-pF capacitor, shunt connected from Terminal 7 to ground, rolls off higher-order harmonics, thereby preventing oscillation at the crystal third-harmonic frequency. A curve of the typical static phase error as a function of the free-running oscillator frequency is shown in Fig. 5. It should be noted that the slope of the curve determines the dc gain of the phase-locked loop, i.e., 40 Hz per degree.

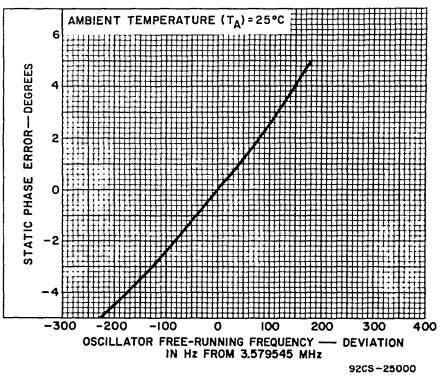


Fig. 5—Static phase error.

Thermal Considerations

The circuit of the CA3126Q is thermally compensated to achieve the optimal operating characteristics over the normal operating temperature range of TV receivers. Figs. 6 and 7 show the oscillator- and chroma-output amplitudes and phases as a function of temperature (Terminals 8 and 15), respectively. Both the oscillator- and chroma-output amplitudes and phases are measured relative to the chroma-input phase. The performance of the oscillator free-running frequency as a function of temperature is shown in Fig. 8. All the temperature plots are characteristic of the test circuit with the indicated component types and values given in Fig. 2.

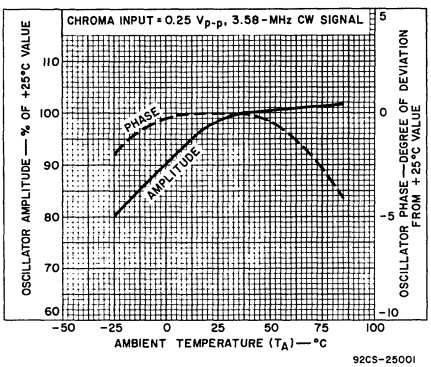


Fig. 6—Amplitude and phase variations of oscillator output vs. temperature.

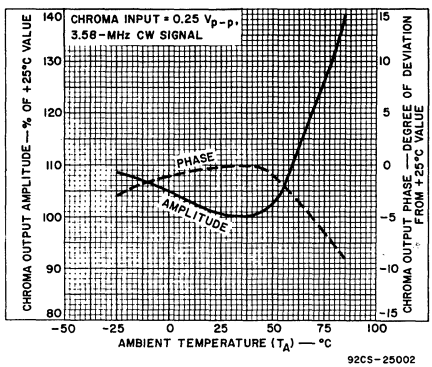


Fig. 7—Amplitude and phase variations of chroma output vs. temperature.

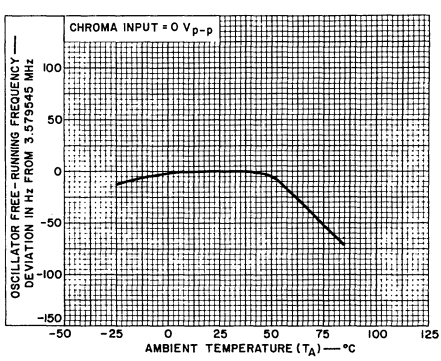


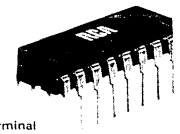
Fig. 8—Variation of oscillator free-running frequency vs. temperature.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3127E



16-Lead Dual-In-Line
Plastic Package H-1622

High-Frequency N-P-N Transistor Array

For Low-Power Applications at Frequencies up to 500 MHz

Features:

- Gain-Bandwidth Product (f_T) > 1 GHz
- Power Gain = 30 dB (typ.) at 100 MHz
- Noise Figure = 3.5 dB (typ.) at 100 MHz
- Five independent transistors on a common substrate

Applications:

- VHF amplifiers
- VHF mixers
- Multifunction combinations – RF/mixer/oscillator
- IF Converter
- IF amplifiers
- Sense amplifiers
- Synthesizers
- Synchronous detectors
- Cascade amplifiers

RCA-CA3127E* consists of five general-purpose silicon n-p-n transistors on a common monolithic substrate. Each of the completely isolated transistors exhibits low 1/f noise and a value of f_T in excess of 1 GHz, making the CA3127E useful from dc to 500 MHz. Access is provided to each of the terminals for the individual transistors and a separate substrate connection has been provided for maximum application flexibility. The monolithic construction of the CA3127E provides close electrical and thermal matching of the five transistors.

The CA3127E is supplied in a 16-lead dual-in-line plastic package and operates over the full military temperature range of −55 to +125°C.

* Formerly RCA Dev. No. TA6206.

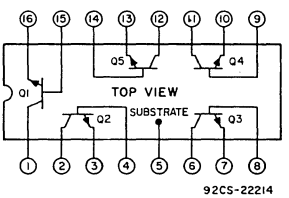


Fig. 1– Schematic diagram of CA3127E.

MAXIMUM RATINGS, *Absolute-Maximum Values, at T_A = 25°C*

POWER DISSIPATION, P_D:	
Any one transistor	85 mW
Total Package:	
For T_A up to 75°C	425 mW
For T_A > 75°C Derate Linearly at	6.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	−55 to +125°C
Storage	−65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C

The following ratings apply for each transistor in the device.

Collector-to-Emitter Voltage, V_{CEO}	15 V
Collector-to-Base Voltage, V_{CBO}	20 V
Collector-to-Substrate Voltage, V_{CIO}*	20 V
Collector Current, I_C	20 mA

*The collector of each transistor of the CA3127E is isolated from the substrate by an integral diode. The substrate (terminal 5) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.

STATIC ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
For Each Transistor:						
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_C = 10\ \mu A, I_E = 0$	20	32	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	$I_C = 1$ mA, $I_B = 0$	15	24	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	$I_{C1} = 10\ \mu A, I_B = 0, I_E = 0$	20	60	–	V
Emitter-to-Base Breakdown Voltage*	$V_{(BR)EBO}$	$I_E = 10\ \mu A, I_C = 0$	4	5.7	–	V
Collector-Cutoff-Current	I_{CEO}	$V_{CE} = 10$ V, $I_B = 0$	–	–	0.5	μA
Collector-Cutoff-Current	I_{CBO}	$V_{CB} = 10$ V, $I_E = 0$	–	–	40	nA
DC Forward-Current Transfer Ratio	h_{FE}	$V_{CE} = 6$ V, $I_C = 5$ mA	35	88	–	
		$V_{CE} = 6$ V, $I_C = 1$ mA	40	90	–	
		$V_{CE} = 6$ V, $I_C = 0.1$ mA	35	85	–	
Base-to-Emitter Voltage	V_{BE}	$V_{CE} = 6$ V, $I_C = 5$ mA	0.71	0.81	0.91	V
		$V_{CE} = 6$ V, $I_C = 1$ mA	0.66	0.76	0.86	
		$V_{CE} = 6$ V, $I_C = 0.1$ mA	0.60	0.70	0.80	
Collector-to-Emitter Saturation Voltage	$V_{CE(sat)}$	$I_C = 10$ mA, $I_B = 1$ mA	–	0.26	0.50	V
Magnitude of Difference in V_{BE}	$\lvert\Delta V_{BE}\rvert$	Q_1 & Q_2 Matched	–	0.5	5	mV
Magnitude of Difference in I_B	$\lvert\Delta I_B\rvert$	$V_{CE} = 6$V, $I_C = 1$ mA	–	0.2	3	μA

*When used as a zener for reference voltage, the device must not be subjected to more than 0.1 millijoule of energy from any possible capacitive or electrostatic discharge in order to prevent degradation of the junction. Maximum operating zener current should be less than 10 mA.

DYNAMIC CHARACTERISTICS at $T_A = 25^oC$

CHARACTERISTICS	SYMBOL	TEST CONDITIONS		LIMITS			UNITS
			Fig. No.	Min.	Typ.	Max.	
I/F Noise Figure	NF	f = 100 kHz, $R_S = 500\ \Omega$, $I_C = 1$ mA	2	–	1.8	–	dB
Gain-Bandwidth Product	f_T	$V_{CE} = 6$ V, $I_C = 5$ mA	4	–	1.15	–	GHz
Collector-to-Base Capacitance	C_{CB}	$V_{CB} = 6$ V, f = 1 MHz	5	–	See	–	pF
Collector-to-Substrate Capacitance	C_{CI}	$V_{CI} = 6$ V, f = 1 MHz	5	–	Fig.	–	pF
Emitter-to-Base Capacitance	C_{EB}	$V_{BE} = 4$ V, f = 1 MHz	5	–	5	–	pF
Voltage Gain	A	$V_{CE} = 6$ V, f = 10 MHz, $R_L = 1\ K\Omega$, $I_C = 1$ mA	6, 18	–	28	–	dB
Power Gain	G_p	Cascode Configuration f = 100 MHz, $V^+ = 12$ V	19, 20	27	30	–	dB
Noise Figure	NF	$I_C = 1$ mA	19, 20	–	3.5	–	dB
Input Resistance	$1/g_{11}$	Common-Emitter	10	–	400	–	Ω
Output Resistance	$1/g_{22}$	Configuration	12	–	4.6	–	kΩ
Input Capacitance	C_{11}	$V_{CE} = 6$ V	10	–	3.7	–	pF
Output Capacitance	C_{22}	$I_C = 1$ mA	12	–	2	–	pF
Magnitude of Forward Transadmittance	$\lvert Y_{21}\rvert$	f = 200 MHz	14, 15	–	24	–	mmho

CHARACTERISTICS CURVES
COMMON-EMITTER CONFIGURATION

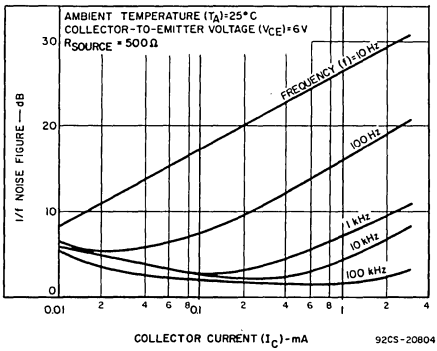


Fig. 2 — 1/f noise figure vs. collector current at R_{SOURCE} = 500 Ω.

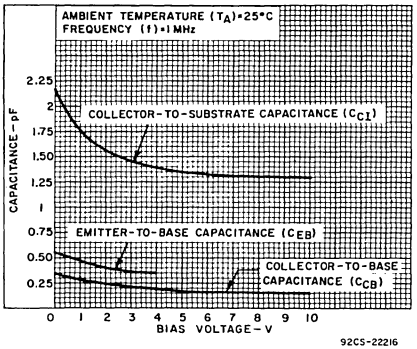


Fig. 5(a) — Capacitance vs. bias voltage for Q_2.

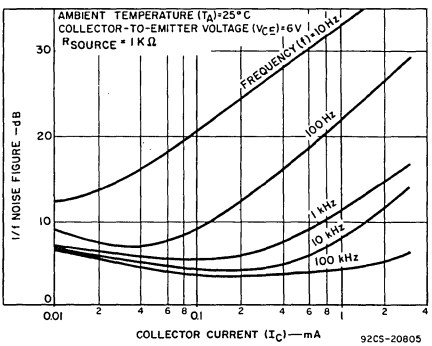


Fig. 3 — 1/f noise figure vs.collector current at R_{SOURCE} = 1 kΩ.

Transistor	Capacitance (pF)							
	C_{CB}		C_{CE}		C_{EB}		C_{CI}	
	Pkg.	Total	Pkg.	Total	Pkg.	Total	Pkg.	Total
Bias Voltage	–	6V	–	6V	–	4V	–	6V
Q1	0.025	0.190	0.090	0.125	0.365	0.610	0.475	1.65
Q2	0.015	0.170	0.225	0.265	0.130	0.360	0.085	1.35
Q3	0.040	0.200	0.215	0.240	0.360	0.625	0.210	1.40
Q4	0.040	0.190	0.225	0.270	0.365	0.610	0.085	1.25
Q5	0.010	0.165	0.095	0.115	0.140	0.365	0.090	1.35

Fig. 5(b) — Typical capacitance values at f = 1 MHz. Three terminal measurement. Guard all terminals except those under test.

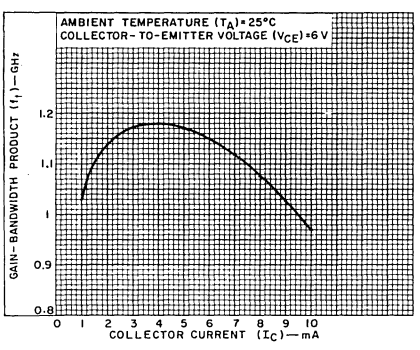


Fig. 4 — Gain-bandwidth product vs. collector current.

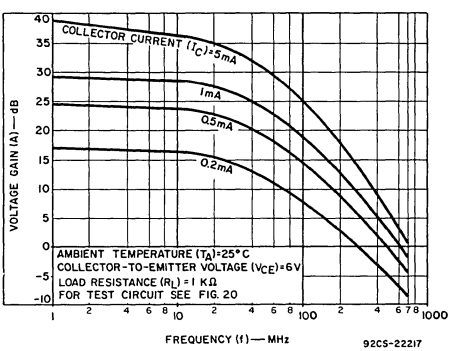


Fig. 6 — Voltage gain vs. frequency at R_L = 1 kΩ.

CHARACTERISTICS CURVES (Cont'd)
COMMON-EMITTER CONFIGURATION

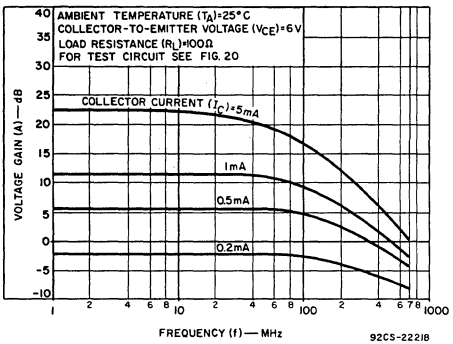


Fig. 7 — Voltage gain vs. frequency at R_L = 100 Ω.

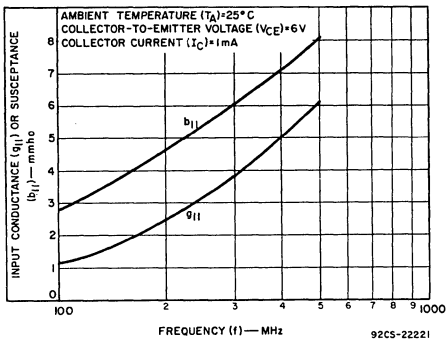


Fig. 10 — Input admittance (Y_{11}) vs. frequency.

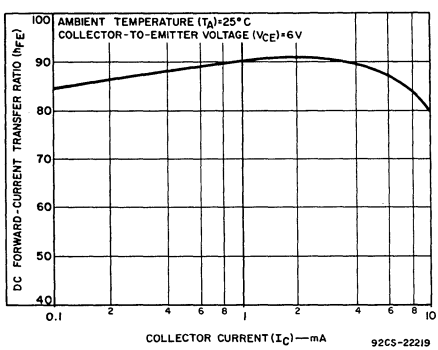


Fig. 8 — DC forward-current transfer ratio vs. collector current.

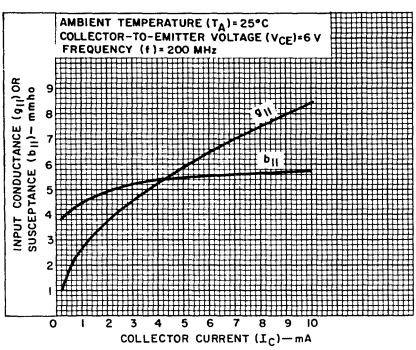


Fig. 11 — Input admittance (Y_{11}) vs. collector current.

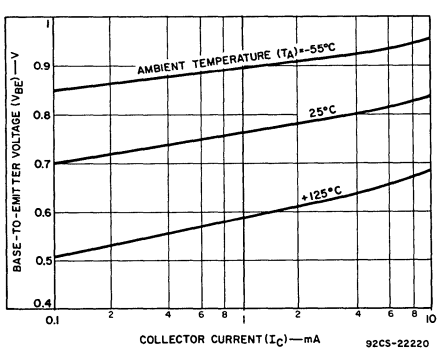


Fig. 9 — Base-to-emitter voltage vs. collector current.

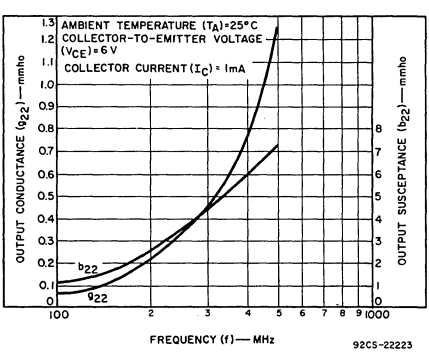


Fig. 12 — Output admittance (Y_{22}) vs. frequency.

CHARACTERISTICS CURVES (Cont'd)
COMMON-EMITTER CONFIGURATION

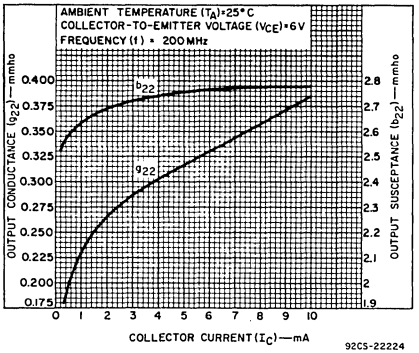


Fig. 13 – Output admittance (Y_{22}) vs. collector current.

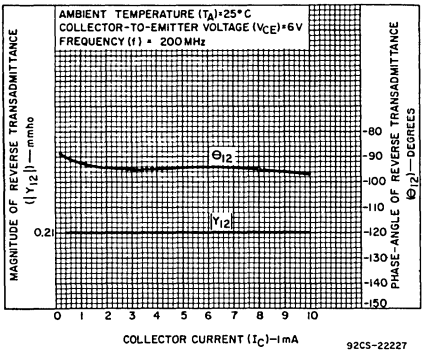


Fig. 16 – Reverse transadmittance (Y_{12}) vs. collector current.

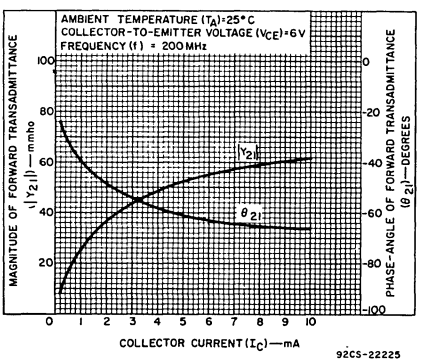


Fig. 14 – Forward transadmittance (Y_{21}) vs. collector current.

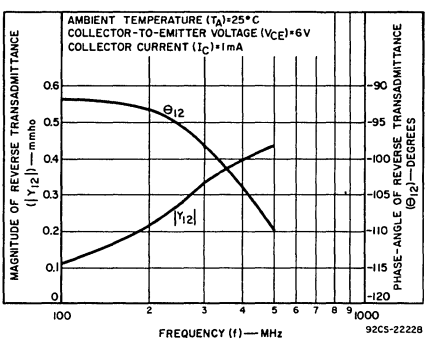


Fig. 17 – Reverse transadmittance (Y_{12}) vs. frequency.

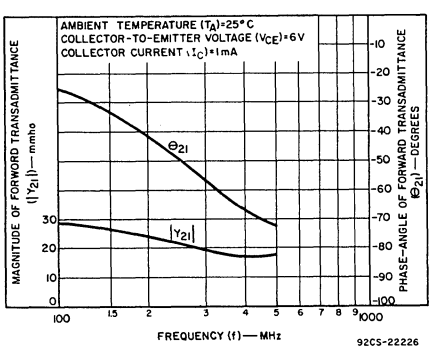


Fig. 15 – Forward transadmittance (Y_{21}) vs. frequency.

TEST CIRCUITS

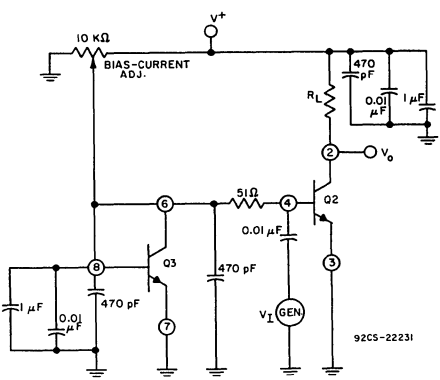
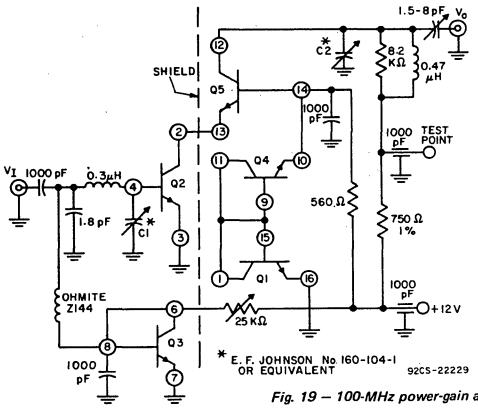


Fig.18 – Voltage-gain test circuit using current-mirror biasing for Q_2.

TEST CIRCUITS (Cont'd)



This circuit was chosen because it conveneietly represents a close approximation in performance to a properly unilateralized single transistor of this type. The use of Q3 in a current-mirror configuration facilitates simplified biasing. The use of the cascode circuit in no way implies that the transistors cannot be used individually.

Fig. 19 — 100-MHz power-gain and noise-figure test circuit.

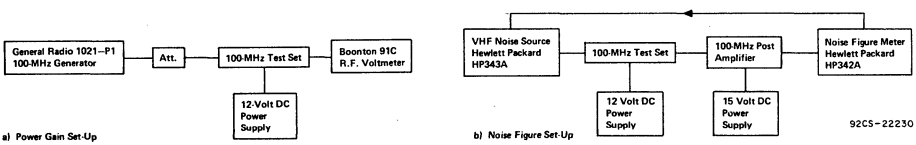
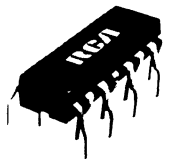


Fig. 20 — Block diagrams of power-gain and noise-figure test set-ups.



16-lead Quad-in-line palstic package

H-1706

TV Chroma Processor for PAL Systems

Features:

- Phase-locked subcarrier regeneration utilizes sample-and-hold techniques in the automatic frequency phase control (AFPC) servo loop
- Automatic chrominance control (ACC)/ killer detector employs sample-and-hold techniques
- Supplementary ACC with an overload detector to prevent oversaturation of the picture tube
- Sinusoidal subcarrier output
- Keyed chroma output
- Emitter-follower buffered outputs for low output impedance
- Linear dc saturation control
- PAL identification output

The RCA-CA3128Q is a monolithic silicon integrated circuit designed primarily for PAL chroma processing applications in color TV receivers. For a circuit description of the CA3128Q and an explanation of this device in PAL systems, refer to "A New Chroma Processing IC Using Sample -and-Hold Techniques" by L. A. Harwood (ST6144).

- Only the initial crystal filter tuning is required . . . no killer and ACC adjustments required at any time
- Few external components required
- Compensation for temperature and supply variations
- All terminals protected against short circuits

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25°C$

DC SUPPLY VOLTAGE (Between Terms. 12 and 5)	13.2	V
DC VOLTAGE (Term. 9):		
Positive Value	+3	V
Negative Value	−5	V
DEVICE DISSIPATION:		
Up to $T_A = 55°C$	750	mW
Above $T_A = 55°C$	derate linearly at 7.9 mW/°C	
AMBIENT TEMPERATURE RANGE:		
Operating	−40 to +85	°C
Storage	−65 to +150	°C
LEAD TEMPERATURE (DURING SOLDERING):		
At a distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265	°C

TYPICAL STATIC CHARACTERISTICS at $T_A = 25°C$:

DC Supply Current (I_{12})with $V_{12} = 11.2$ V dc	25	mA

TYPICAL DYNAMIC CHARACTERISTICS at $T_A = 25°C$ with a Burst-to-Chroma Ratio of 46.5%:

100% Chroma Output Voltage at $V_{I\,(p\text{-}p)} = 0.5$ V	3.5	Vp-p
Oscillator-Level Output Voltage	1	Vp-p
Killer Threshold Input Voltage	0.018	Vp-p
Pull-in Frequency	500	Hz
PAL Identification Output Voltage	1	Vp-p

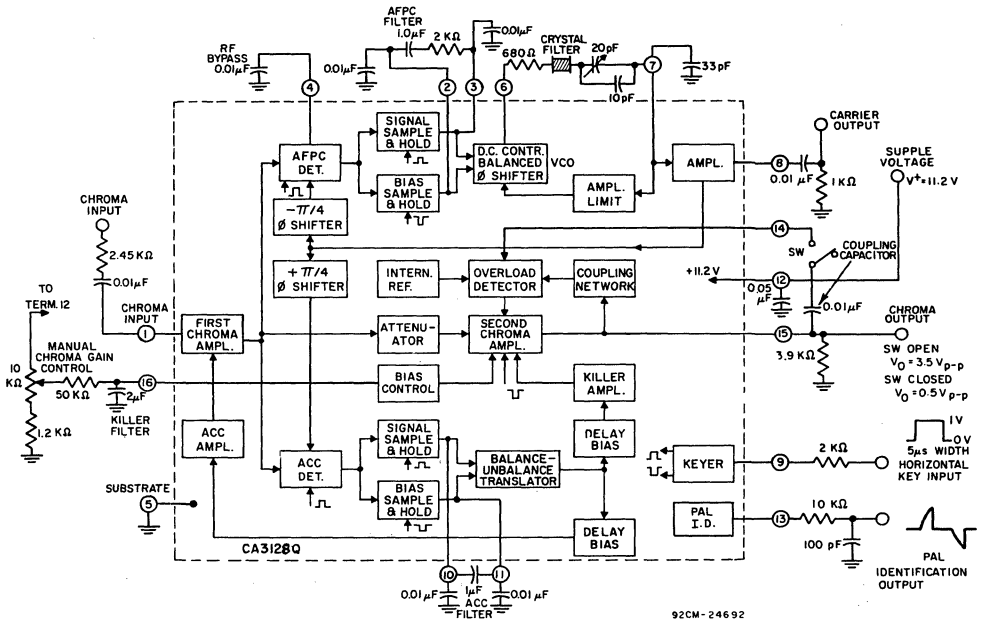


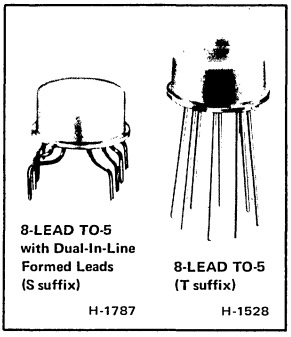
Fig. 1 — Block diagram of CA3128Q TV Chroma Processor.

RCA
Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3130BT, CA3130BS
CA3130AT, CA3130AS
CA3130T, CA3130S



8-LEAD TO-5 with Dual-In-Line Formed Leads (S suffix) H-1787

8-LEAD TO-5 (T suffix) H-1528

COS/MOS Operational Amplifiers

With MOS/FET Input

Features:

- MOS/FET input stage provides:
  very high Z_I = 1.5 TΩ (1.5 x 10^{12} Ω) typ.
  very low I_I = 5 pA typ. at 15 V operation
  2 pA typ. at 5 V operation
- Common-mode input-voltage range includes negative supply rail; input terminals can be swung 0.5 V below negative supply rail
- COS/MOS output stage permits signal swing to either (or both) supply rails

} Ideal for single-supply applications

- Low V_{IO}: 2 mV max. (CA3130B)
- Wide BW: 15 MHz typ. (unity-gain crossover)
- High SR: 10 V/μs typ. (unity-gain follower)
- High output current (I_O): 20 mA typ.
- High A_{OL}: 320,000 (110 dB) typ.
- Compensation with single external capacitor

Applications:

- Ground-referenced single-supply amplifiers
- Fast sample-hold amplifiers
- Long-duration timers/monostables
- High-input-impedance comparators
  (ideal interface with digital COS/MOS)
- High-input-impedance wideband amplifiers
- Voltage followers
  (e.g., follower for single-supply D/A converter)
- Voltage regulators
  (permits control of output voltage down to zero volts)
- Peak detectors
- Single-supply full-wave precision rectifiers
- Photo-diode sensor amplifiers

RCA-CA3130T, CA3130S, CA3130AT, CA3130AS, CA-3130BT, and CA3130BS are integrated-circuit operational amplifiers that combine the advantages of both COS/MOS and bipolar transistors on a monolithic chip.

Gate-protected p-channel MOS/FET (PMOS) transistors are used in the input circuit to provide very-high-input impedance, very-low-input current, and exceptional speed performance. The use of PMOS field-effect transistors in the input stage results in common-mode input-voltage capability down to 0.5 volt below the negative-supply terminal, an important attribute in single-supply applications.

A complementary-symmetry MOS (COS/MOS) transistor pair, capable of swinging the output voltage to within millivolts of either supply-voltage terminal (at very high values of load impedance), is employed as the output circuit.

The CA3130 Series circuits operate at supply voltages ranging from 5 to 16 volts, or ±2.5 to ±8 volts when using split supplies. They can be phase compensated with a single external capacitor, and have terminals for adjustment of offset voltage for applications requiring offset-null capability. Terminal provisions are also made to permit strobing of the output stage.

The CA3130 Series is supplied in either the standard 8-lead TO-5-style package (T suffix) or in the 8-lead dual-in-line formed-lead TO-5-style package "DIL-CAN" (S suffix) and operates over the full military-temperature range of −55°C to +125°C. The CA3130B is intended for applications requiring premium-grade specifications and with limits established for: input current, temperature coefficient of input-offset voltage, and gain over the range of −55°C to +125°C. The CA3130A offers superior input characteristics over those of the CA3130.

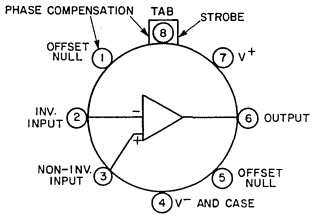


Fig. 1—Functional diagram of the CA3130 Series.

MAXIMUM RATINGS, *Absolute-Maximum Values*

DC SUPPLY VOLTAGE (BETWEEN V^+ AND V^- TERMINALS) ..... 16 V
DIFFERENTIAL-MODE INPUT VOLTAGE ..... ±8 V
COMMON-MODE DC INPUT VOLTAGE ..... V^+ to (V^--0.5 V)
INPUT-TERMINAL CURRENT ..... 1 mA
DEVICE DISSIPATION:
WITHOUT HEAT SINK–
UP TO 55°C ..... 630 mW
ABOVE 55°C ..... Derate linearly 6.67 mW/°C
WITH HEAT SINK–
AT 125°C ..... 418 mW
BELOW 125°C ..... Increase linearly at 16.7 mW/°C
TEMPERATURE RANGE:
OPERATING ..... –55 to +125°C
STORAGE ..... –65 to +150°C
OUTPUT SHORT-CIRCUIT DURATION* ..... INDEFINITE
LEAD TEMPERATURE (DURING SOLDERING):
AT DISTANCE 1/16 ± 1/32 INCH (1.59 ± 0.79 MM)
FROM CASE FOR 10 SECONDS MAX. .....+265°C

*Short circuit may be applied to ground or to either supply.

ELECTRICAL CHARACTERISTICS – For Equipment Design

CHARACTERISTIC	SYMBOL	TEST CONDITIONS V^+=15 V, V^-=0 V, T_A=25°C (Unless Specified Otherwise)	CA3130B			CA3130A			CA3130			UNITS	FIG. NO.
			Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.		
Input Offset Voltage	$\|V_{IO}\|$	$V^{\pm}$=±7.5 V	–	0.8	2	–	2	5	–	8	15	mV	–
Input Offset Current	$\|I_{IO}\|$	$V^{\pm}$=±7.5 V	–	0.5	10	–	0.5	20	–	0.5	30	pA	–
Input Current	I_I	$V^{\pm}$=±7.5 V	–	5	20	–	5	30	–	5	50	pA	–
Large-Signal Voltage Gain	A_{OL}	V_O=10 $V_{p\text{-}p}$	100 k	320 k	–	50 k	320 k	–	50 k	320 k	–	V/V	4,5
		R_L=2 kΩ	100	110	–	94	110	–	94	110	–	dB	
Common-Mode Rejection Ratio	CMRR		86	100	–	80	90	–	70	90	–	dB	–
Common-Mode Input-Voltage Range	V_{ICR}		0	–0.5 to 12	10	0	–0.5 to 12	10	0	–0.5 to 12	10	V	–
Power-Supply Rejection Ratio	$\Delta V_{IO}/\Delta V^+$	$V^{\pm}$=±7.5 V	–	32	100	–	32	150	–	32	320	μV/V	–
	$\Delta V_{IO}/\Delta V^-$		–	32	100	–	32	150	–	32	320		
Maximum Output Voltage	V_{OM}^+	R_L=2 kΩ	12	13.3	–	12	13.3	–	12	13.3	–	V	9
	V_{OM}^-		–	0.002	0.01	–	0.002	0.01	–	0.002	0.01		10
	$\|V_{OM}^+\|$	R_L=∞	14.99	15	–	14.99	15	–	14.99	15	–		9
	$\|V_{OM}^-\|$		–	0	0.01	–	0	0.01	–	0	0.01		10
Maximum Output Current: Source	I_{OM}^+	V_O=0 V	12	22	45	12	22	45	12	22	45	mA	9
Sink	I_{OM}^-	V_O=15 V	12	20	45	12	20	45	12	20	45		10
Supply Current	I^+	V_O=7.5 V, R_L=∞	–	10	15	–	10	15	–	10	15	mA	7,8
		V_O=0 V, R_L=∞	–	2	3	–	2	3	–	2	3		
Input Current	I_I	T_A=–55 to 125°C, $V^{\pm}$=±7.5 V▲, V_O=10 $V_{p\text{-}p}$*, R_L=2 kΩ*	–	Fig. 11	15	–	Fig. 11	–	–	Fig. 11	–	nA	–
Input Offset Voltage Temperature Drift	$\Delta V_{IO}/\Delta T$		–	5	15	–	10	–	–	10	–	μV/°C	–
Large-Signal Voltage Gain	A_{OL}		50 k	320 k	–	–	320 k	–	–	320 k	–	V/V	5
			94	110	–	–	110	–	–	110	–	dB	

* Applies only to A_{OL}.

▲ Applies only to I_I and $\Delta V_{IO}/\Delta T$.

TYPICAL VALUES INTENDED ONLY FOR DESIGN GUIDANCE

CHARACTERISTIC	SYMBOL	TEST CONDITIONS (V^+=+7.5 V, V^-=–7.5 V, T_A=25°C (Unless Specified Otherwise))	CA3130B	CA3130A	CA3130	UNITS	FIG. NO.
Input Offset Voltage Adjustment Range		10 kΩ across Terms. 4 and 5 or 4 and 1	±22	±22	±22	mV	–
Input Resistance	R_I		1.5	1.5	1.5	TΩ	–
Input Capacitance	C_I	f = 1 MHz	4.3	4.3	4.3	pF	–
Equivalent Input Noise	e_n	BW=0.2 MHz R_S=1 MΩ*	23	23	23	μV	14
Unity Gain Crossover Frequency	f_T	C_C = 0	15	15	15	MHz	4,15
		C_C = 47 pF	4	4	4		
Slew Rate: Open Loop	SR	C_C = 0	30	30	30	V/μs	–
Closed Loop		C_C = 56 pF	10	10	10		15
Transient Response: Rise Time	t_r	C_C = 56 pF, C_L = 25 pF, R_L = 2 kΩ (Voltage Follower)	0.09	0.09	0.09	μs	15
Overshoot			10	10	10	%	15
Settling Time (4 Vp-p Input to <0.1%)			1.2	1.2	1.2	μs	15

* Although a 1-MΩ source is used for this test, the equivalent input noise remains constant for sources of R_S up to 10 MΩ.

CHARACTERISTIC	SYMBOL	TEST CONDITIONS (V^+ = 5 V, V^- = 0 V, T_A = 25°C (Unless Specified Otherwise))	CA3130B	CA3130A	CA3130	UNITS	FIG. NO.
Input Offset Voltage	V_{IO}		1	2	8	mV	–
Input Offset Current	I_{IO}		0.1	0.1	0.1	pA	–
Input Current	I_I		2	2	2	pA	–
Common-Mode Rejection Ratio	CMRR		100	90	80	dB	–
Large-Signal Voltage Gain	A_{OL}	V_O = 4 Vp-p, R_L = 5 kΩ	100 k	100 k	100 k	V/V	–
			100	100	100	dB	–
Common-Mode Input Voltage Range	V_{ICR}		0 to 2.8	0 to 2.8	0 to 2.8	V	–
Supply Current	I^+	V_O = 5 V, R_L = ∞	300	300	300	μA	7,8
		V_O=2.5 V, R_L = ∞	500	500	500		
Power Supply Rejection Ratio	$\Delta V_{IO}/\Delta V^+$		200	200	200	μV/V	–

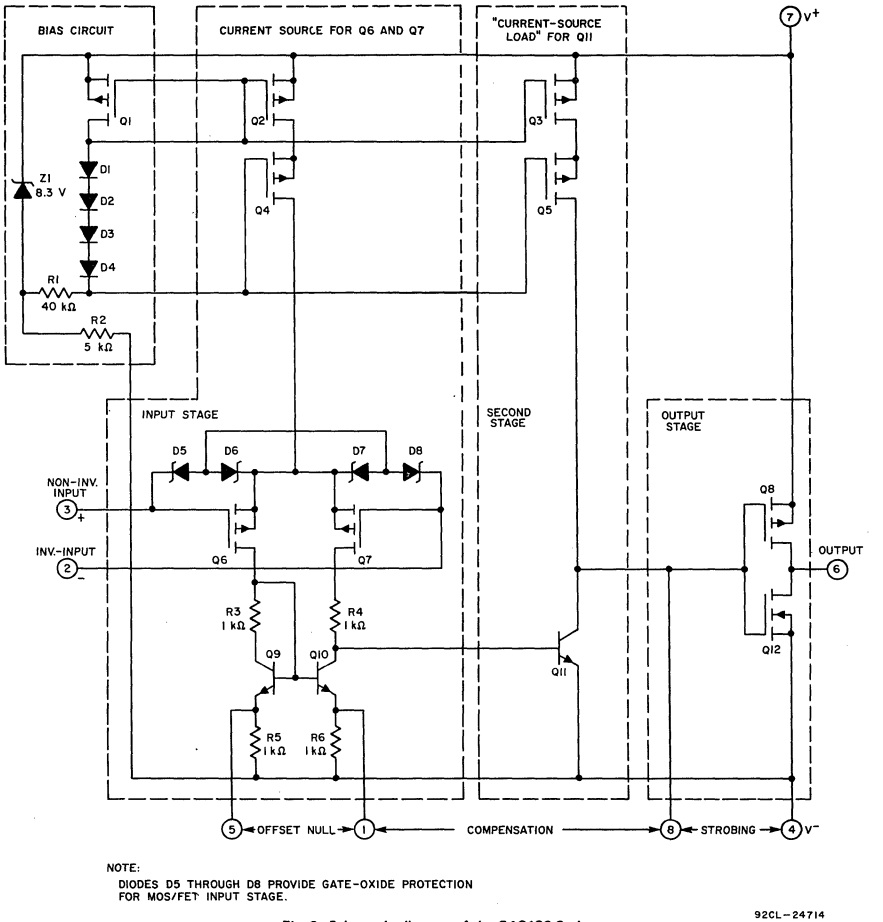


Fig. 2—Schematic diagram of the CA3130 Series.

CIRCUIT DESCRIPTION

Fig. 3 is a block diagram of the CA3130 Series COS/MOS Operational Amplifiers. The input terminals may be operated down to 0.5 V below the negative supply rail, and the output can be swung very close to either supply rail in many applications. Consequently, the CA3130 Series circuits are ideal for single-supply operation. Three Class A amplifier stages, having the individual gain capability and current consumption shown in Fig. 3, provide the total gain of the CA3130. A biasing circuit provides two potentials for common use in the first and second stages. Term. 8 can be used both for phase compensation and to strobe the output stage into quiescence. When Term. 8 is tied to the negative supply rail (Term. 4) by mechanical or electrical means, the output potential at Term. 6 essentially rises to the positive supply-rail potential at Term. 7. This condition of essentially zero current drain in the output stage under the strobed "OFF" condition can only be achieved when the ohmic load resistance presented to the amplifier is very high (e.g., when the amplifier output is used to drive COS/MOS digital circuits in comparator applications).

Input Stages—The circuit of the CA3130 is shown in Fig. 2. It consists of a differential-input stage using PMOS field-effect transistors (Q6, Q7) working into a mirror-pair of bipolar transistors (Q9, Q10) functioning as load resistors together with resistors R3 through R6. The mirror-pair transistors also function as a differential-to-single-ended converter to provide base drive to the second-stage bipolar transistor (Q11). Offset nulling, when desired, can be effected by connecting a 100,000-ohm potentiometer across Terms. 1 and 5 and the potentiometer slider arm to Term. 4. Cascode-connected PMOS

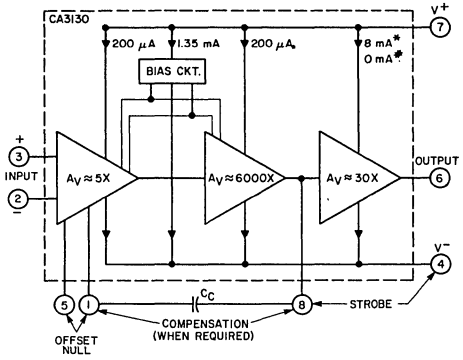


Fig. 3—Block diagram of the CA3130 Series.

transistors Q2, Q4 are the constant-current source for the input stage. The biasing circuit for the constant-current source is subsequently described. The small diodes D5 through D8 provide gate-oxide protection against high-voltage transients, e.g. including static electricity during handling for Q6 and Q7.

Second Stage—Most of the voltage gain in the CA3130 is provided by the second amplifier stage, consisting of bipolar transistor Q11 and its cascode-connected load resistance provided by PMOS transistors Q3 and Q5. The source of bias potentials for these PMOS transistors is subsequently described. Miller-Effect compensation (roll-off) is accomplished by simply connecting a small capacitor between Terms. 1 and 8. A 47-picofarad capacitor provides sufficient compensation for stable unity-gain operation in most applications.

Bias-Source Circuit—At total supply voltages, somewhat above 8.3 volts, resistor R2 and zener diode Z1 serve to establish a voltage of 8.3 volts across the series-connected circuit, consisting of resistor R1, diodes D1 through D4, and PMOS transistor Q1. A tap at the junction of resistor R1 and diode D4 provides a gate-bias potential of about 4.5 volts for PMOS transistors Q4 and Q5 with respect to Term. 7. A potential of about 2.2 volts is developed across diode-connected PMOS transistor Q1 with respect to Term. 7 to provide gate bias for PMOS transistors Q2 and Q3. It should be noted that Q1 is "mirror-connected"† to both Q2 and Q3. Since transistors Q2 and Q3 are twice the size of Q1, the approximate 100-microampere current in Q1 establishes 200-microampere "mirrored" currents in Q2 and Q3 as constant-current sources for the first and second amplifier stages, respectively.

At total supply voltages somewhat less than 8.3 volts, zener diode Z1 becomes non-conductive and the potential, developed across series-connected R1, D1-D4, and Q1, varies directly with variations in supply voltage. Consequently, the gate bias for Q4, Q5 and Q2, Q3 varies in accordance with supply-voltage variations. This variation results in deterioration of the power-supply-rejection ratio (PSRR) at total supply voltages below 8.3 volts. Operation at total supply voltages below about 4.5 volts results in seriously degraded performance.

Output Stage—The output stage consists of a drain-loaded inverting amplifier using COS/MOS transistors operating in the Class A mode. When operating into very high resistance loads, the output can be swung within millivolts of either supply rail. Because the output stage is a drain-loaded amplifier, its gain is dependent upon the load impedance. The transfer characteristics of the output stage for a load returned to the negative supply rail are shown in Fig. 6. Typical op-amp loads are readily driven by the output stage. Because large-signal excursions are non-linear, requiring feedback for good waveform reproduction, transient delays may be encountered. As a voltage follower, the amplifier can achieve 0.01 per cent accuracy levels, including the negative supply rail.

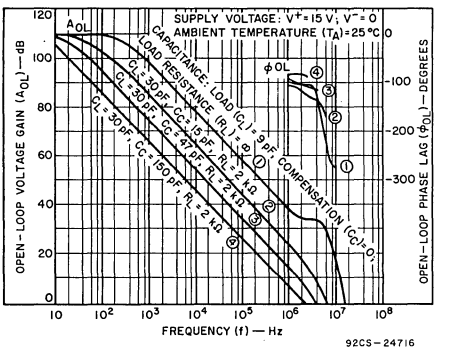


Fig. 4—Open-loop voltage gain and phase shift vs. frequency for various values of C_L, C_C, and R_L.

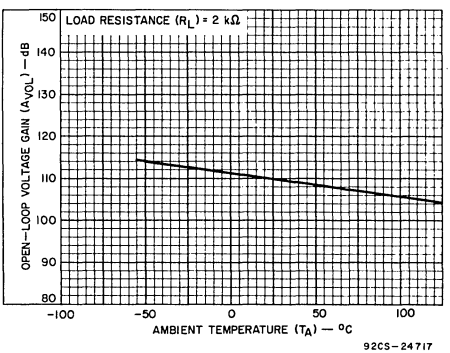


Fig. 5—Open-loop gain vs. temperature.

†For general information on the characteristics of COS/MOS transistor pairs in linear-circuit applications, see File No. 619, data bulletin on CA3600E "COS/MOS Transistor Array."

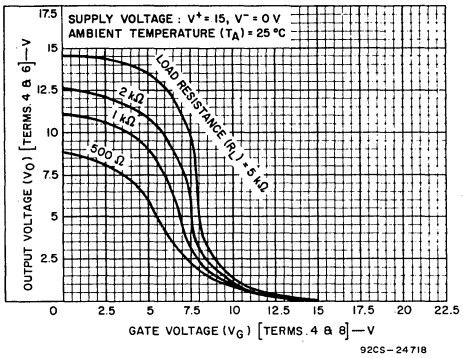


Fig. 6—Voltage transfer characteristics of COS/MOS output stage.

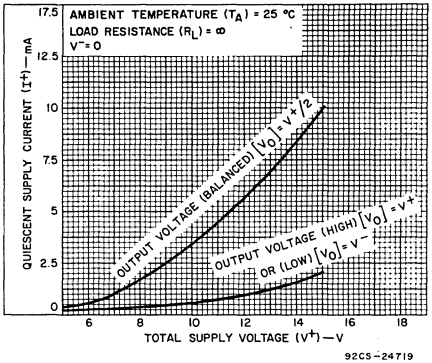


Fig. 7—Quiescent supply current vs. supply voltage.

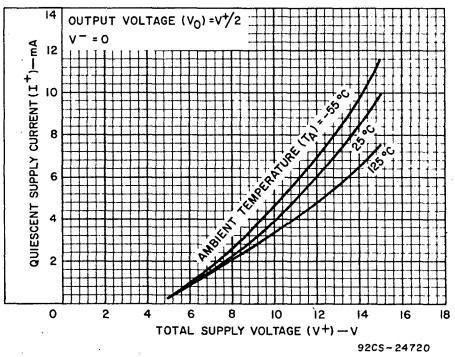


Fig. 8—Quiescent supply current vs. supply voltage at several temperatures.

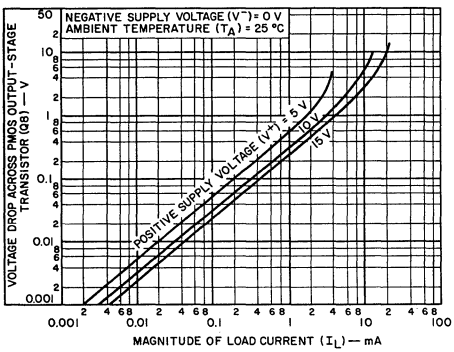


Fig. 9—Voltage across PMOS output transistor (Q8) vs. load current.

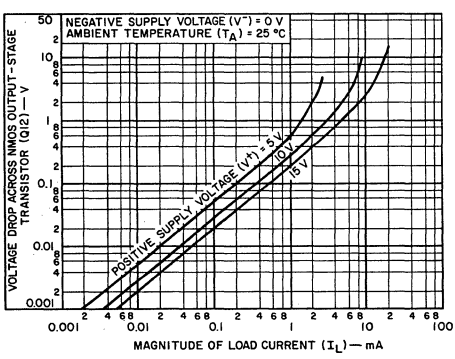


Fig. 10—Voltage across NMOS output transistor (Q12) vs. load current.

HANDLING AND OPERATING CONSIDERATIONS

Handling Considerations

The CA3130 uses MOS field-effect transistors in the input circuit. Because MOS/FET's have extremely high input resistances, they are susceptible to damage when exposed to extremely high static electrical charges. To minimize the possibilities of damaging the input stage transistors, Q6 and Q7, the CA3130 utilizes a protective diode network in the input stage. Nevertheless, it is good practice that the following precautions be observed during handling, testing, and actual operation of the CA3130 devices to minimize exposure to damage-inducing hazards:

1. Soldering-iron tips, metal parts of fixtures, tools, and handling facilities should be grounded.
2. Devices should not be inserted into or removed from circuits with the power ON because transient voltages may cause damage.
3. Signals should not be applied to the input (Terms. 2 and 3) when the device power supply is OFF. Input-terminal currents should not exceed 1 mA.
4. After CA3130 devices have been mounted on circuit boards, proper handling precautions should still be observed if the input terminals are unterminated. It is good practice during board-processing operations to return Terms. 2 and 3 to Term. 4 by jumping the appropriate conductors.

Offset Nulling

Offset-voltage nulling is usually accomplished with a 100,000-ohm potentiometer connected across Terms. 1 and 5 and with the potentiometer slider arm connected to Term. 4. A fine offset-null adjustment usually can be effected with the slider arm positioned in the mid-point of the potentiometer's total range.

Input-Current Variation with Temperature

The input current of the CA3130 Series circuits is typically 5 pA at 25°C. The major portion of this input current is due to leakage current through the gate-protective diodes in the input circuit. As with any semiconductor-junction device, including op amps with a junction-FET input stage, the leakage current approximately doubles for every 10°C increase in temperature. Fig. 11 provides data on the typical variation of input bias current as a function of temperature in the CA3130.

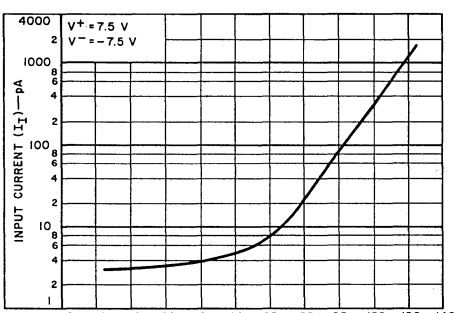


Fig. 11—Input current vs. ambient temperature.

In applications requiring the lowest practical input current and incremental increases in current because of "warm-up" effects, it is suggested that an appropriate heat sink be used with the CA3130. In addition, when "sinking" or "sourcing" significant output current the chip temperature increases, causing an increase in the input current. In such cases, heat-sinking can also very markedly reduce and stabilize input current variations.

Input-Offset-Voltage (V_{IO}) Variation with DC Bias vs. Device Operating Life

It is well known that the characteristics of a MOS/FET device can change slightly when a dc gate-source bias potential is applied to the device for extended time periods. The magnitude of the change is increased at high temperatures. Users of the CA3130 should be alert to the possible impacts of this effect if the application of the device involves extended operation at high temperatures with a significant differential dc bias voltage applied across Terms. 2 and 3. Fig. 12 shows typical data pertinent to shifts in offset voltage encountered with CA3130 devices during life testing. The two-volt dc differential voltage example represents conditions when the amplifier output stage is "toggled", e.g., as in comparator applications.

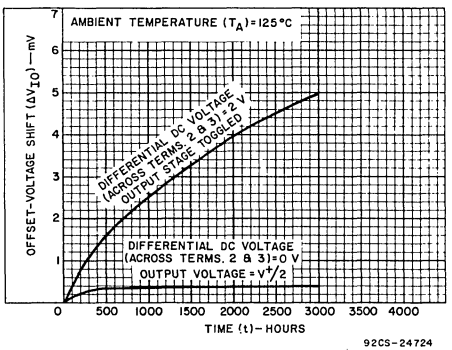


Fig. 12—Typical incremental offset-voltage shift vs. operating life.

Power-Supply Considerations

Because the CA3130 is very useful in single-supply applications, it is pertinent to review some considerations relating to power-supply current consumption under both single and dual-supply service. Figs. 13a and 13b show the CA3130 connected for both dual- and single-supply operation.

Dual-supply operation: When the output voltage at Term. 6 is zero-volts, the currents supplied by the two power supplies are equal. When the gate terminals of Q8 and Q12 are driven increasingly positive with respect to ground, current flow through Q12 (from the negative supply) to the load is increased and current flow through Q8 (from the positive supply) decreases correspondingly. When the gate terminals of Q8 and Q12 are driven increasingly negative with respect to ground, current flow through Q8 is increased and current flow through Q12 is decreased accordingly.

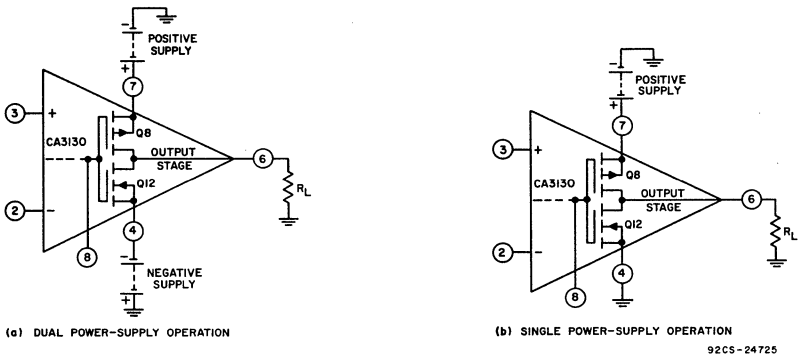


Fig. 13—CA3130 output stage in dual and single power-supply operation.

Single-supply operation: Initially, let it be assumed that the value of R_L is very high (or disconnected), and that the input-terminal bias (Terms. 2 and 3) is such that the output terminal (No. 6) voltage is at $V^+/2$, i.e., the voltage-drops across Q8 and Q12 are of equal magnitude. Fig. 7 shows typical quiescent supply-current vs. supply-voltage for the CA3130 operated under these conditions. Since the output stage is operating as a Class A amplifier, the supply-current will remain constant under dynamic operating conditions as long as the transistors are operated in the linear portion of their voltage-transfer characteristics (see Fig. 6). If either Q8 or Q12 are swung out of their linear regions toward cut-off (a non-linear region), there will be a corresponding reduction in supply-current. In the extreme case, e.g., with Term. 8 swung down to ground potential (or tied to ground), NMOS transistor Q12 is completely cut off and the supply-current to series-connected transistors Q8, Q12 goes essentially to zero. The two preceding stages in the CA3130, however, continue to draw modest supply-current (see the lower curve in Fig. 7) even though the output stage is strobed off. Fig. 13a shows a dual-supply arrangement for the output stage that can also be strobed off, assuming $R_L = \infty$, by pulling the potential of Term. 8 down to that of Term. 4.

Let it now be assumed that a load-resistance of nominal value (e.g., 2 kilohms) is connected between Term. 6 and ground in the circuit of Fig. 13b. Let it further be assumed again that the input-terminal bias (Terms. 2 and 3) is such that the output terminal (No. 6) voltage is at $V^+/2$. Since PMOS transistor Q8 must now supply quiescent current to both R_L and transistor Q12, it should be apparent that under these conditions the supply-current must increase as an inverse function of the R_L magnitude. Fig. 9 shows the voltage-drop across PMOS transistor Q8 as a function of load current at several supply-voltages. Fig. 6 shows the voltage-transfer characteristics of the output stage for several values of load resistance.

Wideband Noise

From the standpoint of low-noise performance considerations, the use of the CA3130 is most advantageous in applications where in the source resistance of the input signal is in the order of 1 megohm or more. In this case, the total input-referred noise voltage is typically only 23 μV when the test-circuit amplifier of Fig. 14 is operated at a total supply voltage of 15 volts. This value of total input-referred noise remains essentially constant, even though the value of source resistance is raised by an order of magnitude. This characteristic is due to the fact that reactance of the input capacitance becomes a significant factor in shunting the source resistance. It should be noted, however, that for values of source resistance very much greater than 1 megohm, the total noise voltage generated can be dominated by the thermal noise contributions of both the feedback and source resistors.

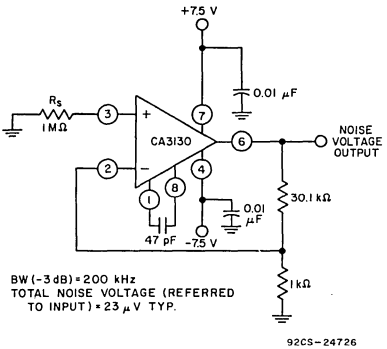
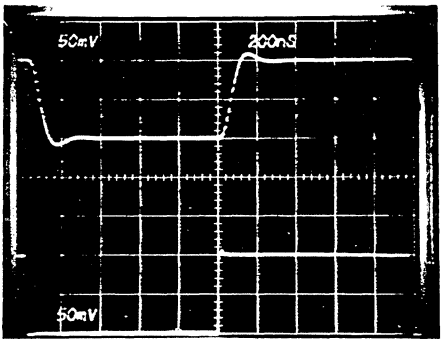
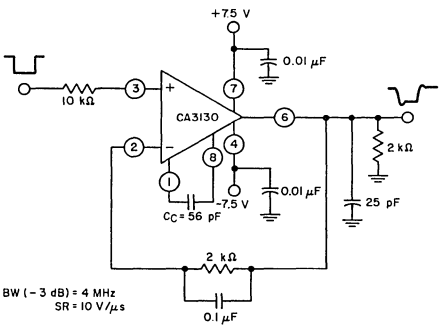


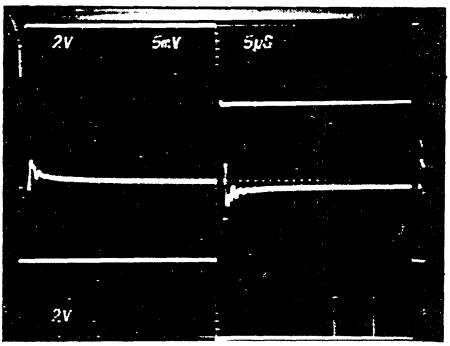
Fig. 14—Test-circuit amplifier (30-dB gain) used for wideband noise measurements.



Top Trace: Output
Bottom Trace: Input

(a) Small-signal response (50 mV/div. and 200 ns/div.)





Top Trace: Output signal (2 V/div. and 5 μs/div.)
Center Trace: Difference signal (5 mV/div. and 5 μs/div.)
Bottom Trace: Input signal (2 V/div. and 5 μs/div.)

(b) Input-output difference signal showing settling time (Measurement made with Tektronix 7A13 differential amplifier)

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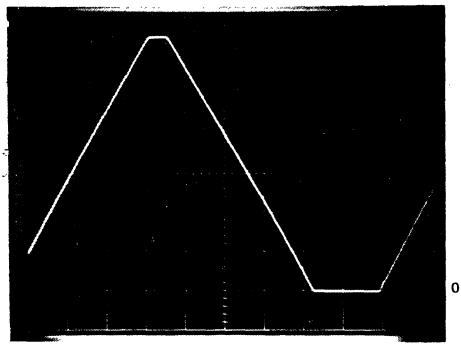
Fig. 15—Split-supply voltage follower with associated waveforms.

TYPICAL APPLICATIONS

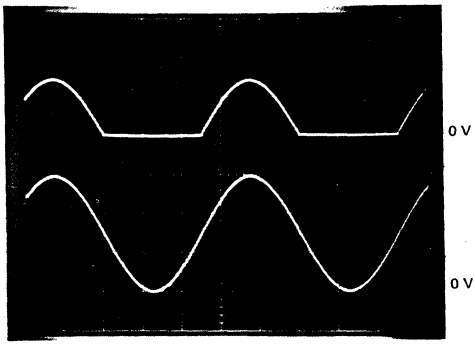
Voltage Followers

Operational amplifiers with very high input resistances, like the CA3130, are particularly suited to service as voltage followers. Fig. 15 shows the circuit of a classical voltage follower, together with pertinent waveforms using the CA3130 in a split-supply configuration.

A voltage follower, operated from a single supply, is shown in Fig. 16, together with related waveforms. This follower circuit is linear over a wide dynamic range, as illustrated by the reproduction of the output waveform in Fig. 16a with input-signal ramping. The waveforms in Fig. 16b show that the follower does not lose its input-to-output phase-sense, even though the input is being swung 7.5 volts below ground potential. This unique characteristic is an important attribute in both operational amplifier and comparator applications. Fig. 16b also shows the manner in which the COS/MOS output stage permits the output signal to swing down to the negative supply-rail potential (i.e., ground in the case shown). The digital-to-analog converter (DAC) circuit, described in the following section, illustrates the practical use of the CA3130 in a single-supply voltage-follower application.



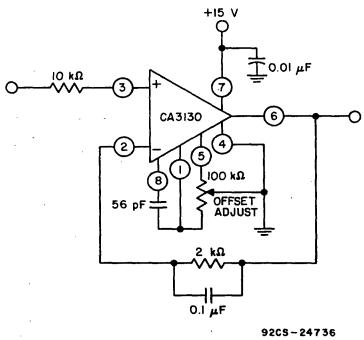
(a) Output-waveform with input-signal ramping (2 V/div. and 500 μs/div.)



Top Trace: Output (5 V/div. and 200 μs/div.)
Bottom Trace: Input (5 V/div. and 200 μs/div.)

(b) Output-waveform with ground-reference sine-wave input

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Fig. 16—Single-supply voltage-follower with associated waveforms. (e.g., for use in single-supply D/A converter; see Fig. 9 in ICAN-6080).

9-BIT COS/MOS DAC

A typical circuit of a 9-bit Digital-to-Analog Converter (DAC)* is shown in Fig. 17. This system combines the concepts of multiple-switch COS/MOS IC's, a low-cost ladder network of discrete metal-oxide-film resistors, a CA3130 op-amp connected as a follower, and an inexpensive monolithic regulator in a simple single power-supply arrangement. An additional feature of the DAC is that it is readily interfaced with COS/MOS input logic, e.g., 10-volt logic levels are used in the circuit of Fig. 17.

The circuit uses an R/2R voltage-ladder network, with the output potential obtained directly by terminating the ladder arms at either the positive or the negative power-supply terminal. Each CD4007A contains three "inverters", each "inverter" functioning as a single-pole double-throw switch to terminate an arm of the R/2R network at either the positive or negative power-supply terminal. The resistor ladder is an assembly of one per cent tolerance metal-oxide film resistors. The five arms requiring the highest accuracy are assembled with series and parallel combinations of 806,000-ohm resistors from the same manufacturing lot.

A single 15-volt supply provides a positive bus for the CA3130 follower amplifier and feeds the CA3085 voltage regulator. A "scale-adjust" function is provided by the regulator output

*"Digital-to-Analog Conversion Using the RCA-CD4007A COS/MOS IC," Application Note ICAN-6080.

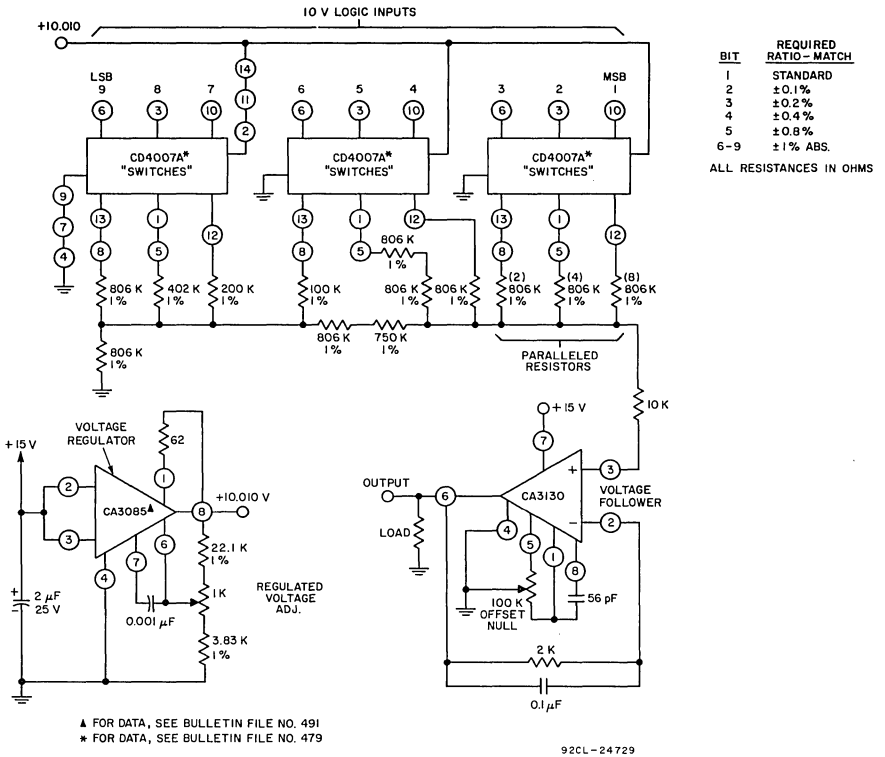


Fig. 17–9-bit DAC using COS/MOS digital switches and CA3130.

control, set to a nominal 10-volt level in this system. The line-voltage regulation (approximately 0.2%) permits a 9-bit accuracy to be maintained with variations of several volts in the supply. The flexibility afforded by the COS/MOS building blocks simplifies the design of DAC systems tailored to particular needs.

Single-Supply, Absolute-Value, Ideal Full-Wave Rectifier

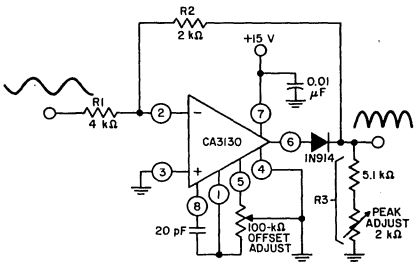
The absolute-value circuit using the CA3130 is shown in Fig. 18. During positive excursions, the input signal is fed through the feedback network directly to the output. Simultaneously, the positive excursion of the input signal also drives the output terminal (No. 6) of the inverting amplifier in a negative-going excursion such that the 1N914 diode effectively disconnects the amplifier from the signal path. During a negative-going excursion of the input signal, the CA3130 functions as a normal inverting amplifier with a gain equal to $-R2/R1$. When the equality of the two equations shown in Fig. 18 is satisfied, the full-wave output is symmetrical.

Peak Detectors

Peak-detector circuits are easily implemented with the CA-3130, as illustrated in Fig. 19 for both the peak-positive and the peak-negative circuit. It should be noted that with large-signal inputs, the bandwidth of the peak-negative circuit is much less than that of the peak-positive circuit. The second stage of the CA3130 limits the bandwidth in this case. Negative-going output-signal excursion requires a positive-going signal excursion at the collector of transistor Q11, which is loaded by the intrinsic capacitance of the associated circuitry in this mode. On the other hand, during a negative-going signal excursion at the collector of Q11, the transistor functions in an active "pull-down" mode so that the intrinsic capacitance can be discharged more expeditiously.

Error-Amplifier in Regulated Power Supplies

The CA3130 is an ideal choice for error-amplifier service in regulated power supplies since it can function as an error-amplifier when the regulated output voltage is required to



$$\text{GAIN} = \frac{R2}{RI} = X = \frac{R3}{RI + R2 + R3}$$

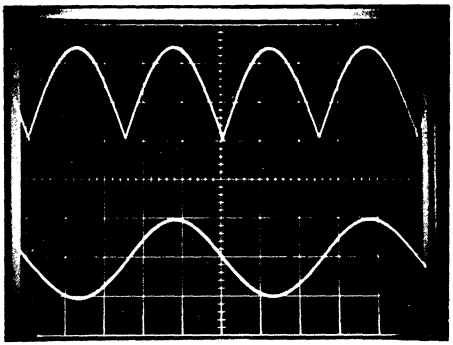
$$R3 = RI\left(\frac{X + X^2}{1 - X}\right)$$

$$\text{FOR } X = 0.5: \frac{2\text{ k}\Omega}{4\text{ k}\Omega} = \frac{R2}{RI}$$

$$R3 = 4\text{ k}\Omega\left(\frac{0.75}{0.5}\right) = 6\text{ k}\Omega$$

20 V p-p INPUT: BW(-3dB) = 230 kHz, DC OUTPUT (AVG.) = 3.2 V
1 VOLT p-p INPUT: BW(-3dB) = 130 kHz, DC OUTPUT (AVG.) = 160 mV

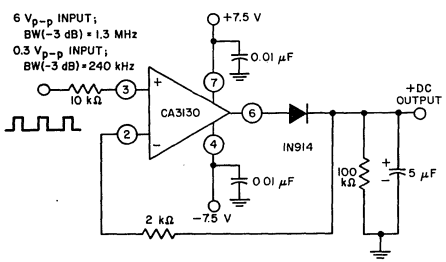
92CS-24730



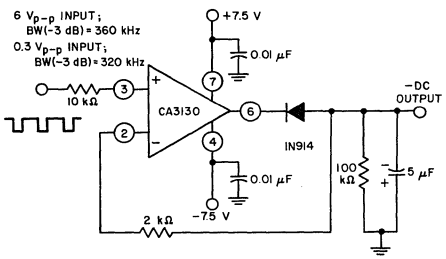
Top Trace: Output signal (2 V/div.)
Bottom Trace: Input signal (10 V/div.)
Time base on both traces: 0.2 ms/div.

92CS-24738

Fig. 18—Single-supply, absolute-value, ideal full-wave rectifier with associated waveforms.



(a) PEAK POSITIVE DETECTOR CIRCUIT



(b) PEAK NEGATIVE DETECTOR CIRCUIT

92CS-24731

Fig. 19—Peak-detector circuits.

approach zero. Fig. 20 shows the schematic diagram of a 40-mA power-supply capable of providing regulated output voltage by continuous adjustment over the range from 0 to 13 volts. Q3 and Q4 in IC2 (a CA3086 transistor-array IC) function as zeners to provide supply-voltage for the CA3130 comparator (IC1). Q1, Q2, and Q5 in IC2 are configured as a low impedance, temperature-compensated source of adjustable reference voltage for the error amplifier. Transistors Q1, Q2, Q3, and Q4 in IC3 (another CA3086 transistor-array IC) are connected in parallel as the series-pass element. Transistor Q5 in IC3 functions as a current-limiting device by diverting base drive from the series-pass transistors, in accordance with the adjustment of resistor R2.

Fig. 21 contains the schematic diagram of a regulated power-supply capable of providing regulated output voltage by continuous adjustment over the range from 0.1 to 50 volts and currents up to 1 ampere. The error amplifier (IC1) and circuitry associated with IC2 function as previously described, although the output of IC1 is boosted by a discrete transistor (Q4) to provide adequate base drive for the Darlington-connected series-pass transistors Q1, Q2. Transistor Q3 functions in the previously described current-limiting circuit.

Multivibrators

The exceptionally high input resistance presented by the CA3130 is an attractive feature for multivibrator circuit design because it permits the use of timing circuits with high R/C ratios. The circuit diagram of a pulse generator (astable multivibrator), with provisions for independent control of the "on" and "off" periods, is shown in Fig. 22. Resistors R1 and R2 are used to bias the CA3130 to the mid-point of the supply-voltage and R3 is the feedback resistor.

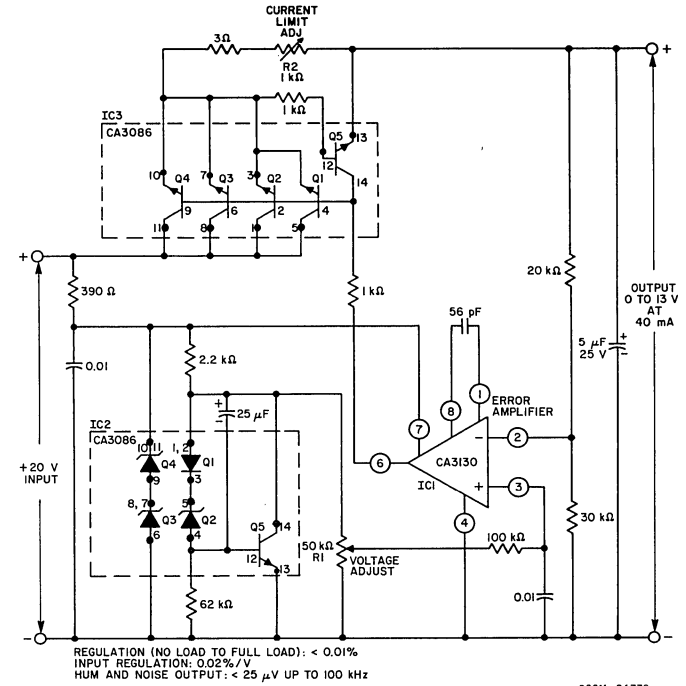


Fig. 20—Voltage regulator circuit (0 to 13 V at 40 mA).

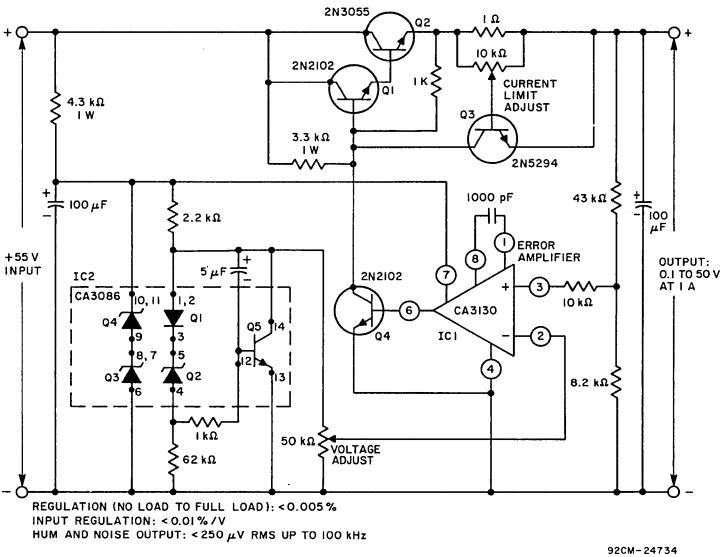


Fig. 21—Voltage regulator circuit (0.1 to 50 V at 1 A).

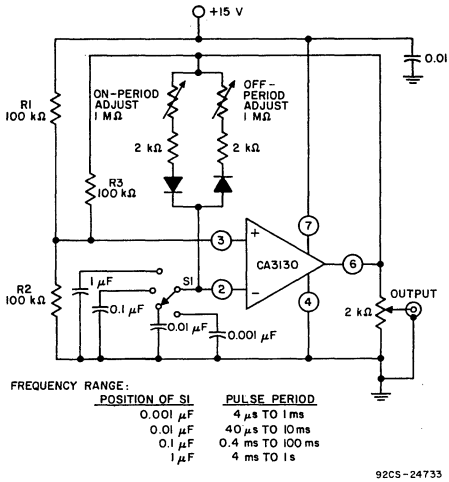


Fig. 22—Pulse generator (astable multivibrator) with provisions for independent control of "ON" and "OFF" periods.

Function Generator

Fig. 23 contains the schematic diagram of a function generator using the CA3130 in the integrator and threshold detector functions. This circuit generates a triangular or square-wave output that can be swept over a 1,000,000:1 range (0.1 Hz to 100 kHz) by means of a single control, R1. A voltage-control input is also available for remote sweep-control.

The heart of the frequency-determining system is an operational-transconductance-amplifier (OTA)*, IC1, operated as a voltage-controlled current-source. The output, I_O, is a current applied directly to the integrating capacitor, C1, in the feedback loop of the integrator IC2, using a CA3130, to provide the triangular-wave output. Potentiometer R2 is used to adjust the circuit for slope symmetry of positive-going and negative-going signal excursions.

Another CA3130, IC3, is used as a controlled switch to set the excursion limits of the triangular output from the integrator circuit. Capacitor C2 is a "peaking adjustment" to optimize the high-frequency square-wave performance of the circuit.

Potentiometer R3 is adjustable to perfect the "amplitude symmetry" of the square-wave output signals. Output from the threshold detector is fed back via resistor R4 to the input of IC1 so as to toggle the current source from plus to minus in generating the linear triangular wave.

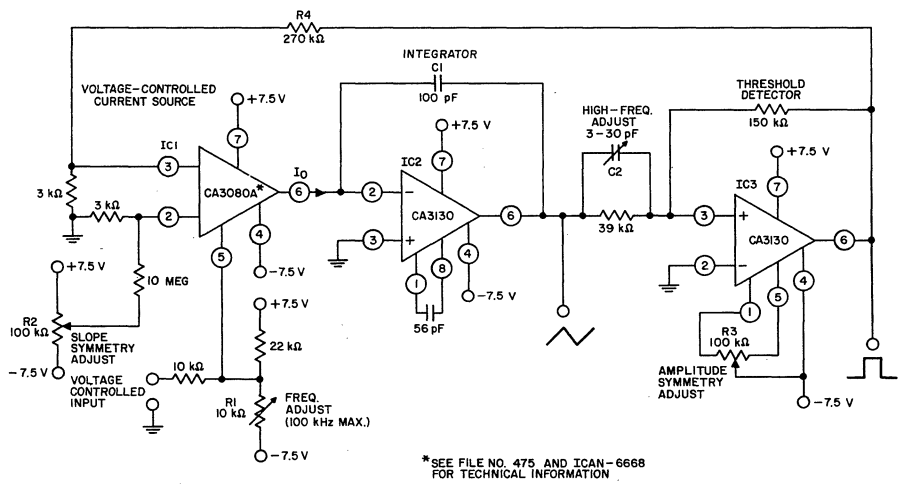


Fig. 23—Function generator (frequency can be varied 1,000,000/1 with a single control).

*See File No. 475 and ICAN-6668.

Operation with Output-Stage Power-Booster

The current-sourcing and -sinking capability of the CA3130 output stage is easily supplemented to provide power-boost capability. In the circuit of Fig. 24, three COS/MOS transistor-pairs in a single CA3600E* IC array are shown parallel connected with the output stage in the CA3130. In the Class A mode of CA3600E shown, a typical device consumes 20 mA of supply current at 15 V operation. This arrangement boosts the current-handling capability of the CA3130 output stage by about 2.5x.

The amplifier circuit in Fig. 24 employs feedback to establish a closed-loop gain of 48 dB. The typical large-signal bandwidth (−3 dB) is 50 kHz.

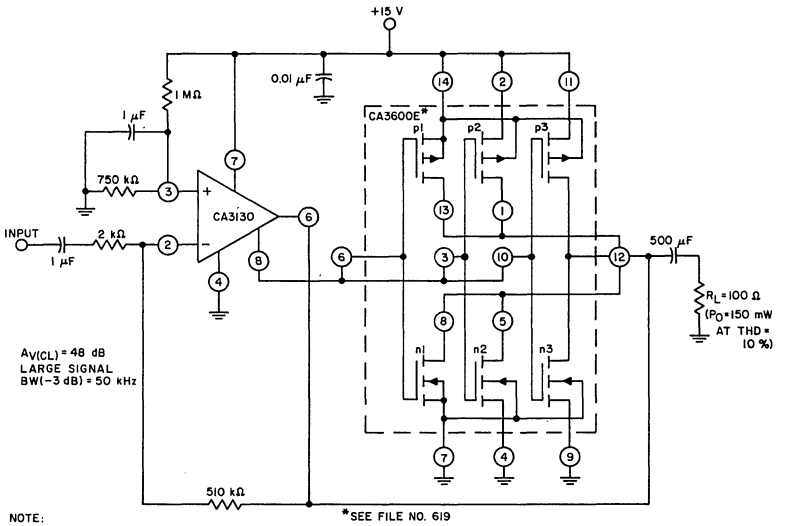


Fig. 24—COS/MOS transistor array (CA3600E) connected as power-booster in the output stage of the CA3130.

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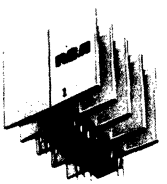
Linear Integrated Circuits

Monolithic Silicon

CA3131EM

CA3132EM

Preliminary Data



Modified
16-Lead Dual-in-Line
Plastic Package With
Heat Sink H-1827

5-Watt Audio Amplifiers

With Integral Heat Sink

Features:

- **Power Output: 4 W min., 5 W typ.**
- **Complete amplifier including: preamplifier stages, power-output amplifier, and integral heat sink**
- **High power-supply rejection ratio**
- **Operating voltage: $V^+ = 24$ V typ.**
- **Available with internal feedback (CA3131EM) or without feedback (CA3132EM)**

RCA-CA3131EM and CA3132EM are audio amplifiers with integral preamplifier stages on single integrated-circuit monolithic chips.

Utilizing a uniquely designed package with an integral heat sink, these devices can provide a power-output signal in excess of five watts at an ambient temperature of 25°C.

The CA3131EM employs an internal feedback network that sets the over-all gain of the amplifier to typically 48 dB.

The CA3132EM omits the internal feedback network. This arrangement offers the circuit designer a wide latitude in the choice of an external feedback network more suitable to a specific application.

Both types are encapsulated in a 16-lead dual-in-line plastic package with 4 center leads removed.

The CA3131EM and CA3132EM are electrically equivalent to and pin compatible with types SN76013 and SN76023, respectively.

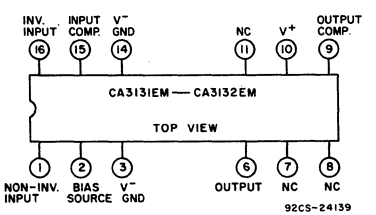


Fig. 1—Terminal assignment of the CA3131EM and CA3132EM.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

SUPPLY VOLTAGE, V^+	28 V
CONTINUOUS OUTPUT POWER, P_O (with $R_L = 8\ \Omega$ and $V^+ = 24$ V)	8 W RMS
MINIMUM RECOMMENDED LOAD IMPEDANCE, R_L	8 Ω
AMBIENT OPERATING TEMPERATURE, T_A (at 6 W RMS Output Power)	70 °C
STORAGE TEMPERATURE RANGE	−55 to +150 °C

ELECTRICAL CHARACTERISTICS at $T_A = 25°C$, $V^+ = 24$ V

Characteristic	Symbol	Conditions	Values		Unit
			Min.	Typ.	
Input Impedance	Z_I		200k	—	Ω
Power Output	P_O	At clipping onset			
		$R_L = 8\ \Omega$	4	—	W
		$R_L = 16\ \Omega$	3	—	W
Closed-Loop Gain — CA3131EM	A	f = 1 kHz	46	48	dB
Supply Current	I^+	Zero signal	—	10	mA
Total Harmonic Distortion	THD	$P_O = 50$ mW—4 W, $R_L = 8\ \Omega$	—	1	%
		$P_O = 50$ mW—3 W, $R_L = 15\ \Omega$	—	1	%
Noise Voltage	V_n	f=20 Hz—20 kHz	—	1.5	mV RMS

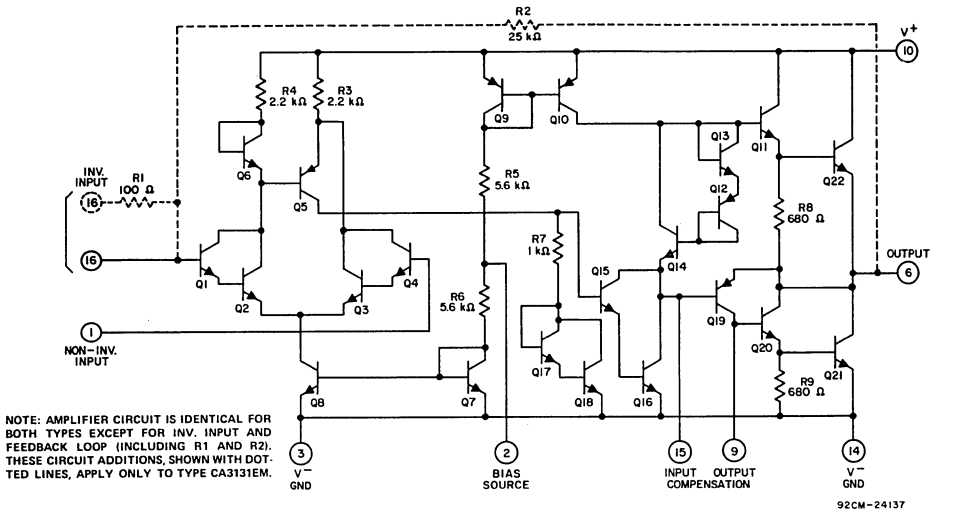
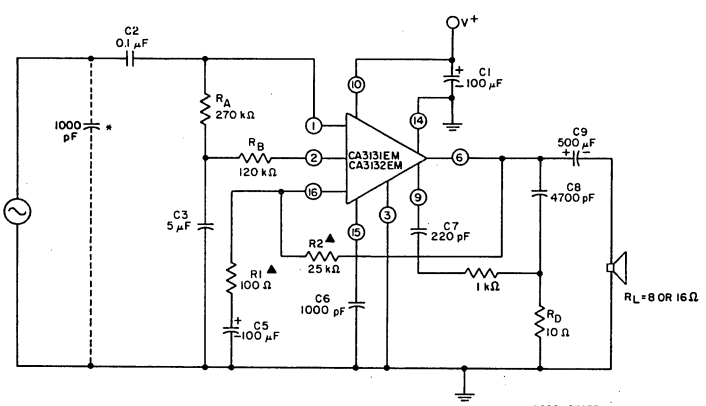


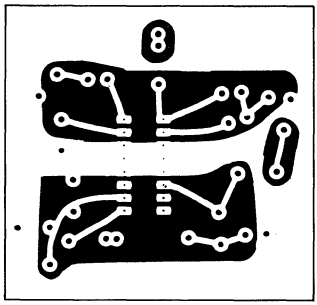
Fig. 2–Schematic diagram of types CA3131EM and CA3132EM.



* A 1000-pF capacitor is required if input has an open circuit.

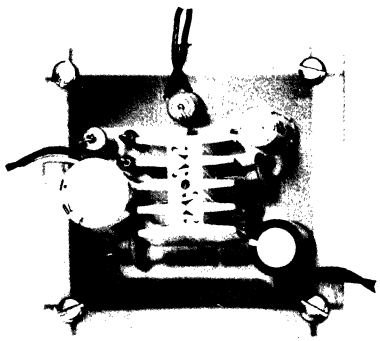
▲ External resistors R1 and R2 are used only with the CA3132EM. When testing the CA3131EM, omit R1 and R2 and connect the (+) termination of C5 to Terminal 16.

Fig. 3–Test circuit for types CA3131EM and CA3132EM.



92CS-24974

a. Bottom view of printed-circuit board.



92CS-24921

b. Component view of printed-circuit board - top view.

Fig. 4—Printed-circuit board containing the test circuit, shown in Fig. 3, for the CA3131EM.

Determining External Component Values (Refer to Figs. 2 & 3)

The dc quiescent output voltage is set by the voltage at Terminal 1. This voltage, in turn, is set by the internal voltage at Terminal 2 less I_I (input current, fixed by $R_A + R_B$, for Q4). The voltage at Terminal 2 is set slightly above half the supply voltage to allow for the voltage drop across $R_A + R_B$. Filter R_BC3 attenuates any ac ripple injected from the supply line and prevents positive feedback to Terminal 1. The rejection of supply voltage is a direct function of the filter attenuation.

The input impedance of the audio amplifiers is a function of the closed-loop gain and the magnitude of the Q8 current. In practice the input impedance is well above 1 megohm. The input signal, applied through C2, sees an impedance equivalent to the resistance of R_A connected in parallel with the amplifier input impedance. Hence, the value of R_A in most cases is dominant in establishing the input signal impedance.

The value of C1 depends on the regulation of the power supply. It is possible for the amplifier to work with a value of C1 as low as 0.1 μF to attenuate high-frequency signals in the supply line. Ideally, C1 should be placed as near Terminal 10 as possible. An electrolytic capacitor should be used for C1 if the power supply is poorly regulated to avoid ripple at the output.

Capacitor C6 at Terminal 15 provides over-all compensation. If a 1000-pF capacitor is used for C6, then the first breakpoint for a 46-dB closed-loop gain occurs at 200 kHz. Higher capacitance values will cause the constant current from Q10 to charge C6 on the positive voltage swing and thus limit the slew rate at high-signal levels. Because p-n-p transistor Q19 has a lower gain-bandwidth product (f_T) than the n-p-n transistors, C7 is connected to Terminal 9 to compensate for gain losses occurring in the negative voltage swings.

The use of the filter networks C8 and R_D at the output Terminal 6 is a standard requirement for class B audio outputs driving reactive speaker loads. Capacitor C8 compensates for the speaker inductance and R_D limits the current surges through C8.

The value of the coupling capacitor C9 to the load determines the low-frequency response of the amplifier.

Closed-Loop Gain

The closed-loop gain for either type is set by the ratio (R1 + R2)/R1. These resistors are included in the CA3131EM circuit and are external when used with the CA3132EM. In either type, the low-frequency value (−3 dB point) is reached when the impedance of C5 equals the value of R1.

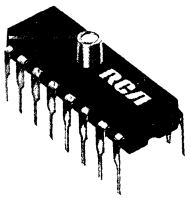
RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

Preliminary Data

CA3134E



16-Lead Plastic "Power-Stud" Dual-In-Line Package (E Suffix) H-1828

TV Sound IF and Audio Output Subsystem

Features:

- Nominal power output: 3 W (with suitable heat sink)
- Power amplifier with current limiting and thermal shutdown
- Wide power-supply range: 12 V to 40 V (30 V nominal)
- Low quiescent current: 30 mA typ.
- 5-kHz deviation sensitivity: 1 W output typ.
- 3-dB limiting sensitivity: 200 μV typ.
- Excellent AM rejection
- Differential peak detector—requires one tuned coil
- Electronic volume control with improved taper
- Optional unattenuated audio output

The RCA-CA3134E* Television Sound System is a monolithic integrated circuit which includes a multistage IF amplifier-limiter, an FM detector, an electronic attenuator, and an audio power amplifier that is designed to drive an 8-, 16-, or 32-ohm speaker. The CA3134E is encapsulated in a 16-lead plastic "power-stud" dual-in-line package that lends itself to a wide variety of techniques for mounting heat sinks.

* Formerly Dev. No. TA6480.

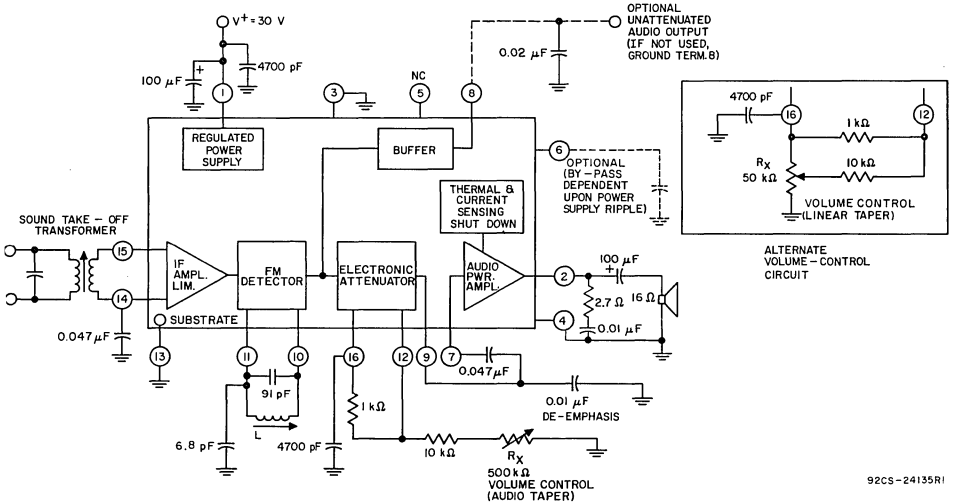


Fig. 1—Block diagram of the CA3134E in a typical circuit application.

MAXIMUM RATINGS, *Absolute-Maximum Values*

DC SUPPLY VOLTAGE (Between Term. 1, V^+ and Terms. 4, audio-output ground and 13, substrate) ..... 40 V

DEVICE DISSIPATION:

With Infinite Heat Sink—

Up to T_A = 70°C ..... 6.5 W

Above T_A = 70°C ..... derate linearly 83.3 mW/°C

With No Heat Sink—

Up to T_A = 25°C ..... 1.4 W

Above T_A = 25°C ..... derate linearly 11.1 mW/°C

THERMAL RESISTANCE (Junction to Stud) ..... 12°C/W

AMBIENT TEMPERATURE RANGE:

Operating ..... −40 to +85°C

Storage ..... −65 to +150°C

LEAD TEMPERATURE (During Soldering):

At a distance 1/16 in. ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max. ..... +265°C

ELECTRICAL CHARACTERISTICS

Test Conditions: T_A = 25°C; V^+ = +30 V (applied to Term. 1); DC Volume Control, R_X = 500 kΩ; R_L = 16 Ω; unless otherwise indicated. Refer to Fig. 1.

CHARACTERISTIC	SYMBOL	SPECIAL TEST CONDITIONS	NOMINAL VALUE	UNITS
Static Characteristics				
Current into Term. 1	I_1	P_O = 0	30	mA
Dynamic Characteristics				
IF Amplifier Input Limiting Voltage (at −3 dB point)	V_{15}(lim)	f_O = 4.5 MHz f_m = 400 Hz Δf = ±25 kHz	200	μV
AM Rejection	AMR	f_O = 4.5 MHz, f_m = 400 Hz, Modulation Index = 0.3, V_{15} = 20 mV	50	dB
Detector Recovered AF Voltage (Term. 9)	V_o(af)	f_O = 4.5 MHz, f_m = 400 Hz,	600	mV
Total Harmonic Distortion	THD	Δf = ±25 kHz, V_{15} = 100 mV	0.8	%
Attenuator Maximum Attenuation		R_X = 0	75●	dB
Unattenuated Audio Recovered AF Voltage (Term. 8)	V_o(af)	f_O = 4.5 MHz, f_m = 400 Hz,	−1.5#	dB
Total Harmonic Distortion	THD	Δf = ±25 kHz, V_{15} = 100 mV	0.8	%
Audio Power Amplifier Voltage Gain	A(af)	f = 1 kHz	35	dB
System Total Harmonic Distortion	THD(System)	P_O = 1 W (I_T = 120 mA typ.) P_O = 2 W (I_T = 165 mA typ.)	1.5 2.2	% %
Power Output	P_O	THD(System) = 10%	3.0*	W
Input Resistance	R_I(af)	f = 1 kHz	100	kΩ

* With suitable heat sink.

● The attenuation range can be increased by substituting lower-valued resistors for the 10-kΩ resistor in the volume-control circuit.

\# With respect to recovered AF voltage at Terminal 9 and volume control, R_X, at maximum resistance position.

OPERATING CONSIDERATIONS

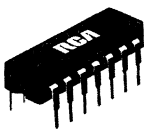
Power measurements were taken with a tin-plated copper-strap-type heat sink attached to the CA3134E. The heat sink, 1/32-inch thick, 1/4-inch wide, 1-7/8-inch long, is "U" shaped and has a 0.107-inch mounting hole to accept the stud on the IC package. The strap was also attached to the top side of the plastic package with epoxy cement to minimize stress on the stud during the mounting and handling operation. The torque, applied to the stud, should be limited to less than 3 in.-lbs. The applied tensile stress should be limited to less than 15 pounds and the compression stress should be limited to less than 100 pounds. Conductive epoxy (such as Dupont 5504A) was used for the interface between the stud and the heat sink; whereas non-conductive epoxy (such as Uniset Structural Adhesive) was used to attach the heat sink to the plastic package. The assembly when soldered to a 3-inch by 4-inch PC board has a typical over-all thermal resistance ($\theta_{\text{Stud to Amb.}}$) of 20°C/W.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3401E



14-Lead Dual-In-Line Plastic Package

H-1517

Quad Single-Supply Operational Amplifier

For Automotive Electronics and Industrial Control Systems

Features:

- **Single-supply operation – +5 V to +18 Vdc**
- **Internally compensated**
- **Wide unity-gain bandwidth – 5 MHz typ.**
- **Low input bias current – 50 nA typ.**
- **High open-loop gain – 2000 V/V typ.**

RCA-CA3401E* is a high-gain monolithic quad operational amplifier designed specifically for applications using a single positive power supply. No external compensation is necessary. Closed-loop stability in each of the four independent amplifiers is maintained by a 3-pF on-chip capacitor. The CA3401E is ideally suited for applications in industrial control systems, automotive electronics, and general-purpose amplifiers, e.g., oscillators, tachometers, active filters, and multichannel amplifiers. The CA3401E is supplied in a 14-lead dual-in-line plastic package and operates over a temperature range of −55 to 125°C. It is a direct replacement for the Motorola MC3401P, is pin-compatible with the Motorola MC3301P, and pin compatible with the National Semiconductor LM3900N.

*Formerly RCA Dev. No. TA6306.

Applications:

- **Automotive**
- **Constant-Current Sources**
- **Multivibrators**
- **Sample and Hold**
- **Square-Wave Generator**
- **Oscillators**
- **Tachometers**
- **Active Filters**
- **Multi-Channel Amplifiers**
- **Summing Amplifiers**

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A=25°C*

DC Supply Voltage	+18	V
Input Signal Current	5	mA
Device Dissipation:		
Up to T_A = 25°C	625	mW
Above T_A = 25°C	Derate linearly 5 mW/°C	
Ambient Temperature Range:		
Operating	−55 to +125°C	
Storage	−65 to +150°C	
Lead Temperature (During soldering):		
At distance 1/16±1/32 inch (1.59±0.79 mm) from case for 10 seconds max.	300	°C

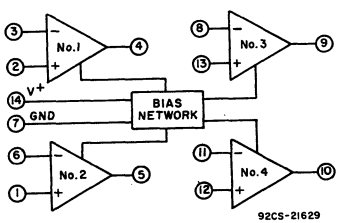


Fig. 1– Block diagram of CA3401E.

ELECTRICAL CHARACTERISTICS AT $T_A = 25^oC$, $V^+ = 15$ V (UNLESS INDICATED OTHERWISE)

CHARACTERISTIC	SYMBOL	TEST CONDITIONS			LIMITS			UNITS
		CIRCUIT		Typical Characteristics Curves				
		Fig. No.		Fig. No.	Min.	Typ.	Max.	
STATIC								
Output Voltage:								
High	V_{OH}	5,6			13.5	14.2	–	V
Low	V_{OL}				–	0.03	0.1	
Max. Undistorted Output Swing	V_{OP-P}		$0^oC < T_A < 75^oC$		10	13.5	–	
Output Current:								
Source	I_{SOURCE}	3		11, 12	5	10	–	mA
Sink	I_{SINK}				0.5	1	–	
Total Quiescent Current:								
Noninverting inputs open	I_Q	4		10	–	6.9	10	mA
Noninverting inputs grounded					–	7.8	14	
Input Bias Current	I_{IB}	3	$R_L = \infty$ $T_A = 25^oC$		–	50	300	nA
			$R_L = \infty$ $0^oC \leq T_A \leq 75^oC$		–	–	500	
DYNAMIC								
Open-Loop Voltage Gain	A_{OL}	3	$T_A = 25^oC$	7, 9	1000	2000	–	V/V
			$0^oC \leq T_A \leq 75^oC$		800	–	–	
Input Resistance	R_I	3			0.1	1	–	MΩ
Slew Rate	SR		$C_L = 100$ pF, $R_L = 5$ kΩ		–	0.6	–	V/μs
Unity Gain Bandwidth	BW				–	5	–	MHz
Phase Margin	ϕ				–	70	–	DEGREES
Power Supply Rejection			f=100 Hz		–	55	–	dB
Channel Separation	e_{01}/e_{02}		f=1 kHz		–	65	–	dB

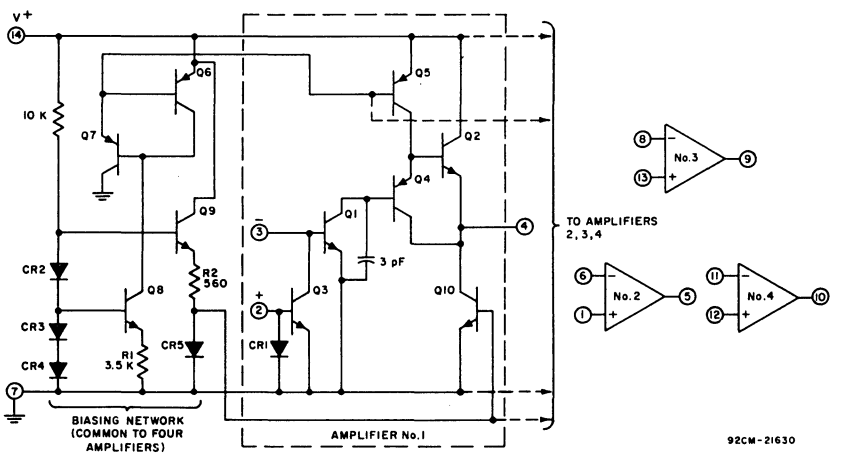


Fig. 2– Schematic diagram of CA3401E.

TEST CIRCUITS

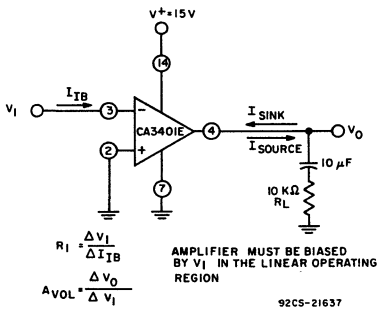


Fig.3 – Open-loop gain and input resistance, input bias current and output current test circuit.

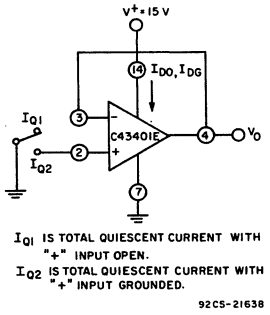


Fig.4 – Quiescent power supply current test circuit.

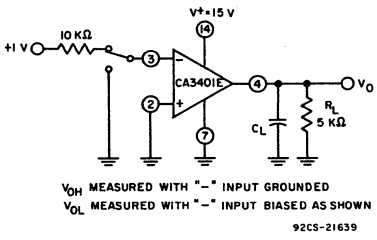


Fig.5 – Output voltage swing test circuit.

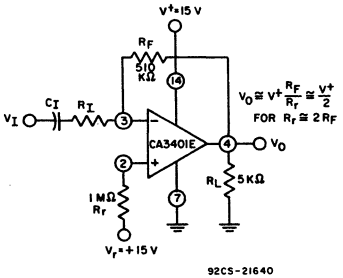


Fig.6 – Peak-to-peak output voltage test circuit.

TYPICAL CHARACTERISTIC CURVES

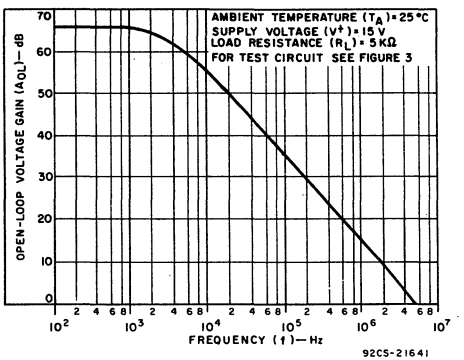


Fig.7 – Open-loop voltage gain vs. frequency.

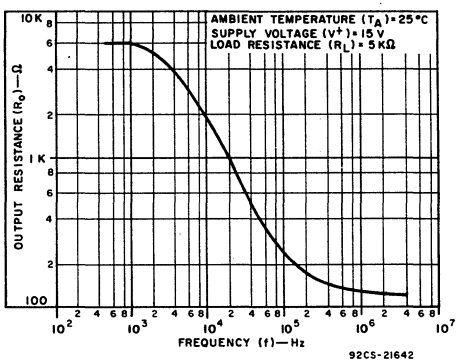


Fig.8 – Output resistance vs. frequency.

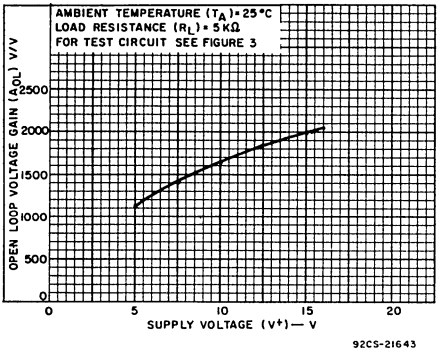


Fig.9 – Open-loop voltage gain vs. supply voltage.

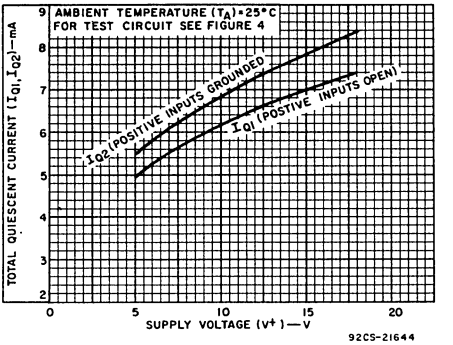


Fig.10 – Supply current vs. supply voltage.

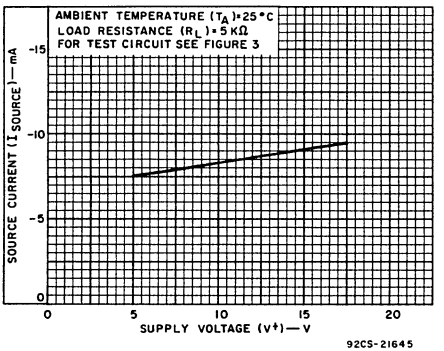


Fig.11 – Source current vs. supply voltage.

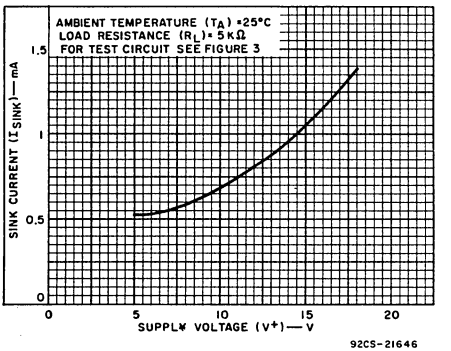


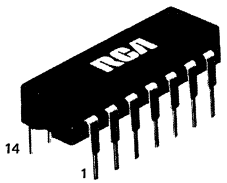
Fig.12 – Sink current vs. supply voltage.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3600E



14-Lead dual-in-line plastic package

H-1517

COS/MOS Transistor Array

For Linear Circuit Applications

Applications:

- High input impedance, general-purpose amplifiers
- Pre-amplifiers
- Differential amplifiers
- Op-amps and comparators
- Constant-current sources and current mirrors
- Micropower amplifiers and oscillators
- Control of lamps, LED's, relays, and thyristors
- Timers
- Choppers
- Mixers

RCA-CA3600E is an array of COmplementary-Symmetry MOS Field-Effect Transistors* on a monolithic silicon substrate. It is comprised of three n-channel and three p-channel enhancement-type MOS transistors arrayed as shown in Fig. 1, and specified and tested for linear circuit operation. These transistors are uniquely suitable for service in complementary-symmetry circuits at supply voltages in the range of 3 to 15 volts and are useful at frequencies up to 5 MHz (untuned). Each transistor in the CA3600E can conduct currents up to 10 mA. This device is supplied in the 14-lead dual-in-line plastic package.

Formerly RCA Dev. No. TA6368.

* The theory and construction of COS/MOS transistors are described in the "RCA COS/MOS Integrated Circuits Manual," RCA Solid State Division Technical Series Publication No. CMS-271.

Features:

- High input resistance . . . . . 100 GΩ (typ.)
- Low gate-terminal current . . . . 10 pA (typ.)
- Matched p-channel pair:
  Gate-voltage differential (I_D = -100 μA) ±20 mV (max.)
- No "Popcorn" (burst) noise
- Stable transfer characteristics over an operating temperature range of −55°C to +125°C when operated in complementary circuit configuration at supply voltages in the 5 to 15 volt range (see Fig. 14)
- Integrated integral gate-protection system (see Fig. 34)
- High voltage gain (see Fig. 11). . . up to 53 dB (typ.) per COS/MOS stage
- Individual MOS transistors have square-law characteristics, superior cross-modulation performance, and greater dynamic range than bipolar transistors

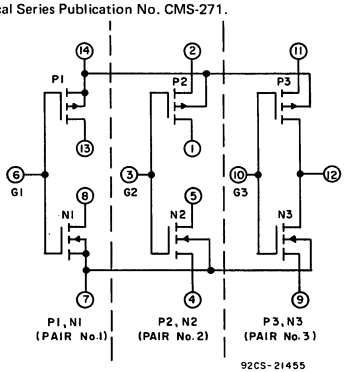


Fig.1 – Schematic diagram for CA3600E COS/MOS transistor array. (See Fig.34 for internal gate-and-channel-protection circuits)

1. Drain terminal, p-channel of pair no. 2
2. Source terminal, p-channel of pair no. 2
3. Common gate terminal of pair no. 2
4. Source terminal, n-channel of pair no. 2
5. Drain terminal, n-channel of pair no. 2
6. Common gate terminal of pair no. 1
7. Source terminal, n-channel of pair no. 1 and substrate connection for all n-channel transistors · · · V_{SS} terminal
8. Drain terminal, n-channel of pair no. 1
9. Source terminal, n-channel of pair no. 3
10. Common gate terminal of pair no. 3
11. Source terminal, p-channel of pair no. 3
12. Common drain terminal of pair no. 3
13. Drain terminal, p-channel of pair no. 1
14. Source terminal, p-channel of pair no. 1 and substrate connection for all p-channel transistors · · · V_{DD} terminal

Terminal Identification for Fig. 1.

MAXIMUM RATINGS, *Absolute-Maximum Values at $T_A = 25^oC$*

DISSIPATION:
- Any one transistor at T_A up to 55°C . . . . . 150 mW
- Total package at T_A up to 55°C . . . . . 750 mW
- Above T_A = 55°C . . . . . derate linearly 6.67 mW/°C

AMBIENT TEMPERATURE RANGE:
- Operating . . . . . −55 to +125°C
- Storage . . . . . −65 to +150°C

LEAD TEMPERATURE (During Soldering)
- At distance not less than 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 s max. . . . . 265°C

The Following Ratings Apply for Each Transistor in the Device:

DRAIN-TO-SOURCE VOLTAGE, V_{DS}:
- n-channel . . . . . +15 V
- p-channel . . . . . −15 V

DRAIN-TO-GATE VOLTAGE, V_{DG}:
- n-channel . . . . . +15 V
- p-channel . . . . . −15 V

SOURCE-TO-SUBSTRATE VOLTAGE, V_{SB}:
- n-channel . . . . . +15 V
- p-channel . . . . . −15 V

GATE-TO-SOURCE VOLTAGE, V_{GS}:
- p-channel transistors (p_1, p_2, p_3) . . . . . 0 V(min.),$-V_D$(max.)
- n-channel transistors (n_1, n_2, n_3) . . . . . 0 V(min.),$+V_D$(max.)
- COS/MOS transistor-pairs (p_1-n_1, p_2-n_2, p_3-n_3) . . . . . 0 V(min.),$+V_{DD}$(max.)

DRAIN CURRENT, $|I_D|$ . . . . . 10 mA

GATE CURRENT, $|I_G|$ . . . . . 100 μA

The Following Rating Applies for Each COS/MOS Transistor-Pair in the Device:

DC SUPPLY VOLTAGE ($V_{DD} - V_{SS}$) . . . . . +15 V

Rules for Maintaining Electrical Isolation Between Transistors and Monolithic Substrate

Terminal No. 14 must be maintained at the most positive potential (or equally positive potential) with respect to any other terminal in the CA3600E.

Terminal No. 7 must be maintained at the most negative potential (or equally negative potential) with respect to any other terminal in the CA3600E.

Violation of these rules will result in improper transistor operation, circuit "latching," and/or possible permanent damage to the CA3600E.

Note: Users should observe the "Considerations in Handling CA3600E Devices", discussed on pages 599 and 600.

ELECTRICAL CHARACTERISTICS, *At T_A = 25°C*

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	TYPICAL CURVE OR CIRCUIT FIG. NO.	LIMITS			UNIT
				Min.	Typ.	Max.	
For Each p-Channel MOS Transistor							
Drain Current	I_D	V_{DS}=−10 V,V_{GS}=−3.6 V	2,3,4	−0.5	−1.1	−2.0	mA
Gate-to-Source Threshold Voltage	$V_{GS(th)}$	I_D=−10 μA	–	–	−1.75	–	V
Gate-to-Source Voltage Differential (p_1 vs. p_2)	$\lvert V_{GS1}-V_{GS2}\rvert$	I_D=−100 μA,V_{DS}=−10 V	5	–	±4	±20	mV
Forward Transconductance	g_{fs}	I_D=−1 mA,f=1 kHz	6	–	920	–	μmho
Low-Frequency Noise Voltage	e_N	I_D=−1 mA,f=1 kHz,R_s=0 Ω	7	–	0.03	–	μV $\sqrt{Hz}$
Low-Frequency Noise Current	i_N	I_D=−1 mA,f=1 kHz,R_s=1 MΩ	7	–	0.2	–	pA $\sqrt{Hz}$
Current-Mirror Transfer Ratio (p_1/p_2)	I_{MTR}	I_1=−100 μA,V_{DS}=−10 V	30	0.7	1.1	1.5	–
Gate-Terminal Current	I_{GT}	V_{DS}=−10 V,V_{GS}=−3.5 V	–	–	±0.015	−40	nA
Input Capacitance	C_I	–	–	–	6.3	–	pF
Output Capacitance	C_O	–	–	–	3	–	pF
Input-to-Output Capacitance	C_{I-O}	–	–	–	0.75	–	pF
For Each n-Channel MOS Transistor							
Drain Current	I_D	V_{DS}=+10 V,V_{GS}=+3.6 V	2,3,4	0.4	0.9	1.6	mA
Gate-to-Source Threshold Voltage	$V_{GS(th)}$	I_D=10 μA	–	–	1.5	–	V
Gate-to-Source Voltage Differential (n_1 vs n_2)	$\lvert V_{GS1}-V_{GS2}\rvert$	I_D=100 μA,V_{DS}=+10 V	5	–	±30	–	mV
Forward Transconductance	g_{fs}	I_D=1 mA,f=1 kHz	6	–	860	–	μmho
Low-Frequency Noise Voltage	e_N	I_D=1 mA,f=1 kHz,R_s=0 Ω	7	–	0.2	–	μV $\sqrt{Hz}$
Low-Frequency Noise Current	i_N	I_D=1 mA,f=1 kHz,R_s=1 MΩ	7	–	0.3	–	pA $\sqrt{Hz}$
Current-Mirror Transfer Ratio (n_1/n_2)	I_{MTR}	I_1=100 μA,V_{DS}=+10 V	29	0.7	1.3	2.0	–
Gate-Terminal Current	I_{GT}	V_{DS}=+10 V,V_{GS}=+3.7 V	–	–	±0.01	+40	nA
Input Capacitance	C_I	–	–	–	5.5	–	pF
Output Capacitance	C_O	–	–	–	2.0	–	pF
Input-to-Output Capacitance	C_{I-O}	–	–	–	0.35	–	pF
For Each COS/MOS Transistor Pair							
Drain Current	I_{DD}	V_{DD}=+10 V	9,10	1.0	2.2	4.0	mA
Drain-to-Source Cutoff Current	$I_{DD(off)}$	V_{DD}=+10 V,V_{SS}=0 V Gate Voltage(V_G)=+10 V or 0 V	8	–	0.5	100	nA
DC Output Voltage	V_O	V_{DD}=+10 V	10	4.2	5.0	5.8	V
Forward Transconductance	g_{fs}	V_{DD}=+10 V, f = 1 kHz	6	–	2300	–	μmho
Slew Rate (Open-Loop)	SR	V_{DD}=+15 V	10	–	95	–	V/μs
Amplifier Voltage Gain	A_{OL}	V_{DD}=+10 V,f=1 kHz,R_b=22 MΩ R_s=50 Ω	10,11	–	32	–	dB
Gate-Terminal Current	I_{GT}	V_{DD}=+10 V	10	–	±0.005	±20	nA
Broadband Output Noise Voltage	E_{ON}	V_{DD}=+10 V,R_b=22 MΩ,R_s=10 kΩ	10,11	–	500	–	μV
Input Capacitance	C_I	–	–	–	11.8		pF
Output Capacitance	C_O	–	–	–	5.0		pF
Input-to-Output Capacitance	C_{I-O}	–	–	–	1.1		pF

TYPICAL CHARACTERISTICS CURVES

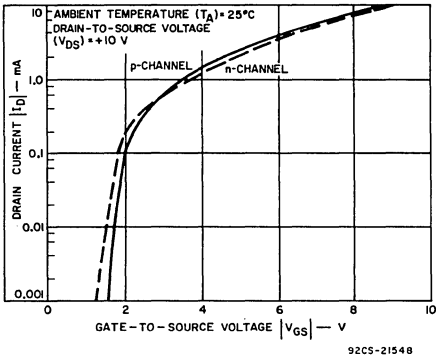


Fig. 2—Drain current vs. gate-to-source voltage.

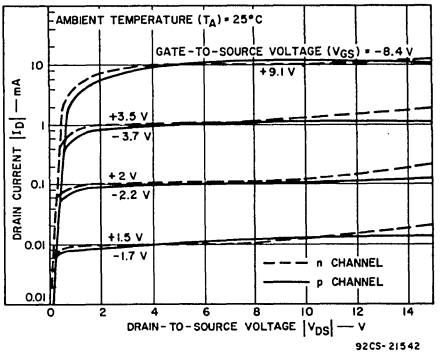


Fig. 3— Drain current vs. drain-to-source voltage.

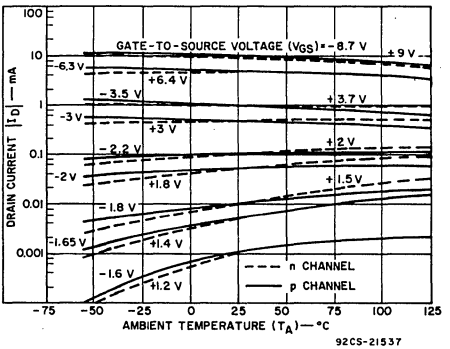


Fig. 4— Drain current vs. ambient temperature.

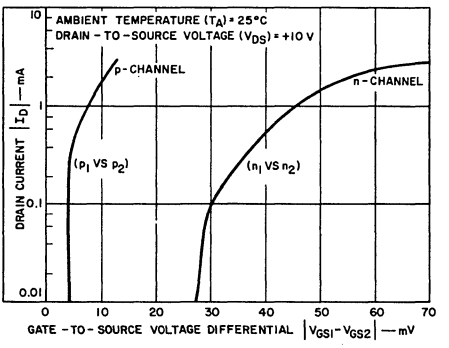


Fig. 5— Gate-to-source voltage differential vs. drain current.

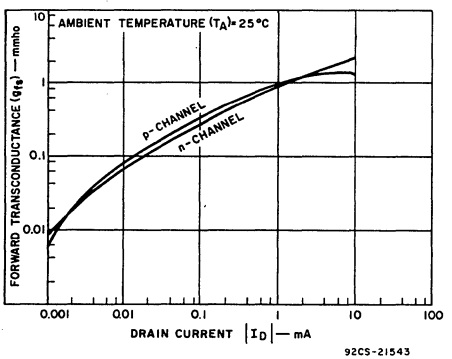


Fig. 6— Forward transconductance vs. drain current.

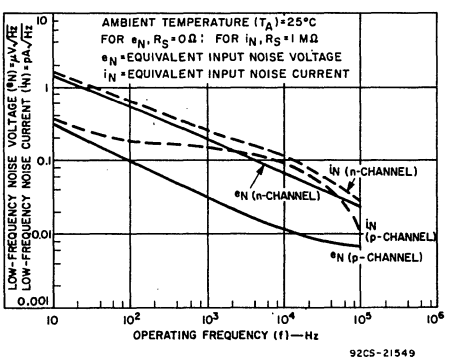


Fig. 7— Noise voltage and noise current vs. operating frequency.

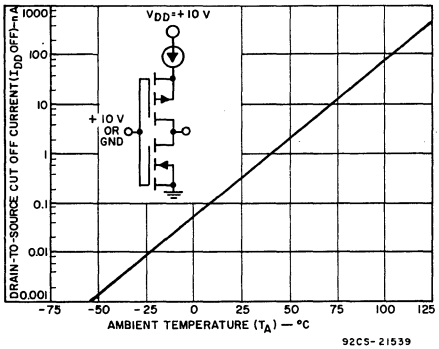


Fig. 8– Drain-to-source cutoff current vs. ambient temperature.

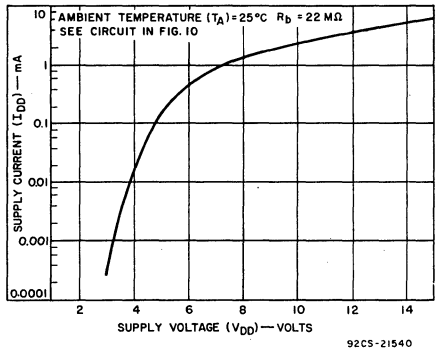


Fig. 9 – Typical V_{DD} vs. I_{DD} characteristics for amplifier circuits of Fig. 10 and Fig. 15.

APPLICATIONS

The Basic COS/MOS Linear Amplifier

P-n-p and n-p-n bipolar transistors have been used for many years in the design of so-called "true-complementary" linear amplifier circuits[1]. Since mutually compatible p-channel and n-channel MOS/FET devices were not generally available, "true-complementary" amplifier circuits using MOS transistors were seldom used. Now, COS/MOS transistor technology[5] has made it possible to supply compatible p-channel/n-channel transistors in monolithic IC form such as the CA3600E COS/MOS transistor array shown in Fig. 1.

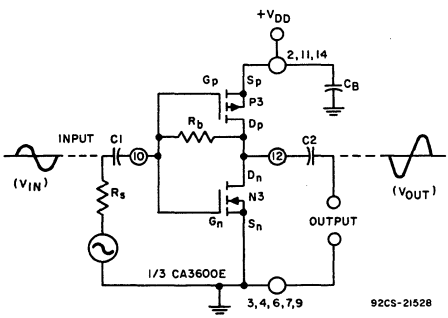


Fig. 10– COS/MOS transistor-pair biased for linear-mode operation.

A "True-Complementary" Linear Amplifier Using COS/MOS Transistors

Fig. 10 shows the schematic diagram of a single-stage "true-complementary" linear amplifier using one pair of the complementary MOS transistors in the CA3600E, connected in a common-source circuit. Resistor R_b is used to bias the complementary pair for Class A operation, as described subsequently, and R_s represents the source resistance of the signal source. This generic amplifier is suitable for operation with a single or split voltage supply in the range of 3 to 15 volts. Fig. 11 shows voltage gain as a function of operating frequency at various supply voltages for the single-stage amplifier. This amplifier is capable of producing very high output-swing voltages (V_{out}); for example, its output voltages can be swung to within several millivolts of either supply-voltage "rail". Fig. 9 shows typical supply voltage (V_{DD}) vs. supply current (I_{DD}) characteristics for the single-stage amplifier. The curves in Fig. 12 show the normalized amplifier supply current as a function of ambient temperature at various supply voltages. When the amplifier is operating at V_{DD} = 3 V, the supply current changes rapidly as a function of temperature because the MOS transistors are operating in the proximity of their individual gate-source threshold voltages.

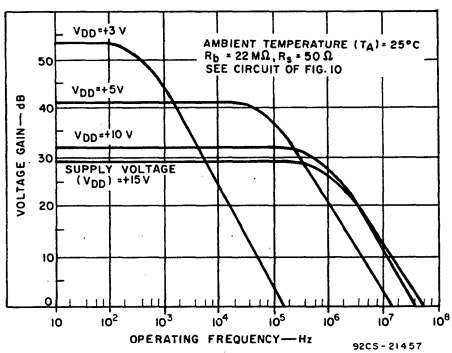


Fig. 11– Typical voltage gain vs. frequency characteristics for amplifier circuit of Fig. 10.

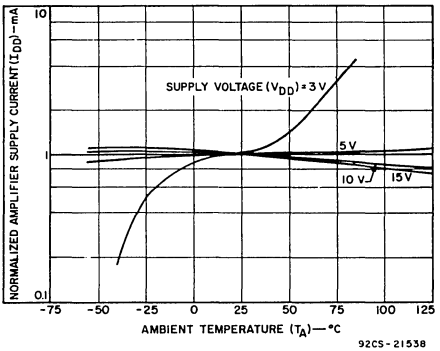


Fig. 12– Normalized amplifier supply current vs. ambient temperature characteristics for amplifier circuit of Fig. 10.

Voltage-Transfer Characteristics

Fig. 13 illustrates a voltage-transfer characteristic curve of a COS/MOS transistor pair connected in the amplifier circuit of Fig. 10, with a biasing resistor (R_b) connected between the drain and gate terminals (10,12). If the p- and n-channel transistors have identical characteristics, their channel resistances are equal, and the biasing method shown establishes a steady-state condition such that terminal 12 is at mid-potential between V_{DD} and ground. Thus, with negligibly small gate-source leakage resistances, under zero-signal conditions, the biasing resistor (R_b) establishes gate potential at the mid-point between V_{DD} and ground, i.e., $V_{in} = V_{out}$. Under these conditions the amplifier is biased for operation about the mid-point ("0") in the linear segment on the steep transition of the voltage-transfer characteristic as shown in Fig. 13. When the input signal (V_{in}) swings in the positive direction, there is a reduction in the instantaneous output voltage (V_{out}) with respect to ground. Negative-going input signals have inverse effects. Thus, phase-inversion occurs in the COS/MOS-pair amplifier. Power-supply current is constant during dynamic linear operation, i.e., Class A amplifier service. When the signal input-voltage level (V_{in}) becomes very large, the output signal (V_{out}) waveforms become distorted because the transistors are driven into the non-linear portions of their voltage-transfer characteristics. If the positive-going input-signal is sufficiently large, for example, the p-channel transistor can be driven to cutoff and the amplifier supply current (I_{DD}) is reduced to essentially zero.

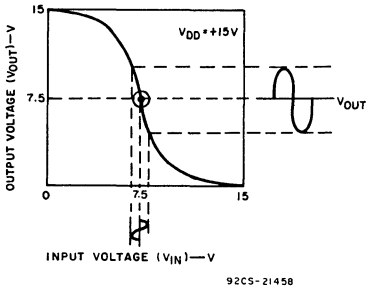


Fig. 13 – Representation of voltage-transfer characteristics for COS/MOS transistor pair.

Fig. 14 shows typical voltage-transfer characteristics of each COS/MOS pair in the CA3600E at several values of V_{DD}. The shape of these transfer characteristics is comparatively constant despite temperature changes from −55 to +125°C.

The biasing arrangement used in the circuit of Fig. 10 provides an easy method of establishing feedback for ac signals in accordance with the R_b/R_s ratio. When the feedback of ac signals is not desirable, the circuit of Fig. 15 may be used. The ac bypass capacitor (C_3) minimizes ac signal feedback.

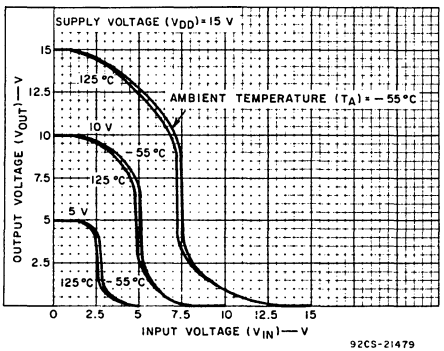


Fig. 14– Voltage transfer characteristics for COS/MOS transistor-pair amplifier in Fig. 10.

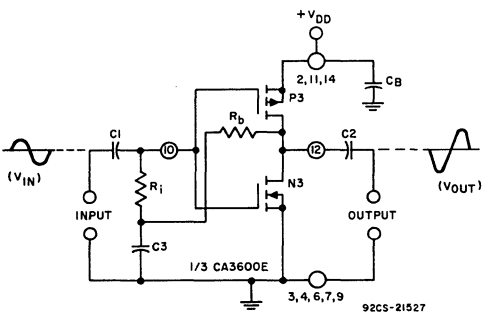


Fig. 15– Alternate method of biasing COS/MOS transistor-pair for linear-mode operation.

Cascading Amplifier Stages of COS/MOS Transistor Pairs

Ultra-high-gain amplifiers can be designed by cascading stages of COS/MOS transistor pairs as shown in Fig. 16. The biasing system used is similar to that described above in connection with Fig. 10. The supply current for the three-stage amplifier shown in Fig. 16 is typically three times the values shown in Fig. 9. Gain and frequency-response characteristics of the amplifier are shown in Fig. 17.

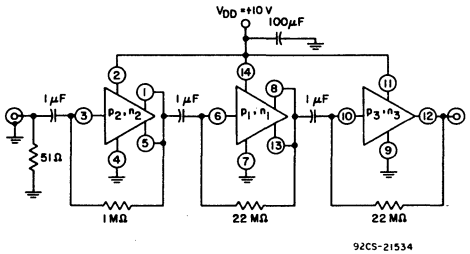


Fig. 16– High-gain amplifier uses cascaded COS/MOS transistor-pair in CA3600E.

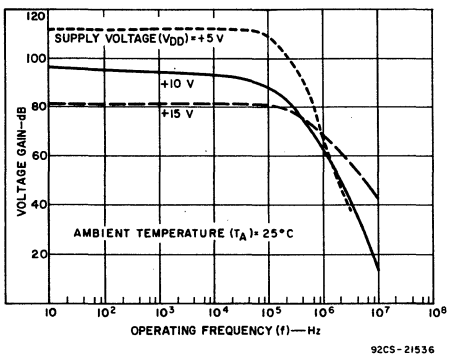


Fig. 17– Typical voltage gain vs. operating frequency characteristics for three-stage COS/MOS transistor-pair amplifier in Fig. 16.

Post-Amplifiers For Op-Amps

COS/MOS transistor-pairs can be advantageously applied as post-amplifiers for op-amps. Because the input impedance of the COS/MOS pair is comparatively high, the op-amp operates under essentially unloaded conditions. Each COS/MOS pair can sink and source output current up to about 10 mA. Additionally, the op-amp output can be directly coupled to bias the COS/MOS pair. A detailed description of the subject has been published previously.[2]

The schematic diagram in Fig. 18 shows a COS/MOS transistor-pair serving as a post-amplifier to an RCA-CA3080 Operational Transconductance Amplifier.[3] The approximate 30-dB gain in a single COS/MOS transistor-pair is an added increment to the 100-dB gain in the CA3080, yielding a total forward gain of about 130 dB. The open-loop slew rate of the circuit in Fig. 19 is approximately 65 V/μs. When compensated for the unity-gain voltage-follower mode shown in Fig. 19, the slew rate is about 1 V/μs. For greater current output, the two remaining transistor pairs of the CA3600E may be connected in parallel with the single stages shown in Figs. 18 and 19.

The use of the two-stage COS/MOS post-amplifier shown in Fig. 20 increases the total open-loop gain of the system to about 160 dB (100,000,000X). Open-loop slew rate remains at about 65 V/μs. A slew rate of about 1 V/μs is maintained with this circuit connected in the unity-gain voltage-follower mode, as shown in Fig. 21. These circuits operate in concert with stability.

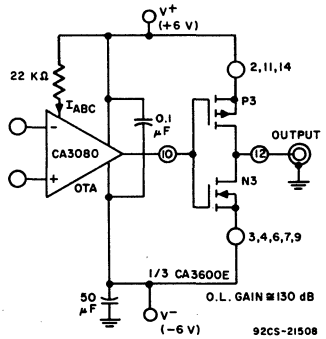


Fig. 18– COS/MOS transistor-pair used as post-amplifier to op-amp in open-loop circuit.

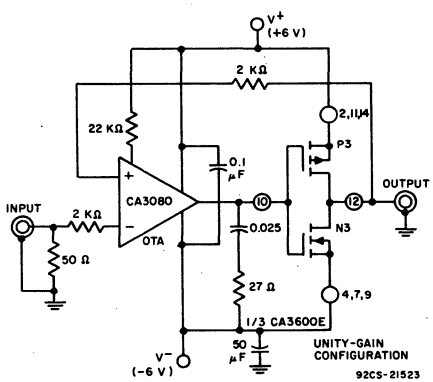


Fig. 19– COS/MOS transistor-pair used as post-amplifier to op-amp in unity-gain circuit.

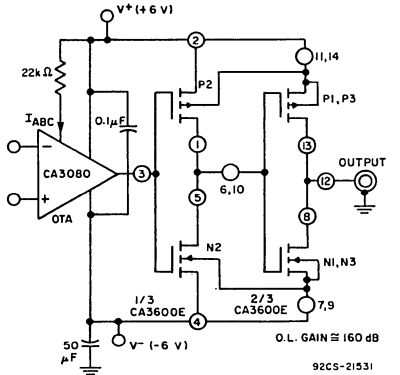


Fig. 20– COS/MOS transistor-pairs used as two-stage post-amplifier to op-amp in open-loop circuit.

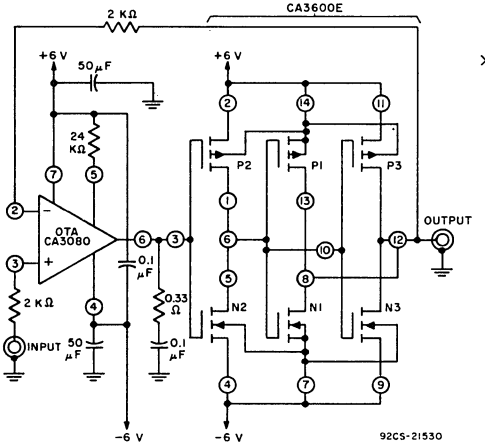


Fig. 21– Unity-gain amplifier uses COS/MOS transistor-pairs as two-stage post-amplifier to op-amp.

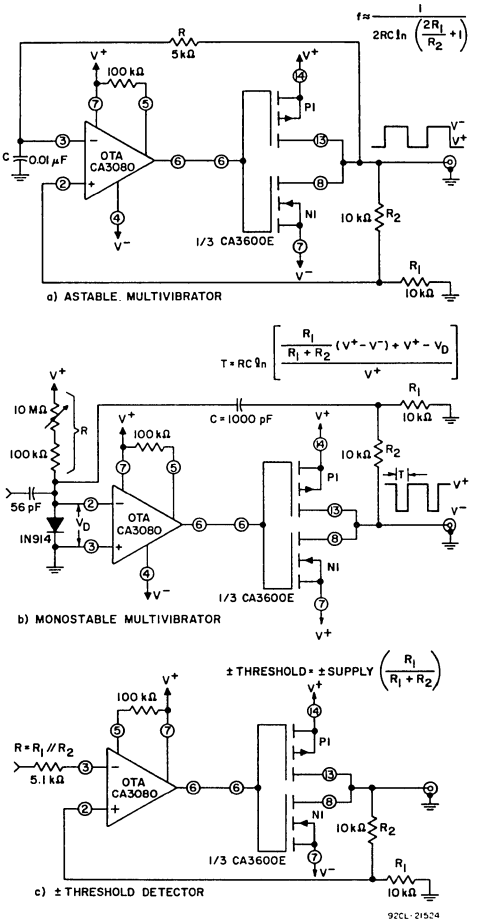


Fig. 22– Multistable circuits using COS/MOS transistor-pairs.

Multivibrators, Threshold Detectors, and Comparators

Descriptions of several circuits using COS/MOS transistor-pairs in both monostable and astable multivibrators have been published.[4,5] The characteristics of COS/MOS pairs are also ideal for mating with micropower op-amps in circuits such as the precision multistable circuits shown in Fig. 22. In these circuits precise timing and thresholds are assured by the stable characteristics of the input differential amplifier in the CA3080 Operational Transconductance Amplifier.[2,3] Moreover, speed vs. power consumption tradeoffs can be made by adjustment of the Amplifier-Bias-Current (I_{ABC}) supplied to terminal 5 of the CA3080. The quiescent power consumption of the circuits shown in Fig. 22 is typically 6 mW, but can be made to operate in the micropower region by suitable modifications.

The schematic diagram of a programmable micropower comparator, shown in Fig. 23 employs the combination of an op-amp (CA3080A) and COS/MOS transistor-pairs in the CA3600E. Quiescent power consumption of the circuit is about 10 μW(typ.). When the comparator is strobed "ON", transistor P1 is driven into conduction and the OTA becomes active. Under these conditions, the circuit consumes 420 μW and responds to a differential-input signal in about 8 μs. By suitably biasing the CA3080A, the circuit response time can be decreased to about 150 ns but the power consumption is increased to 21 mW. The differential amplifier input common-mode range for this circuit is −1 V to +10.5 V. Voltage gain of this micropower comparator is typically 130 dB.

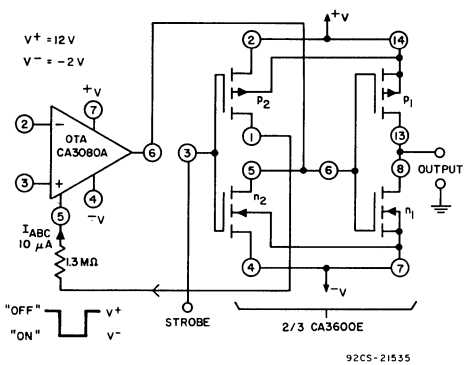


Fig. 23– Programmable micropower comparator.

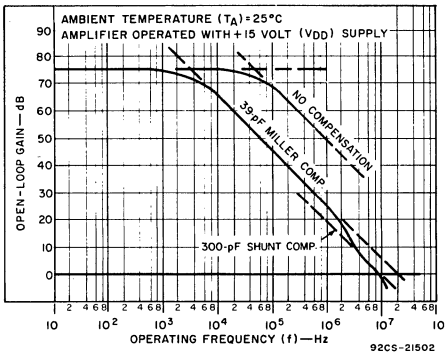


Fig. 25– Open-loop gain characteristic for op-amp in Fig. 24.

Operational Amplifiers

COS/MOS transistor-pairs can be used in conjunction with a bipolar transistor-array IC to build an op-amp as shown in Fig. 24. It is particularly suited for single-supply operation (e.g., mobile and aircraft service). The op-amp is unique in that it is responsive to small-signal ground-referenced inputs and the output stage can easily be driven within 1 mV of ground potential. Its open-loop gain characteristics are shown in Fig. 25; the open-loop slew rate is approximately 30 V/μs.

This circuit is ideal for use as a unity-gain voltage-follower and has been described for operation in connection with a 9-Bit Single-Supply Digital-to-Analog Converter (DAC) using COS/MOS transistors in the resistor-network switches.[6]

The op-amp in Fig. 24 has three stages; its first stage is a differential input circuit using two p-channel transistors (P_4,P_5) in a CA3600E. The second stage is an n-p-n

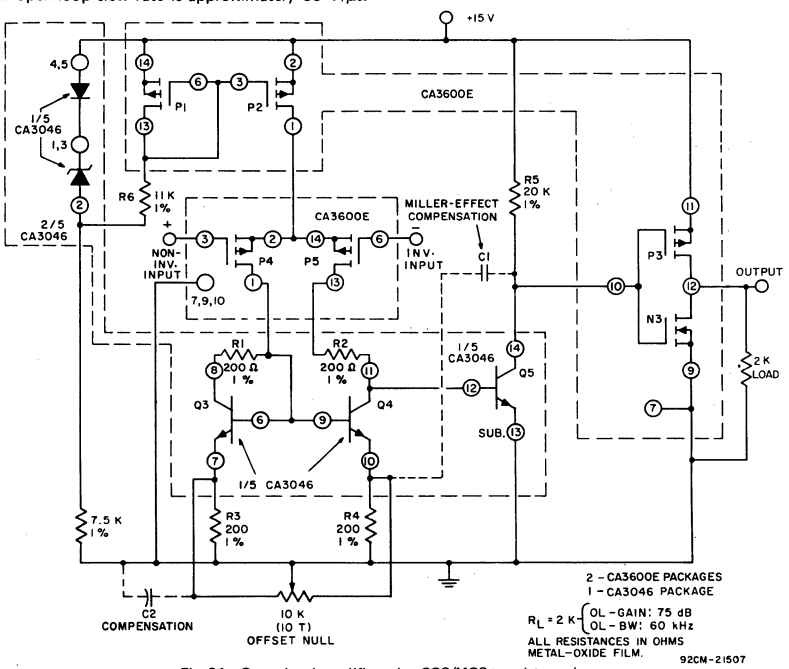


Fig. 24– Operational amplifier using COS/MOS transistor-pairs.

transistor (Q_5) and the output stage is a COS/MOS transistor-pair (P_3,N_3) operating in the manner described above. A constant current of about 400 μA is established in the differential input stage by the zener network in the upper-left portion of Fig. 24. The zener network energizes a current mirror comprised of two p-channel transistors (P_1,P_2) to establish constant-current flow in the differential amplifier stage (P_4,P_5). The drain load for the differential amplifier consists of resistors R_1–R_4 and a current mirror (Q_3 Q_4) to optimize conditions for balanced operation of the differential amplifier. The operating theory of current-mirror circuits has been described in reference [2]. Amplifier voltage-offset is nulled with the 10-kilohm balance potentiometer. The second-stage current is established by R_5, and is selected to approximate the first-stage current level (400 μA), to assure similar positive and negative slew rates. The amplifier is shown driving a 2-kilohm load, a typical value used with monolithic op-amps. Voltage gain varies inversely with the choice of load resistance.

The amplifier can be compensated with a single capacitor (C_1), connected as shown by the dotted lines. However, optimum compensation for the unity-gain non-inverting mode is provided by two capacitors: Miller Effect feedback through a 39-pF capacitor C_1 (connected as shown), and a 300-pF capacitor connected between terminals 7 and 13 of the CA3046 transistor array to shunt one-half the driving current. Fig. 25 shows the open-loop gain characteristics with compensation for unity-gain operation. When the amplifier is operated as a voltage-follower, it is recommended that a 1-kilohm resistor, shunted with a 150-pF capacitor, be connected between the amplifier output terminal and terminal 6 of P_5 to avoid a potential latch situation involving the integral gate-protection network. The circuit can also be latched if either input terminal is driven more than about 0.7 volt below ground potential. This latch situation can be avoided by connecting a 1N914 diode from each input terminal to ground, with the diode anode grounded.

Analog Timer

The CA3600E is useful in the design of analog timer circuits. A typical circuit is shown in Fig. 26. For purposes of explanation, let it be assumed that capacitor C_1 initially is in a completely discharged condition; terminal 10, therefore, is initially at ground potential and transistor N_3 is non-conductive. The circuitry at the left of terminal 10 provides a source of constant-current flow through P_1 to charge capacitor C_1 increasingly positive with respect to ground. After the passage of time (T), capacitor C_1 is charged sufficiently in the positive direction so that transistor N_3 is driven into conduction by its gate and the lamp is lighted to signify the end of the time-delay period. The circuit is reset by momentarily closing switch S_1 to discharge capacitor C_1 through R_4. Resistor-divider network R_1,R_2 establishes the supply voltage to a constant-current network comprised of resistor R_3 and the series-connected COS/MOS pair N_2,P_2, biased for linear operation by resistor R_b as previously described. This combination is connected to the gate terminal (No. 6) of transistor P_1 to form a current mirror, i.e., the current flowing through P_1 to charge C_1 will be essentially equal to the constant-current flow established through R_3, N_2, and P_2. A description of current-mirror operation with MOS transistors is given subsequently.

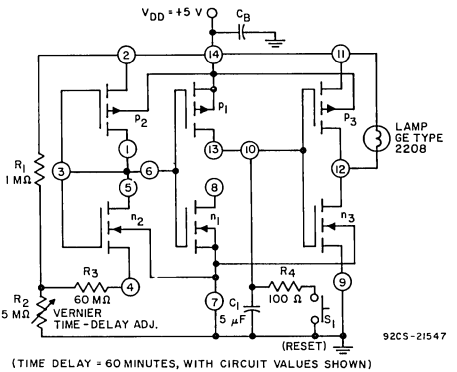


Fig. 26– Analog timer using CA3600E.

Oscillator Circuits

Oscillator circuits using COS/MOS transistor-pairs have been widely used for several years in clock and watch circuits because of their low power consumption and good frequency stability. Details of their operating theory and characteristics have been published.[5,7]

The design of COS/MOS oscillator circuits, like the design of any oscillator circuit, involves the provision of an amplifying section to operate compatibly with an appropriate feedback network. A single stage amplifier using a COS/MOS transistor-pair has already been described. A suitable feedback network to insure stable oscillator performance is easily added, as illustrated in connection with the crystal oscillator circuit shown in Fig. 27. The familiar pi-network has been connected between the input and output terminals, points "D" and "G", to provide the required 180° phase shift for stable oscillator performance. The frequency-determining crystal is an integral part of the pi-network feedback circuit. The resistors R_1 and R_2 decrease the total power consumption of the oscillator at a particular supply voltage and enhance the frequency stability. Variable frequency oscillators can be built by replacing the crystal with an appropriate inductance and tuning the pi-network by conventional means.

Current Mirrors Using MOS Transistors

Monolithic linear IC's using bipolar transistors frequently employ so-called "current-mirror" circuits. The theory and practical applications of current mirrors using bipolar transistors have been described in the literature.[2] As shown in

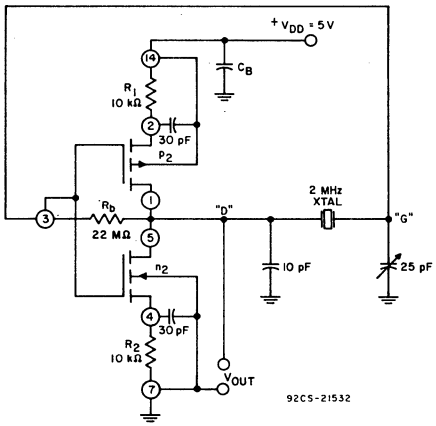


Fig. 27– Typical crystal-oscillator circuit using COS/MOS transistor-pair (1/3 CA3600E).

Fig. 28, a rudimentary form of "current-mirror" consists of a transistor Q_1 with a second transistor Q_2 connected as a diode. When both transistors have identical characteristics, a current I_1 forced to flow through Q_2 produces a current (I_2) of equal magnitude to flow in the collector of Q_1 (provided there is sufficient collector potential for Q_1). In a common form of application, a source of potential is used to force

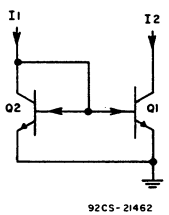


Fig. 28– Current mirror using n-p-n bipolar transistors.

constant-current flow I_1, and thus to establish the flow of constant current I_2 through Q_1. Arrangements of this generic current-mirror type are frequently used when Q_1 acts as the common-emitter impedance in a differential-amplifier circuit.

MOS transistors are also applicable as current mirrors, as shown in Fig. 29. The diode-connected MOS transistor N_2 functions as a transistor with 100 per-cent feedback. Therefore, the gate-to-source voltage (V_{GS}) in N_2 retains control of the drain current as in normal transistor action, i.e., $I_D \cong g_{fs}V_{GS}$, where g_{fs} is the forward transconductance of the device. If a current I_1 is forced into the diode-connected transistor (N_2), the gate-to-source voltage will rise until equilibrium is reached. Thus, a gate-to-source voltage is established in N_2 such that N_2 "sinks" the applied current I_1.

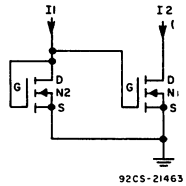


Fig. 29– Current mirror using n-channel MOS transistors.

If the gate and source terminals of another transistor (N_1) are connected in shunt with the gate and source terminals of N_2, as shown in Fig. 29, N_1 is also able to "sink" a mirror current approximately equal to that flowing in the drain lead of the diode-connected transistor N_2. It is assumed that both MOS transistors have identical characteristics, a prerequisite that is essentially established by the monolithic IC fabrication technology used in manufacturing the CA3600E COS/MOS transistor array.

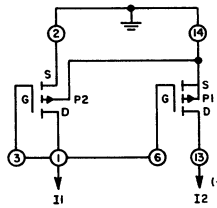


Fig. 30– Current mirror using p-channel MOS transistors in CA3600E.

Current mirrors can also be designed with p-channel MOS transistors as illustrated by the arrangement in Fig. 30 using transistors in the CA3600E. The characteristics of a current mirror using the p-channel transistors in the CA3600E are superior to those which can be achieved with a current mirror using the n-channel transistors because the characteristics of the p-channel transistors are more nearly matched. The data

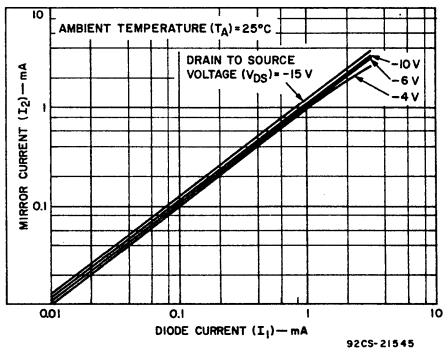


Fig. 31– Characteristics of current mirror circuit of Fig. 30 using p-channel transistors.

contained in Fig. 31 show the high degree of tracking between I_1 and I_2 for several values of drain voltage V_D. Fig. 32 also illustrates the fact that this high degree of tracking between I_1 and I_2 can be maintained to within about one per-cent despite wide variations in ambient temperature.

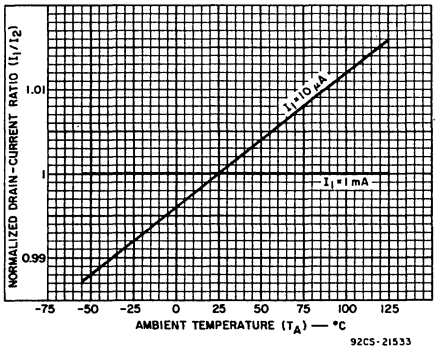


Fig. 32– Normalized drain current ratio vs. ambient temperature for typical current mirror using p-channel transistors (Fig. 30).

The op-amp circuit in Fig. 24 contains an illustrative example of a current-mirror circuit using two p-channel transistors in the CA3600E. Transistor P_2 serves as a constant-current source (≅ 400 μA) for the differential amplifier, consisting of transistors P_4 and P_5 and their drain-load network. Transistor P_2 is in a "mirrored" connection with transistor P_1. A stabilized source of supply potential is developed across the zener diode (terminals 11 and 12 of the CA3083) and drives about 400 μA of current through R_6 and P_1.

Complementary Current Mirrors Using COS/MOS Transistor-Pairs

COS/MOS transistor-pairs can be applied advantageously in the design of Complementary Current-Mirrors, as shown in Fig. 33. Transistors P_1 and N_1 are series-connected and biased for linear operation as previously described, so that there is a current flow I_{D1} through P_1 and N_1. The potential developed between terminals 13 and 14 is applied as gate-source (2,3) voltage for P_2, forcing "mirror" operation of P_2 to produce a current source $I_{D2\text{-}P}$ equal to I_{D1}. Likewise, the potential developed between terminals 7 and 8 is applied as gate-source (3,4) voltage for N_2 forcing "mirror" operation of N_2 to produce a current-sink $I_{D2\text{-}N}$ equal to I_{D1}.

A variant of this complementary current mirror is used in the analog timer circuit shown in Fig. 26. Transistors P_2 and N_2 are series-connected together with a 60-megohm resistor to establish their drain current at 5 nA. The potential developed across terminals 1 and 2 also appears as the gate-source voltage for transistor P_1, thereby establishing a mirror-current source of 5 nA at terminal 13 to charge capacitor C_1 linearily. In this circuit, the "mirrored" current-sink available at terminal 8 (transistor N_1) is unused. This type of current-mirror configuration is exceptionally stable with temperature variations.

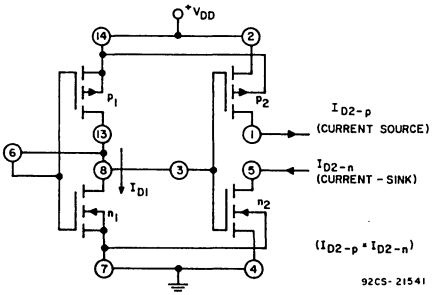


Fig. 33– Complementary current mirrors using COS/MOS transistor-pairs in CA3600E.

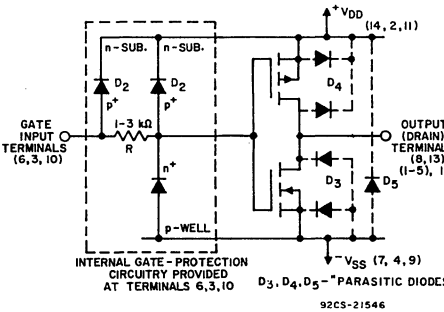


Fig. 34– Integral protection circuits used in CA3600E.

Considerations in Handling CA3600E Devices

Failure of the gate-channel oxide was a persistent problem in early MOS devices. The breakdown of the oxide is generally in the order of 100 volts, and the dc resistance is in the order of 10^{12} ohms. Because of this extremely high resistance, even a very-low-energy source (such as static charge) is capable of developing sufficient voltage to cause damage. Furthermore, the oxide can be punctured and damaged by a single voltage excursion beyond the breakdown limit.

Fig. 34 shows a protection circuit[5,8] which is incorporated at each gate-lead of the CA3600E. A typical value of 1 to 3 kilohms is used for the input resistor R, which functions in combination with the capacitance of the gate and the associated protective diode to integrate and clamp input voltages to a safe level. This circuit also shows the "substrate diodes" (D_3, D_4, and D_5) which provide protection to the MOS channels at the output terminals.

Although the gate-protection system is very effective in guarding against damage due to static charges, it is prudent to observe the following precautions:[5,9]

1. The leads of devices should be in contact with a conductive material, except when being tested or in actual operation. A conductive material such as "ECCOSORB LD26"* or equivalent is suggested for use during storage and/or handling. Devices should not be

inserted in non-conductive containers such as conventional plastic "snow" or trays.

* Trade Mark: Emerson and Cumming, Inc.

2. Soldering-iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.
3. Devices should not be inserted into or removed from circuits with the power on because transient voltages may cause permanent damage.
4. Signals from low-impedance sources should not be applied to the gate terminals while the power supply is off. As a corollary, it follows that the power supply should not be turned off while a signal from a low-impedance source is being applied to any gate terminal. When the V_{DD} supply is off, the positive "back-bias" voltage is removed from the cathode of diode D_2 (see Fig. 34). Consequently, an input signal with positive-going polarity can drive D_2 into conduction. Under these conditions a low-impedance signal source can provide sufficient current to permanently damage D_2 and/or melt aluminum interconnection paths. Therefore, if, in any system design using the CA3600E, any gate input excursion is expected to exceed $+V_{DD}$ or fall below $-V_{SS}$, the current through the input diodes should be limited to 100 μA.
5. All unused gate-input terminals should be connected to V_{SS} (ground). When source terminals (e.g., Nos. 2 and 11) of p-channel transistors are unused in circuitry, they should be connected to terminal No. 14. Likewise, when source terminals (e.g., Nos. 4 and 9) of n-channel transistors are unused, they should be connected to terminal No. 7.
6. After CA3600E units have been mounted on circuit boards, proper handling precautions should still be observed. Until these subassemblies are inserted into a complete system, the board is no more than an extension of the device leads mounted on the board. It is a good practice to place conductive tape or jumpers on circuit-board terminals to "ground" gate terminals.
7. In some applications of the CA3600E separate positive and negative power supplies may be employed (e.g., see Fig.22). In such applications provisions must be made so that the positive supply voltage is applied prior to the application of negative supply voltage and vice versa on shutdown. This precaution is necessary to avoid possible damage due to "latching" involving the substrate and protective diode circuits.

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14-Lead Dual-In-Line
Plastic Package

H-1517

High-Current N-P-N Transistor Arrays

Four Individual Sealed-Junction High-Current N-P-N Transistors

Features:

- High Current – 1 A
- High Breakdown Voltage:
  CA3725G = 80 V dc min. $V_{(BR)CES}$ @ I_C = 10 μA
  CA3724G = 70 V dc min. $V_{(BR)CES}$ @ I_C = 10 μA
- Fast Switching Speeds:
  t_{on} = 30 ns typ. @ I_C = 500 mA
  t_{off} = 36 ns typ. @ I_C = 500 mA
- "Hermetic Chip" Construction
- Silicon Nitride Passivated
- Platinum Silicide Ohmic Contacts
- Gold Chip-Metallization
- Electrically similar and pin compatible with industry types MPQ3724, MPQ3725; FPQ3724, FPQ3725; DH3724, DH3725; SP3724, SP3725 in similar packages

The RCA-CA3724G and -CA3725G are high-current n-p-n transistor arrays each containing 4 individual sealed-junction high-current n-p-n transistors. They are intended for high-current, high-speed switching and driver applications.

These devices are alike except for breakdown voltage ratings.

The CA3724G and CA3725G are supplied in a 14-lead dual-in-line plastic package and operate over the full military temperature range of −55°C to +125°C. The transistor chips used in these packages are of the sealed-junction type to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure.

The semiconductor junctions are sealed by utilizing a silicon nitride passivation layer. A multi-layered, highly corrosion-resistant, terminal-connection system of unique design is employed.

Applications:

- Core-Memory Driver
- High-Speed Switching
- High-Current LED Driver
- High-Voltage Switching
- Relay and Solenoid Driver
- Lamp Driver

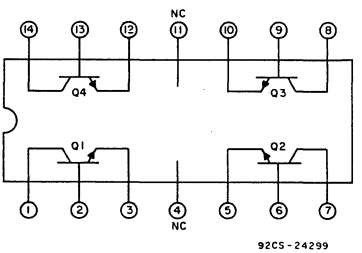


Fig. 1—Terminal diagram (top view).

MAXIMUM RATINGS, Absolute-Maximum Values at $T_A = 25^oC$

		CA3724G	CA3725G	
COLLECTOR-TO-EMITTER VOLTAGE With Base Open	V_{CEO}	40	50	V
COLLECTOR-TO-BASE VOLTAGE With Emitter Open	V_{CBO}	70	80	V
EMITTER-TO-BASE VOLTAGE With Collector Open	V_{EBO}	6	6	V
COLLECTOR CURRENT	I_C	1.0	1.0	A
POWER DISSIPATION:	P_D			
At T_A up to 25°C:				
For Each Transistor		1.0	1.0	W
Total Package		2.0	2.0	W
At T_A above 25°C derate linearly		20		mW/°C
AMBIENT TEMPERATURE RANGE:				
Operating		−55 to +125	−55 to +125	°C
Storage		−65 to +150	−65 to +150	°C
LEAD TEMPERATURE (DURING SOLDERING):				
At distance 1/32" (3.17 mm) from seating plane for 10 s max.		300	300	°C

ELECTRICAL CHARACTERISTICS AT $T_A = 25^oC$

Characteristic	Symbol	Test Conditions	Limits						Units
			CA3724G			CA3725G			
			Min.	Typ.	Max.	Min.	Typ.	Max.	
Collector-to-Emitter Sustaining Voltage *	$V_{CEO}(sus)$	I_C = 10 mA, I_B = 0	40	–	–	50	–	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CES}$	I_C = 10 μA, I_B = 0	70	–	–	80	–	–	V
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	70	–	–	80	–	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	6	–	–	6	–	–	V
Base-to-Emitter Saturation Voltage *	$V_{BE}(sat)$	I_C = 500 mA, I_B = 50 mA	0.75	–	1.0	0.75	–	1.0	V
Collector-to-Emitter Saturation Voltage *	$V_{CE}(sat)$	I_C = 500 mA, I_B = 50 mA	–	–	0.5	–	–	0.5	V
Collector-Cutoff Current	I_{CBO}	V_{CB} = 40 V, I_E = 0	–	–	1.7	–	–	1.7	μA
Static Forward-Current Transfer * Ratio (Beta)	h_{FE}	I_C = 100 mA, V_{CE} = 1.0 V I_C = 500 mA, V_{CE} = 1.0 V I_C = 1 A , V_{CE} = 1.0 V	35 30 20	– – –	– – –	35 30 20	– – –	– – –	
Small-Signal Forward-Current Transfer Ratio	h_{fe}	I_C = 50 mA, V_{CE} = 10 V f = 100 MHz	2.0	–	–	2.0	–	–	
Turn-On Time (See Test Ckt. Fig. 2)	t_{on}	I_C = 500 mA, I_{B1} = 50 mA	–	–	40	–	–	40	ns
Turn-Off Time (See Test Ckt. Fig. 2)	t_{off}	I_C = 500 mA, $I_{B1} = I_{B2}$ = 50 mA	–	–	60	–	–	60	ns
Emitter-to-Base Capacitance	C_{eb}	I_C = 0, V_{EB} = 0.5 V	–	95	–	–	95	–	pF
Collector-to-Base Capacitance	C_{cb}	I_E = 0, V_{CB} = 10 V	–	12	–	–	12	–	pF

* Pulse Conditions: width = 300 μs; duty cycle = 1%

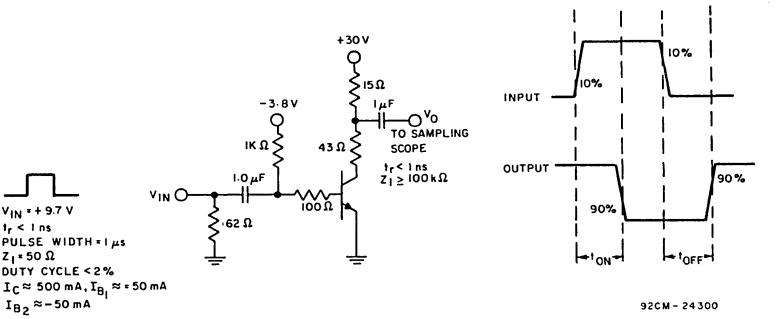


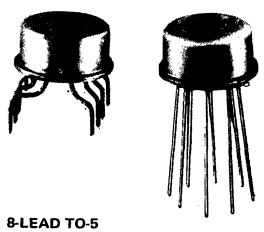
Fig. 2–Switching time test circuit.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

Premium Types CA6078AS, CA6078AT CA6741S, CA6741T



8-LEAD TO-5 with Dual-In-Line Formed Leads H-1787

8-LEAD TO-5 H-1528

Operational Amplifiers

CA6078AT — Micropower Type
CA6741T — General-Purpose Type

For Applications where Low Noise (Burst + 1/f) is a Prime Requirement

Virtually free from "popcorn" (burst) noise: device rejected if any noise burst exceeds 20 μV (peak), referred to input over a 30-second time period.

RCA-CA6078AT and CA6741T* are low-noise linear IC operational amplifiers that are virtually free of "popcorn" (burst) noise.

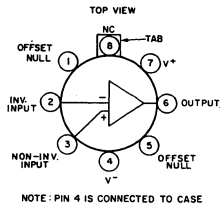
These low-noise versions of the CA3078AT and CA3741T are a result of improved processing developments and rigid burst-noise inspection criteria. A highly selective test circuit (See Fig. 2) assures that each type meets the rigid low-noise standards shown in the data section. This low-burst-noise property also assures excellent performance throughout the 1/f noise spectrum.

In addition the CA6078AT and CA6741T offer the same features incorporated in the CA3078AT and CA3741T respectively, including output short-circuit protection, latch-free operation, wide common-mode and differential-mode signal ranges, and low-offset nulling capability.

For detailed data, characteristics curves, schematic diagram, dimensional outline, and test circuits, refer to the Operational Amplifier Data Bulletins File No. 531 and 535. In addition, for details of considerations in burst-noise measurements, refer to Application Note, ICAN-6732, "Measurement of Burst ("Popcorn") Noise in Linear IC's".

The CA6078AT and CA6741T utilize the hermetically sealed 8-lead TO-5 type package. The CA6078AT and the CA6741T can also be supplied on request with dual-in-line formed leads. These types are identified as the CA6078AS and CA6741S. This formed-lead configuration conforms to that of the 8-lead dual-in-line (Mini—Dip) package. For terminal arrangements, see page 4.

*Formerly Dev. No. TA5807X and TA6029 respectively.



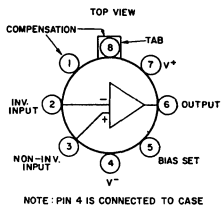
CA6741T

Applications:

- Low-noise AC amplifier
- Narrow-band or band-pass filter
- Integrator or differentiator
- DC amplifier
- Summing amplifier

Features:

- Internal phase compensation
- Input bias current: 500 nA max.
- Input offset current: 200 nA max.
- Open-loop voltage gain: 50,000 (94 dB) min.
- Input offset voltage: 5 mV max.



CA6078AT

Applications:

- Portable electronics
- Medical electronics
- DC amplifier
- Narrow-band or band-pass filter
- Integrator or differentiator
- Instrumentation
- Telemetry
- Summing amplifier

Features:

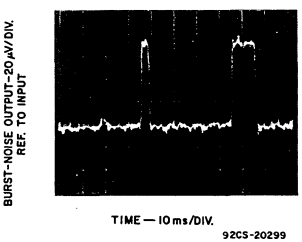
- Open-loop voltage gain: 40,000 (92 dB) min.
- Input offset voltage: 3.5 mV max.
- Operates with low total supply voltage: 1.5 V min. (± 0.75 V)
- Low quiescent operating current: adjustable for application optimization
- Input bias current: adjustable to below 1 nA

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

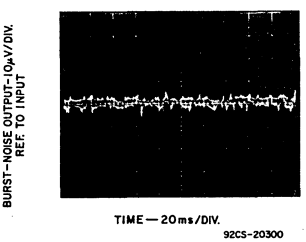
	CA6741T	CA6078AT
DC Supply Voltage (between V^+ and V^- terminals)	44 V	36 V
Differential-Mode Input Voltage	±30 V	±6 V
Common-Mode DC Input Voltage▲	±15 V	V^+ to V^-
Device Dissipation:		
Up to 75°C (CA6741T), Up to 125° (CA6078AT)	500 mW	250 mW
Above 75°C	Derate linearly 5 mW/°C	
Temperature Range:		
Operating	−55 to +125 °C	−55 to +125 °C
Storage	−65 to +150 °C	−65 to +150 °C
Output Short-Circuit Duration●	No limitation	No limitation
Lead Temperature (During soldering):		
At distance 1/16 ±1/32 inch (1.59 ±0.79 mm)		
from case for 10 seconds max.	300 °C	300 °C

▲If Supply Voltage is less than ±15 volts, the Absolute Maximum Input Voltage is equal to the Supply Voltage.

●Short circuit may be applied to ground or to either supply.



a. Typ. device with high-burst-noise characteristic.



b. Typ. device controlled for burst noise.

Fig.1—Typ. waveforms of type with high burst noise and type controlled for burst noise.

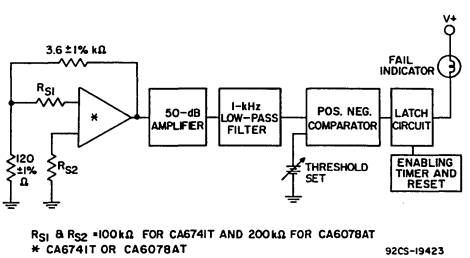


Fig.2—Block diagram of burst-noise "popcorn" test equipment.

ELECTRICAL CHARACTERISTICS – CA6078AT, *For Equipment Design.*

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS Supply Volts: $V^+ = 6$, $V^- = -6$ $T_A = 25^oC$, $I_Q = 20\ \mu A$	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
Noise Characteristic						
"Popcorn" (Burst) Noise		Bandwidth = 1 kHz $R_{S1} = R_{S2} = 200\ k\Omega$	Device is rejected if the total noise voltage (burst + 1/f), referred to input, exceeds 20 μV peak, during a 30-sec. test period.			
Principal Characteristics (For detailed Electrical Characteristics refer to CA3078AT Data Bulletin, File No. 535.)						
Input Offset Voltage	V_{IO}	$R_S \leq 10\ k\Omega$	–	0.7	3.5	mV
Input Offset Current	I_{IO}		–	0.5	2.5	nA
Input Bias Current	I_{IB}		–	7	12	nA
Open-Loop Differential Voltage Gain	A_{OL}	$R_L \geq 10\ k\Omega$ $V_O = \pm 4V$	40,000 92	100,000 100	– –	dB
Common-Mode Input Voltage Range	V_{ICR}	$V^+ = V^- = 15$ V	±14	–	–	V
Common-Mode Rejection Ratio	CMRR	$R_S \leq 10\ k\Omega$	80	115	–	dB
Output Voltage Swing	$V_{O(P-P)}$	$R_L \geq 10\ \Omega$	±13.7	±14.1	–	V
		$R_L \geq 2\ k\Omega$	–	±14	–	
Supply Current	I_Q		–	20	25	μA

ELECTRICAL CHARACTERISTICS – CA6741T, *For Equipment Design.*

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS Supply Volts; $V^+ = 15$, $V^- = -15$ $T_A = 25^oC$	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
Noise Characteristic						
"Popcorn" (Burst) Noise		Bandwidth = 1 kHz $R_{S1} = R_{S2} = 100\ k\Omega$	Device is rejected if the total noise voltage (burst + 1/f), referred to input, exceeds 20 μV peak, during a 30-sec. test period.			
Principal Characteristics (For detailed Electrical Characteristics refer to CA3741T Data Bulletin, File No. 531.)						
Input Offset Voltage	V_{IO}	$R_S \leq 10\ k\Omega$	–	1	5	mV
Input Offset Current	I_{IO}		–	20	200	nA
Input Bias Current	I_{IB}		–	80	500	nA
Open-Loop Differential Voltage Gain	A_{OL}	$R_L \geq 2\ k\Omega$ $V_O = \pm 10$ V	50,000 94	200,000 106	– –	dB
Common-Mode Input Voltage Range	V_{ICR}		±12	±13	–	V
Common-Mode Rejection Ratio	CMRR	$R_S \leq 10\ k\ \Omega$	70	90	–	dB
Output Voltage Swing	$V_{O(P-P)}$	$R_L \geq 10\ k\ \Omega$	±12	±14	–	V
		$R_L \geq 2\ k\ \Omega$	±10	±13		
Supply Current	I_Q		–	1.7	2.8	mA

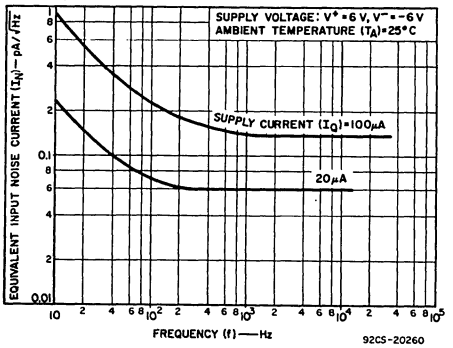


Fig.3—I_N vs. Frequency for CA6078AT.

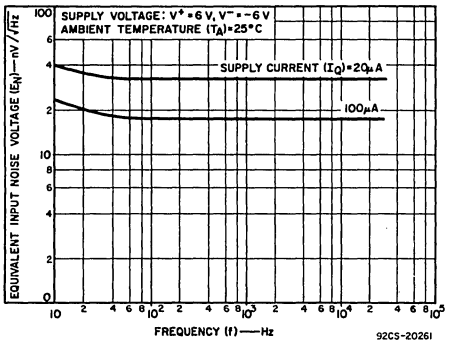


Fig.4—E_N vs. Frequency for CA6078AT.

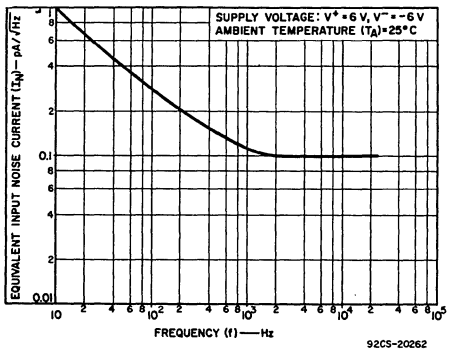


Fig.5—I_N vs. Frequency for CA6741T.

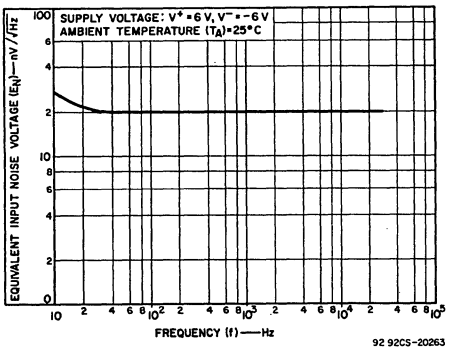


Fig.6—E_N vs. Frequency for CA6741T.

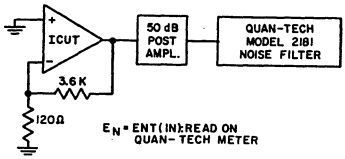


Fig.7—Test block diagram for E_N.

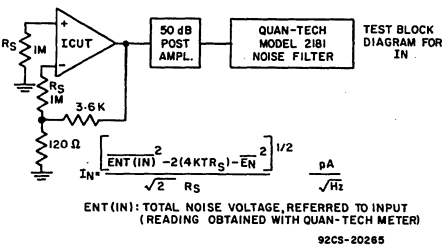


Fig.8—Test block diagram for I_N.

RCA Solid State Division

Digital Integrated Circuits

CD2500E
CD2501E
CD2502E
CD2503E

BCD to 7-Segment Decoder-Drivers

Monolithic Silicon

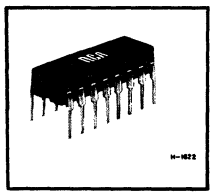
RCA CD2500E series 7-Segment Decoder-Drivers are monolithic MSI integrated circuits which decode BCD (8-4-2-1 code) inputs to 7-line outputs representing a decimal number from 0 to 9 on 7-segment incandescent display devices.

RCA CD2500E and CD2501E are 30 mA per-output-line devices designed for use with incandescent display devices such as the RCA DR2000 and DR2010. The CD2500E, in addition to the outputs for the 7-segment display device, has a decimal point output; the CD2501E also has a special-feature, a terminal to provide for ripple blanking output and intensity control input. The ripple blanking output blanks out all non-significant zeroes in the numerical display. The ripple blanking output terminal is also available for use as an intensity control input from an external variable pulse-width control source, as shown in Fig. 7.

RCA CD2502E and CD2503E are 80 mA-per-line versions of the CD2500E and CD2501E, respectively, and are designed for use with high-current lamps and relays.

RCA CD2500E series devices are supplied in 16-lead dual in-line plastic packages which can be used over the operating temperature range of 0° C to + 75° C.

30mA and 80mA/Segment DECODER-DRIVERS For Use With Low-Voltage Digital Display Devices, Lamps, and Relays



DP_0 = Decimal Point Output
DP_I = Decimal Point Input

DP_I must be supplied from an external source

CD2500E and CD2502E perform the inverter-driver function necessary to energize the decimal point filament in the display device.

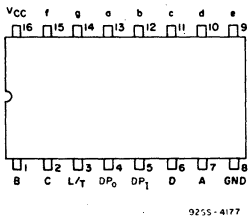


Fig. 1 - CD2500E and CD2502E (with decimal point)

FEATURES:

- High current sinking capability for direct display driving
- Intensity control provision
- BCD inputs are compatible with commercially available DTL & TTL devices
- Lamp test provision
- 5 V power supply
- Clamp diodes on all inputs
- Lamp supply up to +8 volts
- Ripple blanking capability
- Decimal point output
- Over-range detection (automatic blanking of display device when BCD input > 9)

RB_0 = Ripple-Blanking Output & Intensity Control Input
RB_I = Ripple-Blanking Input

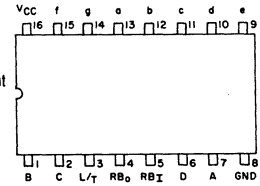


Fig. 2 - CD2501E and CD2503E (with ripple blanking and intensity control provision)

ABSOLUTE MAXIMUM RATINGS at 25°C unless otherwise specified:

Power Supply Voltage:
- Continuous (0°C to +75°C) . . . . . – 0.5 to + 5.5 V
- Pulsed (duration 1 second) . . . . . – 0.5 to + 8 V

Input Voltage . . . . . – 0.5 to + 5.5 V

Output Voltage (open collector transistor) . . – 0.5 to + 8 V

Operating Temperature Range . . . . . 0°C to +75°C

Storage Temperature Range . . . . . – 65°C to + 150°C

Lead Temperature (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. . . . . +265°C

ELECTRICAL CHARACTERISTICS at Ambient Temperature (T_A) Indicated

CHARACTERISTICS	SYMBOLS	MEASUREMENT TERMINALS	TEST CONDITIONS		0°C MIN.	0°C MAX.	+25°C MIN.	+25°C TYP.	+25°C MAX.	+75°C MIN.	+75°C MAX.	UNITS
Input High Voltage (Logic 1)	V_{IH}	1, 2, 5, 6, & 7	Input high threshold voltage		2.0	–	2.0	–	–	2.0	–	V
		3	V_{CC} = 4.75 V, I_{IH} = 0 Ground all other inputs		2.4	–	2.4	–	–	2.4	–	V
Input Low Voltage (Logic 0)	V_{IL}	1, 2, 5, 6, & 7	Input low threshold voltage		–	0.85	–	–	0.85	–	0.85	V
		3			–	0.45	–	–	0.45	–	0.45	
Input Forward Current	I_{IL}	1, 2, 5, 6, & 7	V_F = 0.45 V	V_{CC} = 5.25 V	–	– 1.6	–	– 1.0	– 1.6	–	– 1.6	mA
		3 { CD2501E, CD2503E			–	– 10.0	–	–	– 10.0	–	– 10.0	
		3 { CD2500E, CD2502E			–	– 10.4	–	–	– 10.4	–	– 10.4	
		1, 2, 5, 6, & 7	V_F = 0 Terminal 3 only	V_{CC} = 4.75 V	–	– 1.41	–	–	– 1.41	–	– 1.41	mA
		3 { CD2501E, CD2503E			–	– 9.0	–	–	– 9.0	–	– 9.0	
		3 { CD2500E, CD2502E			–	– 9.4	–	–	– 9.4	–	– 9.4	
Input Reverse Current	I_{IH}	1, 2, 5, 6, & 7	V_{CC} = 5.25 V Terminal 3 grounded	V_R = 4.5 V	–	40	–	–	40	–	60	μA
				V_R = 2.4 V	–	40	–	–	40	–	40	
Output Low Voltage	V_{OL}	9 thru 15 { CD2500E, CD2501E and 4 of CD2500E	V_{CC} = 4.75 V, I_{OL} = 30 mA		–	0.40	–	0.30	0.40	–	0.40	V
		4 { CD2501E, CD2503E	V_{CC} = 5.25, I_{OL} = 3.2 mA		–	0.45	–	0.30	0.45	–	0.45	
			V_{CC} = 4.75, I_{OL} = 2.82 mA		–	0.45	–	0.30	0.45	–	0.45	
		9 thru 15 { CD2502E, CD2503E and 4 of CD2502E	V_{CC} = 4.75 V, I_{OL} = 80 mA		–	1.0	–	0.60	1.0	–	1.0	
Output High Voltage	V_{OH}	9 thru 15 – All types and 4 of { CD2500E, CD2502E	V_{CC} = 5 V, I_{OH} = 200 μA		8.0	–	8.0	–	–	8.0	–	V
		4–CD2501E, CD2503E	V_{CC} = 4.75 V, I_{OH} = – 240 μA		2.4	–	2.4	–	–	2.4	–	
Input Capacitance	C_{IN}	1, 2, 5, 6, & 7	V_{CC} = 5.0 V		–	–	–	3	5	–	–	pF
Power Supply Current Drain (Terminal 16)	I_{CC_L}	CD2501E, CD2503E	V_{CC} = 5.0 V (Segment Output Currents = 0)		–	–	–	48	–	–	–	mA
		CD2500E, CD2502E	Terminal 3 Grounded		–	–	–	50	–	–	–	

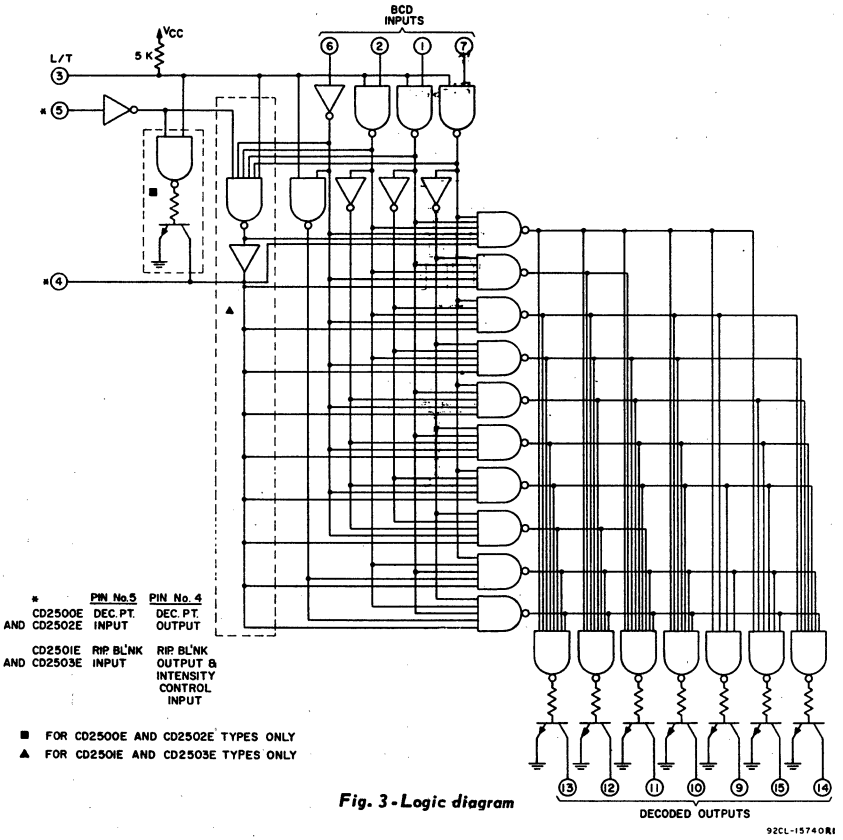
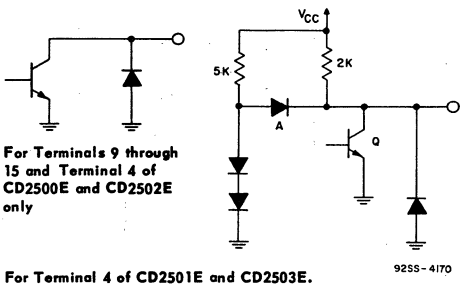


Fig. 3 - Logic diagram



For Terminals 9 through 15 and Terminal 4 of CD2500E and CD2502E only

For Terminal 4 of CD2501E and CD2503E.

Transistor Q is "turned on" when BCD code equals 0 and Terminal 5 is at "0 Level" (Grounded). When BCD code is between 0 and 9, transistor Q is "open". Diode A and transistor are "open" when BCD code is > 9.

Fig. 4 - Equivalent output circuits

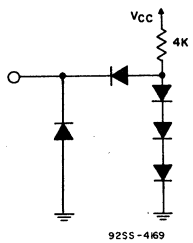


Fig. 5 - Equivalent input circuit for terminals 1, 2, 5, 6 & 7

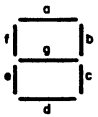
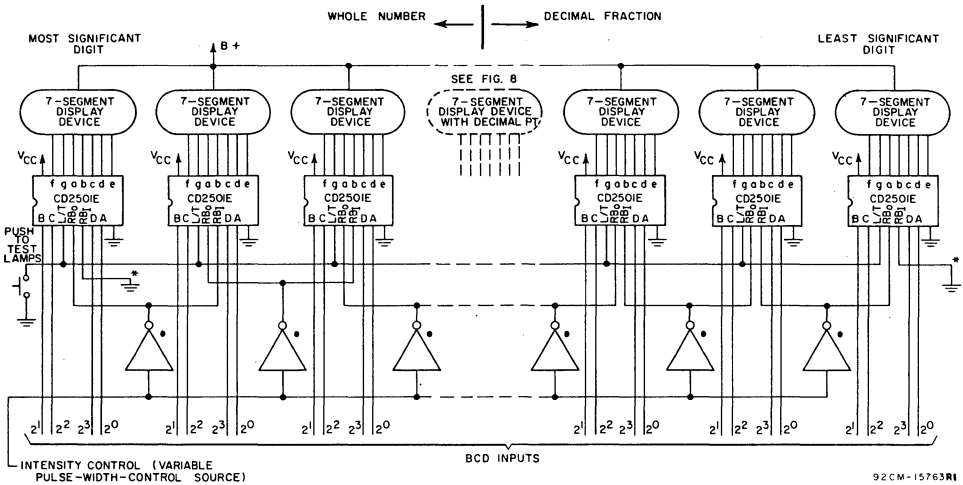


Fig. 6-Digital display device segment designation

TRUTH TABLE

INPUT 0 = Low Level 1 = High Level							OUTPUT 0 = Filament Lit 1 = Filament OUT									TUBE DISPLAY
D	C	B	A	L/T	DP_I	RB_I	a	b	c	d	e	f	g	DP_O	RB_O	
X	X	X	X	0	–	X	0	0	0	0	0	0	0	–	1	8
0	0	0	0	1	–	0	1	1	1	1	1	1	1	–	0	
0	0	0	0	1	–	1	0	0	0	0	0	0	1	–	1	0
0	0	0	1	1	–	X	1	0	0	1	1	1	1	–	1	1
0	0	1	0	1	–	X	0	0	1	0	0	1	0	–	1	2
0	0	1	1	1	–	X	0	0	0	0	1	1	0	–	1	3
0	1	0	0	1	–	X	1	0	0	1	1	0	0	–	1	4
0	1	0	1	1	–	X	0	1	0	0	1	0	0	–	1	5
0	1	1	0	1	–	X	0	1	0	0	0	0	0	–	1	6
0	1	1	1	1	–	X	0	0	0	1	1	1	1	–	1	7
1	0	0	0	1	–	X	0	0	0	0	0	0	0	–	1	8
1	0	0	1	1	–	X	0	0	0	0	1	0	0	–	1	9
1	0	1	0	1	–	X	1	1	1	1	1	1	1	–	1	
1	0	1	1	1	–	X	1	1	1	1	1	1	1	–	1	
1	1	0	0	1	–	X	1	1	1	1	1	1	1	–	1	
1	1	0	1	1	–	X	1	1	1	1	1	1	1	–	1	
1	1	1	0	1	–	X	1	1	1	1	1	1	1	–	1	
1	1	1	1	1	–	X	1	1	1	1	1	1	1	–	1	
–	–	–	–	1	1	–	–	–	–	–	–	–	–	0	–	.
–	–	–	–	1	0	–	–	–	–	–	–	–	–	1	–	
–	–	–	–	0	X	–	–	–	–	–	–	–	–	0	–	.

X = Don't care (0 or 1 entry has no effect)
L/T = Lamp test
RB_I = Ripple Blanking Input
RB_O = Ripple Blanking Output
DP_I = Decimal Point Input
DP_O = Decimal Point Output



• Resistor pull-up output T^2L, DTL, or RTL inverter.

* Suppression of the non-significant zeroes (at both extremes of the display) is accomplished by grounding the RB_1 terminal of the devices associated with the most significant digit of the whole part of the number displayed and the least significant digit of the fractional portion of that number.

Fig. 7 - Typical ripple blanking and intensity control application diagram using RCA CD2501E and display devices DR2000 or equivalents (See Table A)

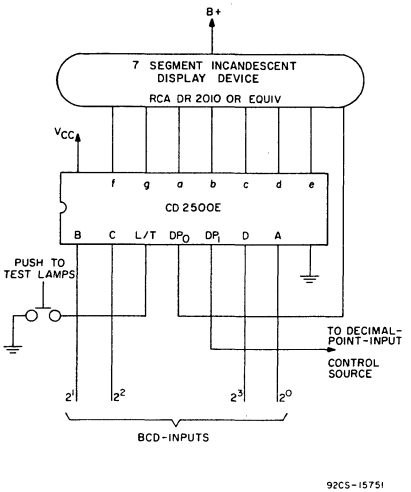


Fig. 8 - Typical decimal point feature application diagram using RCA CD2500E and RCA display device DR2010 (or equivalent)

TABLE A

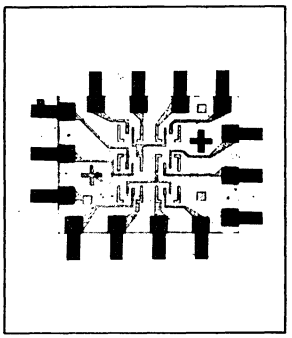
DISPLAY DEVICE TYPE	TYPE OF DISPLAY	CHARACTERISTICS
DR2000		Required Driving Current = 24 ± 2 mA per segment
DR2010	X	0.6" Letter height

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA741L CA3018L CA3045L CA3083L
CA3015L CA3028AL CA3049L CA3084L
CA3039L CA3054L CA3085L



Beam-Lead Devices for Hybrid Circuit Applications

- Transistor Arrays
- Diode Arrays
- Differential Amplifiers
- Operational Amplifiers
- Voltage Regulators

Features

Assembly

- Simplified repairability
- Use of non-hermetic packages possible
- Silicon nitride passivated
- Platinum silicide ohmic contacts
- Batch handling of chips, batch bonding of beam leads and external lead connections

The beam-lead sealed-junction integrated circuits described in this bulletin are fabricated by a technology which involves the utilization of a passivated layer to seal delicate semiconductor junctions and a multilayered interconnection system of unique design which is stable, highly corrosion-resistant, and readily bondable for attachment to a suitable substrate containing thick or thin film wiring.

Beam Lead identifies a structure in which gold beam leads are extended over the semiconductor chip edges as cantilever beams. **Sealed Junction** indicates that the integrated circuit chip is completely protected from the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure.

General Considerations

Conventional IC technology has made very substantial contributions to the reliability of solid-state electronics despite the fact that the conventional IC chip is non-hermetic and employs an aluminum-film interconnection system. These considerations have forced the use of hermetic packages or elaborate bulky plastic packages to guard the integrated circuit chip against even modest amounts of humidity. In addition, connection to the aluminum metallization on the chip is customarily accomplished by the use of tiny wires. The reliability of these wired connections to the chip and its external circuit is dependent on human skill and accuracy to a considerable extent.

The culmination of continuing research and development in the quest for IC's having greater reliability, has led to the development of sealed-junction technology for IC fabrication. The beam-lead, sealed-junction device is a truly hermetic IC chip which is impervious to the deteriorating effects of moisture and other potential contaminants. Furthermore, circuit interconnections on and to the chip are accomplished by the use of gold conductors to further enhance reliability. The precious-metal interconnection system on the chip, which is integral with the chip, is, in turn, connected to tiny gold beams (0.003" x 0.006" x 0.0005") which extend over the edge of the chip to serve as leads to external circuit paths and components.

- Precious metal interconnection metallization
- Precious metal beam leads
- Broad beam leads make interconnect paths less critical; bonds easier to inspect, and defective chips easier to replace
- Batch fabrication techniques provide devices with high reliability at lowest possible cost.

Performance

- Exceptional reliability results from use of sealed-junction beam-lead technology
- Inspectable bonds
- Low-stress, high-strength bonds achieved
- Reliable operation over full military temperature range $-55^{\circ}C$ to $+125^{\circ}C$

The beam lead integrated circuit chip with its gold leads has ideal mechanical characteristics for use in connection with automated handling methods of attachment to film type wiring on a suitable substrate thus making it possible to achieve a higher order of reliability in the interconnection system than has been achieved heretofore.

A brief resume of the manufacturing process used in producing beam lead IC's is included in the APPENDIX following the OPERATING CONSIDERATIONS.

OPERATING CONSIDERATIONS

When a beam lead device is being bonded to a substrate, certain minimal precautions (listed below), with reference to pattern screening must be taken to prevent stress that can result in breakage, or separation of the conductor paths:

1) Do not mount components within the outside dimension of the bonding tool.
2) Do not use any cross-over or insulation within this dimension.
3) Do not use any resistor terminations within this dimension.
4) Use individual pads for bonding leads wherever feasible.

As in any design, adequate cooling must be considered. Temperature rise in a beam-lead device, when mounted in a particular assembly is a direct result of the dissipation within the device, the distribution of other heat sources within the assembly, and the ability of the assembly to dissipate the total heat generated.

Specific factors which govern the heat flow within such assemblies are:

1. Beam-lead width and thickness
2. Number of beam leads
3. Thermal characteristics of the substrate
4. Thermal characteristics of the ambient surrounding the beam-lead device.

Because of these factors it is, therefore, impractical to specify thermal ratings for beam-lead device assemblies. In consideration of these factors, it is recommended that the chip temperature be checked by direct measurement to avoid exceeding a maximum chip junction temperature of 150°C.

TERMINAL LAYOUT DIAGRAMS

RCA beam lead devices will normally be designed utilizing the outline shown in Fig. 1 viewed with the metallization down.

The resistance values included on the schematic diagrams are typical values and have been supplied as a convenience to assist Equipment Manufacturers in optimizing the selection of "outboard" components of equipment designs.

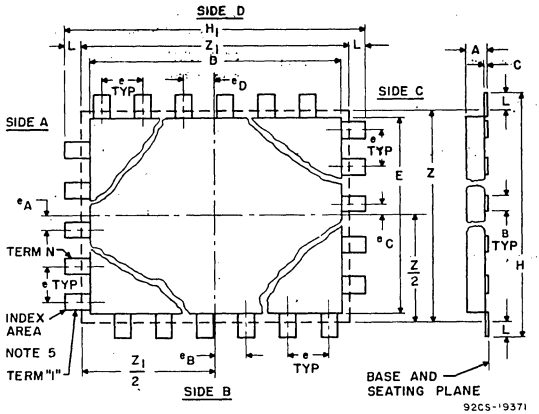
RCA reserves the right to make any changes in the resistance values provided such changes do not adversely affect the published performance characteristics of the device.

APPENDIX

Beam-Lead Manufacturing Processes

An integral passivation layer of silicon nitride protects the beam-lead device from the deteriorating effects of both moisture and contaminants. Low-resistance ohmic contacts to the device junctions are made with platinum silicide which is an extremely stable, non-corrosive intermetallic compound. Gold is used for both the chip interconnections and for the cantilevered beams because it provides high conductivity, is corrosion-resistant, and is readily bondable to a wide variety of substrates and materials. This combination of metallurgically stable components offers the user a chip structure having excellent reliability as compared with the performance of aluminum metallization used in conventional IC designs.

As indicated in the preceding paragraphs, beam-lead technology encompasses a passivating (sealant) layer, a multilayered metal system, and uniquely designed metallization. The metallization consists of a contact of platinum silicide and a layered structure of titanium, platinum, and gold. The metallized pattern which is brought out to the grid, and the subsequent processing are designed to produce a chip in which the attaching leads extend over the edge of the chip. The processing procedure involves the removal of the silicon and the oxide in the grid to leave the beams cantilevered over



SYMBOL	14-LEAD VARIATIONS				16-LEAD VARIATIONS				18-LEAD VARIATIONS				22-LEAD VARIATIONS				NOTES
	INCHES		MILLIMETERS		INCHES		MILLIMETERS		INCHES		MILLIMETERS		INCHES		MILLIMETERS		
	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	
A	.002	.010	.051	.254	.002	.010	.051	.254	.002	.010	.051	.254	.002	.010	.051	.254	
B	.0020	.0045	.0510	.1140	.0020	.0045	.0510	.1140	.0020	.0045	.0510	.1140	.0020	.0045	.0510	.1140	5
C	.0004	.0006	.0102	.0152	.0004	.0006	.0102	.0152	.0004	.0006	.0102	.0152	.0004	.0006	.0102	.0152	
D	–	.045	–	1.14	–	.045	–	1.14	–	.055	–	1.39	–	.065	–	1.65	
E	–	.035	–	.889	–	.045	–	1.14	–	.045	–	1.14	–	.055	–	1.39	
e	.010 TP		.2540TP		.010 TP		.254 TP		.010 TP		.254 TP		.010 TP		.254 TP		4
e_A	.0025TP		.0635TP		.0075TP		.191 TP		.0075TP		.1905TP		.0025TP		.0635TP		2,4
e_B	.0075TP		.1905TP		.0075TP		.191 TP		.0025TP		.0635TP		.0075TP		.1905TP		2,4
e_C	.0025TP		.0635TP		.0075TP		.191 TP		.0075TP		.1905TP		.0025TP		.0635TP		2,4
e_D	.0075TP		.1905TP		.0075TP		.191 TP		.0025TP		.0635TP		.0075TP		.1905TP		2,4
H	.042	.049	1.07	1.25	.052	.059	1.321	1.498	.052	.059	1.32	1.49	.062	.069	1.58	1.75	
H_1	.052	.059	1.32	1.49	.052	.059	1.321	1.498	.062	.069	1.58	1.75	.072	.079	1.83	2.00	
L	.0035	.0070	.0889	.1770	.0035	.0070	.0889	.1770	.0035	.0070	.0889	.1770	.0035	.0070	.0889	.1770	
Z	.035Bsc		.889Bsc		.045Bsc		1.143 Bsc		.045Bsc		1.14Bsc		.055Bsc		1.39Bsc		4
Z_1	.045Bsc		1.14Bsc		.045Bsc		1.143 Bsc		.055Bsc		1.39Bsc		.065Bsc		1.65Bsc		4
N	14		14		16		16		18		18		22		22		6
N_A	3		3		4		4		4		4		5		5		3
N_B	4		4		4		4		5		5		6		6		3
N_C	3		3		4		4		4		4		5		5		3
N_D	4		4		4		4		5		5		6		6		3

NOTES

1. Refer to Rules for Dimensioning Peripheral Lead Outlines.
2. e_A is basic distance from centerline of Z to centerline of first adjacent counter-clockwise beam position, e_B is basic distance from centerline of first adjacent counter-clockwise beam position, etc. Beam position located by e_A, e_B etc. lies at beam pattern if N_A etc. odd; at first counter-clockwise beam position from pattern if N_A etc. even.
3. N_A is the maximum quantity of lead positions on side "A", N_B on side "B" etc. Picture represents case where N = 22, $N_A = N_C = 5$ and $N_B = N_D = 6$. Applicable number of lead positions, both on each side and total, are as tabulated for each variation.
4. Leads shall be held within .001 total of True Position (TP) at Least Material Condition (LMC) and within .002 total of TP at Maximum Material Condition (MMC). Both location requirements to be checked at Z and Z_1 limits.
5. Number one lead position is counter-clockwise end lead position on a side of minimum width. This index lead shall be distinctively marked if existing; if not existing the first clockwise lead shall be so marked. Any projection shall not extend more than .002" over B maximum nor any notch less than .001 under B minimum.
6. N is the maximum quantity of lead positions.

Fig. 1– Terminal layout for all beam-lead devices. See data bulletins for specific information on individual devices. Millimeter dimensions in chart above and shown in parentheses on the terminal layouts for each beam-lead type are derived from the basic inch dimensions as indicated.

the edge of the chip and available for easy attachment to a package or substrates.

RCA's beam lead technology consists of the following processes:

a) deposition of silicon nitride
b) contact openings
c) deposition and formation of conducting paths (contacts and interconnections)
d) circuit separation
e) bonding

A brief description of these processes follows.

deposition of silicon nitride

Silicon nitride which functions as the passivating (sealant) layer is deposited over the surface of the wafer following the diffusion and oxidation steps required to form the individual components of the device.

contact openings

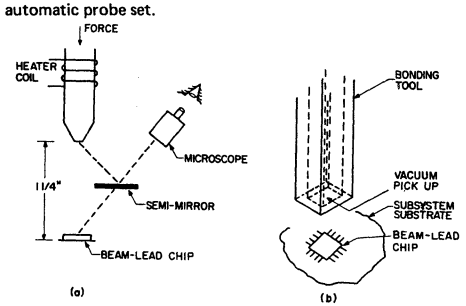
After the entire wafer has been covered with the protective layer of silicon nitride, appropriate windows are opened both in this and the previously formed oxide layer to permit contact with the junction areas of the individual components.

deposition of contacts and interconnections

To integrate the individual components into the circuits, the exposed terminal areas are interconnected with gold leads formed by electroplating. The gold leads are underlaid with titanium, and platinum in that order, over a platinum silicide layer in the contact openings to attain a low-resistance ohmic contact to the silicon. Two electroplating steps are used to form both the gold metallization network and the gold beam leads by means of which appropriate circuit terminals can be connected to external electrical contacts.

circuit separation

A thinning and etching technique is next used to separate the completed circuit chip from the wafer in which they are formed. This separation involves removal of the silicon from the grids between the chips by a very precise chemical etching process which physically separates the circuits from each other but leaves them firmly held in a matrix position. In this position, the individual circuits can be evaluated by an automatic test set operating in conjunction with an automatic probe set.



beam-lead, bonding (See Fig. 2)

The actual bonding of the beam leads to a metallized package or a substrate is performed by a thermocompression technique as follows:

A bonding tool is used to pick up the chip and bond it to the subsystem substrate metallization. The chip and the bonding tool are aligned through the use of a semi-mirror shown in Fig. 2(a). The bonding tool is lowered to the chip. The chip is held firmly by the vacuum [inside the bonding tool, see Fig. 2(b)] and transferred to the bonding station. Another alignment is made [see Fig. 2(a)] by viewing the chip in the tool (through the semi-mirror) and the subsystem substrate metallization. The bonding tool and the chip are then lowered to the subsystem substrate where the heated substrate and the heated bonding tool develop an interface temperature of 300°C between the beam leads and the substrate. Simultaneously, a force is applied to the bonding tool which deforms the ends of the beam leads and completes the thermocompression bond. The bonding time is 2 to 3 seconds.

Any faulty chips can be rebonded. The most significant advantages of the beam lead technology are in this bonding process--

1. Manufacturing the silicon chip beam leads as an integral part of the device eliminates the necessity of bonding to the chip and immediately reduces the number of bonds to be made for an equivalent interconnection.
2. Furthermore, since each lead is an integral part of the contact and not a mechanically-made connection, the reliability of the circuit is greatly enhanced.
3. In addition, the single metal system gold-to-gold employed between contacts and leads not only obviates a reliability factor often associated with bonds with contacts made between dissimilar metals, but also insures a bond completely free from corrosion.
4. And finally, all bonds for a single chip can be made simultaneously providing both technical and economic advantages.

REFERENCES

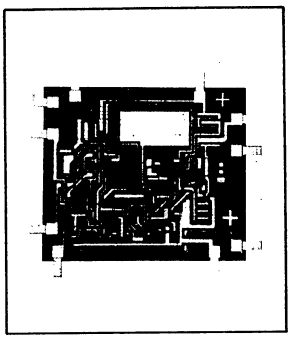
1. The Western Electric Engineer, Dec. 1967.

Fig. 2— a) Alignment to pick up chip and to bond chip to subsystem substrate: force is used only to bond chip;
b) detail of bonding tool to show vacuum pick up.

Linear Integrated Circuits

Monolithic Silicon

CA741L



Beam-Lead Operational Amplifier

High-Gain Operational Amplifier With Internal Phase Compensation

FOR MILITARY, INDUSTRIAL AND CONSUMER APPLICATIONS

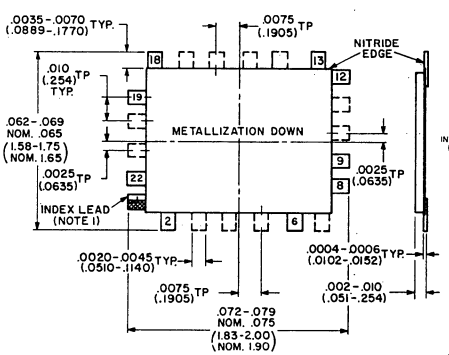
Applications

- Comparator
- Integrator or differentiator
- Summing amplifier
- DC amplifier
- Multivibrator
- Narrow-band or band-pass filter

RCA CA741L is the beam-lead version of the family of CA741, general-purpose high-gain, monolithic operational amplifiers. The CA741L features internal phase compensation, provides output short-circuit protection, and latch-free operation. This type also features large common mode and differential mode signal ranges and has a low offset voltage and nulling capability. The CA741L consists of a differential-input amplifier with an effectively double-ended output that drives a gain and level-shifting stage having a complementary emitter-follower output.

Features

- No external phase compensation required.
- Open-loop voltage gain: 50,000 min.
- Input bias current: 500 nA max.
- Input offset current: 200 nA max.
- Input offset voltage: 5 mV max.
- Operation over the full military temperature range: -55 to +125°C



NOTES:

1. INDEX CONFIGURATION ON BEAM NO.1 OPTIONAL, PROVIDED LEAD MAX. IS NOT INCREASED MORE THAN .002
2. LEADS AT TRUE POSITION (TP) WITHIN .002 TOTAL (REF. USANS Y14.5-1966)
3. SILICON MUST NOT EXTEND BEYOND NITRIDE OVER A BEAM.

Fig. 1-1— Terminal layout for CA741L (22-lead configuration)

CAUTION: Although RCA-CA741L is electrically similar to CA741T, it is not a pin-for-pin replacement.

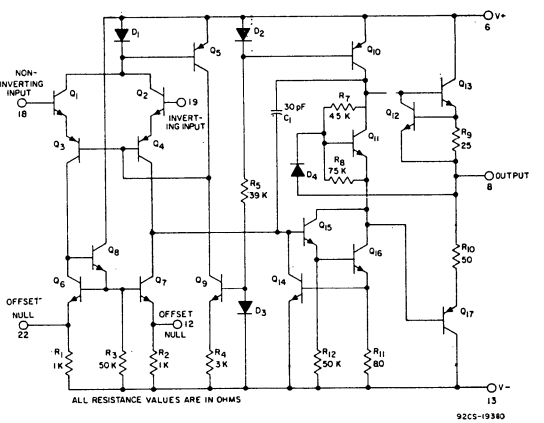


Fig. 1-2— Schematic diagram of CA741L

MAXIMUM RATINGS, *Absolute-Maximum Values at $T_A = 25^oC$*

DC Supply Voltage (between V^+ and V^- terminals) . . . . . 44 V
Differential Input Voltage . . . . . ±30 V
DC Input Voltage* . . . . . ±15 V
Output Short-Circuit Duration . . . . . Indefinite
Voltage between Offset Null and V^- . . . . . ±0.5 V
Temperature Range:
Operating . . . . . −55 to +125°C
Storage . . . . . −65 to +150°C

*If Supply Voltage is less than ±15 volts, the Absolute Maximum Input Voltage is equal to the Supply Voltage.

ELECTRICAL CHARACTERISTICS
For Equipment Design

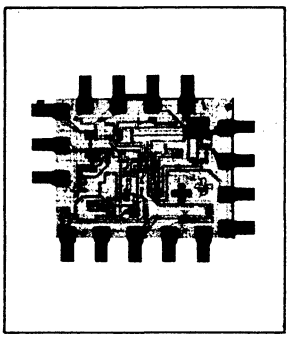
CHARACTERISTICS	SYMBOLS	TEST CONDITIONS SUPPLY VOLTS: $V^+ = 15, V^- = -15$	AMBIENT TEMPERATURE (T_A)	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
Input Offset Voltage	V_{IO}	$R_S \leq 10$ kΩ	25°C	–	1	5	mV
			-55 to +125°C	–	1	6	
Input Offset Current	I_{IO}		25°C	–	20	200	nA
			-55°C	–	85	500	
			+125°C	–	7	200	
Input Bias Current	I_I		25°C	–	80	500	nA
			-55°C	–	300	15000	
			+125°C	–	30	500	
Input Resistance	R_I			0.3	2	–	MΩ
Open-Loop Differential Voltage Gain	A_{OL}	$R_L \geq 2$ kΩ, $V_O = \pm 10$ V	25°C	50,000	200,000	–	
			-55 to +125°C	25,000	–	–	
Common-Mode Input Voltage Range	V_{ICR}		25°C	–	–	–	V
			-55 to +125°C	±12	±13	–	
Common-Mode Rejection Ratio	C_{MRR}	$R_S \leq$ 10 kΩ	25°C	–	–	–	dB
			-55 to +125°C	70	90	–	
Supply Voltage Rejection Ratio	V_{RR}	$R_S \leq$ 10 kΩ	25°C	–	–	–	μV/V
			-55 to +125°C	–	30	150	
Output Voltage Swing	$V_{O(P-P)}$	$R_L \geq$ 10 kΩ	25°C	–	–	–	V
			-55 to +125°C	±12	±14	–	
		$R_L \geq$ 2 kΩ	25°C	–	–	–	
			-55 to +125°C	±10	±13	–	
Supply Current			25°C	–	1.7	2.8	mA
			-55°C	–	2	3.3	
			+125°C	–	1.5	2.5	
Device Dissipation	P_D		25°C	–	50	85	mW
			-55°C	–	60	100	
			+125°C	–	45	75	

When incorporating RCA Solid State Devices in equipment, it is recommended that the designer refer to "Operating Considerations for RCA Solid State Devices", Form No. 1CE-402, available on request from RCA, Commercial Engineering, Harrison, N. J. 07029.

Linear Integrated Circuits

Monolithic Silicon

CA3015L



Beam-Lead Operational Amplifier

Applications

- Narrow-Band and Bandpass Amplifier
- Operational Functions
- Feedback Amplifier
- DC and Video Amplifier
- Multivibrator
- Oscillator
- Comparator
- Servo Driver
- Scaling Adder
- Balanced Modulator-Driver

Features

Open-Loop Voltage Gain	70	dB	typ.
Common-Mode Rejection Ratio	103	dB	typ.
Output Impedance	92	Ω	typ.
Input Offset Voltage	1	mV	typ.
Static Device Dissipation at ±12V	175	mW	typ.
± 6V	30	mW	typ.
± 3V	7	mW	typ.

- Operation over the full military temperature range: −55 to +125°C

The RCA CA3015L is the beam-lead version of the CA3015 operational amplifier family. The beam leads of this device are formed as an integral part of the IC chip during the batch fabrication process.

The CA3015L is particularly suited for applications in hybrid circuits where hermetic packaging, low cost, and reliable operation are prime considerations.

For applications of the CA3015 family of operational amplifiers see the companion Application Notes, ICAN-5290 "Integrated Circuit Operational Amplifiers", ICAN-5213 "Application of the RCA-CA3015, CA3016 Integrated Circuit Operational Amplifiers,"and ICAN-5015 "Application of the RCA-CA3008, CA3010 Integrated Circuit Operational Amplifiers".

CAUTION: ALTHOUGH RCA-CA3015L is electrically similar to CA3015, it is not a pin-for-pin replacement.

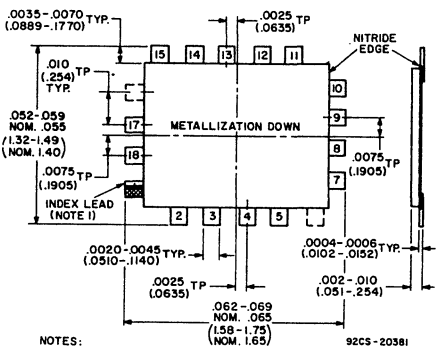


Fig. 2-1— Terminal layout for CA3015L (18 lead configuration).

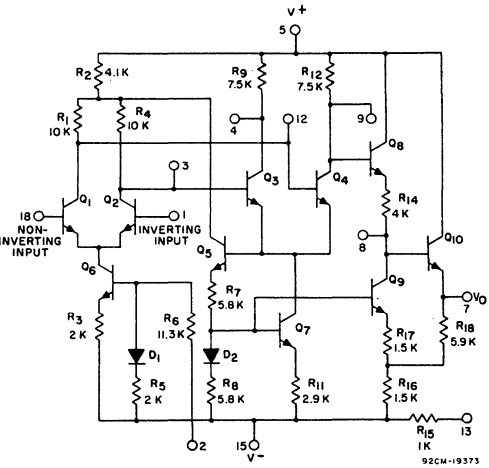


Fig. 2-2— Schematic diagram of CA3015L

MAXIMUM RATINGS,
ABSOLUTE-MAXIMUM VALUES.

OPERATING TEMPERATURE RANGE . . . . . -55°C to +125°C
STORAGE TEMPERATURE RANGE . . . . . -65° to +150°C
SIGNAL VOLTAGE . . . . . –8 V to +1 V

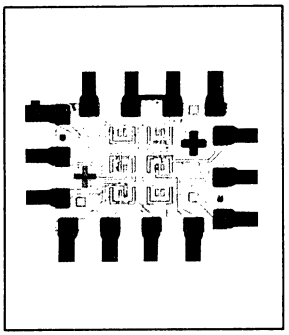
ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOL	LIMITS			UNITS
		MIN.	TYP.	MAX.	
STATIC CHARACTERISTICS: V^+ = +12 V, V^- = -12 V					
Input Offset Voltage	V_{IO}	–	1.37	5	mV
Input Offset Current	I_{IO}	–	1.07	5	μA
Input Bias Current	I_I	–	9.6	24	μA
Input Offset Voltage Sensitivity:					mV/V
Positive	$\Delta V_{IO}/\Delta V^+$	–	0.096	0.5	
Negative	$\Delta V_{IO}/\Delta V^-$	–	0.156	0.5	
Device Dissipation	P_D	–	175	–	mW
DYNAMIC CHARACTERISTICS:					
Open-Loop Differential Voltage Gain	A_{OL}	66	70	–	dB
Common-Mode Rejection Ratio	CMRR	80	103	–	dB
Maximum Output-Voltage Swing	$V_{O(P-P)}$	12	14	–	V_{P-P}
Input Impedance	Z_{IN}	5	7.8	–	kΩ
Output Impedance	Z_{OUT}	–	92	–	Ω
Common-Mode Input-Voltage Range	V_{ICR}	–	+0.65	–	V
		–	-8	–	

Linear Integrated Circuits

Monolithic Silicon

CA3018L



Beam-Lead General-Purpose Transistor Array

Two Isolated Transistors and a Darlington-Connected Transistor Pair

FOR LOW-POWER APPLICATIONS AT FREQUENCIES FROM DC THROUGH THE VHF RANGE

The CA3018L is a beam-lead version of the RCA CA3018 and consists of four general purpose silicon n-p-n transistors on a common monolithic substrate. The beam leads of this device are formed as an integral part of the IC chip during the batch fabrication process.

Two of the four transistors are connected in the Darlington configuration. The substrate is connected to a separate terminal for maximum flexibility.

The CA3018L is particularly suited for applications in hybrid circuits where hermetic packaging, low costs, and reliable operation are prime considerations. For applications of the general purpose transistors see RCA Application Note, ICAN-5296 "Application of the RCA CA3018 Integrated-Circuit Transistor Array".

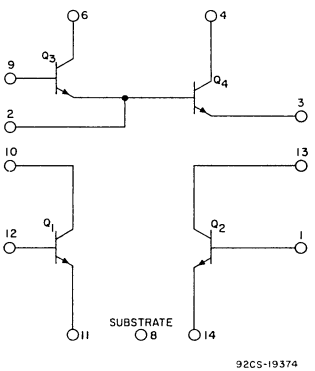


Fig. 3-1— Schematic diagram of CA3018L

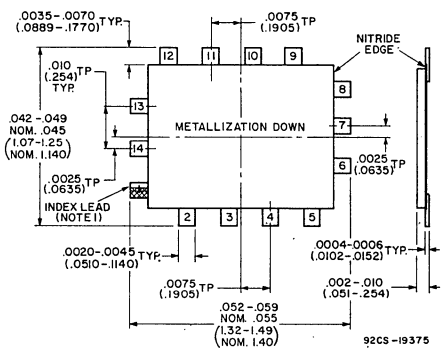
Applications

- General use in signal processing systems in DC through VHF range
- Custom designed differential amplifiers
- Temperature compensated amplifiers

Features

- Matched monolithic general purpose transistors
- h_{FE} matched ±10%
- V_{BE} matched ±5 mV
- Operation from DC to 120 MHz
- Wide operating current range
- Low noise figure — 3.4 dB typical at 1 kHz
- Operation over the full military temperature range: -55 to +125°C

CAUTION: Although RCA-CA3018L is electrically similar to CA3018, it is not a pin-for-pin replacement.



NOTES:
1. INDEX CONFIGURATION ON BEAM NO.1 OPTIONAL, PROVIDED LEAD MAX. IS NOT INCREASED MORE THAN .002
2. LEADS AT TRUE POSITION (TP) WITHIN .002 TOTAL (REF. USANS Y14.5-1966)
3. SILICON MUST NOT EXTEND BEYOND NITRIDE OVER A BEAM.

Fig. 3-2— Terminal layout for CA3018L (14-lead configuration)

MAXIMUM RATINGS, *Absolute-Maximum Values, at $T_A = 25^oC$*

The following ratings apply for each transistor in the device:

Temperature Range:
- Operating ..... -55 to +125°C
- Storage ..... -65 to +150°C

Collector-to-Emitter Voltage, V_{CEO} ..... 15 V
Collector-to-Base Voltage, V_{CBO} ..... 20 V
Collector-to-Substrate Voltage, V_{CIO}* ..... 20 V
Emitter-to-Base Voltage, V_{EBO} ..... 5 V
Collector Current, I_C ..... 50 mA

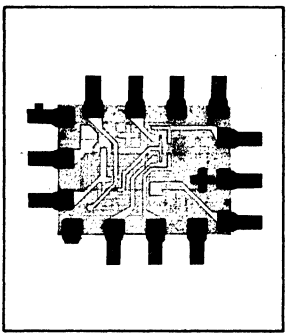
*The collector of each transistor of CA3018L is isolated from the substrate by an integral diode. *The substrate (terminal 8) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.*

ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$ FOR EACH TRANSISTOR	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
STATIC CHARACTERISTICS						
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10V, I_E = 0	–	0.002	100	nA
Collector-Cutoff Current	I_{CEO}	V_{CE} = 10V, I_B = 0	–	–	5	μA
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	15	24	–	V
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10μA, I_E = 0	20	60	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10μA, I_C = 0	5	7	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10μA, I_{CI} = 0	20	60	–	V
Collector-to-Emitter Saturation Voltage	V_{CES}	I_B = 1 mA, I_C = 10 mA	–	0.23	–	V
Static Forward Current Transfer Ratio	h_{FE}	V_{CE} = 3V, I_C = 10mA	–	100	–	–
		V_{CE} = 3V, I_C = 1mA	30	100	–	–
		V_{CE} = 3V, I_C = 10μA	–	54	–	–
Magnitude of Static-Beta Ratio (Isolated Transistors Q_1 and Q_2)		V_{CE} = 3V, $I_{C1} = I_{C2}$ = 1 mA	0.9	0.97	–	–
Static Forward Current Transfer Ratio Darlington Pair (Q_3 and Q_4)	h_{FED}	V_{CE} = 3V, I_C = 1 mA	1500	5400	–	–
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3V, I_E = 1 mA	–	0.715	–	V
		V_{CE} = 3V, I_E = 10mA	–	0.800	–	
Input Offset Voltage	$\lvert V_{BE_1} - V_{BE_2} \rvert$	V_{CE} = 3V, I_E = 1 mA	–	0.48	5	mV
Temperature Coefficient: Base-to-Emitter Voltage Q_1, Q_2	$\frac{\lvert \Delta V_{BE} \rvert}{\Delta T}$	V_{CE} = 3V, I_E = 1 mA	–	-1.9	–	mV/°C
Base (Q_3)-to-Emitter (Q_4) Voltage Darlington Pair	$V_{BED(V_{9-1})}$	V_{CE} = 3 V, I_E = 10 mA	–	1.46	–	V
		V_{CE} = 3 V, I_E = 1 mA	–	1.32	–	
Temperature Coefficient: Base-to-Emitter Voltage Darlington Pair - Q_3, Q_4	$\frac{\lvert \Delta V_{BED} \rvert}{\Delta T}$	V_{CE} = 3V, I_E = 1 mA	–	4.4	–	mV/°C
Temperature Coefficient: Magnitude of Input-Offset Voltage	$\frac{\lvert V_{BE_1} - V_{BE_2} \rvert}{\Delta T}$	V_{CC} = +6V, V_{EE} = -6V, $I_{C_1} = I_{C_2}$ = 1 mA	–	1	–	μV/°C

Linear Integrated Circuits

Monolithic Silicon

CA3028AL



Beam-Lead Differential/Cascode Amplifier

FOR COMMUNICATIONS AND INDUSTRIAL EQUIPMENT AT FREQUENCIES FROM DC to 120 MHz

Applications

- RF and IF Amplifiers (Differential or Cascode)
- DC, Audio, and Sense Amplifiers
- Converter in the Commercial FM Band
- Oscillator • Mixer • Limiter

RCA CA3028AL is the beam-lead version of the CA3028A family of differential/cascode amplifiers designed for use in communications and industrial equipment operating at frequencies from dc to 120 MHz. The beam leads of this device are formed as an integral part of the IC chip during the batch fabrication process.

The CA3028AL is particularly suited for applications in hybrid circuits where hermetic packaging, low cost, and reliable operation are prime considerations.

For applications of the CA3028AL see the companion Application Note ICAN-5337 "Application of the RCA CA3028 Integrated Circuit Amplifier in the HF and VHF Ranges".

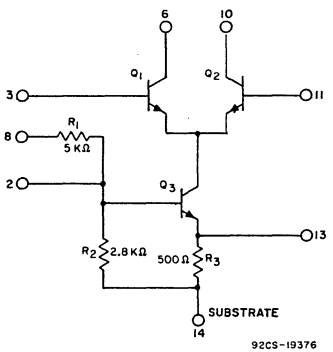
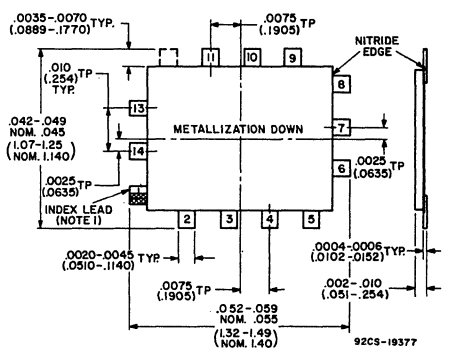


Fig. 4-1— Schematic diagram of CA3028AL

Features

- Controlled for input bias current
- Balanced differential amplifier configuration with controlled constant-current source to provide unexcelled versatility
- Single- and dual-ended operation
- Operation from dc to 120 MHz
- Balanced-AGC capability
- Wide operating-current range
- Operation over the full military temperature range: -55 to +125°C



NOTES:
1. INDEX CONFIGURATION ON BEAM NO.1 OPTIONAL, PROVIDED LEAD MAX. IS NOT INCREASED MORE THAN .002
2. LEADS AT TRUE POSITION (TP) WITHIN .002 TOTAL (REF. USANS Y14.5-1966)
3. SILICON MUST NOT EXTEND BEYOND NITRIDE OVER A BEAM.

Fig. 4-2— Terminal layout for CA3028AL (14-lead configuration)

CAUTION: Although RCA-CA3028AL is electrically similar to CA3028A, it is not a pin-for-pin replacement.

MAXIMUM RATINGS, *Absolute-Maximum Ratings at* $T_A = 25^oC$

TEMPERATURE RANGE:

Operating . . . . . -55°C to +125°C
Storage . . . . . -65°C to +150°C

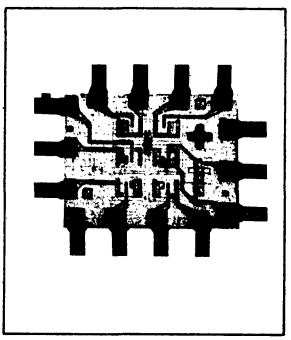
ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$

CHARACTERISTIC	SYMBOL	SPECIAL TEST CONDITIONS		LIMITS			UNITS
				MIN.	TYP.	MAX.	
STATIC CHARACTERISTICS							
		V^+	V^-				
Input Bias Current	I_I	6V 12V	6V 12V	– –	16.6 36	70 106	μA
Quiescent Operating Current	I_6 or I_{10}	6V 12V	6V 12V	0.8 2	1.25 3.3	2 5	mA
Input Current (Term. No. 8)	I_8	6V 12V	6V 12V	0.5 1	0.85 1.65	1 2.1	mA
Device Dissipation	P_T	6V 12V	6V 12V	24 120	36 175	54 260	mW

Linear Integrated Circuits

Monolithic Silicon

CA3039L



Beam-Lead Diode Array

6 Matched Ultra-Fast Low-Capacitance Diodes

FOR APPLICATIONS IN COMMUNICATIONS AND SWITCHING SYSTEMS

Applications

- **Balanced modulators or demodulators**
- **Ring modulators**
- **High-speed diode gates**
- **Analog switches**

RCA CA3039L is the beam-lead version of the CA3039 which consists of six ultra-fast, low capacitance diodes on a common monolithic substrate. The beam leads of the device are formed as an integral part of the IC chip during the batch fabrication process.

CA3039L is particularly suited for applications in hybrid circuits where hermetic packaging, low cost and reliable operation are prime considerations.

Five of the diodes are independently accessible, the sixth shares a common terminal with the substrate.

For applications such as balanced modulators or ring modulators where capacitive balance is important, the substrate should be returned to a dc potential which is significantly more negative (with respect to the active diodes) than the peak signal applied.

Features

- **Excellent reverse recovery time: 1 ns typ.**
- **Matched monolithic construction: V_F matched ±5 mV**
- **Low diode capacitance: C_D = 0.65 pF typ. at V_R = −2 V**
- **Operation over the full military temperature range: −55 to +125°C**

CAUTION: Although RCA-CA3039L is electrically similar to CA3039, it is not a pin-for-pin replacement.

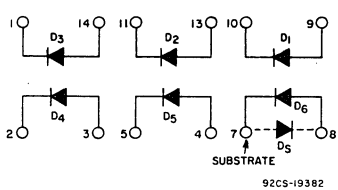
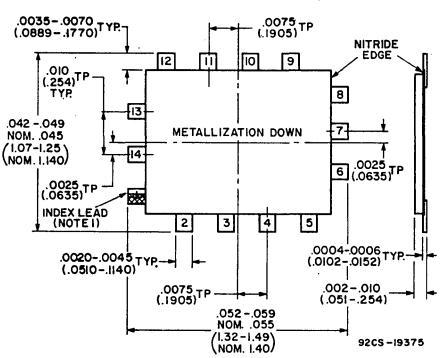


Fig. 5-1— Schematic diagram of CA3039L



NOTES:

1. INDEX CONFIGURATION ON BEAM NO.1 OPTIONAL, PROVIDED LEAD MAX. IS NOT INCREASED MORE THAN .002
2. LEADS AT TRUE POSITION (TP) WITHIN .002 TOTAL (REF. USANS Y14.5-1966)
3. SILICON MUST NOT EXTEND BEYOND NITRIDE OVER A BEAM.

Fig. 5-2— Terminal layout for CA3039L (14-lead configuration)

MAXIMUM RATINGS, *Absolute-Maximum Ratings at* $T_A = 25^oC$

Peak Inverse Voltage V_{RM} for: D_1-D_5 ..... 5V
D_6 ..... 0.5V

TEMPERATURE RANGE:
Operating ..... -55 to +125°C
Storage ..... -65 to +150°C

Peak Diode-to-Substrate Voltage, V_{DI}
for D_1-D_5 (term. 3, 4, 9, 13 or 14 to term. 7) +20, -1 V
DC Forward Current, I_F ..... 25 mA
Peak Recurrent Forward Current I_{FRM} ..... 100 mA
Peak Forward Surge Current I_{FSM} ..... 100 mA

ELECTRICAL CHARACTERISTICS, at $T_A = 25^oC$

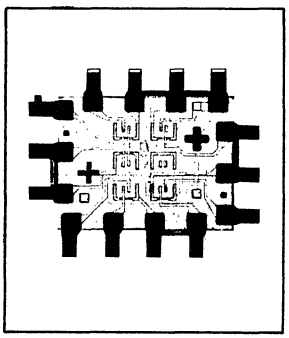
Characteristics apply for each diode unit, unless otherwise specified.

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS			UNITS
			MIN.	TYP.	MAX.	
DC Forward Voltage Drop	V_F	I_F = 50 μA	–	0.65	0.69	V
		1 mA	–	0.73	0.78	
		3 mA	–	0.76	0.80	
		10 mA	–	0.81	0.90	
DC Reverse Breakdown Voltage	$V_{(BR)R}$	I_R = -10 μA	5	7	–	V
DC Reverse Breakdown Voltage Between any Diode Unit and Substrate	$V_{(BR)R}$	I_R = -10 μA	20	–	–	V
DC Reverse (Leakage) Current	I_R	V_R = -4V	–	0.016	100	nA
DC Reverse (Leakage) Current Between any Diode Unit and Substrate	I_R	V_R = -10V	–	0.022	100	nA
Magnitude of Diode Offset Voltage (Difference in DC Forward Voltage Drops of any Two Diode Units)	$\lvert V_{F_1} - V_{F_2} \rvert$	I_F = 1 mA	–	0.5	5	mV
Temperature Coefficient of $\lvert V_{F_1} - V_{F_2} \rvert$	$\frac{\Delta \lvert V_{F_1} - V_{F_2} \rvert}{\Delta T}$	I_F = 1 mA	–	1	–	μV/°C
Temperature Coefficient of Forward Drop	$\frac{\Delta V_F}{\Delta T}$	I_F = 1 mA	–	-1.9	–	mV/°C
DC Forward Voltage Drop for Anode-to-Substrate Diode (D_S)	V_F	I_F = 1 mA	–	0.65	–	V
Reverse Recovery Time	t_{rr}	I_F = 10 mA, I_R = 10 mA	–	1	–	ns
Diode Capacitance	C_D	V_R = -2 V, I_F = 0	–	0.65	–	pF
Diode-to-Substrate Capacitance	C_{DI}	V_{DI} = +4 V, I_F = 0	–	3.2	–	pF

Linear Integrated Circuits

Monolithic Silicon

CA3045L



Beam-Lead General-Purpose N-P-N Transistor Array

Three Isolated Transistors and One Differentially Connected Transistor Pair.

FOR LOW-POWER APPLICATIONS AT FREQUENCIES FROM DC THROUGH THE VHF RANGE

Applications

- General use in various types of signal processing systems operating anywhere in the frequency range from DC to VHF
- Custom designed differential amplifiers
- Temperature compensated amplifiers

Features

- Two matched transistor pairs: V_{BE} matched ±5 mV, Input offset current 2 μA max. at I_C = 1 mA
- 5 general-purpose monolithic transistors
- Operation from DC to more than 120 MHz
- Wide operating current range
- h_{FE} (each transistor)=100 typ. at V_{CE}=3 V, I_C=1 mA
- Low-noise figure: 3.2 dB typ. at 1 kHz
- Operation over the full military temperature range: -55 to +125°C

RCA CA3045L is a beam-lead version of the CA3045 and contains an array of general-purpose transistors for use in signal-level applications at frequencies up to more than 120 MHz. The beam leads of this device are formed as an integral part of the IC chip during the batch fabrication process.

The CA3045L is particularly suited for use in hybrid type construction where compactness, hermeticity, ultra-reliability, and low cost are prime requirements. For suggested applications of transistor arrays, see RCA Application Note, ICAN-5296 "Application of the RCA-CA3018 Integrated-Circuit Transistor Array".

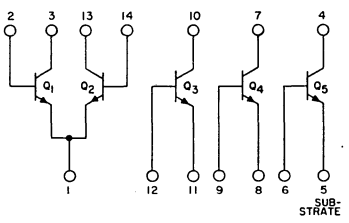
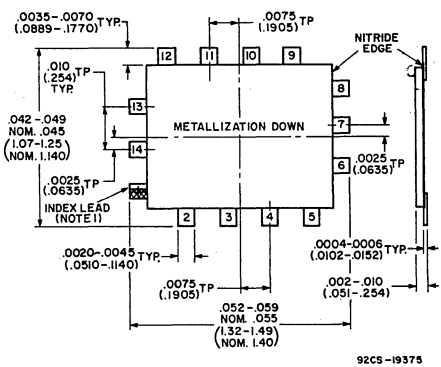


Fig. 6-1— Schematic diagram of CA3045L



NOTES:

1. INDEX CONFIGURATION ON BEAM NO.1 OPTIONAL, PROVIDED LEAD MAX. IS NOT INCREASED MORE THAN .002
2. LEADS AT TRUE POSITION (TP) WITHIN .002 TOTAL (REF. USANS Y14.5-1966)
3. SILICON MUST NOT EXTEND BEYOND NITRIDE OVER A BEAM.

Fig. 6-2— Terminal layout for CA3045L (14-lead configuration)

CAUTION: Although RCA-CA3045L is electrically similar to CA3045, it is not a pin-for-pin replacement.

MAXIMUM RATINGS, *Absolute-Maximum Values at $T_A = 25^oC$*

Collector-to-Emitter Voltage, V_{CEO} . . . . . 15 V
Collector-to-Base Voltage, V_{CBO} . . . . . 20 V
Collector-to-Substrate Voltage, V_{CIO}* . . . . . 20 V
Emitter-to-Base Voltage, V_{EBO} . . . . . 5 V
Collector Current, I_C . . . . . 50 mA

Temperature Range:
Operating . . . . . -55 to +125°C
Storage . . . . . -65 to +150°C

*The collector of each transistor is isolated from the substrate by an integral diode. *The substrate (terminal 5) must be more negative than all collectors to maintain isolation between transistors and to provide for normal transistor action.*

ELECTRICAL CHARACTERISTICS, at $T_A = 25^oC$

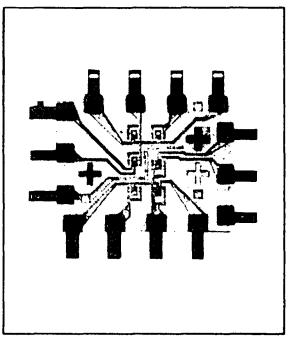
Characteristics apply for each transistors

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS			UNITS
			MIN.	TYP.	MAX.	
STATIC CHARACTERISTICS						
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_C = 10\,\mu A$, $I_E = 0$	20	60	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	$I_C = 1$ mA, $I_B = 0$	15	24	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	$I_C = 10\,\mu A$, $I_{CI} = 0$	20	60	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	$I_E = 10\,\mu A$, $I_C = 0$	5	7	–	V
Collector-Cutoff Current	I_{CBO}	$V_{CB} = 10$ V, $I_E = 0$	–	0.002	40	nA
Collector-Cutoff Current	I_{CEO}	$V_{CE} = 10$ V, $I_B = 0$	–		0.5	μA
Static Forward Current Transfer Ratio (Static Beta)	h_{FE}	$V_{CE} = 3V$, $I_C = 10$ mA	–	100	–	–
		$V_{CE} = 3V$, $I_C = 1$ mA	40	100	–	–
		$V_{CE} = 3V$, $I_C = 10\,\mu A$	–	54	–	–
Input Offset Current for Matched Pair Q_1 and Q_2	$\lvert I_{IO_1} - I_{IO_2} \rvert$	$V_{CE} = 3V$, $I_C = 1$ mA	–	0.3	2	μA
Base-to-Emitter Voltage	V_{BE}	$V_{CE} = 3V$, $I_E = 1$ mA	–	0.715	–	V
		$V_{CE} = 3V$, $I_E = 10$ mA	–	0.800	–	V
Magnitude of Input Offset Voltage for Differential Pair	$\lvert V_{IO_1} - V_{IO_2} \rvert$	$V_{CE} = 3V$, $I_C = 1$ mA	–	0.45	5	mV
Magnitude of Input Offset Voltage for Isolated Transistors	$\lvert V_{IO_3} - V_{IO_4} \rvert$, $\lvert V_{IO_4} - V_{IO_5} \rvert$, $\lvert V_{IO_5} - V_{IO_3} \rvert$	$V_{CE} = 3V$, $I_C = 1$ mA	–	0.45	5	mV
Temperature Coefficient: Magnitude of Input-Offset Voltage	$\frac{\lvert \Delta V_{IO} \rvert}{\Delta T}$	$V_{CE} = 3V$, $I_C = 1$ mA	–	1.1	–	$\mu V/^oC$
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	$V_{CE} = 3V$, $I_C = 1$ mA	–	−1.9	–	$mV/^oC$
Collector-to-Emitter Saturation Voltage	V_{CES}	$I_B = 1$ mA, $I_C = 10$ mA	–	0.23	–	V

Linear Integrated Circuits

Monolithic Silicon

CA3049L



Beam-Lead Dual Independent Differential Amplifiers

For Low-Power Applications at Frequencies up to 500 MHz

Applications

- VHF amplifiers
- VHF mixers
- Multifunction combinations — RF/Mixer/Oscillator; Converter/IF
- IF amplifiers (differential and/or cascode)
- Product detectors
- Doubly balanced modulators and demodulators
- Balanced quadrature detectors
- Cascade limiters
- Synchronous detectors
- Balanced mixers
- Synthesizers
- Balanced (push-pull) cascode amplifiers
- Sense amplifiers

CA3049L is the beam-lead version of the CA3049 and consists of two independent differential amplifiers with associated constant-current transistors on a common monolithic substrate. The six n-p-n transistors which comprise the amplifiers are general-purpose high-frequency devices which exhibit a value of f_T in excess of 1000 MHz. These features make the CA3049L useful to 500 MHz. Bias and load resistors have been omitted to provide maximum application flexibility.

The CA3049L is particularly suited for applications in hybrid circuits where hermetic packaging, low cost, and reliable operation are prime considerations.

The monolithic construction of the CA3049L provides close electrical and thermal matching of the amplifiers. This feature makes this device particularly useful in dual-channel applications where matched performance of the two channels is required.

Features

- Power Gain 23 dB (typ.) at 200 MHz
- Noise Figure 4.6 dB (typ.) at 200 MHz
- Two differential amplifiers on a common substrate
- Independently accessible inputs and outputs
- Full military temperature range capability — -55°C to +125°C

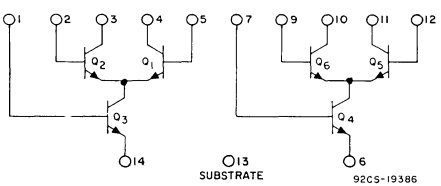


Fig. 7-1— Schematic diagram of CA3049L

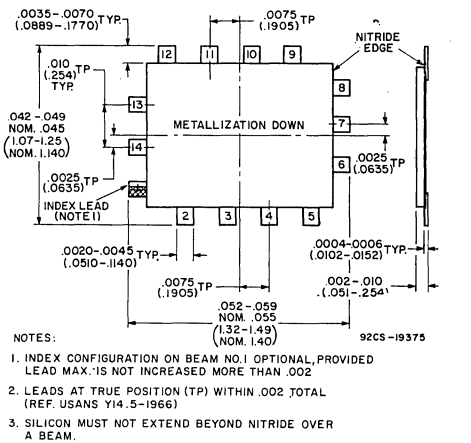


Fig. 7-2—Terminal layout for CA3049L (14-lead configuration)

CAUTION: Substrate MUST be maintained negative with respect to all collector terminals of this device.

Although RCA-CA3049L is electrically similar to CA3049, it is not a pin-for-pin replacement.

MAXIMUM RATINGS, *Absolute-Maximum Values, at $T_A = 25°C$*

Temperature Range:

Operating	−55 to +125	°C
Storage	−65 to +150	°C

The following ratings apply for each transistor in the device:

Collector-to-Emitter Voltage, V_{CEO}	15	V
Collector-to-Base Voltage, V_{CBO}	20	V
Collector-to-Substrate Voltage, V_{CIO}*	20	V
Emitter-to-Base Voltage, V_{EBO}	5	V
Collector Current, I_C	50	mA

*The collector of each transistor of the CA3049L is isolated from the substrate by an integral diode. *The substrate (terminal 13) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.*

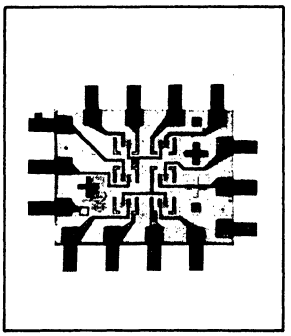
ELECTRICAL CHARACTERISTICS, *at $T_A = 25°C$*

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	CA3049L LIMITS			
			MIN.	TYP.	MAX.	UNITS
STATIC CHARACTERISTICS (for each transistor)						
Input Bias Current	I_{IB}	$V_{CE} = 3$ V, $I_C = 1$ mA	–	10	33	μA
Collector-Cutoff Current	I_{CBO}	$V_{CB} = 10$ V, $I_E = 0$	–	–	100	nA
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	$I_C = 1$ mA, $I_B = 0$	15	–	–	V
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_C = 10$ μA, $I_E = 0$	20	–	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	$I_C = 10$ μA, $I_{CI} = 0$	20	–	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	$I_E = 10$ μA, $I_C = 0$	5	–	–	V

Linear Integrated Circuits

Monolithic Silicon

CA3054L



Beam-Lead Dual Independent Differential Amplifiers

FOR LOW-POWER APPLICATIONS AT FREQUENCIES FROM DC TO 120 MHz

Applications

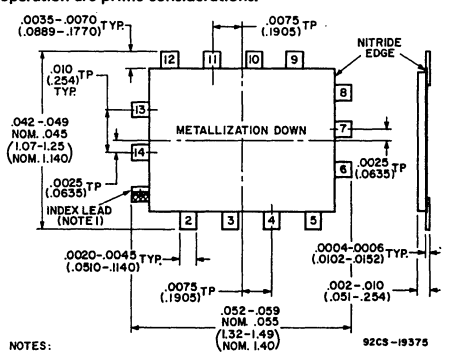
- Dual sense amplifiers
- Dual Schmitt triggers
- Multifunction combinations — RF/Mixer/Oscillators; Converter/IF
- IF amplifiers (differential and/or cascode)
- Product detectors
- Doubly-balanced modulators and demodulators
- Balanced quadrature detectors
- Cascade limiters
- Synchronous detectors
- Pairs of balanced mixers
- Synthesizer mixers
- Balanced (push-pull) cascode amplifiers

The RCA CA3054L is the beam-lead version of the CA3054, and consists of two independent differential amplifiers with associated constant-current transistors on a common monolithic substrate. The beam leads of this device are formed as an integral part of the IC chip during the batch fabrication process.

The CA3054L is particularly suited for applications in hybrid circuits where hermetic packaging, low cost, and reliable operation are prime considerations.

Features

- Two differential amplifiers on a common substrate
- Independently accessible inputs and outputs
- Maximum input offset voltage — ±5 mV
- Operation over the full military temperature range: -55 to +125°C



NOTES:
1. INDEX CONFIGURATION ON BEAM NO.1 OPTIONAL, PROVIDED LEAD MAX. IS NOT INCREASED MORE THAN .002
2. LEADS AT TRUE POSITION (TP) WITHIN .002 TOTAL (REF. USANS Y14.5-1966)
3. SILICON MUST NOT EXTEND BEYOND NITRIDE OVER A BEAM.

Fig. 8-1— Terminal layout for CA3054L (14-lead configuration)

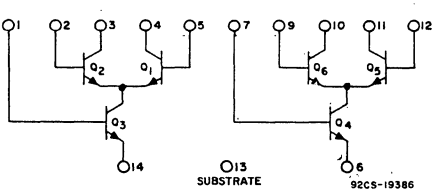


Fig. 8-2— Schematic diagram of CA3054L

CAUTION: Although RCA-CA3054L is electrically similar to CA3054, it is not a pin-for-pin replacement.

CAUTION: Substrate MUST be maintained negative with respect to all collector terminals of this device.

MAXIMUM RATINGS, ***Absolute-Maximum Values, at $T_A = 25^oC$***

The following ratings apply for each transistor in the device:

Collector-to-Emitter Voltage, V_{CEO} ..... 15 V
Collector-to-Base Voltage, V_{CBO} ..... 20 V
Collector-to-Substrate Voltage, V_{CIO}* ..... 20 V

Emitter-to-Base Voltage, V_{EBO} ..... 5 V
Collector Current, I_C ..... 50 mA
Temperature Range:
Operating ..... -55 to +125°C
Storage ..... -65 to +150°C

*The collector of each transistor of the CA3054L is isolated from the substrate by an integral diode. *The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide for normal transistor action. The substrate should be maintained at signal (AC) ground by means of a suitable grounding capacitor, to avoid undesired coupling between transistors.*

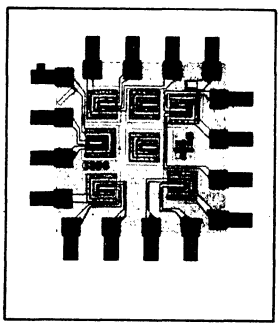
ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
STATIC CHARACTERISTICS						
For Each Differential Amplifier:						
Input Offset Voltage	V_{IO}	V_{CB} = 3 V, $I_{E(Q3)} = I_{E(Q4)}$ = 2 mA	–	0.45	5	mV
Input Offset Current	I_{IO}		–	0.3	2	μA
Input Bias Current	I_I		–	10	24	μA
Quiescent Operating Current Ratio	$\frac{I_{C(Q_1)}}{I_{C(Q_2)}}$ or $\frac{I_{C(Q_5)}}{I_{C(Q_6)}}$		–	0.98 to 1.02	–	–
Temperature Coefficient Magnitude of Input-Offset Voltage	$\frac{\Delta\lvert V_{IO}\rvert}{\Delta T}$		–	1.1	–	μV/°C
For Each Transistor:						
DC Base-to-Emitter Voltage	V_{BE}	V_{CB} = 3 V, I_C = 50 μA	–	0.630	0.700	V
		1 mA	–	0.715	0.800	V
		3 mA	–	0.750	0.850	
		10 mA	–	0.800	0.900	
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	V_{CB} = 3V, I_C = 1 mA	–	-1.9	–	mV/°C
Collector-Cutoff Current	I_{CBO}	V_{CB} = 3V, I_E = 0	–	0.002	100	nA
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	15	24	–	V
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10μA, I_E = 0	20	60	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10μA, I_{CI} = 0	20	60	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10μA, I_C = 0	5	7	–	V

Linear Integrated Circuits

Monolithic Silicon

CA3083L



General-Purpose High-Current N-P-N Transistor Array

Applications

- Signal processing and switching systems operating from DC to VHF
- Lamp and relay driver
- Differential amplifier
- Temperature-compensated amplifier
- Thyristor firing
- See RCA Application Note, ICAN-5296 "Application of the RCA-CA3018 Circuit Transistor Array" for suggested applications

RCA-CA3083L is the beam-lead version of the CA3083. It consists of a versatile array of five high-current (to 100 mA) n-p-n transistors on a common monolithic substrate. Independent connections for each transistor plus a separate terminal for the substrate permit maximum flexibility in circuit design.

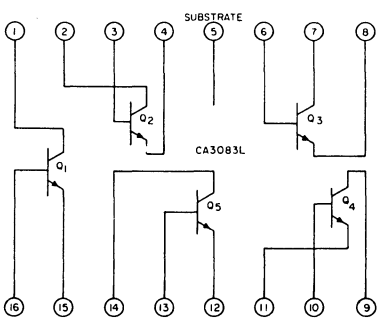


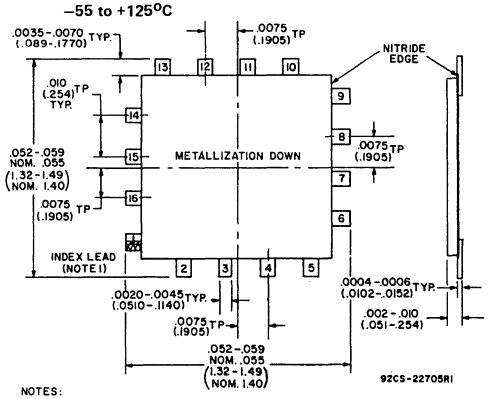
Fig. 9-1– Schematic diagram of CA3083L.

CAUTION: Although RCA-CA3083L is electrically similar to CA3083, it is not a pin-for-pin replacement.

CAUTION: Substrate MUST be maintained negative with respect to all collector terminals of this device.

Features

- High I_C: 100 mA max.
- Low V_{CEsat} (at 50 mA): 0.4 V typ.
- Transistor pair (Q1 and Q2):
  V_{IO} (ΔV_{BE}): 1.2 mV typ.
  I_{IO} 0.7 μA typ.
- 5 independent transistors plus separate substrate connection
- Operation over the full military temperature range:
  −55 to +125°C



NOTES:
1. INDEX CONFIGURATION ON BEAM NO.1 OPTIONAL, PROVIDED LEAD MAX. IS NOT INCREASED MORE THAN .002
2. LEADS AT TRUE POSITION (TP) WITHIN .002 TOTAL (REF. USANS Y14.5-1966)
3. SILICON MUST NOT EXTEND BEYOND NITRIDE OVER A BEAM.

Fig. 9-2– Terminal layout for CA3083L (16-lead configuration).

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

Ambient Temperature Range:

Operating	−55 to +125	°C
Storage	−65 to + 150	°C

The following ratings apply for each transistor in the device:

Collector-to-Emitter Voltage (V_{CEO})	15	V
Collector-to-Base Voltage (V_{CBO})	20	V
Collector-to-Substrate Voltage (V_{CIO})■	20	V
Emitter-to-Base Voltage (V_{EBO})	5	V
Collector Current (I_C)	100	mA
Base Current (I_B)	20	mA

■ The collector of each transistor of the CA3083L is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal (5) should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.

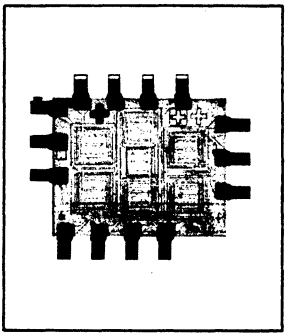
ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOL	TEST CONDITIONS		LIMITS			UNITS
				Min.	Typ.	Max.	
For Each Transistor:							
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 100μA, I_E = 0		20	60	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1mA, I_B = 0		15	24	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI} = 100μA, I_B = 0, I_E = 0		20	60	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 500μA, I_C = 0		5	6.9	–	V
Collector-Cutoff-Current	I_{CEO}	V_{CE} = 10V, I_B = 0		–	–	10	μA
Collector-Cutoff-Current	I_{CBO}	V_{CB} = 10V, I_E = 0		–	–	1	μA
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE} = 3V	I_C = 10mA	40	76	–	
			I_C = 50mA	40	75	–	
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3V, I_C = 10mA		0.65	0.74	0.85	V

Linear Integrated Circuits

Monolithic Silicon

CA3084L



Beam-Lead General-Purpose P-N-P Transistor Array

Applications

- General use in signal processing systems having low-power and low-frequency requirements
- Differential amplifiers
- Temperature compensated amplifiers
- Active loads for differential amplifiers using n-p-n transistors
- Complementary uses with RCA n-p-n transistor arrays

RCA CA3084L is the beam lead version of the CA3084, a general-purpose silicon p-n-p transistor array incorporating two independent transistors, a Darlington circuit, and a current-mirror pair with a shared diode. The beam leads of this device are formed as an integral part of the IC chip during the batch fabrication process.

The CA3084L is particularly suited for applications in hybrid circuits where hermetic packaging, low cost, and reliable operation are prime considerations.

The two independent transistors in the array may be used in a variety of circuit applications. The Darlington pair may be used as the equivalent of a single high-beta transistor. The current-mirror pair is well suited for constant-current applications and can also be used as the active loads in a differential amplifier which uses n-p-n transistors.

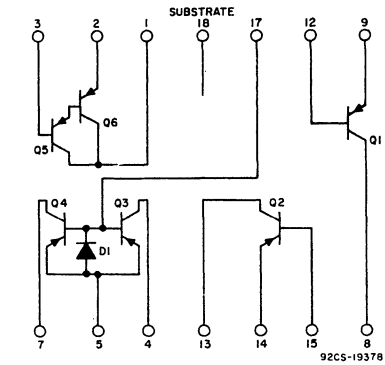


Fig. 10-1— Schematic diagram of CA3084L

CAUTION: Substrate must be maintained negative with respect to all collector terminals of this device.

Features

- A Darlington circuit (Q5 and Q6)
- A current-mirror pair with diode (Q3, Q4, and D1)
- Matched transistor pair (Q1 and Q2)
  V_{IO} (V_{BE} matched): ±6.0 mV max.
  I_{IO} (at 100 μA): ±0.6 μA
- Wide operating current range
- Low noise figure — 3.2 dB typ. at 1 kHz
- Operation over the full military temperature range: -55 to +125°C

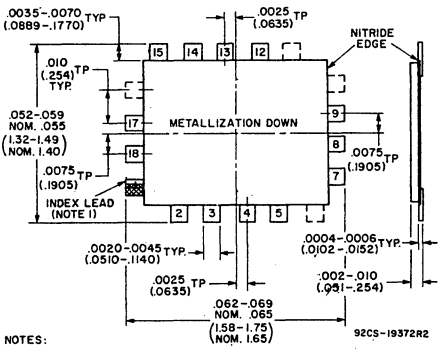


Fig. 10-2— Terminal layout for CA3084L (18-lead configuration)

CAUTION: Although RCA-CA3084L is electrically similar to CA3084, it is not a pin-for-pin replacement.

MAXIMUM RATINGS, *Absolute-Maximum Values at $T_A = 25^oC$*

The following ratings apply for each transistor in the device:

Collector-to-Emitter Voltage (V_{CEO}) . . . . . −40 V
Ambient Temperature Range:
Operating . . . . . -55° to +125°C
Storage . . . . . -65 to +150°C
Collector-to-Base Voltage (V_{CBO}) . . . . . −40V
Base-to-Substrate Voltage (V_{BIO})* . . . . . −40 V.
Emitter-to-Base Voltage (V_{EBO}) . . . . . −40 V
Collector Current (I_C) . . . . . −10 mA

*The base of each transistor of the CA3084L is isolated from the substrate by an integral diode. *The substrate must be connected to a voltage which is more negative than any base voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal 18 should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.*

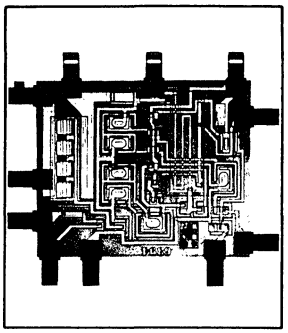
ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$
For Equipment Design

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	LIMITS			UNITS
			MIN.	TYP.	MAX.	
For Each Transistor:						
Collector-Cutoff Current	I_{CBO}	$V_{CB} = -10V$, $I_E = 0$	—	-0.055	-100	nA
Collector-Cutoff Current	I_{CEO}	$V_{CE} = -10V$, $I_B = 0$	—	-0.12	-100	nA
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	$I_{CE} = -100\mu A$, $I_B = 0$	-40	-70	—	V
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_{CB} = -100\mu A$, $I_E = 0$	-40	-80	—	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	$I_{EB} = -100\mu A$, $I_C = 0$	-40	-100	—	V
Emitter-to-Substrate Breakdown Voltage	$V_{(BR)EIO}$	$I_{EI} = 100\mu A$	-40	-100	—	V
Collector-to-Emitter Saturation Voltage	V_{CEsat}	$I_E = 1mA$, $I_B = 100\mu A$	—	-0.125	-0.25	V
Base-to-Emitter Voltage	V_{BE}	$I_E = 100\mu A$, $V_{CE} = -10V$	-0.50	-0.59	-0.68	V
DC Forward-Current Transfer Ratio	h_{FE}		15	40	—	
For Transistors Q1 and Q2 (As a Differential Amplifier):						
Magnitude of Input Offset Voltage	$\lvert V_{IO} \rvert$	$I_E = 100\mu A$, $V_{CE} = -10V$	—	0.422	6	mV
Input Offset Current	I_{IO}		-0.6	0	0.6	μA
For Transistors Q3 and Q4 (Current-Mirror Configuration):						
Collector Current Normalized	I_C/I_{17}	$V_{CE} = -5V$, $V_{CIO} = -5V$	0.85	1.00	1.15	—
Magnitude of Collector Current Ratio	$\lvert I_{C(Q3)}/I_{C(Q4)} \rvert$	Term. 5 = Gnd. $I_{17} = -100\mu A$	0.90	1.00	1.10	
For Transistors Q5 and Q6 (Darlington Configuration):						
Collector-Cutoff Current	I_{CEO}	$V_{CE} = -10V$, $I_B = 0$	—	—	-1.0	μA
Base-to-Emitter Voltage	V_{BE}	$I_E = 100\mu A$, $V_{CE} = -10V$	0.92	1.07	1.20	V
DC Forward-Current Transfer Ratio	h_{FE}		100	1230	—	

Linear Integrated Circuits

Monolithic Silicon

CA3085L



Positive Voltage Regulator

Applications

- Shunt voltage regulator
- Current regulator
- Switching voltage regulator
- High-current voltage regulator
- Combination positive and negative voltage regulator
- Dual tracking regulator
- Operation over the full military temperature range: −55 to +125°C

RCA-CA3085L is the beam-lead version of the CA3085. It is designed specifically for service as a voltage regulator at output voltages ranging from 1.8 to 26 volts at currents up to 12 mA without the use of external pass transistors. However the CA3085L can provide voltage regulation at load currents greater than 100 mA with the use of suitable external pass transistors. The voltage regulators provide important features such as: frequency compensation, short-circuit protection, temperature-compensated reference voltage, current limiting, and booster input. These devices are useful in a wide range of applications for regulating high-current, switching, shunt, and positive and negative voltages. They are also applicable for current and dual-tracking regulation.

Features

- Excellent temperature coefficient
- Input and output short-circuit protection
- Load and line regulation: 0.025%
- Adjustable output voltage

It should be noted that the CA3085L chip has eleven active circuit-access terminals, whereas the CA3085 Series packaged units have only eight access terminals. The additional terminals in the CA3085L provide greater flexibility in circuit applications.

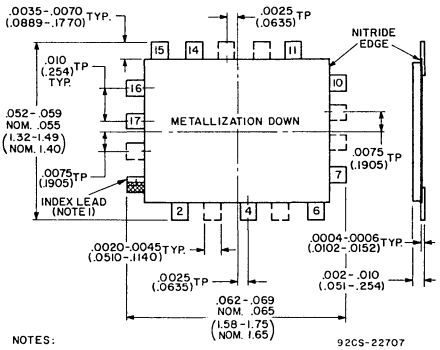
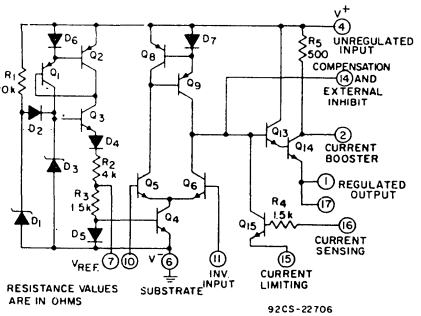


Fig. 11-1— Terminal layout for CA3085L (18-lead configuration).



Caution: Although RCA-CA3085L is electrically similar to the CA3085, it is not a pin-for-pin replacement.

Fig. 11-2—Schematic diagram of CA3085L.

MAXIMUM RATINGS, *Absolute-Maximum Values, at* $T_A = 25^oC$

Temperature Range:

Operating . . . . . −55 to +125°C

Storage . . . . . −65 to +150°C

Unregulated Input Voltage . . . . . 30 V

Maximum Voltage Ratings:

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical Terminal No. 14 and horizontal Terminal No. 1 and 17 is +3 to −10 volts.

MAXIMUM VOLTAGE RATINGS

TERMINAL No.	7	10	11	14	15	16	1&17	2	4	6
7	–	*	*	*	*	*	*	*	*	+10 0
10	–	–	+5 −5	*	*	*	*	*	*	*
11	–	–	–	*	*	*	*	*	*	*
14	–	–	–	–	+3 −10	+10 −1	+3 −10	*	*	+30 0
15	–	–	–	–	–	+5 −1	*	*	*	*
16	–	–	–	–	–	–	+3 −10	*	*	*
1&17	–	–	–	–	–	–	–	+10 −30	0 −30	+30 0
2	–	–	–	–	–	–	–	–	–	+30 0
4	–	–	–	–	–	–	–	–	–	+30 0
6	–	–	–	–	–	–	–	–	–	Substrate

* Voltages are not normally applied between these terminals; however, voltages appearing between these terminals are safe, if the specified voltage limits between all other terminals are not exceeded.

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
7	10	1
10	1	-0.1
11	1	−0.1
14	1	−1
15	0.1	10
16	–	–
1&17	20	150
2	150	60
4	150	60
6	–	–

CHARACTERISTIC	SYMBOL	TEST CONDITIONS T_A = 25°C (Unless indicated otherwise)		LIMITS MIN.	LIMITS TYP.	LIMITS MAX.
Reference Voltage	V_{REF}	V^+_{IN} = 15 V		1.4	1.6	1.8
Quiescent Regulator Current	$I_{quiescent}$	V^+_{IN} = 30 V		–	3.3	4.5
Input Voltage Range	V_{IN} (range)	–		7.5	–	30
Maximum Output Voltage	V_O (max.)	V^+_{IN} = 30 R_L = 365 Ω, Term. No. 6 to Gnd.		26	27	–
Minimum Output Voltage	V_O (min.)	V^+_{IN} = 30 V		–	1.6	1.8
Input-Output Voltage Differential	V_{IN}-V_{OUT}	–		4	–	28
Limiting Current	I_{LIM}	V^+_{IN} = 16 V, V^+_{OUT} = 10 V R_{SCP}* = 6 Ω		–	96	120
Load Regulation●	–	I_L = 1 to 12 mA, R_{SCP} = 0		–	0.003	0.1
Line Regulation▲	–	I_L = 1 mA, R_{SCP} = 0		–	0.025	0.1
		I_L = 1 mA, R_{SCP} = 0 T_A = 0°C to +70°C		–	0.04	0.15
Equivalent Noise Output Voltage	V_{NOISE}	V^+_{IN}=25 V	C_{REF} = 0	–	0.5	–
			C_{REF} = 0.22 μF	–	0.3	–
Ripple Rejection	–	V^+_{IN}=25 V	C_{REF} = 0	–	50	–
		f = 1 kHz	C_{REF} = 2 μF	–	56	–
Output Resistance	r_O	V^+_{IN} = 25 V, f = 1 kHz		–	0.075	1.1
Temperature Coefficient of Reference and Output Voltages	ΔV_{REF}, ΔV_O	I_L = 0, V_{REF} = 1.6 V		–	0.0035	–
Load Transient Recovery Time: Turn On	t_{ON}	V^+_{IN} = 25 V, +50 mA Step		–	1	–
Turn Off	t_{OFF}	V^+_{IN} = 25 V, –50 mA Step		–	3	–
Line Transient Recovery Time: Turn On	t_{ON}	V^+_{IN}=25 V, f=1 kHz, 2 V Step		–	0.8	–
Turn Off	t_{OFF}			–	0.4	–

*R_{SCP}: Short-circuit protection resistance

●Load Regulation = $\frac{\Delta V_{OUT}}{V_{OUT\ (initial)}}$ X 100%

▲Line Regulation = $\frac{(\Delta V_{OUT})}{[V_{OUT\ (initial)}]\ (\Delta V_{IN})}$ X 100%

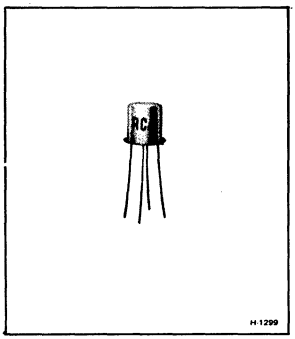
Technical Data-DMOS Devices

RCA Solid State Division

MOS Field-Effect Transistors

N-Channel Depletion Types

3N128
3N143



Silicon MOS Transistors

For Amplifier, Mixer, & Oscillator Applications in Military & Industrial VHF Communications Equipment Operating up to 250 MHz

Applications

- **VHF amplifiers, mixers, converters and if-amplifiers in communication receivers.**
- **High-impedance timing circuits**
- **Detectors, oscillators, frequency multipliers, phase splitters, pulse stretchers and current limiters**
- **Electrometer amplifiers**
- **Voltage-controlled attenuators**
- **High impedance differential amplifiers**

RCA-3N128 and 3N143 are N-channel depletion-type silicon insulated-gate field-effect transistors utilizing the MOS* construction. The 3N128 is intended primarily for VHF amplifier service in military and industrial applications. It also is extremely well suited for use in dc and low-frequency amplifier applications requiring a transistor having high power gain, very high input impedance, and low gate leakage.

The 3N143 is designed for use as a VHF mixer and oscillator. Because of their improved transfer characteristic and increased dynamic range the 3N128 and 3N143 provide substantially better cross-modulation performance in linear amplifier applications than conventional (bipolar) transistors and are free from diode-current loading common to junction type FET's. These transistors are hermetically sealed in JEDEC TO-72 metal packages.

Application data for RCA-3N128, including biasing requirements, basic circuit configurations, selection of optimum operating point, and methods for automatic gain control are given in RCA Application Note AN-3193, "Application Considerations for the RCA-3N128 VHF MOS Field-Effect Transistor".

* Metal-Oxide-Semiconductor.

Performance Features

- **Large dynamic range**
- **Greatly reduces spurious responses in rceiver front ends**
- **Permits use of vacuum-tube biasing techniques**
- **Excellent thermal stability**
- **Superior crossmodulation capability**

Device Features

- **Low noise figure (3N128) – 3.5 dB typ. at 200 MHz**
- **High VHF amplifier gain (3N128) – 16 dB typ. at 200 MHz**
- **Low input capacitance – 5.5 pF typ.**
- **High transconductance – 7500 μmho typ.**
- **High input resistance – 10^{14} Ω typ.**
- **High conversion gain (3N143, mixer) – 13.5 dB typ. at 200 MHz**

Maximum Ratings, *Absolute-Maximum Values at* T_A = 25° C:

*DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20	V
*DRAIN-TO-GATE VOLTAGE, V_{DG}	+20	V
*GATE-TO-SOURCE VOLTAGE, V_{GS}:		
Continuous dc	+1, -8	V
Peak ac	±15	V
*DRAIN CURRENT, I_D	50	mA
*TRANSISTOR DISSIPATION, P_T:		
At Ambient up to 25°C	330	mW
Temperatures above 25°	Derate 2.2	mW/°C
*AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
*LEAD TEMPERATURE (During soldering):		
At distances not closer than 1/32 inch to seating surface for 10 seconds maximum	265	°C

*In accordance with Jedec Registration Data Format JS9-RDF11B.

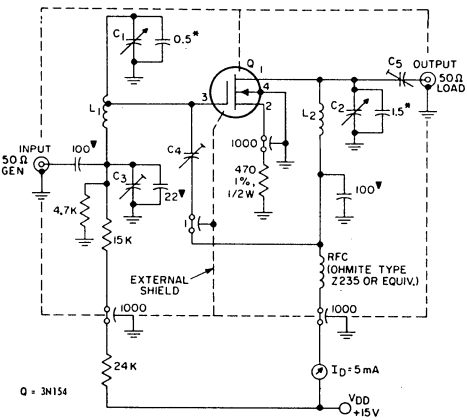
11-67

ELECTRICAL CHARACTERISTICS: *(At T_A = 25°C)*

Measured with Substrate Connected to Source Unless Otherwise Specified.

	CHARACTERISTIC	SYMBOL	CONDITIONS	LIMITS 3N128 MIN.	3N128 TYP.	3N128 MAX.	3N143 MIN.	3N143 TYP.	3N143 MAX.	UNITS
*	Gate Leakage Current	I_{GSS}	V_{DS} = 0, V_{GS} = −8 V T_A = 25°C	–	0.1	50	–	0.1	1000	pA
			V_{DS} = 0, V_{GS} = −8 V T_A = 125°C	–	–	5	–	–	200	nA
*	Zero-Bias Drain Current	I_{DSS}	V_{DS} = 15 V, V_{GS} = 0	5	15	25	5	15	30	mA
*	Drain-to-Source Cutoff Current	$I_{D(off)}$	V_{DS} = 20 V, V_{GS} = −8 V	–	–	50	–	–	50	μA
*	Gate-to-Source Cutoff Voltage	$V_{GS(off)}$	V_{DS} = 15 V, I_D = 50 μA	−0.5	−3	−8	−0.5	−3	−8	V
*	Forward Transconductance	g_{fs}	V_{DS} = 15 V, I_D = 5 mA, f = 1 kHz	5,000	7,500	12,000	5,000	7,500	12,000	μmho
*	Drain-to-Source Channel Resistance	$r_{DS(on)}$	V_{DS} = 0, V_{GS} = 0, f = 1 kHz	–	200	–	–	200	–	Ω
*	Small-Signal Short-Circuit Reverse Transfer Capacitance▲	C_{rss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	0.15	0.25	0.35	0.12	0.25	0.38	pF
*	Small-Signal Short-Circuit Input Capacitance	C_{iss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	–	5.5	7	–	5.5	7	pF
			Common-Source Configuration							
*	Input Admittance	Y_{is}	f = 200 MHz	–	0.4 + J7.3	–	–	–	–	mmho
*	Forward Transfer Admittance	Y_{ss}	V_{os} = 15 Volts	–	7 − J2	–	–	–	–	mmho
*	Output Admittance	Y_{os}	I_D = 5 mA	–	0.28 + J1.8	–	–	–	–	mmho
*	Maximum Available Power Gain	MAG	V_{DS} = 15 V, I_D = 5 mA, f = 200 MHz		21	–	–	–	–	dB
*	Insertion Power Gain (Fixed Neutralization) See Fig. 1	G_{PS}		13.5	16	–	–	–	–	dB
	Power Gain (Conversion (See Fig. 3)	$G_{PS(c)}$	V_{DS} = 15 V, I_D = 1 mA, f_{in} = 200 MHz, f_{out} = 30 MHz	–	–	–	10	13.5	–	dB
	Noise Figure (See Fig. 1 & 2)	NF	V_{DS} = 15 V, I_D = 5 mA, f = 200 MHz	–	3.5	5	–	–	–	–

*In accordance with **JEDEC** Registration Data Format JS9-RDF-11B.
▲Three-Terminal Measurement: Source Returned to Guard Terminal.



C_1, C_2: 1.5-5 pF variable air capacitor: E. F. Johnson Type 160-102 or equivalent

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent

C_4, C_5: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent

L_1: 5 turns silver-plated 0.02" thick, 0.07"-0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding

L_2: Same as L_1 except winding length approx. 0.7"; no tap.

Fig. 1 - Test circuit used to measure 200-MHz maximum usable power gain and noise figure for 3N128

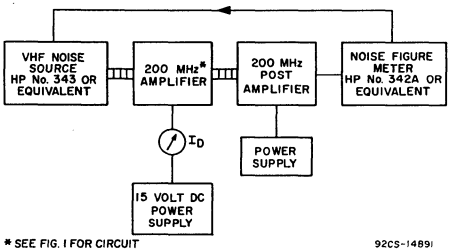


Fig. 2 - Noise figure measurement setup for 3N128

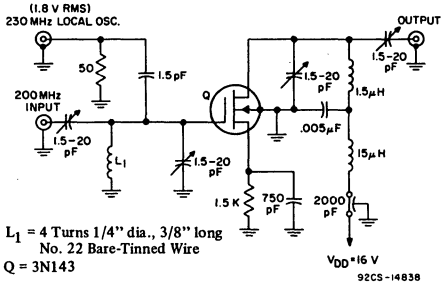


Fig. 3 - Conversion power gain test circuit for 3N143

Typical Characteristics for Types 3N128 and 3N143

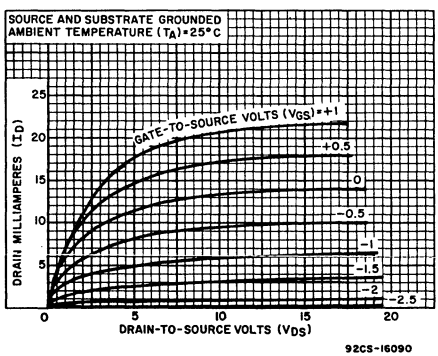


Fig. 4 - Drain current vs. drain-to-source voltage

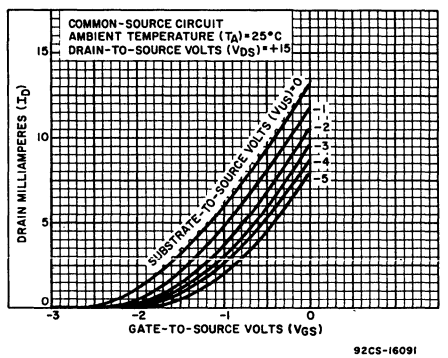


Fig. 5 - Drain current vs. gate-to-source voltage (V_{GS})

Typical Y-Parameters for Types 3N128 and 3N143

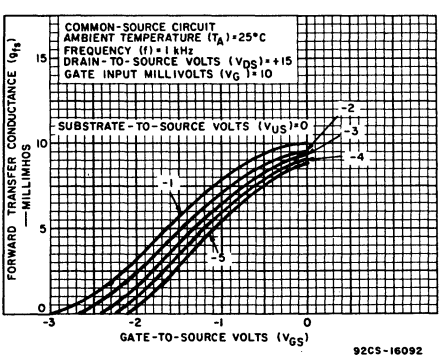


Fig. 6 - Forward transconductance vs. gate bias voltage

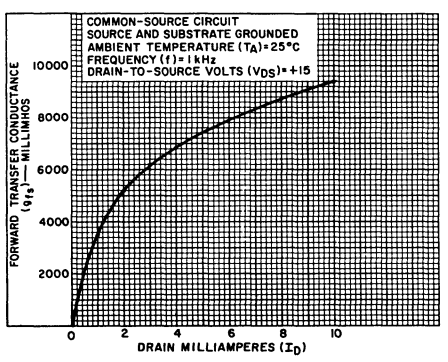


Fig. 7 - Forward transconductance vs. drain current

Typical Y-Parameters for Types 3N128 and 3N143

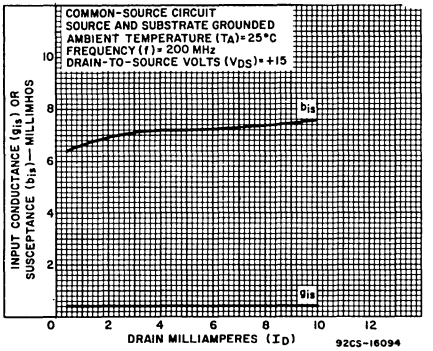


Fig. 8 - Input admittance vs. drain current

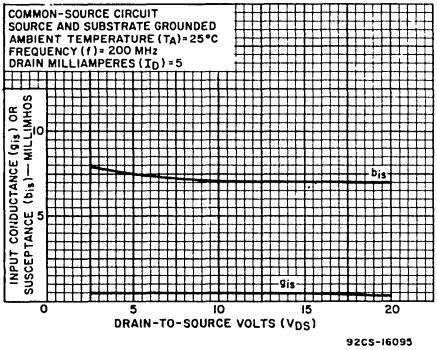


Fig. 9 - Input admittance vs. drain-to-source voltage

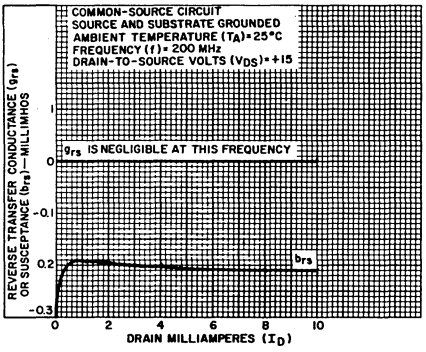


Fig. 10 - Reverse transadmittance vs. drain current

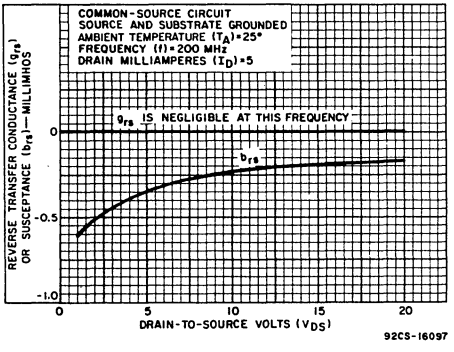


Fig. 11 - Reverse transadmittance vs. drain-to-source voltage

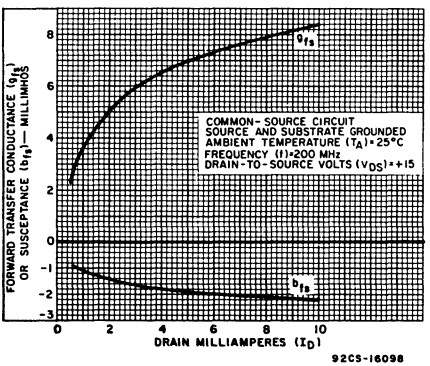


Fig. 12 - Forward transadmittance vs. drain current

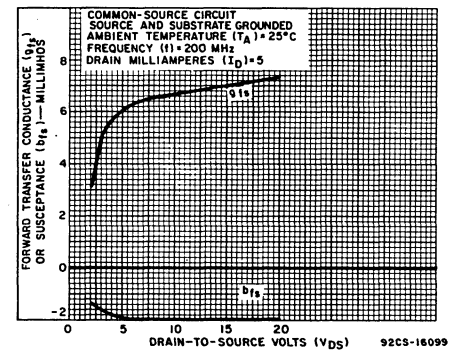


Fig. 13 - Forward transadmittance vs. drain-to-source voltage

Typical Characteristics for Types 3N128 and 3N143

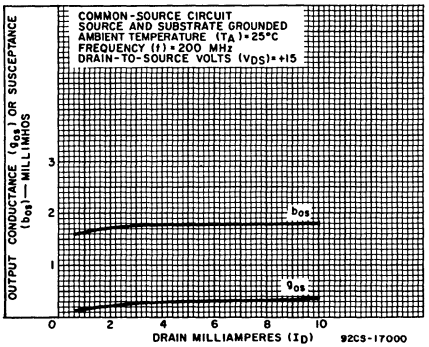


Fig. 14 - Output admittance vs. drain current

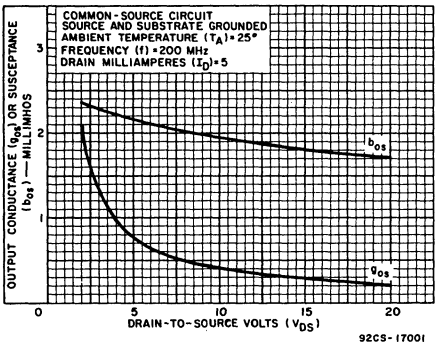


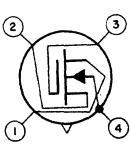
Fig. 15 - Output admittance vs. drain-to-source voltage

OPERATING CONSIDERATIONS

The flexible leads of the 3N128 and 3N143 are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons should be grounded, and appropriate precautions should be taken to protect the devices against high electric fields.

This device should not be connected into or disconnected from circuits with the power on because high transient voltages may cause permanent danage to the device.

TERMINAL DIAGRAM



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

RCA Solid State Division

MOS Field-Effect Transistors

N-Channel Depletion Type

3N138

Applications

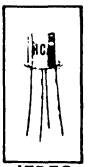
- Servo Amplifiers
- Telemetry Amplifiers
- Computer Operational Amplifiers
- Sampling Circuits
- Electrometer Amplifiers

Features

- excellent thermal stability
- zero inherent offset voltage
- low leakage current: 10 pA max.
- low "on" resistance — $r_{DS}(on)$ = 240Ω typ. (V_{GS} = 0V)
- high "off" resistance — $R_{DS}(off)$ = 10^{10}Ω typ.
- low feedback capacitance — C_{rss} = 0.18pF typ.
- low input capacitance — C_{iss} = 3pF typ.

SILICON INSULATED-GATE FIELD-EFFECT TRANSISTOR N-Channel Depletion Type

For Critical Chopper Applications and Multiplex Service up to 60 MHz: in Military Communications, Navigation, and Instrumentation Equipment in Industrial Instrumentation and Control Circuits



JEDEC TO-72

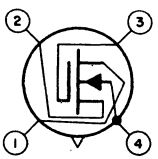
RCA-3N138 is a silicon, insulated-gate field-effect transistor of the N-channel depletion type, utilizing the MOS* construction. It is intended primarily for critical chopper and multiplex applications up to 60MHz.

The insulated gate provides a very high value of input resistance (10^{14} ohms typ.) which is relatively insensitive to temperature and is independent of gate-bias conditions (positive, negative, or zero bias). The 3N138 also features extremely low feed-through capacitance (0.18pF typ.) and zero inherent offset voltage.

The 3N138 is hermetically sealed in the JEDEC TO-72 package and features a gate metallization that covers the entire source-to-drain channel.

* Metal-Oxide-Semiconductor.

TERMINAL DIAGRAM



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

Maximum Ratings, Absolute-Maximum Values:

(Substrate connected to source unless otherwise specified)

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+35 max.	V
DRAIN-TO-SUBSTRATE VOLTAGE, V_{DB}	+35, −0.3 max.	V
SOURCE-TO-SUBSTRATE VOLTAGE, V_{SB}	+35, −0.3 max.	V
DC GATE-TO-SOURCE VOLTAGE, V_{GS}	±10 max.	V
PEAK GATE-TO-SOURCE VOLTAGE, V_{GS}	±14 max.	V
PEAK VOLTAGE, GATE-TO-ALL OTHER TERMINALS: V_{GS}, V_{GD}, V_{GB}, non-repetitive	±45 max.	V
DRAIN CURRENT, I_D (Pulse duration 20 ms, duty factor ≤ 0.10)	50 max.	mA
TRANSISTOR DISSIPATION, P_T: At ambient temperatures up to 25°C	330 max.	mW
above 25°C	Derate linearly at 2.2	mW/°C
AMBIENT TEMPERATURE RANGE: Storage	−65 to +150	°C
Operating	−65 to +125	°C
LEAD TEMPERATURE (During Soldering): At distances ≥ 1/32" to seating surface for 10 seconds max.	265 max.	°C

ELECTRICAL CHARACTERISTICS, at T_A = 25° C, Unless Otherwise Specified. Substrate Connected to Source.

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS Type 3N138			UNITS
			Min.	Typ.	Max.	
Gate-Leakage Current	I_{GSS}	$V_{GS} = \pm 10$, $V_{DS} = 0$, $T_A = 25°C$ $V_{GS} = \pm 10$, $V_{DS} = 0$, $T_A = 125°C$	— —	0.1 20	10 200	pA pA
Drain-to-Source "ON" Resistance	$r_{DS(on)}$	$V_{GS} = 0$, $V_{DS} = 0$, $f = 1$ KHz, $T_A = 25°C$ $V_{GS} = +10$, $V_{DS} = 0$, $f = 1$ KHz, $T_A = 25°C$ $V_{GS} = 0$, $V_{DS} = 0$, $f = 1$ KHz, $T_A = 125°C$	— — —	240 135 350	350 — —	Ω Ω Ω
Drain-to-Source "OFF" Resistance	$R_{DS(off)}$	$V_{GS} = -10$, $V_{DS} = +1$	2×10^8	10^{10}	—	Ω
Drain-to-Source Cutoff Current	$I_{D(off)}$	$V_{GS} = -10$, $V_{DS} = +1$, $T_A = 25°C$ $V_{GS} = -10$, $V_{DS} = +1$, $T_A = 125°C$	— —	0.01 0.01	5 0.5	nA μA
Small-Signal, Short-Circuit, Reverse Transfer Capacitance	C_{rss}	$V_{GS} = -10$, $V_{DS} = 0$, $f = 1$ MHz	—	0.25	0.4	pF
Small-Signal, Short-Circuit, Input Capacitance	C_{iss}	$V_{GS} = -10$, $V_{DS} = 0$, $f = 1$ MHz	—	3	5	pF
Zero-Gate-Bias Forward Transconductance	g_{fs}	$V_{DS} = 12$, $I_D = 5$ mA	—	6000	—	μmho
Offset Voltage	V_O	$V_{GS} = \pm 10$, $V_{DS} = 0$	—	0*	—	V

* In measurements of Offset Voltage, thermocouple effects and contact potentials in the measurement setup may cause erroneous readings of 1 microvolt or more. These errors may be minimized by the use of solder having a low thermal e.m.f. such as Leeds & Northrup No.107-1.0.1, or equivalent.

OPERATING CONSIDERATIONS

The flexible leads of the 3N138 are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons should be grounded, and appropriate precautions should be taken to protect the device against high electric fields.

This device should not be connected into or disconnected from circuits with the power on because high transient voltages may cause permanent damage to the device.

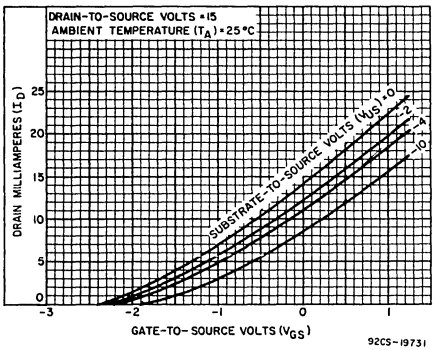


Fig. 1 — Drain Current vs Gate-to-Source Voltage

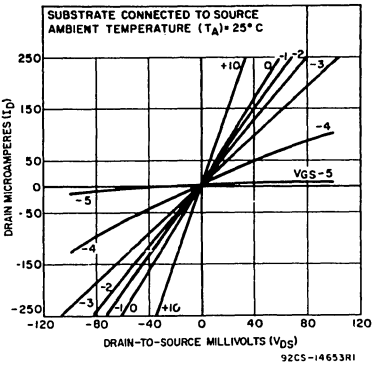


Fig. 2 — Low-Level Drain Current vs Drain-to-Source Voltage

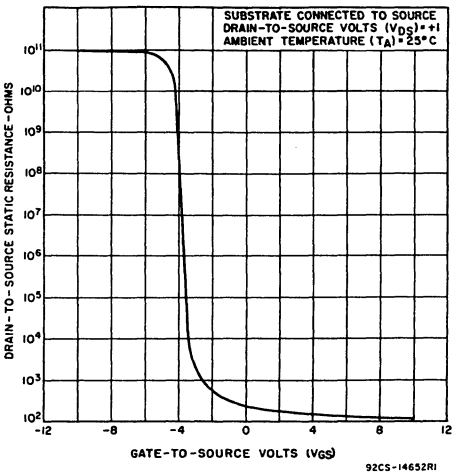


Fig. 3 — Drain-to-Source Static Resistance vs Gate-to-Source Voltage

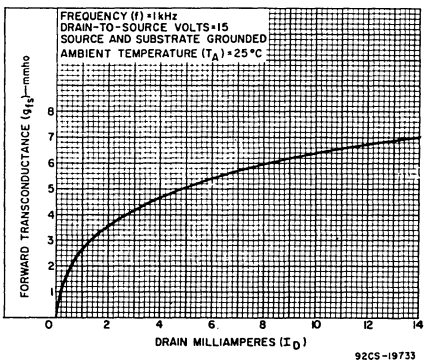


Fig. 4 — 1 KHz forward transconductance vs drain current

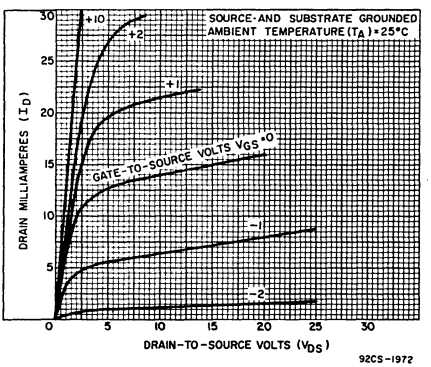


Fig. 5 — Drain Current vs Drain Voltage

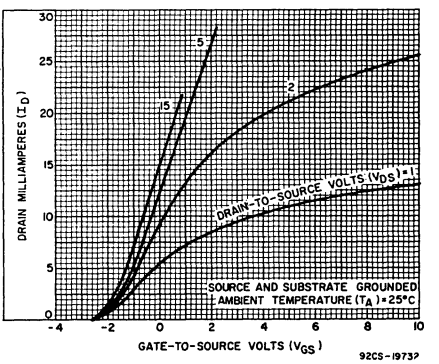


Fig. 6 — Drain Current vs Gate-to-Source Voltage

RCA Solid State Division

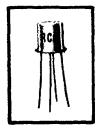
MOS Field-Effect Transistors

N-Channel Depletion Type

3N139

SILICON MOS TRANSISTOR

For Audio, Video, and RF Amplifier Applications



JEDEC TO-72

in Military Communications, Instrumentation, & Navigation Equipment

in Mobile and Fixed Communication Equipment

in Industrial Instrumentation and Control Circuits

RCA 3N139 is a silicon, insulated-gate field-effect transistor of the N-channel depletion type, utilizing the MOS* construction. It is a general purpose transistor especially suited for audio, video, and rf applications, and for wide-band amplifier designs. The insulated gate provides a very high input resistance (10^{14} Ω typ.) which is relatively insensitive to temperature and is independent of gate-bias conditions (positive, negative, or zero bias). The 3N139 also has a high transconductance, a low value of input capacitance (3 pF typ.), and a very low feedback capacitance (0.19 pF typ.).

The 3N139 is hermetically sealed in the standard 4-lead JEDEC TO-72 package.

Maximum Ratings, *Absolute-Maximum Values:*

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+35 max.	V
DRAIN-TO-SUBSTRATE VOLTAGE, V_{DB}	+35, -0.3 max.	V
SOURCE-TO-SUBSTRATE VOLTAGE, V_{SB}	+35, -0.3 max.	V
DC GATE-TO-SOURCE VOLTAGE, V_{GS}	±10 max.	V
PEAK GATE-TO-SOURCE VOLTAGE, V_{GS}	±14 max.	V
PEAK VOLTAGE, GATE-TO-ALL OTHER TERMINALS; V_{GS}, V_{GD}, V_{GB}, non-repetitive	±42 max.	V
DRAIN CURRENT, I_D	50 max.	mA
TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	330	mW
above 25°C	Derate linearly at 2.2 mW/°C	
AMBIENT TEMPERATURE RANGE:		
Storage	-65 to +175	°C
Operating	-65 to +175	°C
LEAD TEMPERATURE (During Soldering):		
At distance not closer than 1/32 inch to seating surface for 10 seconds max.	265 max.	°C

* Metal-Oxide-Semiconductor

FEATURES

- high input resistance
  $R_{GS} = 10^{14}$ Ω typ.
- low input capacitance
  $C_{iss} = 3$ pF typ.
- low feed back capacitance
  $C_{rss} = 0.2$ pF typ.
- low gate leakage current
  $I_{GSS} = 0.1$ nA typ.
- high drain-to-source voltage: +35 max. V

ELECTRICAL CHARACTERISTICS, at T_A = 25° C Unless Otherwise Specified. Bulk (Substrate) Connected to Source

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS				LIMITS			UNITS
		FREQUENCY f	DC DRAIN-TO-SOURCE VOLTAGE V_{DS}	DC GATE-TO-SOURCE VOLTAGE V_{GS}	DC DRAIN CURRENT I_D				
		MHz	V	V	mA	Min.	Typ.	Max.	
Drain-to-Source Cutoff Current	$I_{D(OFF)}$		15	−8		—	—	50	μA
Zero-Bias Drain Current*	I_{DSS}		15	0		5	15	25	mA
Gate Reverse Current	I_{GSS}	T_A = 25°C	0	±10		—	—	1	nA
		T_A = 100°C	0	±10		—	—	100	nA
Gate-to-Source Cutoff Voltage	$V_{GS(OFF)}$		15		0.05	−2	−4	−6	V
Small-Signal, Short-Circuit Reverse-Transfer Capacitance (Drain-to-Gate)	C_{rss}	1	15		5	0.05	0.2	0.4	pF
Input Resistance	r_{is}	100	15		5		12	—	kΩ
Input Capacitance	C_{iss}	100	15		5	—	3	10	pF
Output Resistance	r_{os}	100	15		5		6	—	kΩ
Output Capacitance	C_{oss}	100	15		5	—	1.4	—	pF
Forward Transconductance	g_{fs}	1 kHz	15		5		5	—	mmho

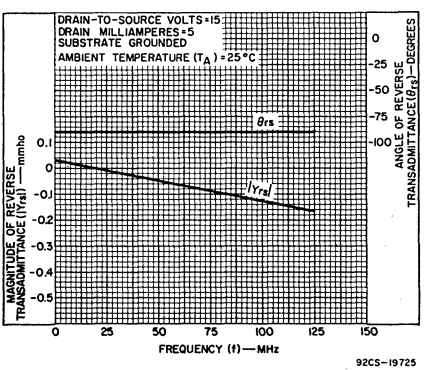


Fig. 1 – Reverse Transadmittance vs Frequency

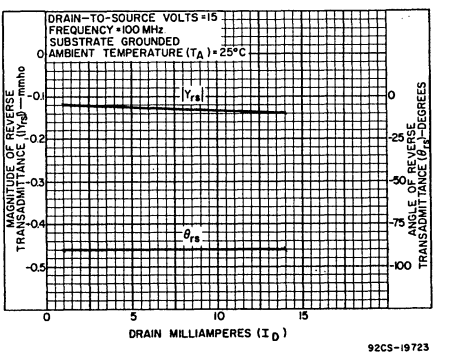


Fig. 2 – Reverse Transadmittance vs Drain Current

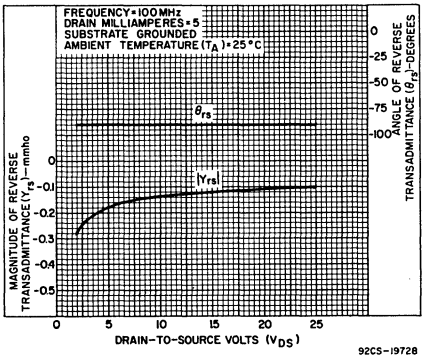


Fig. 3 — Reverse Transadmittance vs Drain-Source Voltage

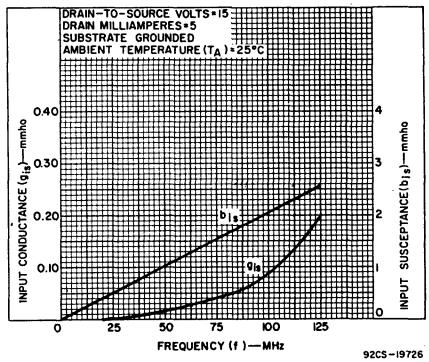


Fig. 4 — Input Admittance vs Frequency

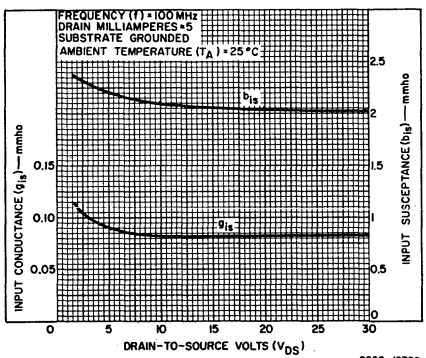


Fig. 5 — Input Admittance vs Drain-Source Voltage

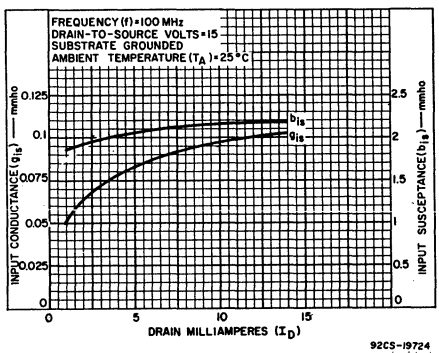


Fig. 6 — Input Admittance vs Drain Current

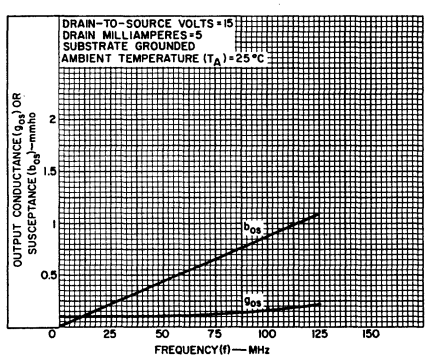


Fig. 7 — Output Conductance vs Frequency

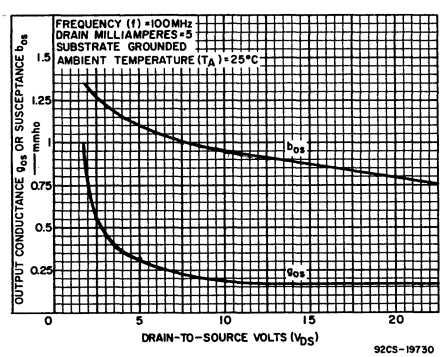


Fig. 8 — Output Admittance vs Drain-Source Voltage

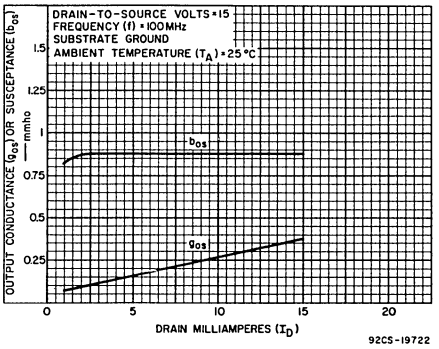


Fig. 9 — Output Admittance vs Drain Current

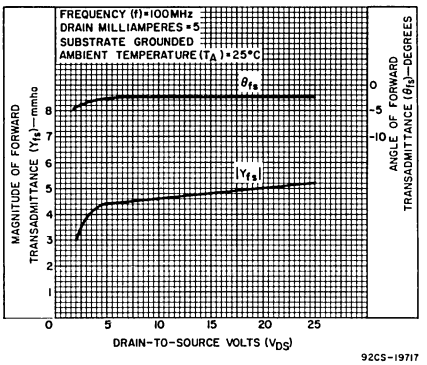


Fig. 10 — Forward Transadmittance vs Drain-Source Voltage

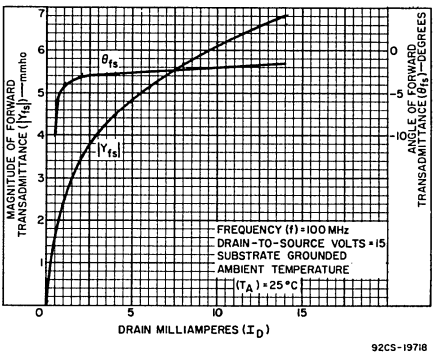


Fig. 11 — Forward Transadmittance vs Drain Current

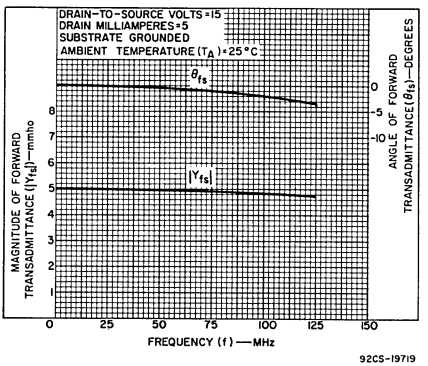


Fig. 12 — Forward Transadmittance vs Frequency

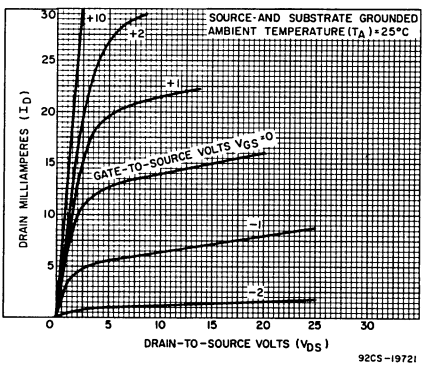


Fig. 13 — Drain Current vs Drain Voltage

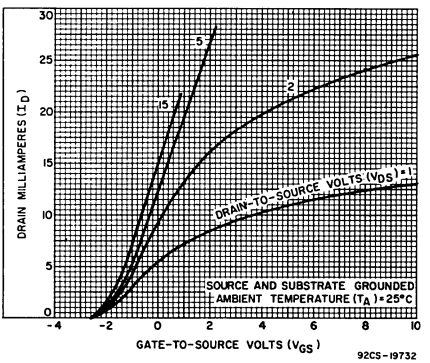


Fig. 14 — Drain Current vs Gate-to-Source Voltage

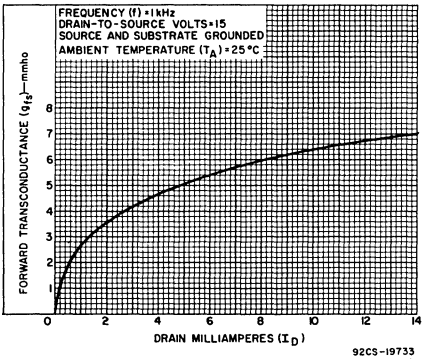


Fig. 15 — 1 KHz forward transconductance vs drain current

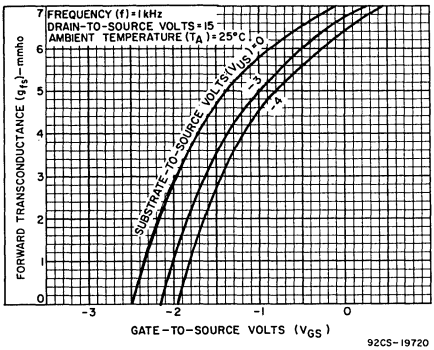
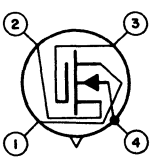


Fig. 16 — 1 KHz forward transconductance vs gate-to-source voltage

TERMINAL ARRANGEMENT



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

RCA Solid State Division

MOS Field-Effect Transistors

N-Channel Depletion Types

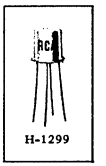
3N140

3N141

SILICON DUAL INSULATED-GATE FIELD-EFFECT TRANSISTORS

N-Channel Depletion Types

For Military and Industrial Amplifier and Mixer Applications Up to 300 MHz



JEDEC TO-72

RCA-3N140 and 3N141* are n-channel silicon, depletion type, dual insulated-gate, field-effect transistors utilizing the MOS** construction. They have exceptional characteristics for rf-amplifier and mixer applications at frequencies up to 300 MHz. These transistors feature a series arrangement of two separate channels, each channel having an independent control gate.

The 3N140, used in a common-source configuration in which gate No.2 is ac grounded, reduces oscillator feedthrough to the antenna thereby minimizing oscillator radiation. The 3N141 provides excellent isolation between the oscillator and rf signals because each of the two signal frequencies being mixed has its own control element.

The mixing function performed by the 3N141 is unique in that the signal applied to gate No.2 is used to modulate the input-gate (gate No.1) transfer characteristic. This technique is superior to conventional "square law" mixing, which can only be accomplished in the nonlinear region of the device transfer characteristic.

The use of the 3N141 as described provides high useful conversion gains at all vhf frequencies, and the reduction in spurious responses is substantial and easily obtainable in simple circuits.

The 3N140 and 3N141 are hermetically sealed in metal JEDEC TO-72 packages.

* Formerly Dev. Nos. TA2644 and TA7274, respectively.

** Metal-Oxide-Semiconductor.

APPLICATIONS

- RF amplifier and mixer in military and industrial communications equipment
- aircraft and marine vehicular receivers
- CATV and MATV equipment
- telemetry and multiplex equipment

PERFORMANCE FEATURES

- wide dynamic range permits large-signal handling before overload
- dual-gate permits simplified agc circuitry
- virtually no agc power required
- greatly reduces spurious responses in fm receivers
- permits use of vacuum-tube biasing techniques
- excellent thermal stability
- superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET's

DEVICE FEATURES

- low gate leakage currents – – I_{G1SS} & I_{G2SS} = 1 nA max. at T_A = 25°C
- high forward transconductance – – g_{fs} = 6000 μmho min.
- high unneutralized RF power gain – – G_{ps} = 16 dB min. at 200 MHz
- low VHF noise figure – – 4.5 dB max. at 200 MHz

Maximum Ratings, *Absolute-Maximum Values, at* T_A = 25°C

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	0 to +20	V
GATE No.1-TO-SOURCE VOLTAGE, V_{G1S}:		
Continuous (dc)	-8 to +1	V
Peak ac	-8 to +20	V
GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:		
Continuous (dc)	-8 to 40% of V_{DS}	V
Peak ac	-8 to +20	V
DRAIN-TO-GATE VOLTAGE, V_{DG1} OR V_{DG2}	+20	V
DRAIN CURRENT, I_D (Pulsed): Pulse duration ≤ 20 ms, duty factor ≤ 0.15	50	mA
TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	400	mW
At ambient temperatures above 25°C	derate linearly at 2.67 mW/°C	
AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
LEAD TEMPERATURE (During soldering):		
At distances ≥ 1/32 inch from seating surface for 10 seconds max.	265	°C

ELECTRICAL CHARACTERISTICS, at T_A = 25°C Unless Otherwise Specified. Common-Source Circuit.

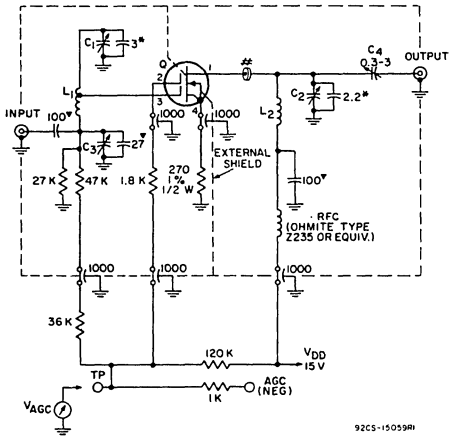
CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS						UNITS
			TYPE 3N140 RF AMPLIFIER			TYPE 3N141 MIXER			
			MIN.	TYP.	MAX.	MIN.	TYP.	MAX.	
Gate No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +16V, I_D = 200 μA V_{G2S} = +4V	-	-2	-4	-	-2	-4	V
Gate No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +16V, I_D = 200 μA V_{G1S} = 0	-	-2	-4	-	-2	-4	V
Gate No.1 Leakage Current	I_{G1SS}	V_{G1S} = -20V, V_{G2S} = 0 V_{DS} = 0, T_A = 25°C	-	-	1	-	-	1	nA
		V_{G1S} = +1V, V_{G2S} = 0 V_{DS} = 0, T_A = 25°C	-	-	1	-	-	1	nA
		V_{G1S} = -20V, V_{G2S} = 0 V_{DS} = 0, T_A = 125°C	-	-	0.2	-	-	0.2	μA
Gate No.2 Leakage Current	I_{G2SS}	V_{G2S} = -20V, V_{G1S} = 0 V_{DS} = 0, T_A = 25°C	-	-	1	-	-	1	nA
		V_{G2S} = +1V V_{DS} = 0, V_{G1S} = 0, T_A = 25°C	-	-	1	-	-	1	nA
		V_{G2S} = -20V, V_{G1S} = 0 V_{DS} = 0, T_A = 125°C	-	-	0.2	-	-	0.2	μA
Zero-Bias Drain Current	I_{DSS}*	V_{DD} = +14V, V_{G1S} = 0, V_{G2S} = +4	5	18	30	5	18	30	mA
Forward Transconductance (Gate No.1 to Drain)	g_{fs}	V_{DD} = +14V, I_D = 10 mA V_{G2S} = +4V, f = 1 kHz	6000	10000	18000	6000	10000	18000	μmho
Cutoff Forward Transconductance (Gate No.1 to Drain)	$g_{fs(off)}$	V_{DD} = +14V, V_{G1S} = -0.5V V_{G2S} = -2V, f = 1 kHz	-	-	100	-	-	-	μmho
Small-Signal, Short-Circuit Input Capacitance♦	C_{iss}	V_{DS} = +13V, I_D = 10 mA V_{G2S} = +4V, f = 1 MHz	3	5.5	7	3	5.5	7	pF
Small-Signal, Short-Circuit Reverse Transfer Capacitance (Drain to Gate No.1)▲	C_{rss}	V_{DS} = +13V, I_D = 10 mA V_{G2S} = +4V, f = 1 MHz	0.01	0.02	0.03	0.01	0.02	0.03	pF
Small-Signal Short-Circuit Output Capacitance	C_{oss}	V_{DS} = +13V, I_D = 10 mA V_{G2S} = +4V, f = 1 MHz	-	2.2	-	-	2.2	-	pF
Power Gain (See Fig.1 for Measurement Circuit)	G_{ps}	V_{DD} = +15V, R_S = 270Ω f = 200 MHz, R_G = 50Ω	16	18	-	-	-	-	dB
Conversion Power Gain (See Fig.2 for Measurement Circuit)	G_{psc}	V_{DD} = +15V, R_S = 120Ω, f_{IN} = 200 MHz, f_{OUT} = 30 MHz Oscillator injection voltage● = 2.5 V (rms)	-	-	-	13	17	-	dB
Measured Noise Figure (See Fig.1 for Measurement Circuit)	NF	V_{DD} = +15V, R_S = 270Ω f = 200 MHz, R_G = 50Ω	-	3.5	4.5	-	-	-	dB

* Pusle test: Pulse duration ≤ 20 ms, duty factor ≤ 0.15.

♦ Capacitance between Gate No.1 and all other terminals.

▲ Three-Terminal Measurement with Gate No.2 and Source Returned to Guard Terminal.

● Measured from gate No.2 to source.



Q = 3N140.

▼Disc ceramic.

* Tubular ceramic.

Ferrite bead (1/2 used); Indiana General No.H1742C-(A-147), F-1157-1-H

All resistors in ohms

All capacitors in pF

C_1, C_2: 1.5-5 pF variable air capacitor: E.F. Johnson Type 160-102 or equivalent.

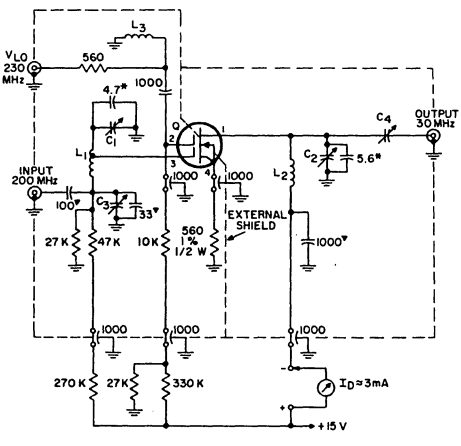
C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent.

C_4: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent.

L_1: 5 turns silver-plated 0.02" thick, 0.07"±0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding.

L_2: Same as L_1 except winding length approx. 0.7"; no tap.

Fig.1 - 200 MHz power gain and noise figure test circuit for type 3N140.



Q = 3N141.

▼Disc ceramic.

* Tubular ceramic.

All resistors in ohms

All capacitors in pF

C_1, C_2: 1.5-5 pF variable air capacitor: E.F. Johnson Type 160-102 or equivalent.

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent.

C_4: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent.

L_1: 5 turns silver-plated 0.02" thick, 0.07"±0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding.

L_2: Ohmite Z-144 RF choke or equivalent.

L_3: J.W. Miller Co. #4580 0.1 μH RF choke or equivalent.

Note: If 50Ω meter is used in place of sweep detector, a low pass filter must be provided to eliminate local oscillator voltage from load.

Fig.2 - Conversion power gain test circuit for type 3N141.

OPERATING CONSIDERATIONS

The flexible leads of the 3N140 and 3N141 are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons should be grounded, and appropriate precautions should be taken to protect the devices against high electric fields.

These devices should not be connected into or disconnected from circuits with the power on because high transient voltages may cause permanent damage to the devices.

TYPICAL CHARACTERISTICS FOR TYPES 3N140, 3N141

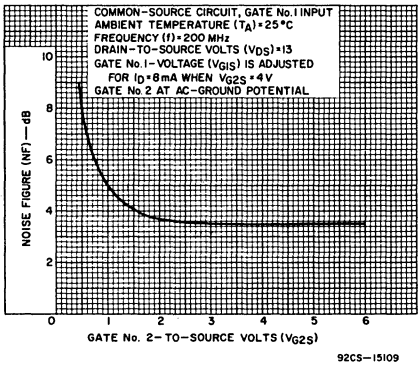


Fig.3 - NF vs V_{G2S}.

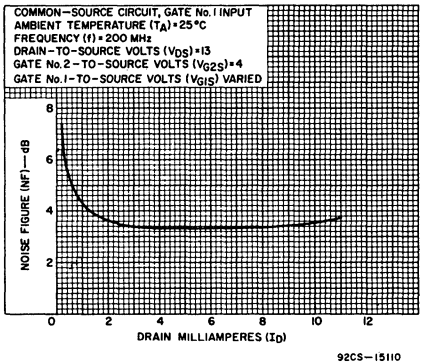


Fig.4 - NF vs I_D.

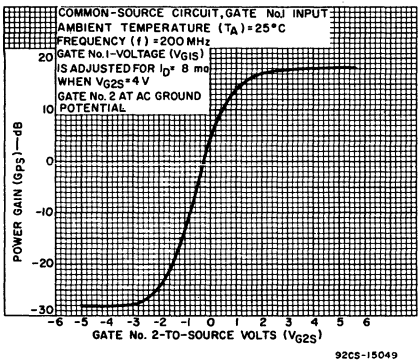


Fig.5 - G_{PS} vs V_{G2S} (For 3N140).

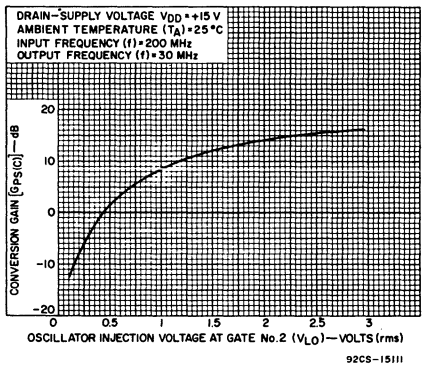


Fig.6 - $G_{PS(C)}$ vs V_{LO} (For 3N141).

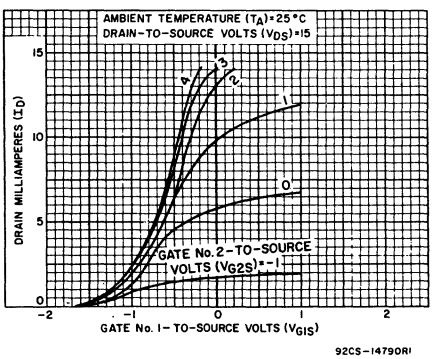


Fig.7 - I_D vs V_{G1S}.

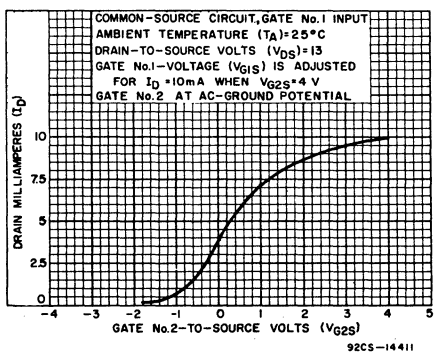


Fig.8 - I_D vs V_{G2S}.

TYPICAL CHARACTERISTICS FOR TYPES 3N140, 3N141

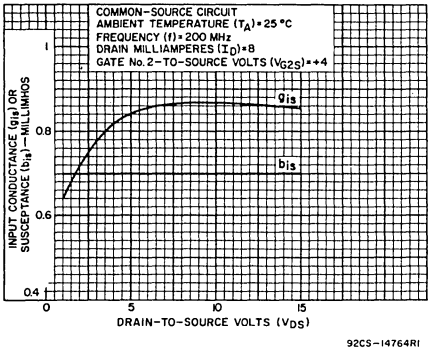


Fig.9 - y_{is} vs V_{DS}.

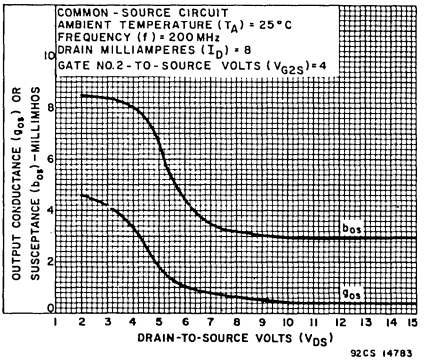


Fig.10 - y_{os} vs V_{DS}.

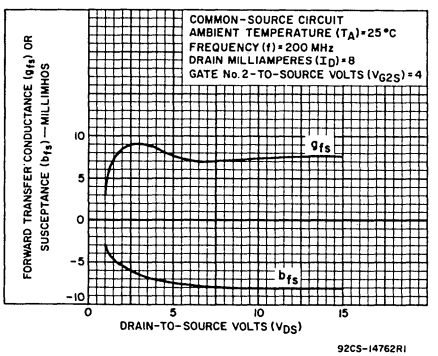


Fig.11 - y_{fs} vs V_{DS}.

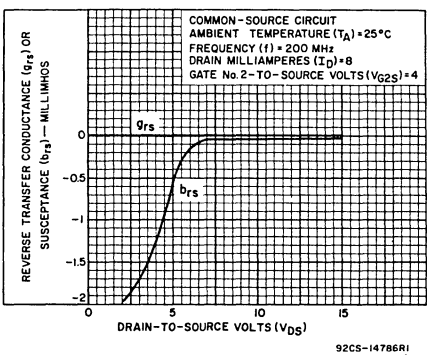


Fig.12 - y_{rs} vs V_{DS}.

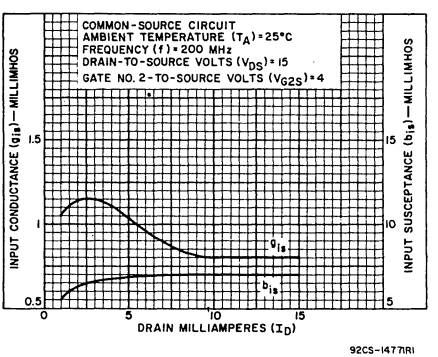


Fig.13 - y_{is} vs I_D.

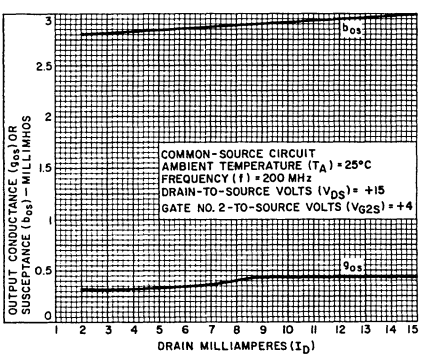


Fig.14 - y_{os} vs I_D.

TYPICAL CHARACTERISTICS FOR TYPES 3N140, 3N141

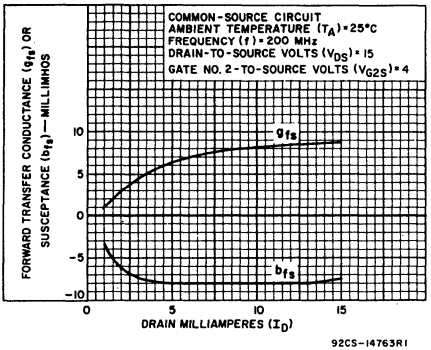


Fig.15 - y_{fs} vs I_D.

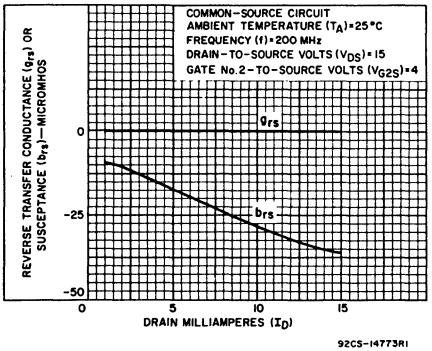


Fig.16 - y_{rs} vs I_D.

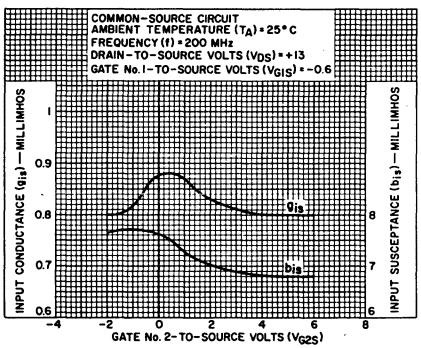


Fig.17 - y_{is} vs V_{G2S}.

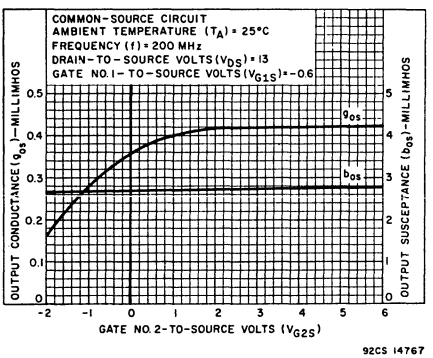


Fig.18 - y_{os} vs V_{G2S}.

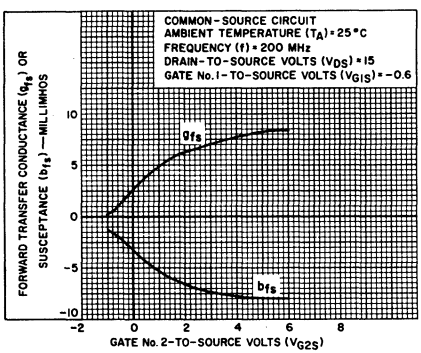


Fig.19 - y_{fs} vs V_{G2S}.

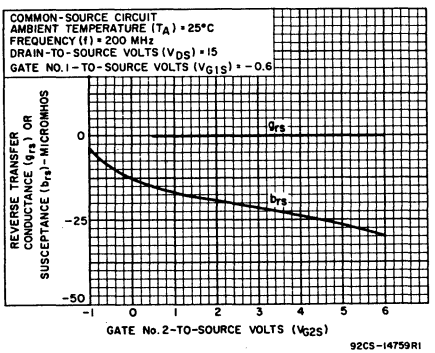


Fig.20 - y_{rs} vs V_{G2S}.

TYPICAL CHARACTERISTICS FOR TYPES 3N140, 3N141

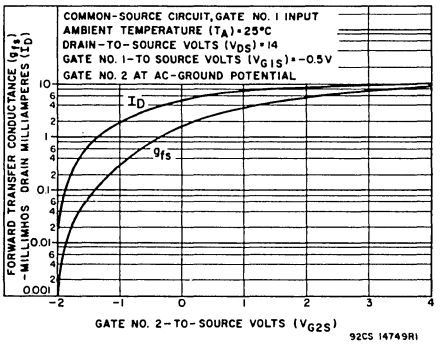


Fig.21 - g_{fs} and I_D vs V_{G2S}.

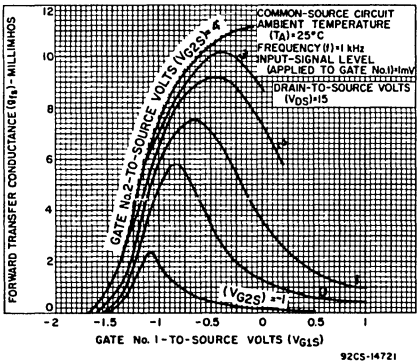


Fig.22 - g_{fs} vs V_{G1S}.

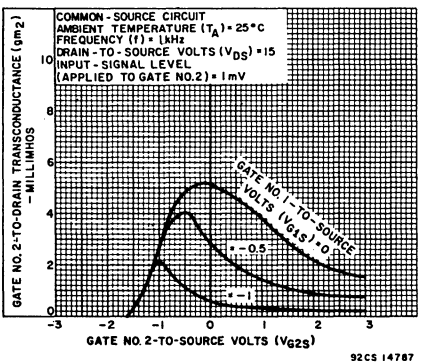
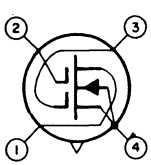


Fig.23 - g_{fs2} vs V_{G2S}.

TERMINAL DIAGRAM



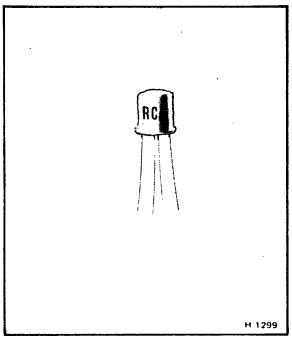
LEAD 1 - DRAIN
LEAD 2 - GATE No.2
LEAD 3 - GATE No.1
LEAD 4 - SOURCE, SUBSTRATE AND CASE

RCA
Solid State
Division

MOS Field-Effect Transistors

N-Channel Depletion Type

3N142



Silicon MOS Transistor

For Industrial and Military Applications to 175 MHz

Applications

- RF amplifier, Mixer, and Oscillator in:
  CB and Mobile Communication Receivers
  Aircraft and Marine Receivers
  CATV and MATV Equipment
- Industrial Control Circuits
- Variable Attenuators
- Current Limiters
- Instrumentation Equipment
- High-Impedance Timing Circuits

The 3N142 is a silicon, insulated-gate field-effect transistor of the N-channel depletion type utilizing the MOS■ construction.

The 3N142 is intended primarily for use as the rf amplifier in FM receivers and general amplifier applications at frequencies up to 175 MHz.

The wide dynamic range of the 3N142 reduces cross-modulation effects in AM receivers and minimizes the generation of spurious responses in FM receivers.

■ Metal-Oxide-Semiconductor

Performance Features

- Large dynamic range
- Enhanced signal-handling capability for low cross-modulation
- Dual-polarity gate permits positive and negative swing without degradation of input impedance
- Reduced spurious responses in FM receivers
- Permits use of vacuum-tube biasing techniques
- Excellent thermal stability for critical oscillator designs

Device Features

- High input resistance – 1000 megohms
- Low feedback capacitance – 0.35 pF max.
- Low noise figure – 2.5 dB typ.
- High useful power gain – neutralized – 16 dB min. at 100 MHz
- Hermetically sealed TO – 72 metal package

Maximum Ratings, *Absolute-Maximum Values at* T_A = 25° C

*DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20	V
*DRAIN-TO-GATE VOLTAGE, V_{DG}	+20	V
*GATE-TO-SOURCE VOLTAGE, V_{GS}:		
Continuous	+1 to −8	V
Peak ac	±15	V
*DRAIN CURRENT, I_D	50	mA
*TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	330	mW
At ambient temperatures above 25°C	Derate at 2.2	mW/°C
*AMBIENT TEMPERATURE RANGE:		
Storage	−65 to +175	°C
Operating	−65 to +175	°C
*LEAD TEMPERATURE (During Soldering): At distances ≥ 1/32" from seating surface for 10 seconds max.	265	°C

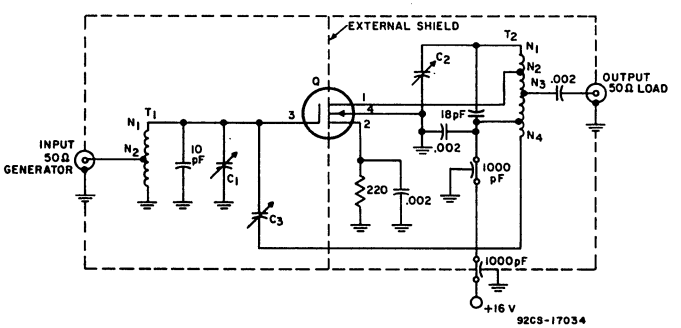
* In accordance with JEDEC Registration Data Format JS-9 RDF11-B

ELECTRICAL CHARACTERISTICS: (At T_A = 25° C)

Measured with Substrate Connected to Source Unless Otherwise Specified.

	CHARACTERISTICS	SYMBOLS	CONDITIONS	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
*	Gate Leakage Current	I_{GSS}	V_{DS} = 0, V_{GS} = -8 V, T_A = 25° C	-	0.0001	1	nA
			V_{DS} = 0, V_{GS} = -8 V, T_A = 125° C	-	-	200	nA
			V_{DS} = 0, V_{GS} = +1, T_A = 25° C	-	0.0001	1	nA
			V_{DS} = 0, V_{GS} = +1, T_A = 125° C	-	-	200	nA
*	Zero-Bias Drain Current**	I_{DSS}	V_{DS} = 15 V, V_{GS} = 0	5	15	25	mA
*	Drain-to-Source Cutoff Current	$I_{D(off)}$	V_{DS} = 20 V, V_{GS} = -8 V	-	-	50	μA
*	Gate-to-Source Cutoff Voltage	$V_{GS(off)}$	V_{DS} = 15 V, I_D = 50 μA	-0.5	-3	-8	V
*	Forward Transconductance	g_{fs}	V_{DS} = 15 V, I_D = 5 mA, f = 1 kHz	5000	7500	12,000	μmho
*	Drain-to-Source Channel Resistance	$r_{DS(on)}$	V_{DS} = 0, V_{GS} = 0, f = 1 kHz	-	200	-	Ω
*	Small-Signal Short-Circuit Reverse Transfer Capacitance‡	C_{rss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	0.10	0.22	0.35	pF
*	Small-Signal Short-Circuit Input Capacitance	C_{iss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	-	5.5	7	pF
*	Input Admittance	Y_{is}	Common Source Configuration f = 100 MHz V_{DS} = 15V I_D = 5 mA	-	0.155+J3.45	-	mmho
*	Forward Transfer Admittance	Y_{fs}		-	7.5-J0.9	-	mmho
*	Output Admittance	Y_{os}		-	0.21+J0.9	-	mmho
*	Maximum Available Power Gain	MAG	V_{DS} = 15 V, I_D = 5 mA, f = 100 MHz	-	26	-	dB
	Maximum Usable Power Gain (Fixed Neutralization)	MUG		-	17	-	
*	Insertion Power Gain** (Fixed Neutralization)	G_{ps}		16	-	-	dB
*	Noise Figure**	NF	V_{DS} = 15 V, I_D = 5 mA, f = 100 MHz	-	2.5	4	dB

* In accordance with JEDEC Registration Data Format JS-9 RDF-11B ‡ Three-Terminal Measurement: Source Returned to Guard Terminal
**See Fig. 1



T_1 N_1 = 6 Turns#20 Tinned Copper Wire; ¼" I.D. ½" Long
Q_o = 205, N_1/N_2 = 4.85

T_2 $N_1 + N_4$ = 6¾ Turns#20 Tinned Copper Wire ¼" I.D. $^9/_{16}$" Long
Q_o = 190 N_1/N_2 = 1.9 N_1/N_3 = 12.3 N_1/N_4 = 8

C_1 = 10 pF Variable Air Capacitor (Hammarlund Mac-10 or Equivalent)
C_2 = 5 pF Variable Air Capacitor (Hammarlund Mac-5 or Equivalent)
C_3 = 0.7-3 pF Piston-Type Variable Air Capacitor (Erie 535C or Equivalent)
Q = 3N142

Fig. 1 - Test Set Up for 100 MHz Insertion Power Gain and Noise Figure

TYPICAL CHARACTERISTICS

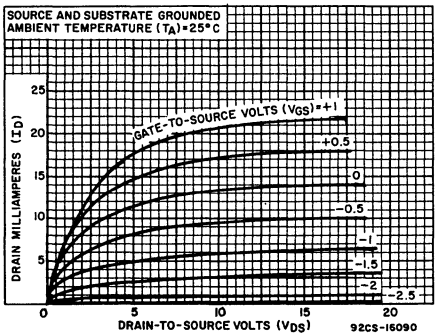


Fig. 2 - Drain Current vs Drain-to-Source Voltage.

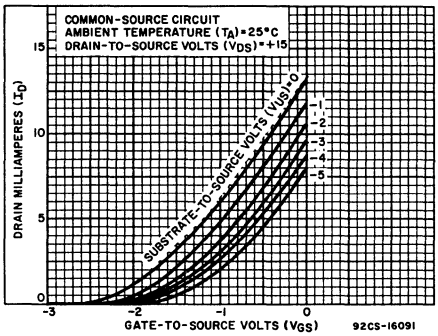


Fig. 3 - Drain Current vs Gate-to-Source Voltage (V_{gs}).

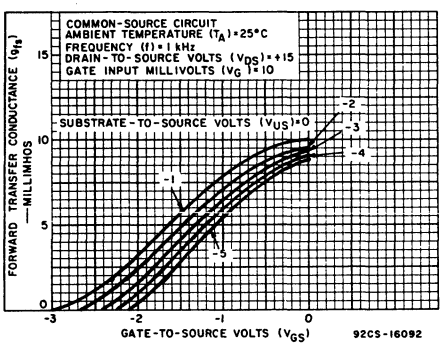


Fig. 4 - Forward Transconductance vs Gate Bias Voltage.

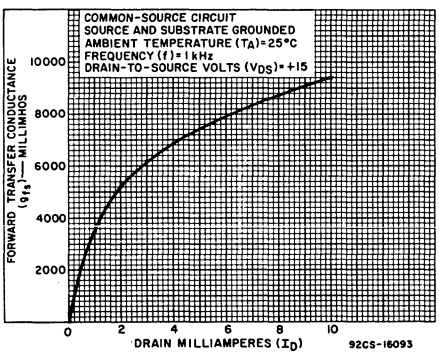


Fig. 5 - Forward Transconductance vs Drain Current.

TYPICAL y PARAMETER CHARACTERISTICS

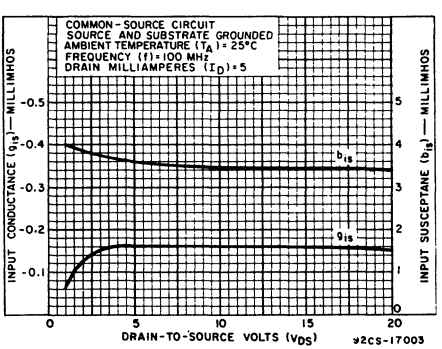


Fig. 6 - Input Admittance vs. Drain-to-Source Voltage

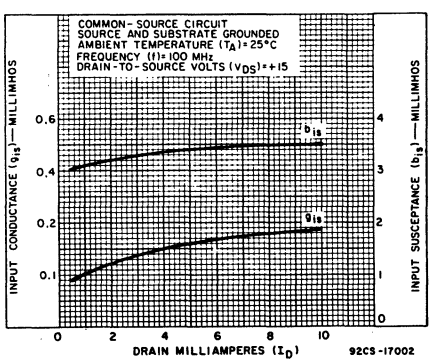


Fig. 7 - Input Admittance vs. Drain Current

TYPICAL y PARAMETER CHARACTERISTICS (Cont'd)

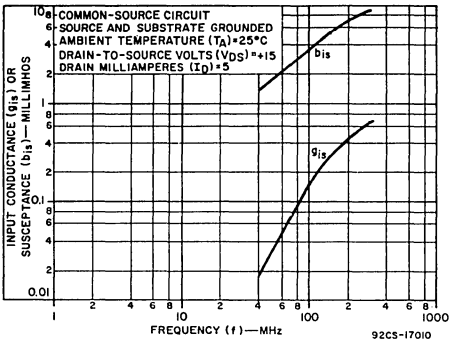


Fig. 8 - Input Admittance vs. Frequency

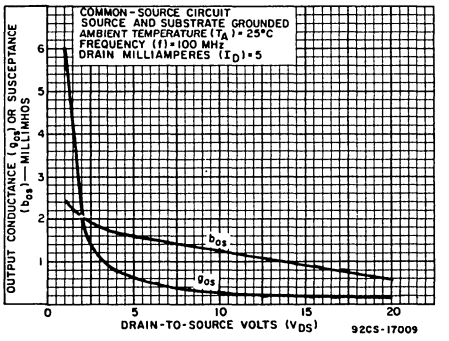


Fig. 9 - Output Admittance vs. Drain-to-Source Voltage

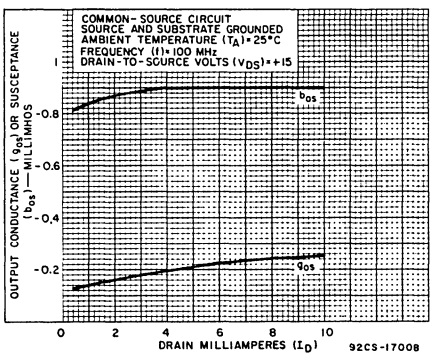


Fig. 10 - Output Admittance vs. Drain Current

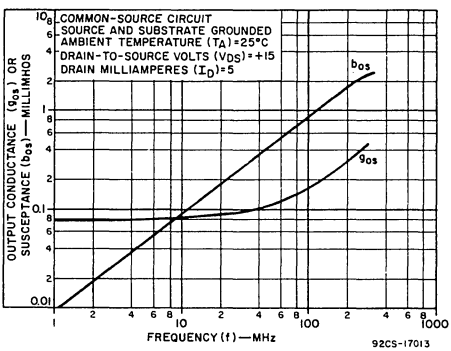


Fig. 11 - Output Admittance vs. Frequency

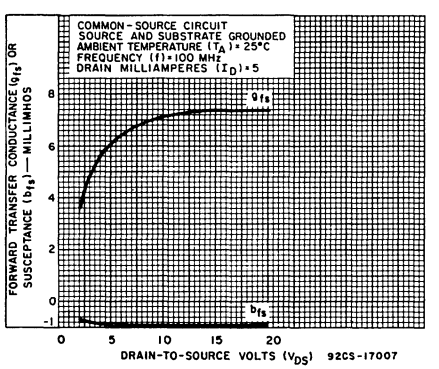


Fig. 12 - Forward Transadmittance vs. Drain-to-Source Voltage

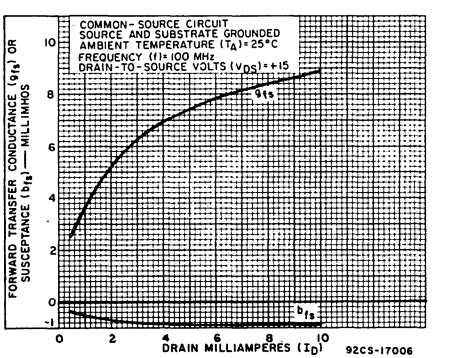


Fig. 13 - Forward Transadmittance vs. Drain Current

TYPICAL y PARAMETER CHARACTERISTICS

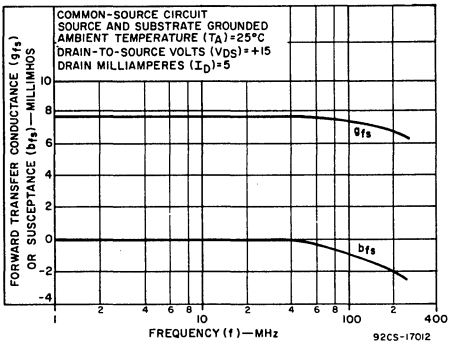


Fig. 14 - Forward Transadmittance vs. Frequency

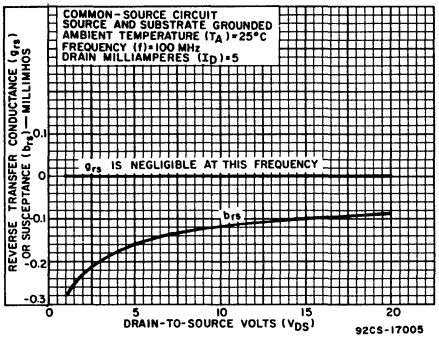


Fig. 15 - Reverse Transadmittance vs. Drain-to-Source Voltage

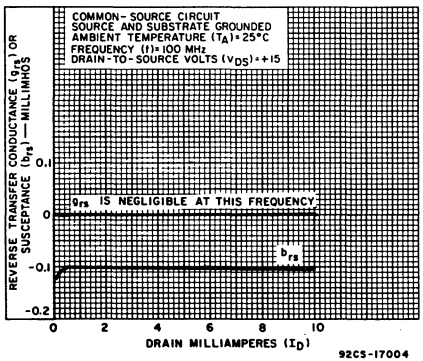


Fig. 16 - Reverse Transadmittance vs. Drain Current

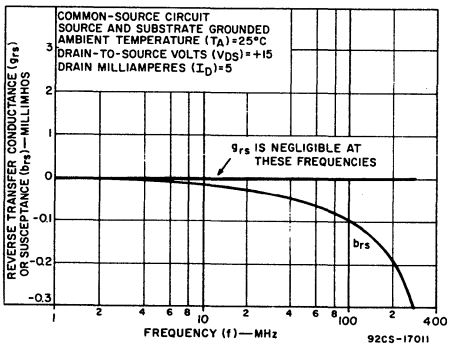
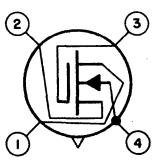


Fig. 17 - Reverse Transadmittance vs. Frequency

TERMINAL DIAGRAM



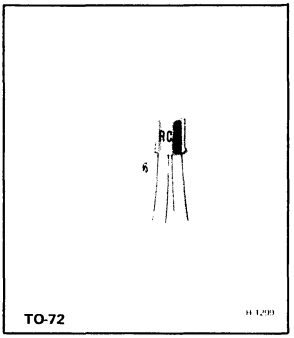
LEAD 1 - DRAIN
LEAD 2 - SOURCE
LEAD 3 - INSULATED GATE
LEAD 4 - BULK (SUBSTRATE) AND CASE

RCA
Solid State
Division

MOS Field-Effect Transistors

N-Channel Depletion Type

3N152



Silicon MOS Transistor

For Low-Noise RF Applications in Military & Industrial VHF Communications Equipment Operating up to 250 MHz

RCA-3N152 is an N-channel depletion-type silicon insulated gate field-effect transistor utilizing the MOS■ construction. It is intended primarily for VHF amplifier applications up to 250 MHz in military and industrial equipment.

Because of its improved transfer characteristic and exceptionally wide dynamic range, the 3N152 with the substrate in the reversed bias mode can provide substantially better cross-modulation performance in linear amplifier applications than conventional bipolar transistors. The insulated gate with its extremely low reverse (leakage) current eliminates the problem of diode-current loading of the input circuit under strong input conditions, which is common to junction-type FET's. These features in addition to low feedback capacitance permit the design of circuits providing superior high-frequency operation and high gain without neutralization. The 3N152 utilizes full-gate construction and is hermetically sealed in a JEDEC TO-72 metal package.

■ Metal-Oxide-Semiconductor.

Maximum Ratings, *Absolute-Maximum Values at* $T_A = 25^oC$:

* DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20 max.	V
* DRAIN-TO-GATE VOLTAGE, V_{DG}	+20	V
* GATE-TO-SOURCE VOLTAGE, V_{GS}:		
* CONTINUOUS (dc)	+1, -8 max.	V
* PEAK ac	±15 max.	V
* DRAIN CURRENT, I_D	50 max.	mA
TRANSISTOR DISSIPATION:		
At ambient } up to 25°C	330 max.	mW
temperatures } above 25°C	derate at 2.2 mW/°C	
* AMBIENT TEMPERATURE RANGE:		
Storage	-65 to +175	°C
Operating	-65 to +175	°C
* LEAD TEMPERATURE (During Soldering):		
At distances not closer than 1/32 inch to seating surface for 10 seconds maximum	265 max.	°C

* In accordance with Jedec Registration Data Format JS-9 RDF 11-B.

Features

- Low gate leakage current — I_{GSS} = 0.1 pA typ.
- Low feedback capacitance — C_{rss} = 0.25 pF typ.
- High forward transconductance — g_{fs} = 7500 μmho typ.
- High vhf power gain — G_{PS} = 16 dB typ. at 200 MHz
- Low vhf noise figure — NF = 2.5 dB typ. at 200 MHz
- Exceptionally good cross-modulation characteristics

Performance

- Large dynamic range
- Greatly reduced spurious responses
- Permits use of vacuum-tube biasing techniques
- Excellent thermal stability
- Superior cross-modulation performance and greater dynamic range than bipolar transistors

ELECTRICAL CHARACTERISTICS AT $T_A = 25^oC$

Measured with Substrate Connected to Source Unless Otherwise Specified

	CHARACTERISTICS	SYMBOLS	CONDITIONS	LIMITS 3N152 Min.	Typ.	Max.	UNITS
*	Gate Leakage Current	I_{GSS}	$V_{DS} = 0, V_{GS} = -8V, T_A = 25^oC$	-	0.0001	1	nA
			$V_{DS} = 0, V_{GS} = -8\ V, T_A = 125^oC$	-	-	200	nA
*	Zero-Bias Drain Current	I_{DSS}	$V_{DS} = 15\ V, V_{GS} = 0$	5	15	30	mA
	Drain-to-Source Cutoff Current	$I_{D(off)}$	$V_{DS} = 20\ V, V_{GS} = -8V$	-	-	50	μA
*	Gate-to-Source-Cutoff Voltage	$V_{GS(off)}$	$V_{DS} = 15\ V, I_D = 50\ \mu A$	-0.5	-3	-8	V
*	Forward Transconductance	g_{fs}	$V_{DS} = 15\ V, I_D = 5\ mA, f = 1\ kHz$	5000	7500	12,000	μmho
	Drain-to-Source Channel Resistance	$r_{DS(on)}$	$V_{DS} = 0, V_{GS} = 0, f = 1\ kHz$	-	200	-	Ω
*	Small-Signal Short-Circuit Reverse Transfer Capacitance▲	C_{rss}	$V_{DS} = 15\ V, I_D = 5\ mA, f = 0.1\ to\ 1\ MHz$	0.15	0.25	0.35	pF
	Small-Signal Short-Circuit Input Capacitance	C_{iss}	$V_{DS} = 15\ V, I_D = 5\ mA, f = 0.1\ to\ 1\ MHz$	-	5.5	7	pF
	Input Admittance	Y_{is}	Common Source Configuration $f = 200\ MHz$	-	0.4 + J7.3	-	mmho
	Forward Transfer Admittance	Y_{fs}	$V_{DS} = 15\ V,$	-	7-J2	-	mmho
	Output Admittance	Y_{os}	$I_D = 5\ mA$	-	0.28 + J1.8	-	mmho
	Power Gain Maximum Available Gain	MAG	$V_{DS} = 15\ V, I_D = 5\ mA, f = 200\ MHz$	-	21	-	dB
	Insertion Power Gain (Fixed Neutralization) See Fig.1	G_{PS}		14.5	16	-	dB
	Noise Figure (See Figs. 1 & 2)	NF	$V_{DS} = 15\ V, I_D = 5\ mA, f = 200\ MHz$	-	2.5	3.5	dB

▲ Three-Terminal Measurement. Source Returned to Guard Terminal.

* In accordance with JEDEC Registration Data Format JS-9 RDF-11B.

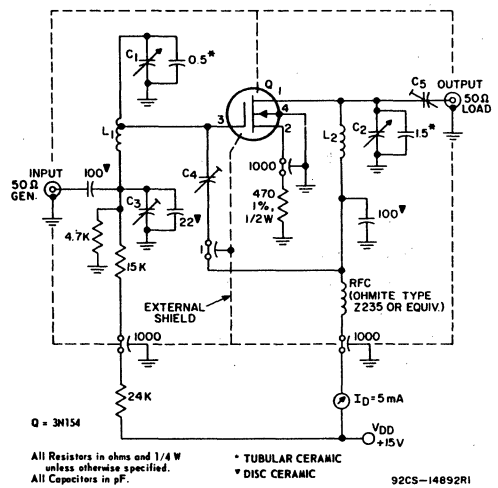


Fig. 1 - Test circuit used to measure 200-MHz maximum usable power gain and noise Figure.

C_1, C_2: 1.5-5 pF variable air capacitor: E. F. Johnson Type 160-102 or equivalent

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent

C_4, C_5: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent

L_1: 5 turns silver-plated 0.02" thick, 0.07"-0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding

L_2: Same as L_1 except winding length approx. 0.7"; no tap

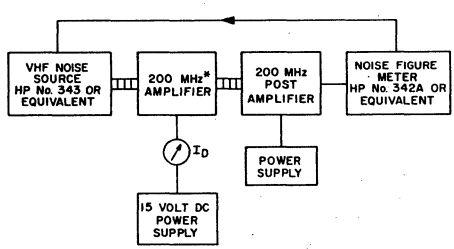


Fig. 2 - Noise figure measurement setup.

TEST SETUP AND TYPICAL CHARACTERISTICS

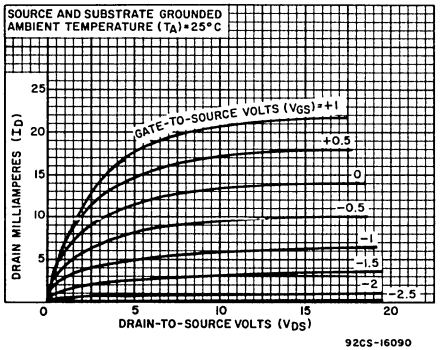


Fig. 3 - Drain Current vs Drain-to-Source Voltage.

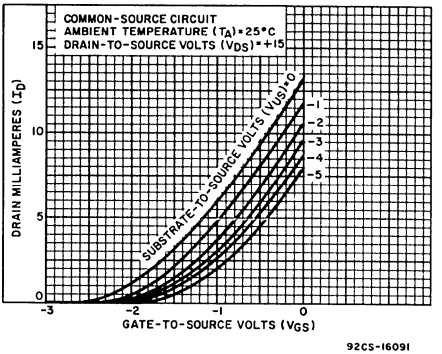


Fig. 4 - Drain Current vs Gate-to-Source Voltage .

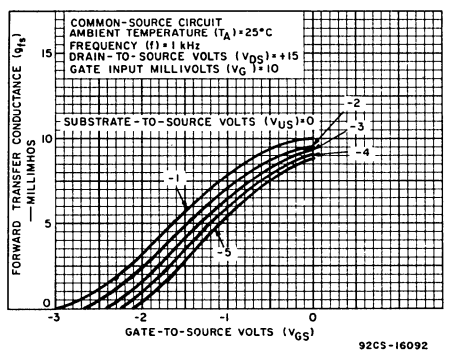


Fig. 5 - Forward Transconductance vs Gate Bias Voltage.

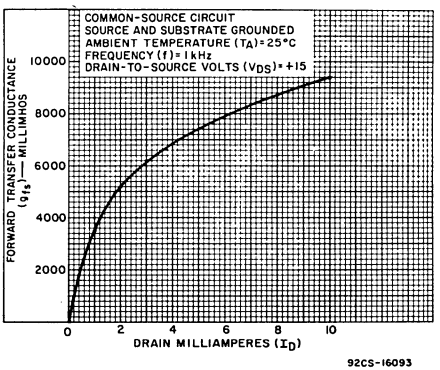


Fig. 6 - Forward Transconductance vs Drain Current.

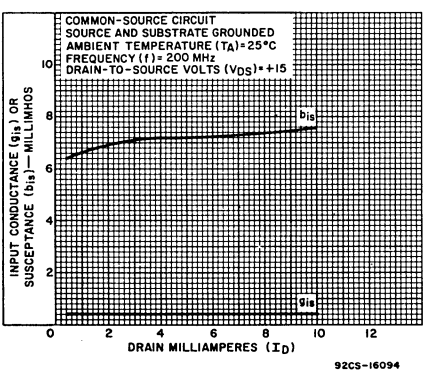


Fig. 7 - Input Admittance vs Drain Current.

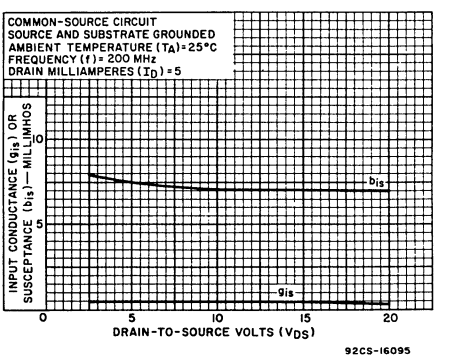


***Fig. 8** - Input Admittance vs Drain-to-Source Voltage.*

TYPICAL 200 MHz COMMON-SOURCE ADMITTANCE (Y) COMPONENTS

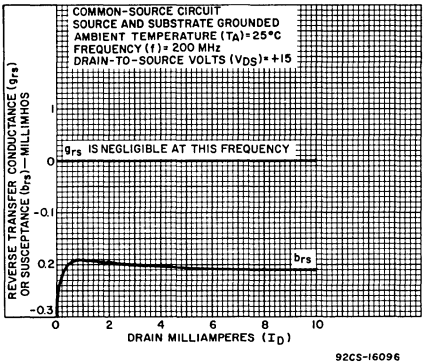


Fig. 9 - Reverse Transadmittance vs Drain Current.

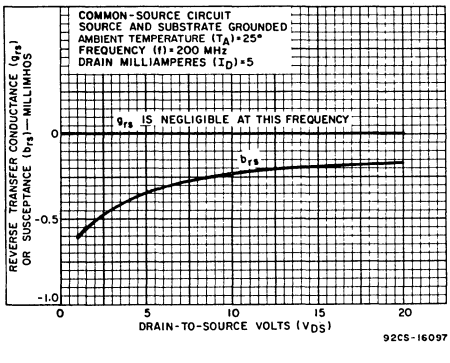


Fig. 10 - Reverse Transadmittance vs Drain-to-Source Voltage.

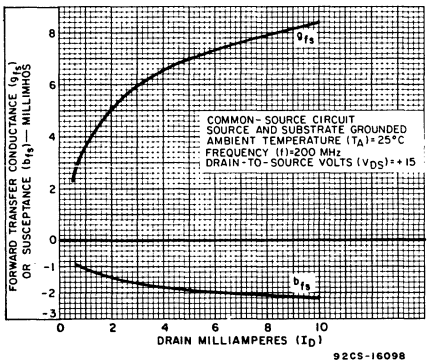


Fig. 11 - Forward Transadmittance vs Drain Current.

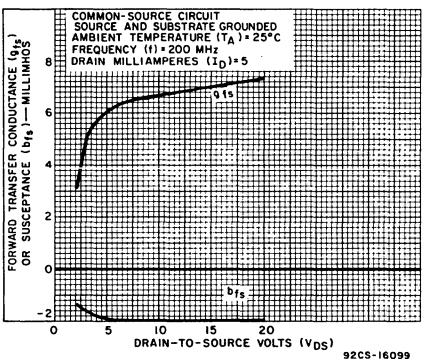


Fig. 12 - Forward Transadmittance vs Drain-to-Source Voltage.

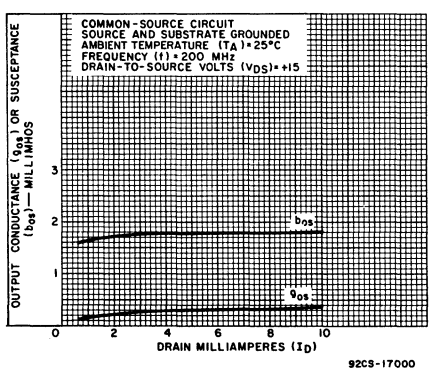


Fig. 13 - Output Admittance vs Drain Current.

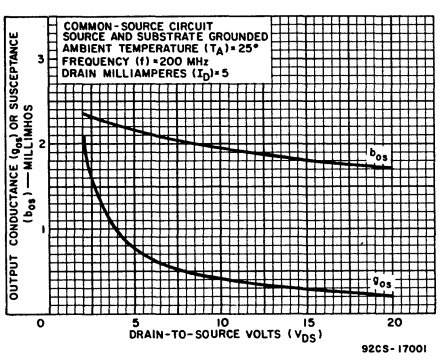


Fig. 14 - Output Admittance vs Drain-to-Source Voltage.

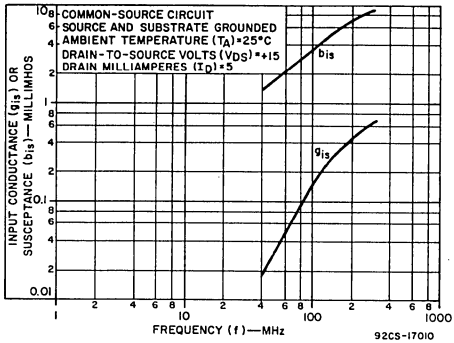


Fig. 15 - Input Admittance vs Frequency.

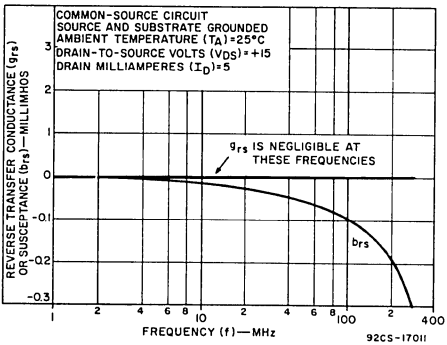


Fig. 16 - Reverse Transadmittance vs Frequency.

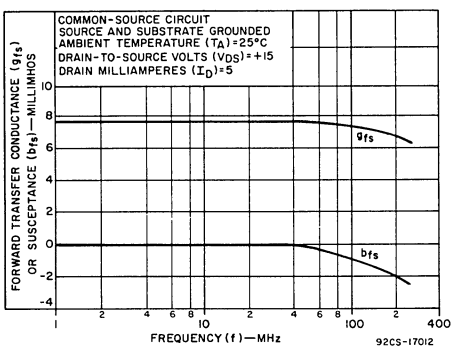


Fig. 17 - Forward Transadmittance vs Frequency.

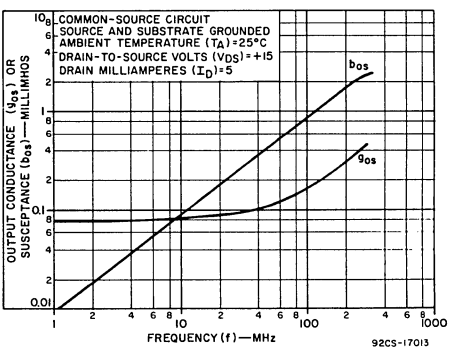


Fig. 18 - Output Admittance vs Frequency.

RCA Solid State Division

MOS Field-Effect Transistors

N-Channel Depletion Type

3N153

RCA 3N153 is a silicon, insulated-gate field-effect transistor of the N-channel depletion type, utilizing the MOS* construction. It is intended primarily for critical chopper and multiplex applications up to 60 MHz.

The insulated gate provides a very high value of input resistance (10^{10} ohms typ) which is relatively insensitive to temperature and is independent of gate-bias conditions (positive, negative, or zero bias). The 3N153 also features extremely low feedback capacitance (0.34 pF typ) and virtually zero inherent offset voltage.

This transistor features a Terminal Arrangement in which the gate and source connections are interchanged to provide maximum isolation between the output (drain) and the input (gate) terminals. Although this new basing configuration does not appreciably change the measured device feedback capacitance, it permits the use of external inter-terminal shields to reduce the feedback due to external capacitances, particularly on printed circuit boards. This feature makes it possible to minimize feedthrough capacitance.

The 3N153 is hermetically sealed in the JEDEC TO-72 package and features a gate metallization that covers the entire source-to-drain channel.

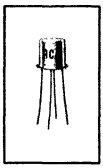
* Metal-Oxide-Semiconductor

Maximum Ratings, *Absolute-Maximum Values:*

(Substrate connected to source unless otherwise specified)

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20	max.	V
DRAIN-TO-SUBSTRATE VOLTAGE, V_{DB}	+20, -0.3	max.	V
SOURCE-TO-SUBSTRATE VOLTAGE, V_{SB}	+20, -0.3	max.	V
DC GATE-TO-SOURCE VOLTAGE, V_{GS}	+6, -8	max.	V
PEAK GATE-TO-SOURCE VOLTAGE, v_{GS}	±14	max.	V
DRAIN CURRENT, I_D (Pulse duration 20 ms, duty factor ≤ 0.10)	50	max.	mA
TRANSISTOR DISSIPATION, P_T:			
At ambient temperatures from -65 to +25°C	400	max.	mW
above 25°C	derate linearly at 2.67 mW/°C		
AMBIENT TEMPERATURE RANGE:			
Storage	-65 to +175		°C
Operating	-65 to +175		°C
LEAD TEMPERATURE (During soldering):			
At distance ≥ 1/32" to seating surface for 10 seconds max.	265	max.	°C

SILICON INSULATED GATE FIELD-EFFECT TRANSISTOR

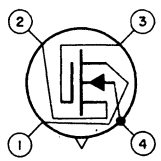


JEDEC TO-72

APPLICATIONS

- Choppers
- Multiplexers
- Servo Amplifiers
- Computer Operational Amplifiers
- Sampling Circuits
- Electrometer Amplifiers

TERMINAL DIAGRAM



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

FEATURES

- excellent thermal stability
- virtually zero inherent offset voltage
- low leakage current: 50 pA max.
- low "on" resistance — $r_{DS(on)}$ = 200 Ω typ.
- high "off" resistance — $R_{DS(off)}$ = 10^{10} Ω typ.
- low feedback capacitance — C_{rss} = 0.34 pF typ.
- low input capacitance — C_{iss} = 6 pF typ.

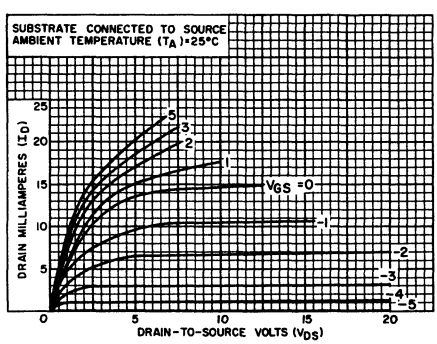


Fig.1 - Drain current vs. drain-to-source voltage.

ELECTRICAL CHARACTERISTICS, at T_A = 25°C, Unless Otherwise Specified. Substrate Connected to Source.

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS Type 3N153			UNITS
			Min.	Typ.	Max.	
Gate-Leakage Current	I_{GSS}	V_{GS}=+6,-8V; V_{DS}=0V; T_A= 25°C V_{GS}=+6,-8V; V_{DS}=0V; T_A=125°C	- -	0.1 -	50 1	pA nA
Static Drain-to-Source "ON" Resistance	$r_{DS(on)}$	V_{GS} = 0V, V_{DS} = 0V	-	200	300	Ω
Drain-to-Source "OFF" Resistance	$R_{DS(off)}$	V_{GS} = -8V, V_{DS} = +1V	10^9	10^{10}	-	Ω
Drain-to-Source Cutoff Current	$I_{D(off)}$	V_{GS}=-8V, V_{DS}=+1V, T_A = 25°C V_{GS}=-8V, V_{DS}=+1V, T_A =125°C	- -	0.1 0.1	1 1	nA μA
Small-Signal, Short-Circuit, Reverse Transfer Capacitance	C_{rss}	V_{GS} =-8V, V_{DS} =0V, f = 1 MHz V_{DS} =15V, I_D = 5 mA, f = 1 MHz	- -	0.34 0.25	0.5 0.38	pF pF
Small-Signal, Short-Circuit, Input Capacitance	C_{iss}	V_{GS} = -8V, V_{DS} = 0V, f = 1 MHz	-	6	8	pF
Small-Signal, Drain-to-Source Capacitance	C_{ds}	V_{DS} = 0V, V_{GS} = -8V, f = 1 MHz	-	-	3	pF
Zero-Gate-Bias Forward Transconductance	g_{fs}	V_{GS} = 0V, V_{DS} = +15V	-	10,000	-	μmho
Offset Voltage	V_0	V_{GS} = +6,-8V; V_{DS} = 0V	-	0*	-	V

* In measurements of Offset Voltage, thermocouple effects and contact potentials in the measurement setup may cause erroneous readings of 1 microvolt or more. These errors may be minimized by the use of solder having a low thermal e.m.f., such as Leeds & Northrup No.107-1.0.1, or equivalent.

OPERATING CONSIDERATIONS

The flexible leads of the 3N153 are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons should be grounded, and appropriate precautions should be taken to protect the device against high electric fields.

This device should not be connected into or disconnected from circuits with the power on because high transient voltages may cause permanent damage to the device.

TYPICAL CHARACTERISTICS

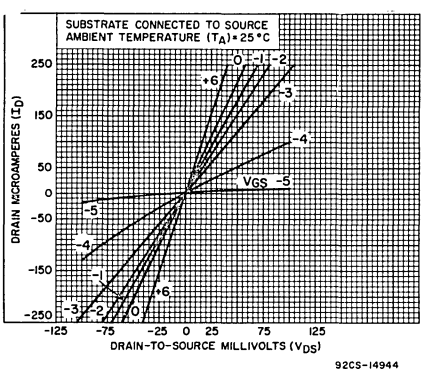


Fig.2 - Low-level drain current vs. drain-to-source voltage.

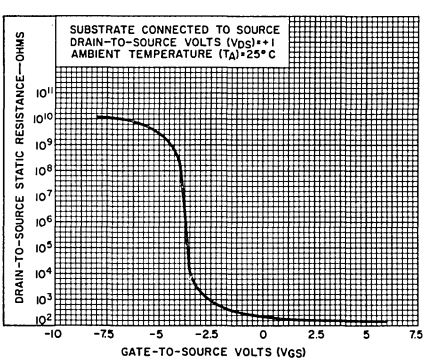


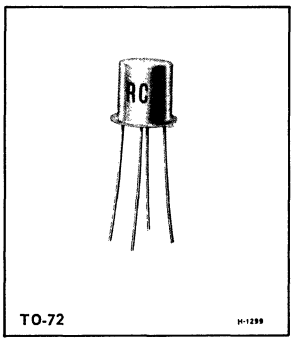
Fig.3 - Drain-to-source static resistance vs. gate-to-source voltage.

RCA Solid State Division

MOS Field-Effect Transistor

N-Channel Depletion Type

3N154



Silicon MOS Transistor

For Critical Amplifier Applications in Military & Industrial VHF Communications Equipment Operating up to 250 MHz

Device Feature:

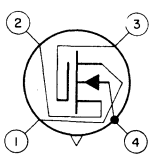
- **Closely controlled I_{DSS} — 10 to 25 mA**
- **Low gate leakage current — I_{GSS} = 0.1 pA typ.**
- **Low feedback capacitance — C_{rss} = 0.25 pF typ.**
- **High forward transconductance — g_{fs} = 7500 μmho typ.**
- **High vhf power gain — G_{PS} = 16 dB typ. at 200 MHz**
- **Low vhf noise figure — NF = 3.5 dB typ. at 200 MHz**
- **Exceptionally good cross-modulation characteristics**

RCA 3N154 is an n-channel depletion-type silicon insulated-gate field-effect transistor utilizing the MOS■ construction. It is intended primarily for vhf amplifier applications up to 250 MHz in military and industrial equipment.

Because of its improved transfer characteristic and exceptionally wide dynamic range, the 3N154 can provide substantially better crossmodulation performance in linear amplifier applications than conventional bipolar transistors. The extremely low gate leakage current eliminates diode-current loading of the input circuit under strong signal conditions, a problem which is common to junction-type FET's. These features, in addition to low feedback capacitance, permit the design of circuits providing superior high-frequency operation and high gain without neutralization. The 3N154 utilizes full-gate construction and is hermetically sealed in a JEDEC TO-72 metal package.

■ Metal-Oxide-Semiconductor

TERMINAL DIAGRAM



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

Performance Features

- **Large dynamic range**
- **Greatly reduced spurious responses**
- **Permits use of vacuum-tube biasing techniques**
- **Excellent thermal stability**
- **Superior cross-modulation performance and greater dynamic range than bipolar transistors**

Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

*DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20	V
*DRAIN-TO-GATE VOLTAGE, V_{DG}	+20	V
*GATE-TO-SOURCE VOLTAGE, V_{GS}:		
* CONTINUOUS (dc)	+1, -8	V
* PEAK ac	±15	V
*DRAIN CURRENT, I_D▲	50	mA
*TRANSISTOR DISSIPATION:		
At ambient temperatures up to 25°C	330	mW
At ambient temperatures above 25°C	derate at 2.2	mW/°C
*AMBIENT TEMPERATURE RANGE:		
Storage	-65 to +175	°C
Operating	-65 to +175	°C
LEAD TEMPERATURE (During Soldering):		
At distances not closer than 1/32 inch to seating surface for 10 seconds maximum	265	°C

In accordance with JEDEC Registration Data Format JS9-RDF-11B

▲ Pulsed:
Pulse duration ≤ 20 ms
Duty factor ≤ 0.15

ELECTRICAL CHARACTERISTICS: (At T_A = 25° C)

Measured with Substrate Connected to Source Unless Otherwise Specified.

	CHARACTERISTICS	SYMBOLS	CONDITIONS	LIMITS 3N154 Min.	Typ.	Max.	UNITS
*	Gate Leakage Current	I_{GSS}	V_{DS} = 0, V_{GS} = -8 V, T_A = 25° C	-	0.0001	0.05	nA
			V_{DS} = 0, V_{GS} = -8 V, T_A = 125° C	-	-	5	nA
			V_{DS} = 0, V_{GS} = +1, T_A = 25° C	-	0.0001	0.05	nA
			V_{DS} = 0, V_{GS} = +1, T_A = 125° C	-	-	5	nA
*	Zero-Bias Drain Current	I_{DSS}	V_{DS} = 15 V, V_{GS} = 0	10	15	25	mA
	Drain-to-Source Cutoff Current	$I_{D(off)}$	V_{DS} = 20 V, V_{GS} = -8 V	-	-	50	μA
*	Gate-to-Source Cutoff Voltage	$V_{GS(off)}$	V_{DS} = 15 V, I_D = 50 μA	-0.5	-3	-8	V
	Forward Transconductance	g_{fs}	V_{DS} = 15 V, I_D = 5 mA, f = 1 kHz	5000	7500	12,000	μmho
	Drain-to-Source Channel Resistance	$r_{DS(on)}$	V_{DS} = 0, V_{GS} = 0, f = 1 kHz	-	200	-	Ω
*	Small-Signal Short-Circuit Reverse Transfer Capacitance	C_{rss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	0.15	0.25	0.35	pF
	Small-Signal Short-Circuit Input Capacitance ▲	C_{iss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	-	5.5	7	pF
	Input Admittance	Y_{is}	Common Source Configuration f = 200 MHz, V_{DS} = 15 V, I_D = 5 mA	-	0.4 + J7.3	-	mmho
	Forward Transfer Admittance	Y_{fs}		-	7 − J2	-	mmho
	Output Admittance	Y_{os}		-	0.28 + J1.8	-	mmho
	Maximum Available Power Gain	MAG	V_{DS} = 15 V, I_D = 5 mA, f = 200 MHz	-	21	-	dB
*	Insertion Power Gain (Fixed Neutralization) (see Fig. 1)	G_{PS}		13.5	16	-	dB
*	Noise Figure (see Figs. 1 & 2)	NF	V_{DS} = 15 V, I_D = 5 mA, f = 200 MHz	-	3.5	5	dB

* In Accordance with JEDEC Registration Data Format JS-9 RDF-11B

▲ Three-Terminal Measurement: Source Returned to Guard Terminal

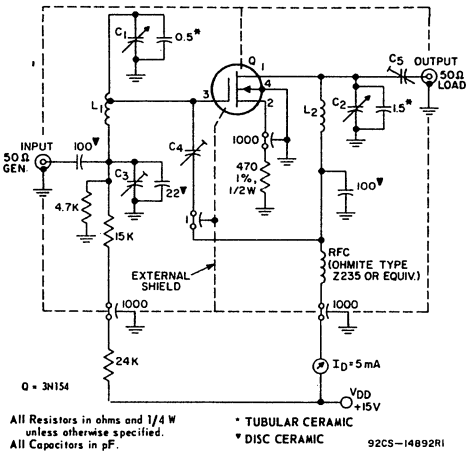


Fig. 1 - Test circuit used to measure 200-MHz maximum usable power gain and noise figure

C_1, C_2: 1.5-5 pF variable air capacitor: E. F. Johnson Type 160-102 or equivalent

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent

C_4, C_5: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent

Q = 3N154

L_1: 5 turns silver-plated 0.02" thick, 0.07"-0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding

L_2: Same as L_1 except winding length approx. 0.7"; no tap.

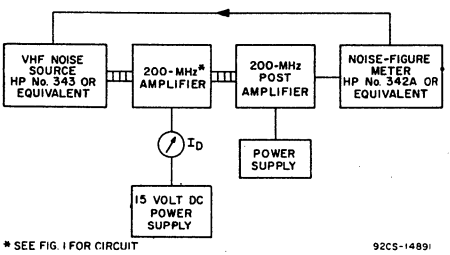


Fig. 2 - Noise figure measurement setup

TYPICAL CHARACTERISTICS

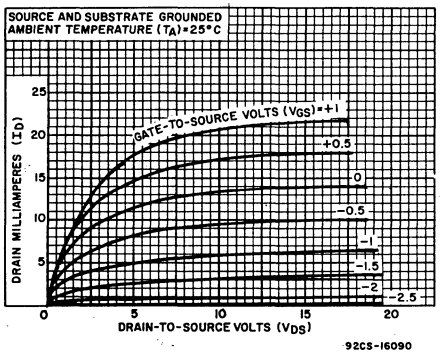


Fig. 3 - Drain current vs drain-to-source voltage

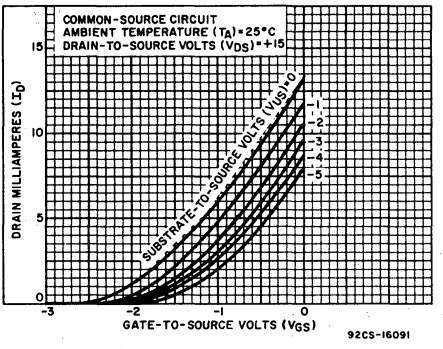


Fig. 4 - Drain current vs gate-to-source voltage

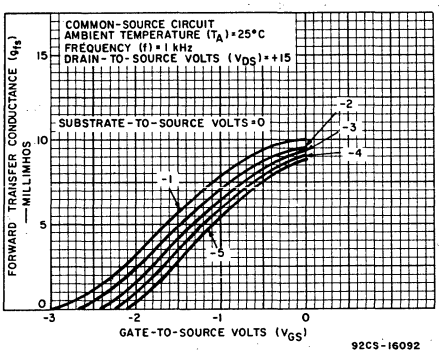


Fig. 5 - Forward transconductance vs gate-to-source voltage

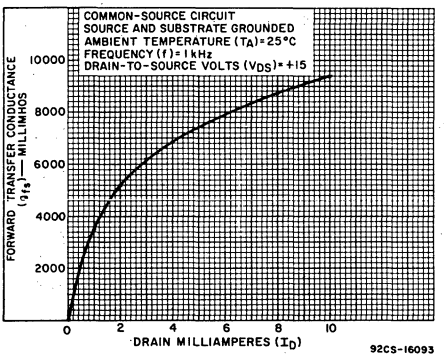


Fig. 6 - Forward transconductance vs drain current

TYPICAL 200 MHz COMMON-SOURCE ADMITTANCE (Y) COMPONENTS

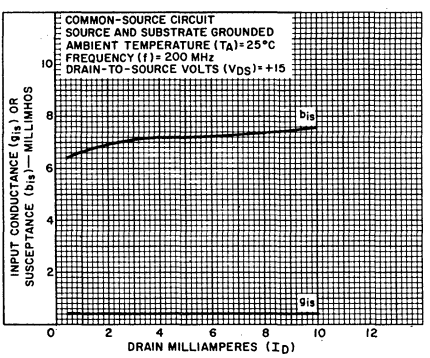


Fig. 7 - Input admittance vs drain current

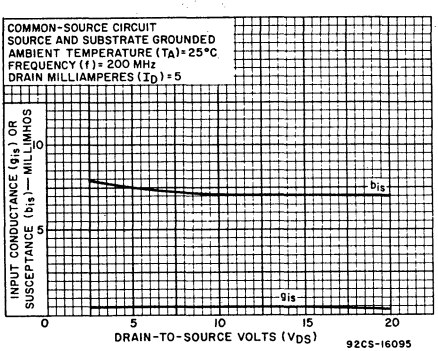


Fig. 8 - Input admittance vs drain-to-source voltage

TYPICAL 200 MHz COMMON-SOURCE ADMITTANCE (Y) COMPONENTS

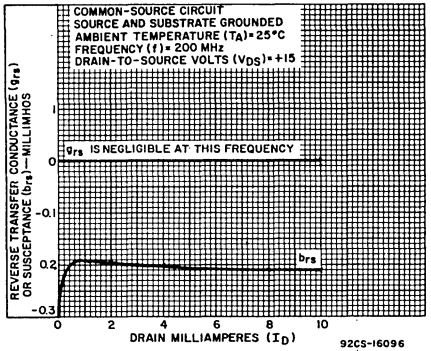


Fig. 9 - Reverse transadmittance vs drain current

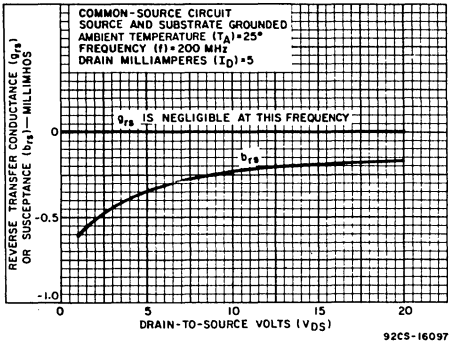


Fig. 10 - Reverse transadmittance vs drain-to-source voltage

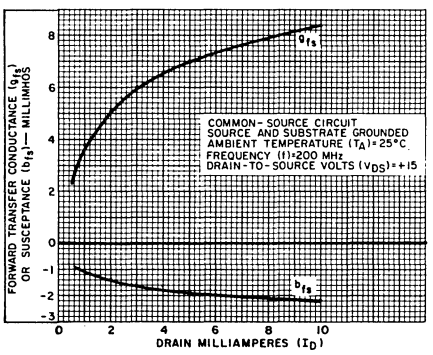


Fig. 11 - Forward transadmittance vs drain current

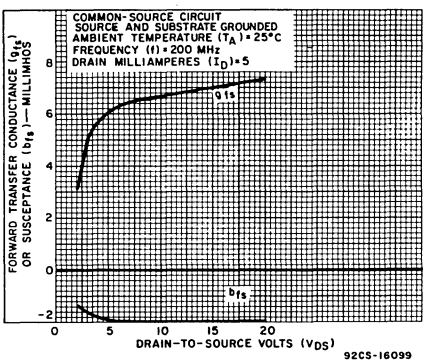


Fig. 12 - Forward transadmittance vs drain-to-source- voltage

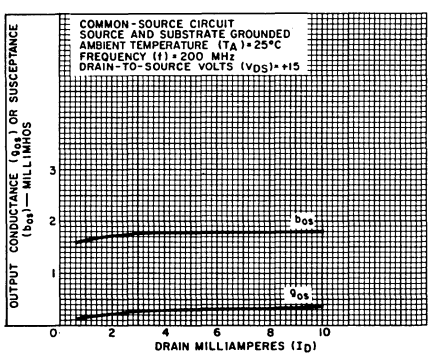


Fig. 13 - Output admittance vs drain current

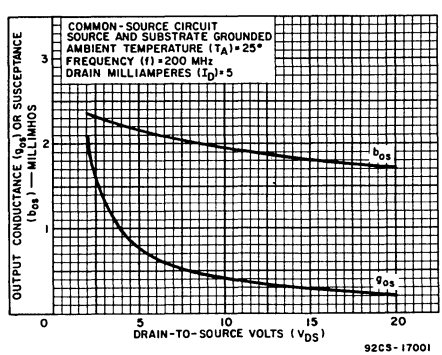


Fig. 14 - Output admittance vs drain-to-source voltage

RCA Solid State Division

MOS Field-Effect Transistors

N-Channel Depletion Type

3N159

The 3N159 is an n-channel silicon, depletion type, dual insulated-gate, field-effect transistor utilizing the MOS** construction. It has exceptional characteristics for rf-amplifier applications at frequencies up to 300 MHz. This transistor features a series arrangement of two separate channels, each channel having an independent control gate.

Type 3N159 has an exceptionally low-noise figure, which makes this type particularly suitable for critical vhf applications. When used in a common-source configuration in which gate No.2 is ac grounded, this device reduces oscillator feedthrough to the antenna thereby minimizing oscillator radiation.

The 3N159 is hermetically sealed in the metal JEDEC TO-72 package.

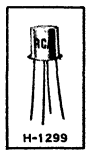
** Metal-Oxide-Semiconductor.

Maximum Ratings, *Absolute-Maximum Values:*
at T_A = 25°C

DRAIN-TO-SOURCE VOLTAGE, V_{DS} . . . . . 0 to +20 V
GATE-No.1-TO-SOURCE VOLTAGE, V_{G1S}:
Continuous (dc) . . . . . -8 to +1 V
Peak ac . . . . . -8 to +20 V
GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:
Continuous (dc) . . . . . -8 to 40% of V_{DS} V
Peak ac . . . . . -8 to +20 V
DRAIN-TO-GATE VOLTAGE:
V_{DG1} or V_{DG2} . . . . . +20 V
DRAIN CURRENT, I_D
Pulsed: Pulse duration ≤ 20 ms,
duty factor ≤ 0.15 . . . . . 50 mA
TRANSISTOR DISSIPATION, P_T:
At ambient temperatures up to 25°C . . . . . 400 mW
above 25°C . . . . . derate linearly at 2.67 mW/°C
AMBIENT TEMPERATURE RANGE:
Storage and Operating . . . . . -65 to +175 °C
LEAD TEMPERATURE (During soldering):
At distances ≥ 1/32 inch from seating surface for 10 seconds max. . . . . 265 °C

SILICON DUAL INSULATED-GATE FIELD-EFFECT TRANSISTOR N-Channel Depletion Type

For Military and Industrial Low-Noise RF-Amplifier Applications Up to 300 MHz



TO-72

APPLICATIONS

- RF amplifier in military and industrial communications equipment
- aircraft, marine and vehicular receivers
- CATV and MATV equipment
- telemetry and multiplex equipment

PERFORMANCE FEATURES

- wide dynamic range permits large-signal handling before overload
- dual-gate permits simplified agc circuitry
- virtually no agc power required
- greatly reduces spurious responses in FM receivers
- permits use of vacuum-tube biasing techniques
- excellent thermal stability
- superior cross-modulation performance and greater dynamic range than bipolar or single-gate field-effect transistors

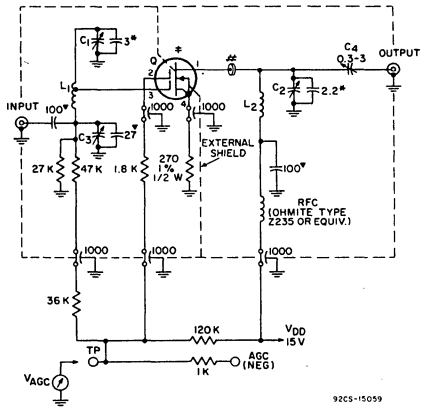
DEVICE FEATURES

- low gate leakage currents — — I_{G1SS} & I_{G2SS} = 1 nA max.
- high forward transconductance — — g_{fs} = 7000 µmho min.
- high unneutralized RF power gain — — G_{ps} = 16 dB min. at 200 MHz
- low vhf noise figure — — NF = 3.5 dB max. at 200 MHz

ELECTRICAL CHARACTERISTICS, at T_A = 25°C unless otherwise specified

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS 3N159 Min.	LIMITS 3N159 Typ.	LIMITS 3N159 Max.	UNITS
Gate-No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +16V, I_D = 200 μA V_{G2S} = +4V	-	-2	-4	V
Gate-No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +16V, I_D = 200 μA V_{G1S} = 0	-	-2	-4	V
Gate-No.1-Leakage Current	I_{G1SS}	V_{G1S} = -20V, V_{G2S} = 0 V_{DS} = 0, T_A = 25°C	-	-	1	nA
		V_{G1S} = +1V, V_{G2S} = 0 V_{DS} = 0, T_A = 25°C	-	-	1	nA
		V_{G1S} = -20V, V_{G2S} = 0 V_{DS} = 0, T_A = 125°C	-	-	0.2	μA
Gate-No.2-Leakage Current	I_{G2SS}	V_{G2S} = -20V, V_{G1S} = 0 V_{DS} = 0, T_A = 25°C	-	-	1	nA
		V_{G2S} = +1, V_{DS} = 0 V_{G1S} = 0, T_A = 25°C	-	-	1	nA
		V_{G2S} = -20V, V_{G1S} = 0 V_{DS} = 0, T_A = 125°C	-	-	0.2	μA
Zero-Bias Drain Current	I_{DSS}*	V_{DD} = +14V, V_{G1S} = 0 V_{G2S} = +4V	5	18	30	mA
Forward Transconductance (Gate-No.1-to-Drain)	g_{fs}	V_{DD} = +14V, I_D = 10 mA V_{G2S} = +4V, f = 1 kHz	7000	10,000	18,000	μmho
Cutoff Forward Transconductance (Gate-No.1-to-Drain)	$g_{fs(off)}$	V_{DD} = +14V, V_{G1S} = -0.5V V_{G2S} = -2V, f = 1 kHz	-	-	100	μmho
Small-Signal, Short-Circuit Input Capacitance▲	C_{iss}	V_{DS} = +13V, I_D = 10 mA V_{G2S} = +4V, f = 1 MHz	3	5.5	7	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate No.1)♦	C_{rss}	V_{DS} = +13V, I_D = 10 mA V_{G2S} = +4V, f = 1 MHz	0.01	0.02	0.03	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}	V_{DS} = +13V, I_D = 10 mA V_{G2S} = +4V, f = 1 MHz	-	2.2	-	pF
Maximum Usable Power Gain (See Fig.1 for Measurement Circuit)	MUG	V_{DD} = +15V, R_S = 270Ω R_G = 50Ω, f = 200 MHz	16	18	22	dB
Measured Noise Figure (See Fig.1 for Measurement Circuit)	NF	V_{DD} = +15V, R_S = 270Ω f = 200 MHz, R_G = 50Ω	-	2.5	3.5	dB

* Pulse Test: Pulse duration ≤ 20 ms, duty factor ≤ 0.15.
▲ Capacitance between Gate No.1 and all other terminals.
♦ Three-Terminal Measurement with Gate No.2 and Source Returned to Guard Terminal.



* Tubular ceramic

▼ Disc ceramic

\# Ferrite bead (1/2 used); Indiana General No. H 1742C-(A-147) or F1157-1-H or equivalent.

‡ VHF plug in socket Jettron CD72-148 and CD72149 (part No.7977-1) or equivalent.

C_1, C_2: 1.5-5 pF variable air capacitor: E. F. Johnson Type 160-102 or equivalent.

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent.

C_4: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent.

L_1: 5 turns silver-plated 0.02" thick, 0.07"-0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding.

L_2: Same as L_1 except winding length approx. 0.7"; no tap.

Fig.1 - 200-MHz power gain and noise-figure test circuit for type 3N159.

TYPICAL CHARACTERISTICS

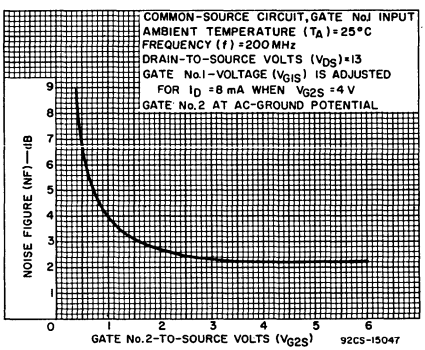


Fig.2 - Noise figure vs gate No.2-to-source voltage.

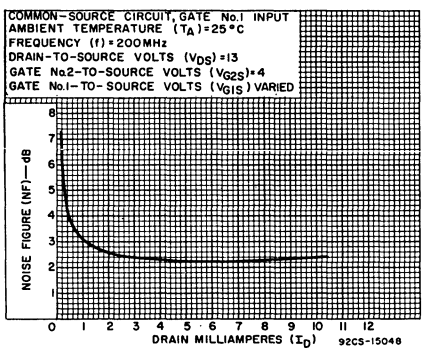


Fig.3 - Noise figure vs drain current.

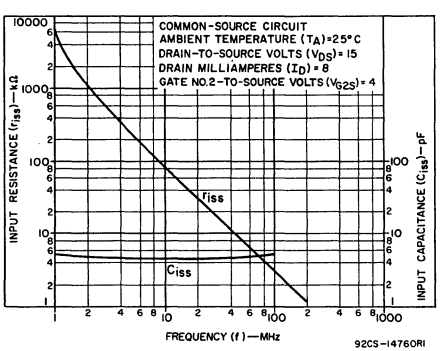


Fig.4 - Input resistance and capacitance vs frequency.

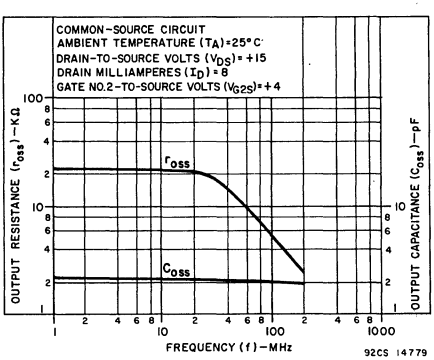


Fig.5 - Output resistance and capacitance vs frequency.

TYPICAL SMALL-SIGNAL y PARAMETERS at 200 MHz

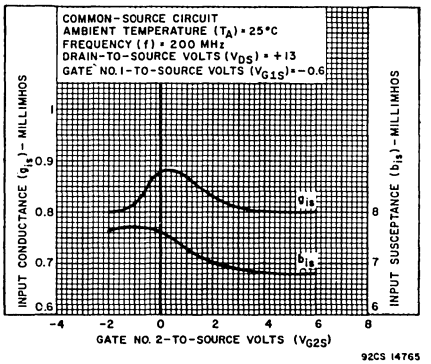


Fig.6 - Input conductance and susceptance vs gate No.2-to-source voltage.

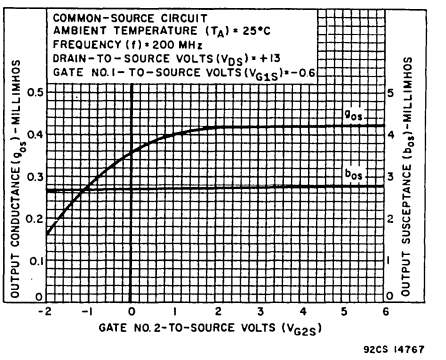


Fig.7 - Output conductance and susceptance vs gate No.2-to-source voltage.

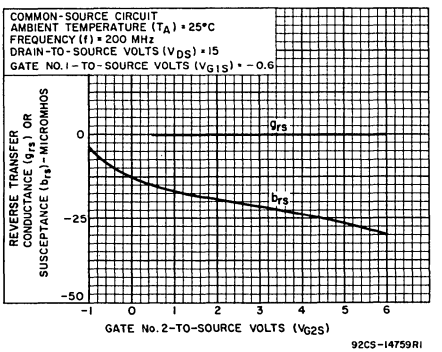


Fig.8 - Reverse transfer conductance or susceptance vs gate No.2-to-source voltage.

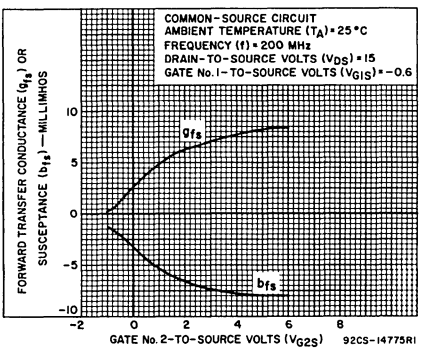


Fig.9 - Forward transfer conductance or susceptance vs gate No.2-to-source voltage.

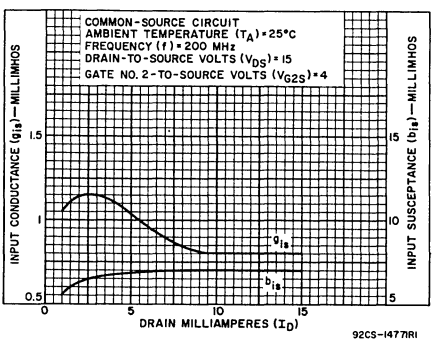


Fig.10 - Input conductance and susceptance vs drain milliamperes.

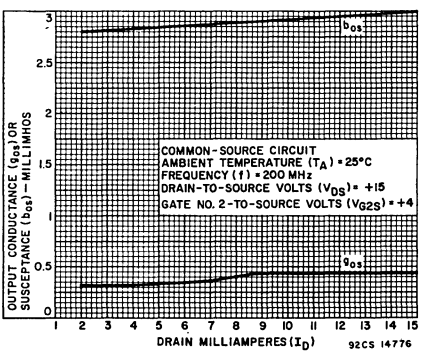


Fig.11 - Output conductance and susceptance vs drain milliamperes.

TYPICAL SMALL-SIGNAL y PARAMETERS at 200 MHz

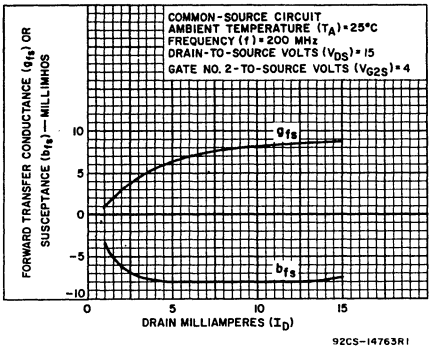


Fig.12 - Forward transfer conductance and susceptance vs drain current.

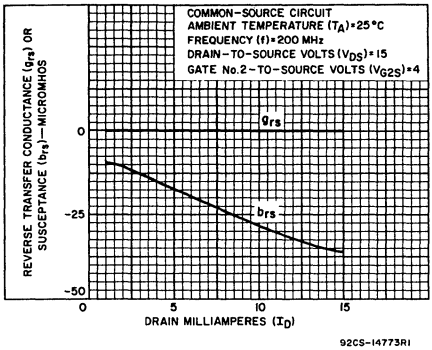


Fig.13 - Reverse transfer conductance and susceptance vs drain current.

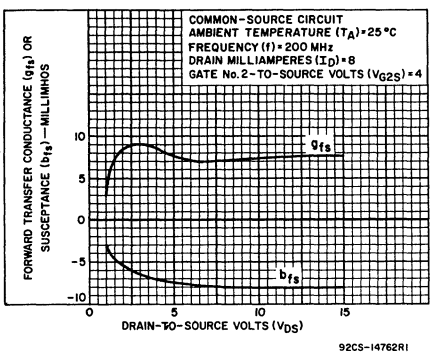


Fig.14 - Forward transfer conductance and susceptance vs drain-to-source voltage.

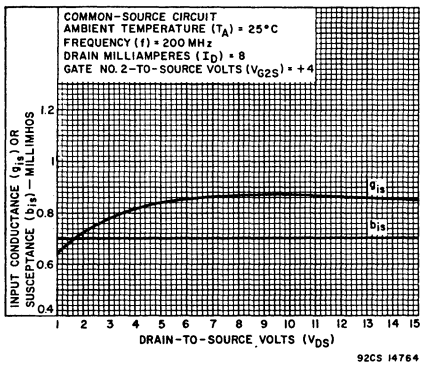


Fig.15 - Input conductance and susceptance vs drain-to-source voltage.

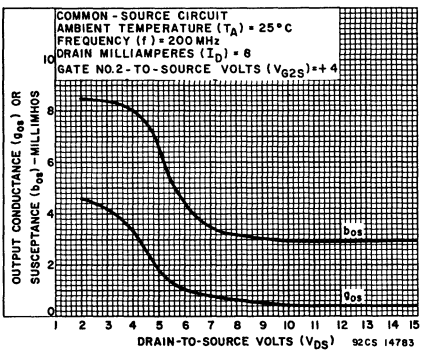


Fig.16 - Output conductance and susceptance vs drain-to-source voltage.

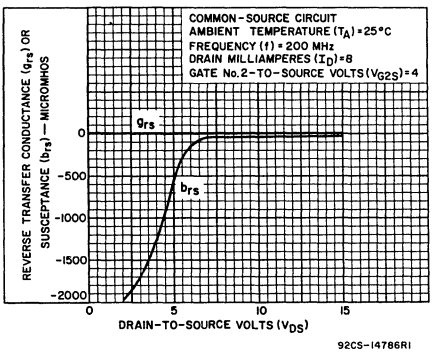


Fig.17 - Reverse transfer conductance and susceptance vs drain-to-source voltage.

TYPICAL CHARACTERISTICS

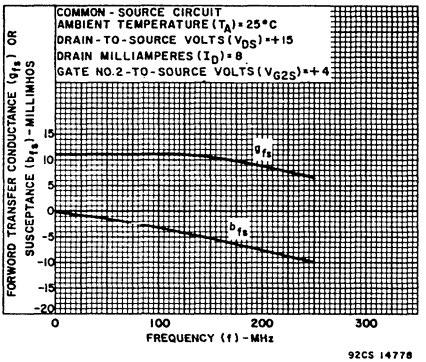


Fig.18 - Forward transfer conductance and susceptance vs frequency.

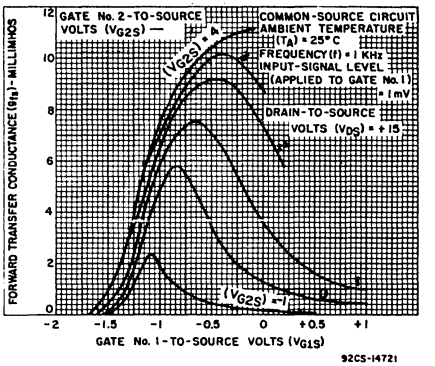


Fig.19 - Forward transfer conductance vs gate No.1-to-source voltage.

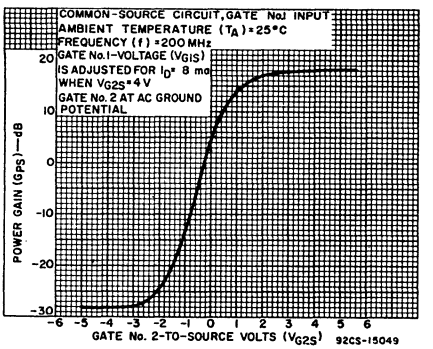
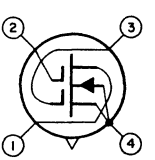


Fig.20 - Power gain vs gate No.2-to-source voltage.

TERMINAL DIAGRAM



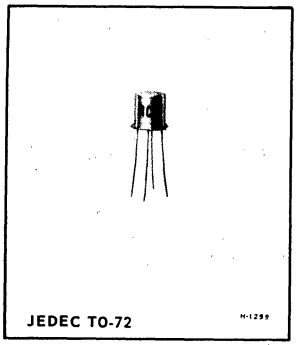
LEAD 1 - DRAIN
LEAD 2 - GATE No.2
LEAD 3 - GATE No.1
LEAD 4 - SOURCE, SUBSTRATE AND CASE

RCA Solid State Division

MOS Field-Effect Transistors

N-Channel Depletion Types

3N187



Silicon Dual Insulated-Gate Field-Effect Transistor

With Integrated Gate-Protection Circuits

For Military and Industrial Applications up to 300 MHz

Device Features

- Back-to-back diodes protect each gate against handling and in-circuit transients
- High forward transconductance — g_{fs} = 12,000 μmho (typ.)
- High unneutralized RF power gain — G_{ps} = 18 dB(typ.) at 200 MHz
- Low VHF noise figure — 3.5 dB(typ.) at 200 MHz

RCA-3N187 is an n-channel silicon, depletion type, dual insulated-gate field-effect transistor.

Special back-to-back diodes are diffused directly into the MOS▲ pellet and are electrically connected between each insulated gate and the FET's source. The diodes effectively bypass any voltage transients which exceed approximately ±10 volts. This protects the gates against damage in all normal handling and usage.

A feature of the back-to-back diode configuration is that it allows the 3N187 to retain the wide input signal dynamic range inherent in the MOSFET. In addition, the junction capacitance of these diodes adds little to the total capacitance shunting the signal gate.

The excellent overall performance characteristics of the RCA-3N187 make it useful for a wide variety of rf-amplifier applications at frequencies up to 300 MHz. The two serially-connected channels with independent control gates make possible a greater dynamic range and lower cross-modulation than is normally achieved using devices having only a single control element.

The two-gate arrangement of the 3N187 also makes possible a desirable reduction in feedback capacitance by operating in the common-source configuration and ac-grounding Gate No. 2. The reduced capacitance allows operation at maximum gain *without neutralization;* and, of special importance in rf-amplifiers, it reduces local oscillator feedthrough to the antenna.

The 3N187 is hermetically sealed in the metal JEDEC TO-72 package.

▲ Metal-Oxide-Semiconductor

Applications

- RF amplifier, mixer, and IF amplifier in military, and industrial communications equipment
- Aircraft and marine vehicular receivers
- CATV and MATV equipment
- Telemetry and multiplex equipment

Performance Features

- Superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET's
- Wide dynamic range permits large-signal handling before overload
- Virtually no agc power required
- Greatly reduces spurious responses in FM receivers

Maximum Ratings,

Absolute-Maximum Values, at T_A = 25°C

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	-0.2 to +20	V
GATE No. 1-TO-SOURCE VOLTAGE, V_{G1S}:		
Continuous (dc)	-6 to +3	V
Peak ac	-6 to +6	V
GATE No. 2-TO-SOURCE VOLTAGE, V_{G2S}:		
Continuous (dc)	-6 to 30% of V_{DS}	V
Peak ac	-6 to +6	V
*DRAIN-TO-GATE VOLTAGE, V_{DG1} OR V_{DG2}	+20	V
*DRAIN CURRENT, I_D	50	mA
*TRANSISTOR DISSIPATION P_T:		
At ambient temperatures up to 25°C	330	mW
At ambient temperatures above 25°C	derate linearly at 2.2 mW/°C	
*AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
*LEAD TEMPERATURE (During Soldering):		
At distances ≥ 1/32 inch from seating surface for 10 seconds max.	265	°C

*In accordance with JEDEC Registration Data Format JS-9 RDF-19A

ELECTRICAL CHARACTERISTICS, at T_A = 25° C unless otherwise specified

	CHARACTERISTICS	SYMBOL	TEST CONDITIONS		LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
*	Gate No. 1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +15 V, I_D = 50 μA V_{G2S} = +4 V		-0.5	-2	-4	V
*	Gate No. 2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +15 V, I_D = 50 μA V_{G1S} = 0		-0.5	-2	-4	V
*	Gate No. 1-Terminal Forward Current	I_{G1SSF}	V_{G1S} = +1 V $V_{G2S} = V_{DS}$ = 0	T_A = 25° C T_A = 100° C	– –	– –	50 5	nA μA
*	Gate No. 1-Terminal Reverse Current	I_{G1SSR}	V_{G1S} = -6 V $V_{G2S} = V_{DS}$=0	T_A = 25° C T_A = 100° C	– –	– –	50 5	nA μA
*	Gate No. 2-Terminal Forward Current	I_{G2SSF}	V_{G2S} = +6 V $V_{G1S} = V_{DS}$=0	T_A = 25° C T_A = 100° C	– –	– –	50 5	nA μA
*	Gate No. 2-Terminal Reverse Current	I_{G2SSR}	V_{G2S} = -6 V $V_{G1S} = V_{DS}$=0	T_A = 25° C T_A = 100° C	– –	– –	50 5	nA μA
*	Zero-Bias Drain Current	I_{DS}	V_{DS} = +15 V V_{G2S} = +4 V V_{G1S} = 0		5	15	30	mA
	Forward Transconductance (Gate No. 1-to-Drain)	g_{fs}	V_{DS} = +15 V, I_D = 10 mA V_{G2S} = +4 V, f = 1 kHz		7000	12,000	18,000	μmho
*	Small-Signal, Short-Circuit Input Capacitance†	C_{iss}	V_{DS} = +15 V, I_D = 10 mA V_{G2S} = +4 V, f = 1 MHz		4.0	6.0	8.5	pF
*	Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate No. 1)♦	C_{rss}			0.005	0.02	0.03	pF
*	Small-Signal, Short-Circuit Output Capacitanc	C_{oss}			–	2.0	–	pF
	Power Gain (see Fig. 1)	G_{PS}	V_{DS} = +15 V, I_D = 10 mA V_{G2S} = +4 V, f = 200 MHz		16	18	22	dB
	Maximum Available Power Gain	MAG			–	20	–	dB
	Maximum Usable Power Gain (unneutralized)	MUG			–	20▲	–	dB
	Noise Figure (see Fig. 1)	NF			–	3.5	4.5	dB
*	Magnitude of Forward Transadmittance	$\|Y_{fs}\|$			–	12,000	–	μmho
*	Phase Angle of Forward Transadmittance	θ			–	-35	–	Degrees
	Magnitude of Reverse Transadmittance	$\|Y_{rs}\|$			–	25	–	μmho
	Angle of Reverse Transadmittance	θ_{rs}			–	-25	–	Degrees
*	Input Resistance	r_{iss}			–	1.0	–	kΩ
*	Output Resistance	r_{oss}			–	2.8	–	kΩ
*	Gate-to-Source Forward Breakdown Voltage: Gate No. 1 Gate No. 2	 $V_{(BR)G1SSF}$ $V_{(BR)G2SSF}$	$I_{G1SSF} = I_{G2SSF}$ = 100 μA		6.5	10	–	V
*	Gate-to-Source Reverse Breakdown Voltage: Gate No. 1 Gate No. 2	 $V_{(BR)G1SSR}$ $V_{(BR)G2SSR}$	$I_{G1SSR} = I_{G2SSR}$ = -100 μA		-6.5	-10	–	V

▲Limited only by practical design considerations.

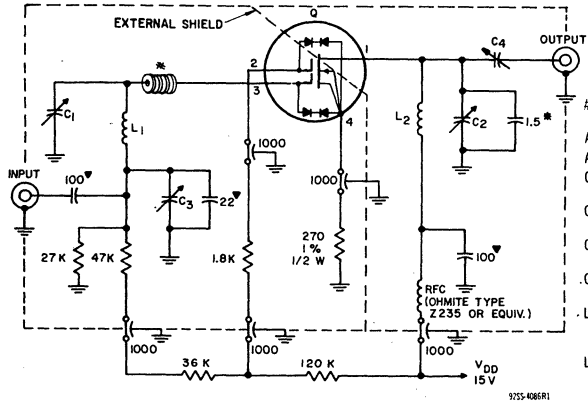
† Capacitance between Gate No. 1 and all other terminals

♦ Three-terminal measurement with Gate No. 2 and Source returned to ground terminal.

* In accordance with JEDEC Registration Data Format JS-9 RDF-19A

OPERATING CONSIDERATIONS

The flexible leads of the 3N187 are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons MUST be grounded.



Ferrite bead (4); Pyroferric Co. "Carbonyl J" 0.09 in. OD; 0.03 in. ID; 0.063 in. thickness.

All resistors in ohms

All capacitors in pF

Q = 3N187

▼ Disc ceramic.

* Tubular ceramic.

C_1: 1.8–8.7 pF variable air capacitor: E.F. Johnson Type 160-104, or equivalent.

C_2: 1.5–5 pF variable air capacitor: E.F. Johnson Type 160-102, or equivalent.

C_3: 1–10 pF piston-type variable air capacitor: JFD Type VAM-010; Johanson Type 4335, or equivalent.

C_4: 0.8–4.5 pF piston type variable air capacitor: Erie 560-013 or equivalent.

L_1: 4 turns silver-plated 0.02-in. thick, 0.075-0.085-in. wide, copper ribbon. Internal diameter of winding = 0.25 in, winding length approx. 0.08 in.

L_2: 4½ turns silver-plated 0.02-in thick, 0.085-0.095-in. wide, 5/16-in. ID. Coil ≈.90 in. long.

Fig. 1 - 200-MHz Power gain and noise-figure test circuit

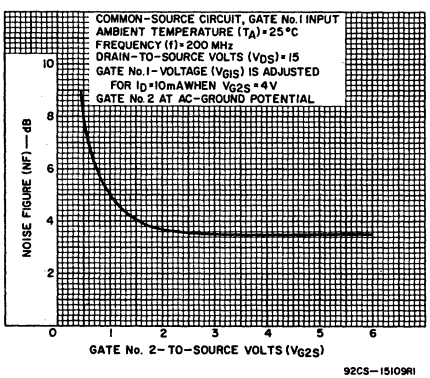


Fig. 2 - NF vs. V_{G2S}

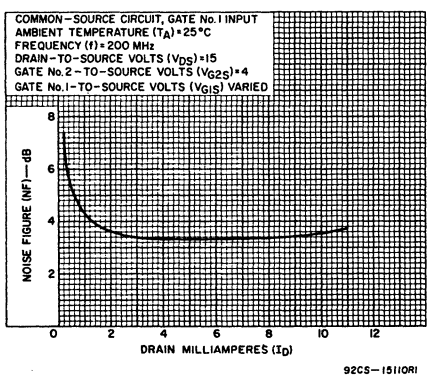


Fig. 3 - NF vs. I_D

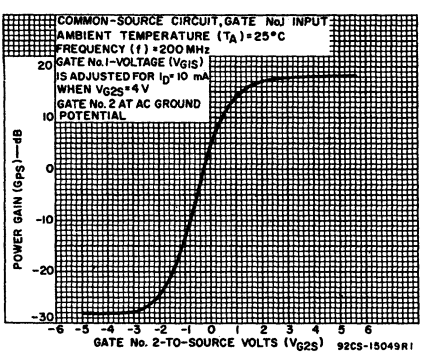


Fig. 4 - G_{PS} vs. V_{G2S}

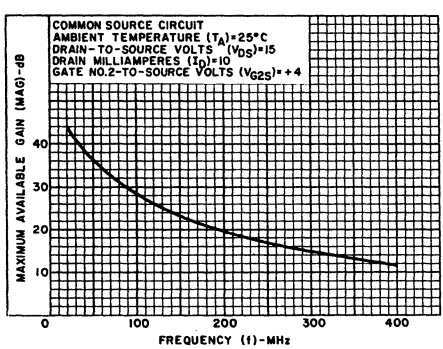


Fig. 5 - MAG. vs. f

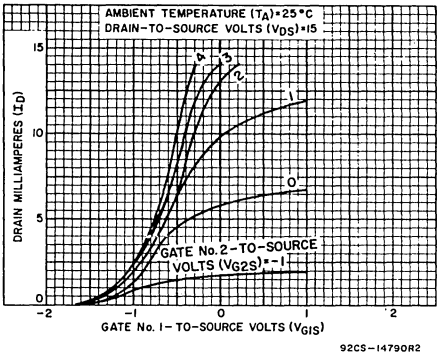


Fig. 6 - I_D vs. V_{G1S}

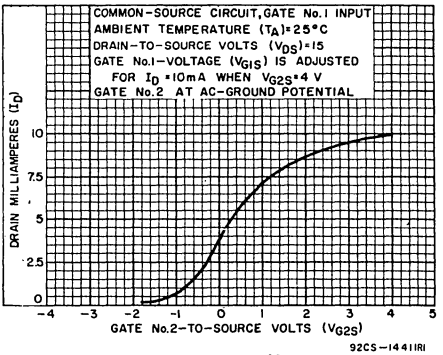


Fig. 7 - I_D vs. V_{G2S}

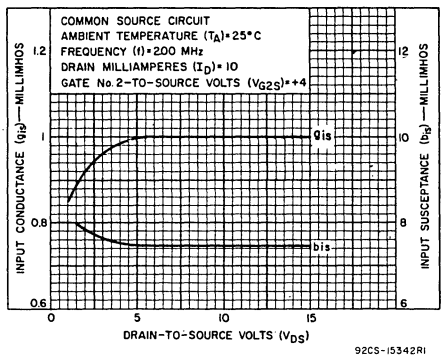


Fig. 8 - y_{is} vs. V_{DS}

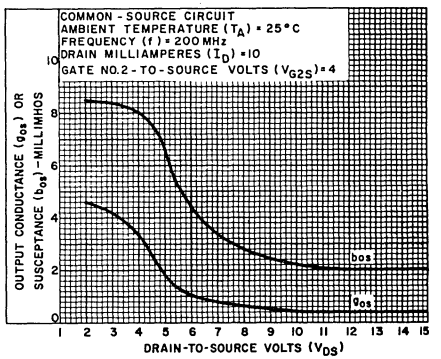


Fig. 9 - y_{os} vs. V_{DS}

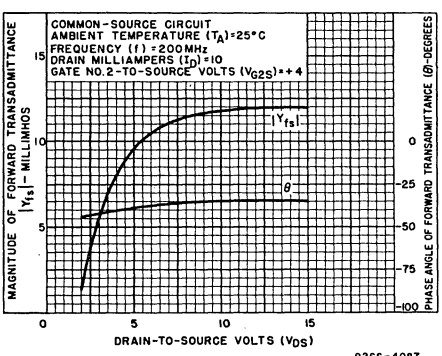


Fig. 10 - y_{fs} vs. V_{DS}

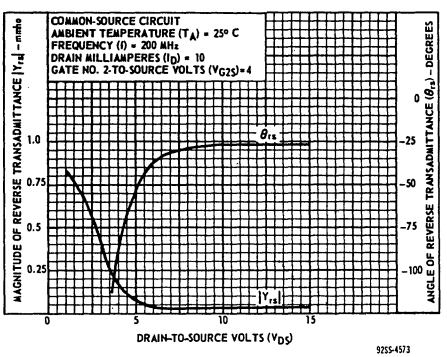


Fig. 11 - y_{rs} vs. V_{DS}

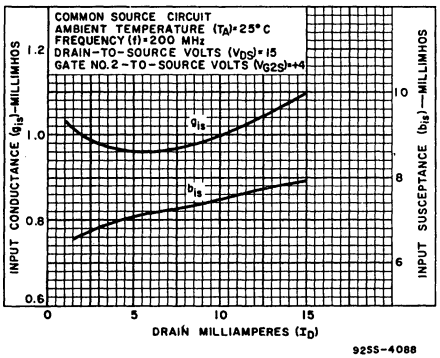


Fig. 12 - y_{is} vs. I_D

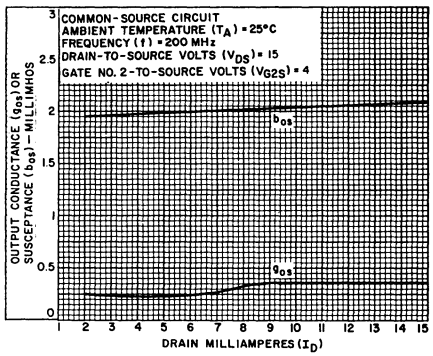


Fig. 13 - y_{os} vs. I_D

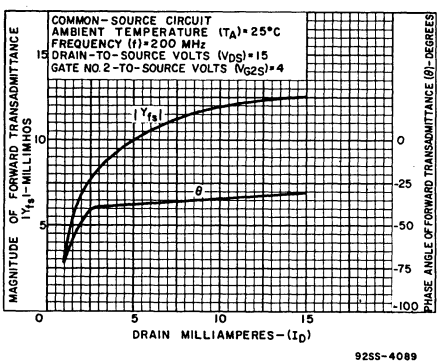


Fig. 14 - y_{fs} vs. I_D

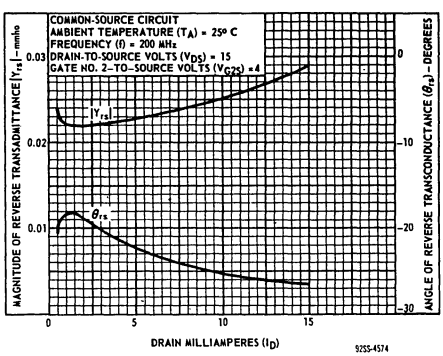


Fig. 15 - y_{rs} vs. I_D

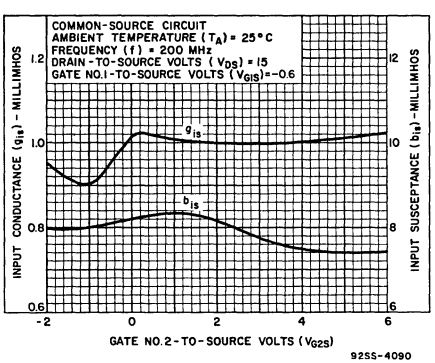


Fig. 16 - y_{is} vs. V_{G2S}

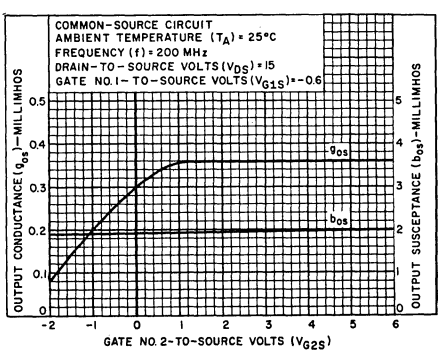


Fig. 17 - y_{os} vs. V_{G2S}

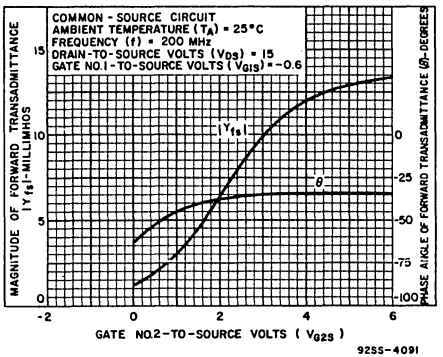


Fig. 18 - y_{fs} vs. V_{G2S}

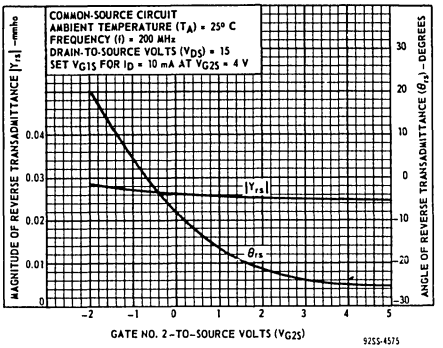


Fig. 19 - y_{rs} vs. V_{G2S}

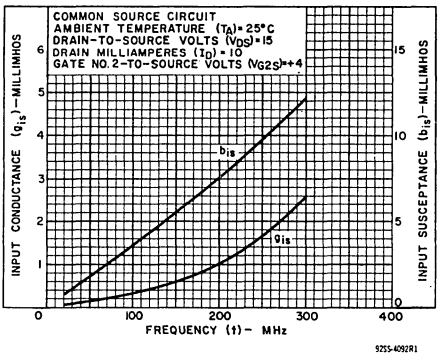


Fig. 20 - y_{is} vs. frequency

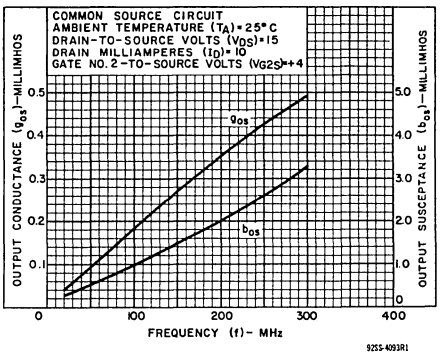


Fig. 21 - y_{os} vs. frequency

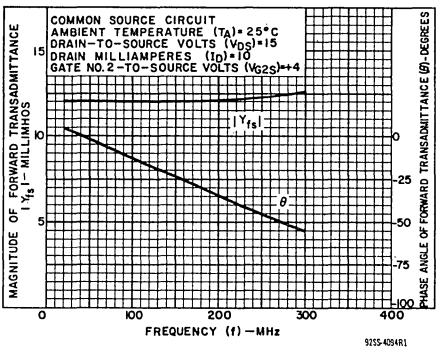


Fig. 22 - y_{fs} vs. frequency

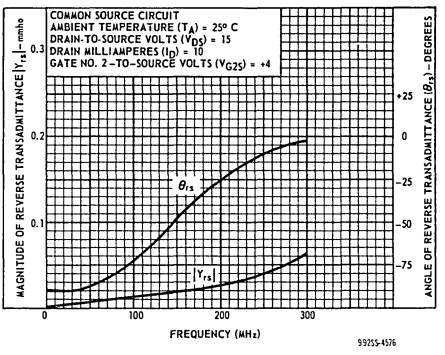


Fig. 23. - y_{rs} vs. frequency

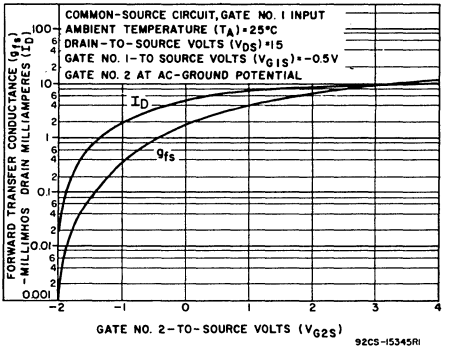


Fig. 24 - g_{fs} and I_D vs. V_{G2S}

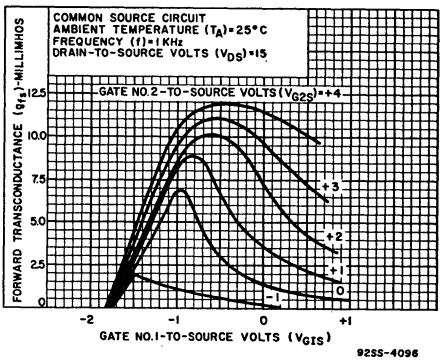


Fig. 25. - g_{fs} vs. V_{G1S}

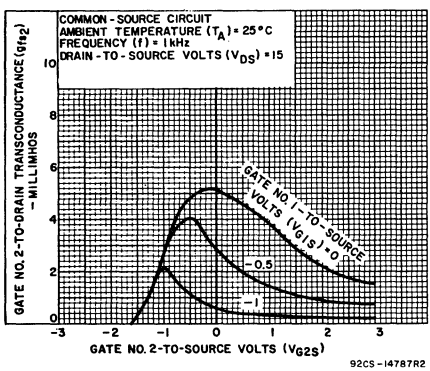
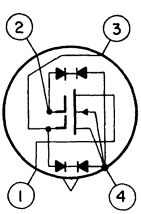


Fig. 26 - g_{fs_2} vs. V_{G2S}

TERMINAL DIAGRAM



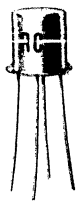
LEAD 1 - DRAIN
LEAD 2 - GATE No. 2
LEAD 3 - GATE No. 1
LEAD 4 - SOURCE, SUBSTRATE AND CASE

RCA Solid State Division

MOS Field-Effect Transistors

N-Channel Depletion Types

3N200



JEDEC TO-72

Silicon Dual Insulated-Gate Field-Effect Transistor

With Integrated Gate-Protection Circuits

For Military and Industrial Applications up to 500 MHz

Applications

- RF amplifier, mixer, and IF amplifier in military and industrial communications equipment
- Aircraft and marine vehicular receivers
- CATV and MATV equipment
- Telemetry and multiplex equipment

RCA-3N200 is an n-channel silicon, depletion type, dual insulated-gate field-effect transistor.

Special back-to-back diodes are diffused directly into the MOS pellet and are electrically connected between each insulated gate and the FET's source. The diodes effectively bypass any voltage transients which exceed approximately ±10 volts. This protects the gates against damage in all normal handling and usage.

A feature of the back-to-back diode configuration is that it allows the 3N200 to retain the wide input signal dynamic range inherent in the MOSFET. In addition, the low junction capacitance of these diodes adds little to the total capacitance shunting the signal gate.

The excellent overall performance characteristics of the RCA-3N200 make it useful for a wide variety of rf-amplifier applications at frequencies up to 500 MHz. The two serially-connected channels with independent control gates make possible a greater dynamic range and lower cross-modulation than is normally achieved using devices having only a single control element.

The two-gate arrangement of the 3N200 also makes possible a desirable reduction in feedback capacitance by operating in the common-source configuration and ac-grounding Gate No. 2. The reduced capacitance allows operation at maximum gain *without neutralization;* and, of special importance in rf-amplifiers, it reduces local oscillator feedthrough to the antenna.

The 3N200 is hermetically sealed in the metal JEDEC TO-72 package.

Maximum Ratings, *Absolute-Maximum Values, at T_A = 25°C*

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	-0.2 to +20	V
GATE No.1-TO-SOURCE VOLTAGE, V_{G1S}:		
Continuous (dc)	-6 to +3	V
Peak ac	-6 to +6	V
GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:		
Continuous (dc)	-6 to 30% of V_{DS}	V
Peak ac	-6 to +6	V
* DRAIN-TO-GATE VOLTAGE, V_{DG1} OR V_{DG2}	+20	V
* DRAIN CURRENT, I_D	50	mA
* TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	330	mW
At ambient temperatures above 25°C	derate linearly at 2.2 mW/°C	
* AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
* LEAD TEMPERATURE (During soldering):		
At distances ≥1/32 inch from seating surface for 10 seconds max.	265	°C

*In accordance with JEDEC registration data format (JS-9 RDF-19A)

Performance Features

- Superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET s
- Wide dynamic range permits large-signal handling before overload
- Dual-gate permits simplified agc circuitry
- Virtually no agc power required
- Greatly reduces spurious responses in FM receivers

Device Features

- Back-to-back diodes protect each gate against handling and in-circuit transients
- High forward transconductance — g_{fs} = 15,000 μmho (typ.)
- High unneutralized RF power gain — G_{ps} = 12.5 dB (typ.) at 400 MHz
  = 19 dB (typ.) at 200 MHz
- Low VHF noise figure — 3.9 dB (typ.) at 400 MHz
  3.0 dB (typ.) at 200 MHz

	ELECTRICAL CHARACTERISTICS *at T_A = 25°C unless otherwise specified*		SYMBOLS	TEST CONDITIONS		LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
*	Gate No. 1-to-Source Cutoff Voltage		$V_{G1S(off)}$	V_{DS} = +15 V, I_D = 50 μA V_{G2S} = +4 V		-0.1	-1	-3	V
*	Gate No. 2-to-Source Cutoff Voltage		$V_{G2S(off)}$	V_{DS} = +15 V, I_D = 50 μA V_{G1S} = 0		-0.1	-1	-3	V
*	Gate No. 1-Terminal Forward Current		I_{G1SSF}	V_{G1S} = +1 V $V_{G2S} = V_{DS} = 0$	T_A = 25°C T_A = 100°C	– –	– –	50 5	nA μA
*	Gate No. 1-Terminal Reverse Current		I_{G1SSR}	V_{G1S} = -6 V $V_{G2S} = V_{DS} = 0$	T_A = 25°C T_A = 100°C	– –	– –	50 5	nA μA
*	Gate No. 2-Terminal Forward Current		I_{G2SSF}	V_{G2S} = +6 V $V_{G1S} = V_{DS} = 0$	T_A = 25°C T_A = 100°C	– –	– –	50 5	nA μA
*	Gate No. 2-Terminal Reverse Current		I_{G2SSR}	V_{G2S} = -6 V $V_{G1S} = V_{DS} = 0$	T_A = 25°C T_A = 100°C	– –	– –	50 5	nA μA
*	Zero-Bias Drain Current		I_{DS}	V_{DS} = +15 V, V_{G1S} = 0 V_{G2S} = +4 V		0.5	5.0	12	mA
*	Forward Transconductance (Gate No. 1-to-Drain)		g_{fs}	V_{DS} = +15 V I_D = 10 mA V_{G2S} = +4 V	f = 1 kHz	10,000	15,000	20,000	μmho
	Small-Signal, Short-Circuit Input Capacitance†		C_{iss}		f = 1 MHz	4.0	6.0	8.5	pF
*	Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate-No. 1)♦		C_{rss}			0.005	0.02	0.03	pF
	Small-Signal, Short-Circuit Output Capacitance		C_{oss}			–	2.0	–	pF
*	Power Gain (see Fig. 1)		G_{PS}		f = 400 MHz	10	12.5	–	dB
	Noise Figure (see Fig. 1)		NF			–	3.9	6.0	dB
*	Bandwidth		BW			28	–	38	MHz
*	Gate-to-Source Forward Breakdown Voltage	Gate No. 1	$V_{(BR)G1SSF}$	I_{G1SSF} = I_{G2SSF} = 100 μA	$V_{G2S} = V_{DS} = 0$	6.5	–	13	V
		Gate No. 2	$V_{(BR)G2SSF}$		$V_{G1S} = V_{DS} = 0$				
*	Gate-to-Source Reverse Breakdown Voltage	Gate No. 1	$V_{(BR)G1SSR}$	I_{G1SSR} = I_{G2SSR} = 100 μA	$V_{G2S} = V_{DS} = 0$	-6.5	–	-13	V
		Gate No. 2	$V_{(BR)G2SSR}$		$V_{G1S} = V_{DS} = 0$				

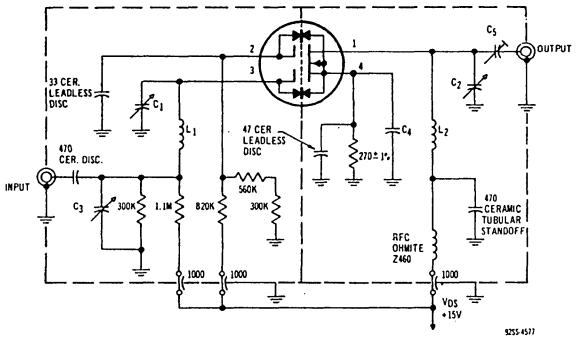
*In accordance with JEDEC registration data format (JS-9 RDF-19A)

†Capacitance between Gate No. 1 and all other terminals.

♦Three-terminal measurement with Gate No. 2 and Source returned to guard terminal.

OPERATING CONSIDERATIONS

The flexible leads of the 3N200 are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons MUST be grounded.



All resistances in ohms
All capacitances in pF

C_1, C_2: 1.3-5.4 pF variable air capacitor: Hammerland Mac 5 type or equivalent

C_3: 1.9-13.8 pF variable air capacitor: Hammerland Mac 15 type or equivalent

C_4: Approx. 300 pF-capacitance formed between socket cover & chassis

C_5: 0.8-4.5 pF piston type variable air capacitor: Erie 560-013 or equivalent

L_1, L_2: Inductance to tune circuit

Fig. 1 - 400 MHz power gain and noise figure test circuit

Typical Characteristics

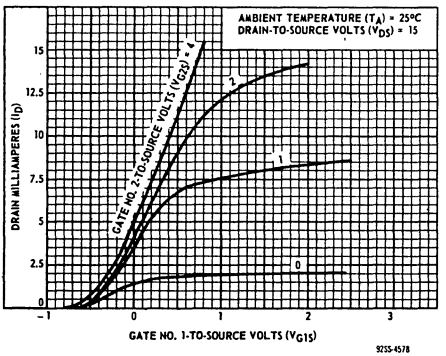


Fig. 2 - I_D vs. V_{G1S}

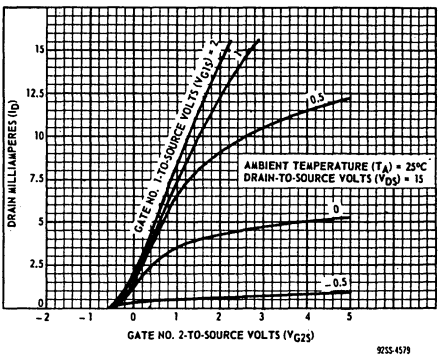


Fig. 3 - I_D vs. V_{G2S}

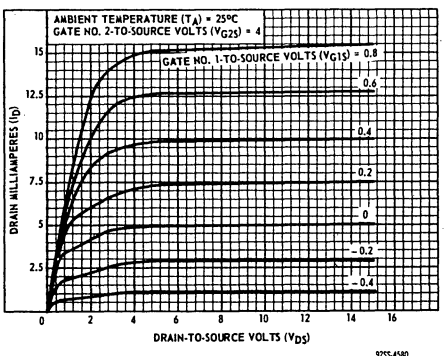


Fig. 4 - I_D vs. V_{DS}

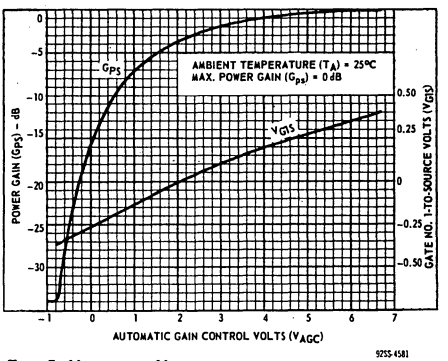


Fig. 5 - V_{AGC} vs. V_{G1S}

y and s Parameters vs. Frequency

TEST CONDITIONS: Drain-to-Source Volts (V_{DS}) = 15, Drain Milliamperes (I_D) = 10, Gate No. 2-to-Source Volts (V_{G2S}) = 4

CHARACTERISTICS	SYMBOL	FREQUENCY (MHz)					UNITS
		100	200	300	400	500	
Maximum Available Power Gain	MAG	32	24	17.5	13	10	dB
Maximum Usable Power Gain (Unneutralized)*	MUG	32	24	17.5	13	10	dB
Y Parameters							
Input Conductance	g_{is}	0.25	0.8	2.0	3.6	6.2	mmho
Input Susceptance	b_{is}	3.4	5.8	8.5	11.2	15.5	mmho
Magnitude of Forward Transadmittance	$\lvert y_{fs} \rvert$	15.3	15.3	15.4	15.5	16.3	mmho
Angle of Forward Transadmittance	$\angle y_{fs}$	−15	−25	−35	−47	−60	degrees
Output Conductance	g_{os}	0.15	0.3	0.5	0.8	1.1	mmho
Output Susceptance	b_{os}	1.5	2.7	3.6	4.25	5.0	mmho
Magnitude of Reverse Transadmittance	$\lvert y_{rs} \rvert$	0.012	0.025	0.06	0.14	0.26	mmho
Angle of Reverse Transadmittance	$\angle y_{rs}$	−60	−25	0	14	20	degrees
S Parameters							
Magnitude of Input Reflection Coeff.	$\lvert s_{is} \rvert$	0.97	0.90	0.84	0.78	0.70	
Angle of Input Reflection Coeff.	$\angle s_{is}$	−20	−32	−55	−68	−82	degrees
Magnitude of Forward Transmission Coeff.	$\lvert s_{fs} \rvert$	1.50	1.40	1.25	1.1	0.9	
Angle of Forward Transmission Coeff.	$\angle s_{fs}$	153	133	112	90	70	degrees
Mangitude of Output Reflection Coeff.	$\lvert s_{os} \rvert$	0.985	0.95	0.93	0.92	0.91	
Angle of Output Reflection Coeff.	$\angle s_{os}$	−7.5	−16	−22	−28	−34	degrees
Magnitude of Reverse Transmission Coeff.	$\lvert s_{rs} \rvert$	0.001	0.0025	0.005	0.010	0.0165	
Angle of Reverse Transmission Coeff.	$\angle s_{rs}$	100	125	141	150	142	degrees

*Limited only by practical design considerations

Typical y Parameters vs. V_{DS}

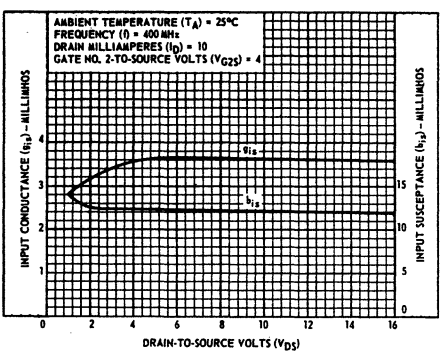


Fig. 6 - y_{is} vs. V_{DS}

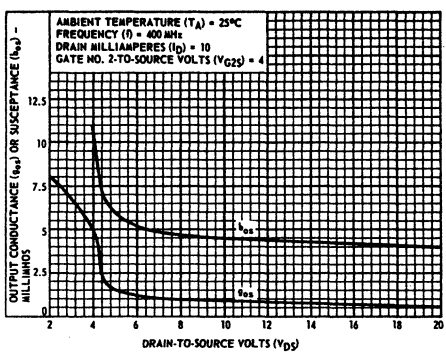


Fig. 7 - y_{os} vs. V_{DS}

Typical y Parameters vs. V_{DS} (cont'd)

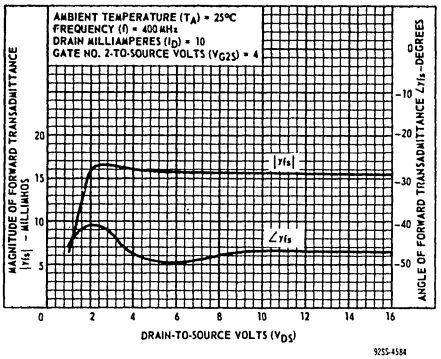


Fig. 8 - y_{fs} vs. V_{DS}

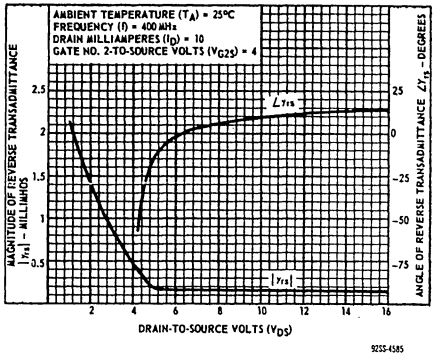


Fig. 9 - y_{rs} vs. V_{DS}

Typical y Parameters vs I_D

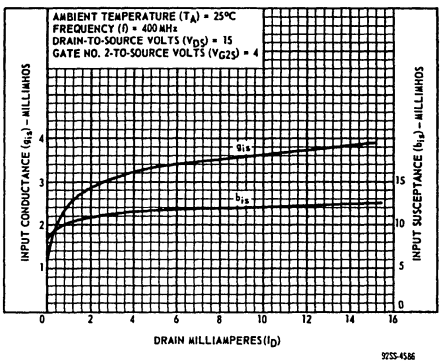


Fig. 10 - y_{is} vs. I_D

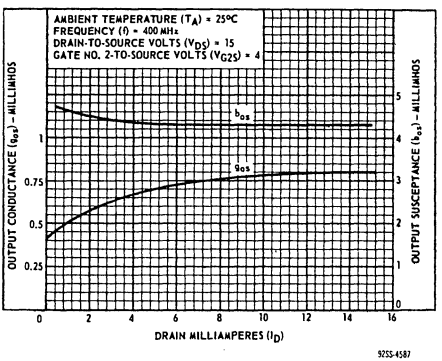


Fig. 11 - y_{os} vs. I_D

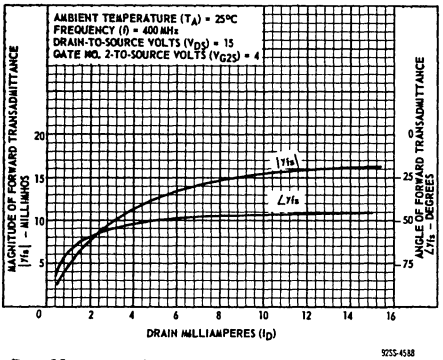


Fig. 12 - y_{fs} vs. I_D

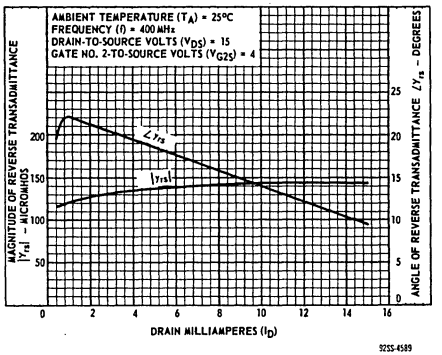


Fig. 13 - y_{rs} vs. I_D

Typical y Parameters vs. V_{G2S}

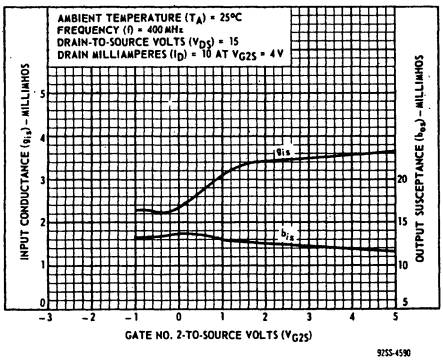


Fig. 14 - y_{is} vs. V_{G2S}

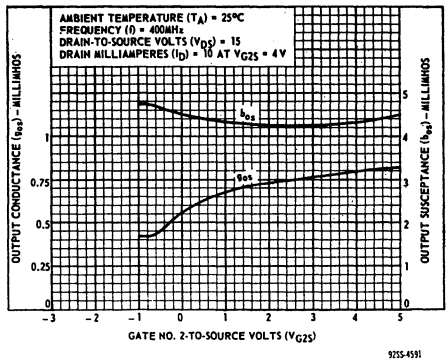


Fig. 15 - y_{os} vs. V_{G2S}

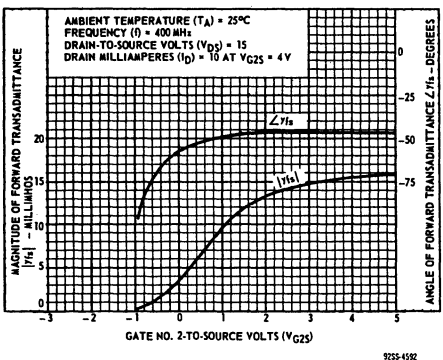


Fig. 16 - y_{fs} vs. V_{G2S}

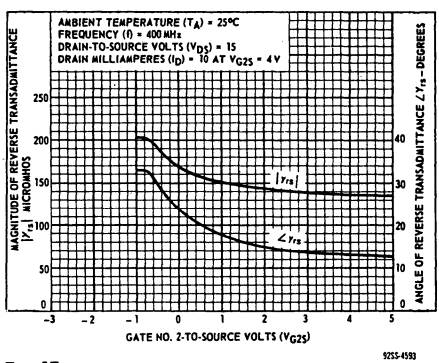


Fig. 17 - y_{rs} vs. V_{G2S}

Typical Characteristics

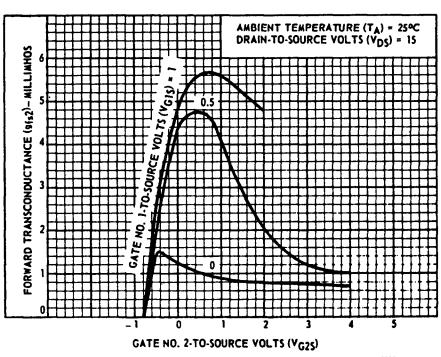


Fig. 18 - g_{fs2} vs. V_{G2S}

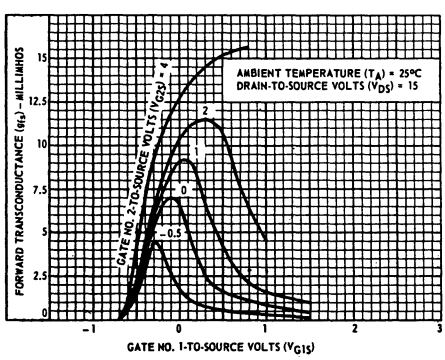


Fig. 19 - g_{fs} vs. V_{G1S}

Typical Characteristics (Cont'd)

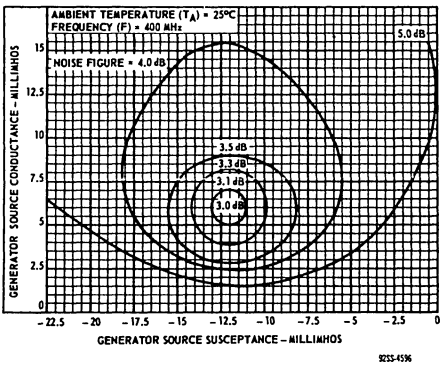
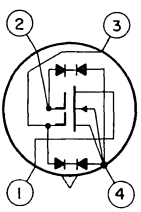


Fig. 20 - Noise figure vs. generator source admittance

TERMINAL DIAGRAM



LEAD 1 - DRAIN
LEAD 2 - GATE No. 2
LEAD 3 - GATE No. 1
LEAD 4 - SOURCE, SUBSTRATE AND CASE

RCA
Solid State
Division

MOS Field-Effect Transistors

N-Channel Depletion Type

40467A



TO-72 H 1299

Silicon MOS Transistor

For VHF Tuners and Other VHF Amplifier Applications in Industrial & Commercial Electronic Equipment Operating up to 220 MHz

Device Features:

- **Low feedback capacitance - C_{rss} = 0.25 pF typ.**
- **High forward transconductance - g_{fs} = 7500 μmho typ.**
- **High vhf power gain - G_{PS} = 16 dB typ at 200 MHz**
- **Low vhf noise figure - NF = 3.5 dB typ at 200 MHz**
- **Exceptionally good cross-modulation characteristics**

Performance Features:

- **Large dynamic range**
- **Greatly reduced spurious responses**
- **Permits use of vacuum-tube biasing techniques**
- **Excellent thermal stability**
- **Superior cross-modulation performance and greater dynamic range than bipolar transistors**

RCA-40467A is an n-channel depletion-type silicon insulated-gate field-effect transistor utilizing the MOS■ construction. It is intended primarily for vhf-amplifier applications in industrial and commercial electronic equipment.

The 40467A is useful in vhf applications requiring devices capable of providing high useful power gains at frequencies up to approximately 220 MHz.

The 40467A features high forward transconductance, high dc gate-to-source resistance, and low feedback capacitance. Because of the improved transfer characteristic and increased dynamic range, the 40467A provides substantially better cross-modulation performance in linear-amplifier applications than conventional (bipolar) transistors and is free from diode-current loading, a problem that exists in junction type FET s. This device is hermetically sealed in the TO-72 metal case and utilizes full-gate construction.

■ Metal-Oxide Semiconductor

Maximum Ratings, *Absolute-Maximum Values at* $T_A = 25^oC$:

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20	V
DRAIN-TO-GATE VOLTAGE, V_{DG}	+20	V
GATE-TO-SOURCE VOLTAGE, V_{GS}:		
CONTINUOUS (dc)	+1, -8	V
PEAK ac	±15	V
DRAIN CURRENT, I_D	50	mA
TRANSISTOR DISSIPATION:		
At ambient temperatures up to 25°C	330	mW
above 25°C	derate at 2.2	mW/°C
AMBIENT TEMPERATURE RANGE:		
Storage	-65 to +175	°C
Operating	-65 to +175	°C
LEAD TEMPERATURE (During Soldering):		
At distances not closer than 1/32 inch to seating surface for 10 seconds maximum	265	°C

ELECTRICAL CHARACTERISTICS AT T_C = 25°C WITH BULK (SUBSTRATE) CONNECTED TO SOURCE

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS			LIMITS			UNITS
		FREQUENCY f	DC DRAIN-TO-SOURCE VOLTAGE V_{DS}	DC DRAIN CURRENT I_D	RCA 40467A			
		MHz	V	mA	Min	Typ.	Max.	
Gate-to-Source Cutoff Voltage	$V_{GS}(off)$		12	0.1	-	-	-8	V
Gate Leakage Current	I_{GSS}		0	V_{GS} = +1V	-	-	1	nA
			0	V_{GS} = -8V	-	-	1	nA
Zero-Bias Drain Current	I_{DSS}		15	V_{GS} = 0	5	15	30	mA
Small-Signal, Short-Circuit Forward Transconductance	g_{fs}	1 KHz	15	5	4000	7500	-	μmho
Small-Signal, Short-Circuit Reverse-Transfer Capacitance (Drain-to-Gate)	C_{rss}	1	15	5	0.12	0.25	0.35	pF
Small Signal Short-Circuit Input Capacitance	C_{iss}	1	15	5	-	5.5	-	pF
Input Admittance	Y_{is}	Common Source Configuration f = 200 mHz V_{DS} = 15V I_D = 5 mA			-	0.4 + j7.3	-	
Forward Transfer Admittance	Y_{fs}				-	7 - J2	-	
Output Admittance	Y_{os}				-	0.28 +J1.8	-	
Maximum Available Power Gain	MAG	200	15	5	-	21	-	dB
Maximum Usable Power Gain (unneutralized)	MUG	200	15	5	-	12	-	dB
Maximum Usable Power Gain (neutralization)	MUG	200	15	5	12	16	-	dB
Noise Figure	NF	200	15	5	-	3.5	5	dB

TYPICAL CHARACTERISTICS

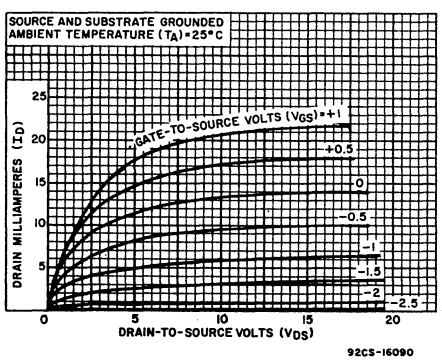


Fig. 1

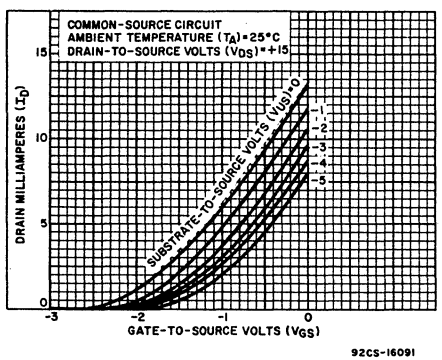


Fig. 2

TYPICAL ADMITTANCE CHARACTERISTICS

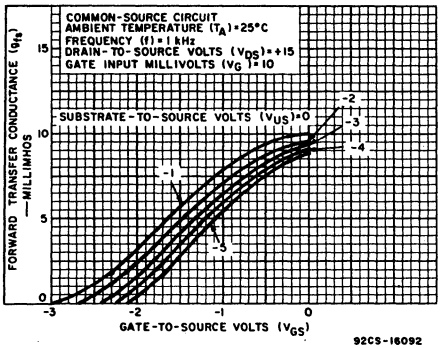


Fig. 3

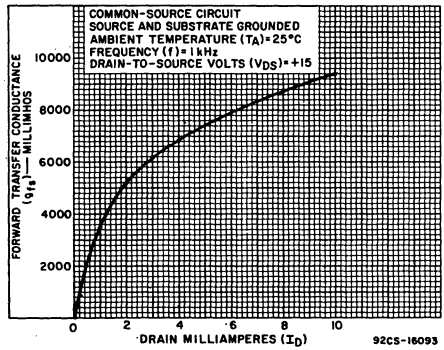


Fig. 4

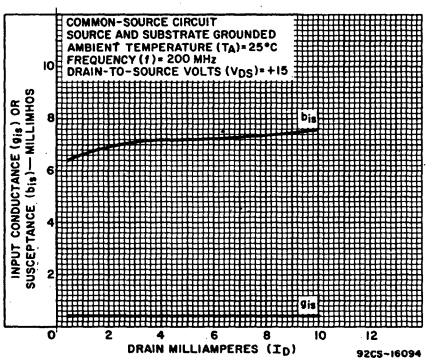


Fig. 5

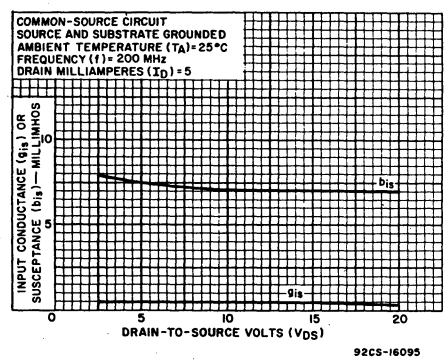


Fig. 6

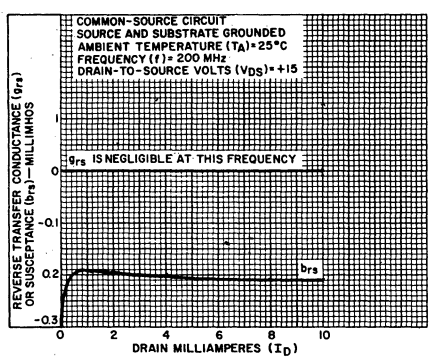


Fig. 7

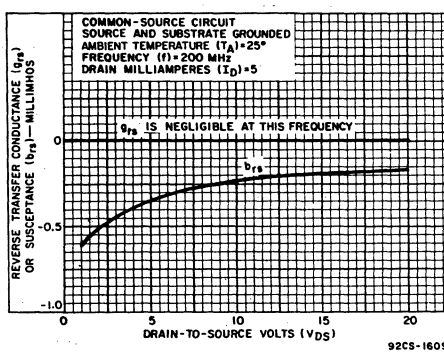


Fig. 8

TYPICAL ADMITTANCE CHARACTERISTICS (cont'd)

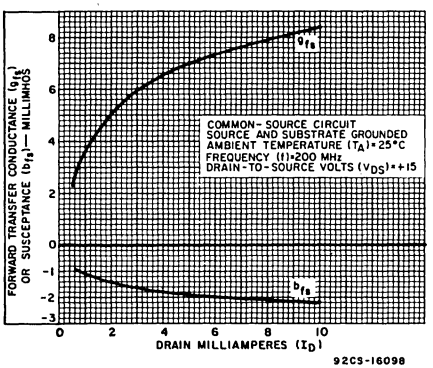


Fig. 9

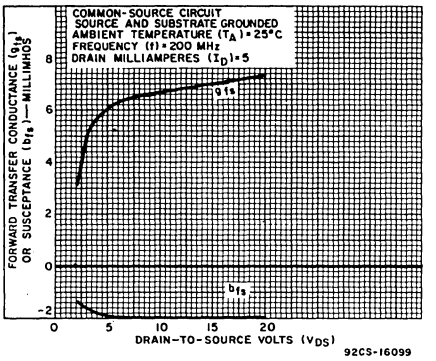


Fig. 10

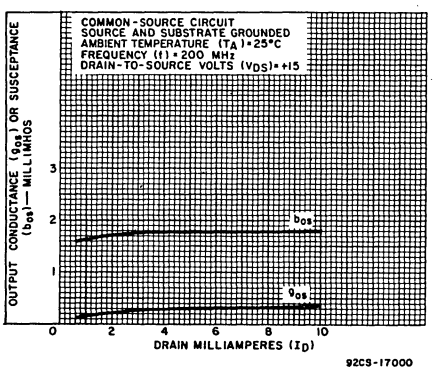


Fig. 11

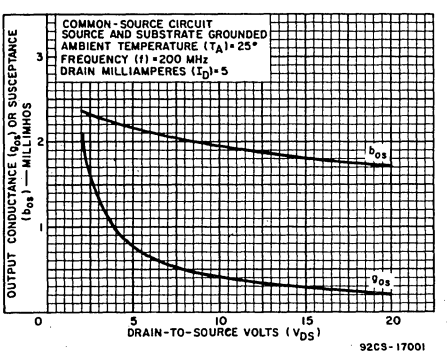
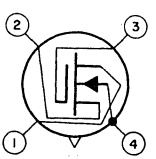


Fig. 12

TERMINAL DIAGRAM



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

OPERATING CONSIDERATIONS

The flexible leads of the 40467A are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons should be grounded, and appropriate precautions should be taken to protect the devices against high electric fields.

This device should not be connected into or disconnected from circuits with the power on because high transient voltages may cause permanent damage to the devices.

RCA
Solid State
Division

MOS Field-Effect Transistors

N-Channel Depletion Types

40468A
40559A



TO-72

MOS Silicon Transistors

For RF Amplifier and Mixer Applications in FM and AM/FM Receivers

Device Features:

- **high forward transconductance - -**
  gfs = 7500 μmho typ. for 40468A
- **low feedback capacitance - -**
  C_{rss} = 0.35 pF max. for 40468A
  0.38 pF max. for 40559A
- **high useful power gains - -**
  neutralized - 17 dB typ.
  unneutralized - 14 dB typ.
- **hermetically sealed in TO-72 metal package**

RCA-40468A and 40559A are silicon insulated-gate field-effect transistors of the n-channel depletion type utilizing the MOS* construction. They are intended primarily for use as the rf amplifier and mixer, respectively, in FM receivers covering the 88 to 108 MHz band, but can be used for general amplifier applications at frequencies up to 125 MHz. For circuit design and typical performance data refer to RCA Application Note AN3535 "An FM Tuner Using Single-Gate MOS Field-Effect Transistors as RF Amplifier and Mixer".

The wide dynamic range of these transistors reduces cross-modulation effects in AM receivers and minimizes the generation of spurious responses in FM receivers.

Operating as a neutralized amplifier at 100 MHz, the 40468A can provide a power gain of 17 dB (typ.). A power gain of 14 dB (typ.) can be realized without neutralization.

* Metal-Oxide-Semiconductor.

Performance Features:

- **reduced spurious responses in FM tuners**
- **reverse bias on substrate improves linearity**
- **reduced cross-modulation effects in AM receivers**

Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20	V
DRAIN-TO-GATE VOLTAGE, V_{DG}	+20	V
GATE-TO-SOURCE VOLTAGE, V_{GS}:		
CONTINUOUS (dc)	+1, -8	V
PEAK ac	±15	V
DRAIN CURRENT, I_D	25	mA
TRANSISTOR DISSIPATION:		
At ambient temperatures up to 25°C	330	mW
above 25°C	derate at 2.2	mW/°C
AMBIENT TEMPERATURE RANGE:		
Storage	-65 to +175	°C
Operating	-65 to +175	°C
LEAD TEMPERATURE (During Soldering):		
At distances not closer than 1/32 inch to seating surface for 10 seconds maximum	265	°C

ELECTRICAL CHARACTERISTICS, at T_A = 25°C

With Bulk (Substrate) Connected to Source Unless Otherwise Specified

Characteristics	Symbols	TEST CONDITIONS			LIMITS						Units
		Frequency f	DC Drain-to-Source V_{DS}	DC Drain Current I_D	RCA-40468A RF Amplifier			RCA-40559A Mixer			
		MHz	V	mA	Min.	Typ.	Max.	Min.	Typ.	Max.	
Drain-to-Source Cutoff Current	$I_{D(off)}$	-	12	V_{GS} = -8V	-	-	100	-	-	500	μA
Gate Leakage Current	I_{GSS}	- -	0 0	V_{GS} = -8V V_{GS} = +1V	- -	- -	1 1	- -	- -	1 1	nA nA
Zero-Bias Drain Current	I_{DSS}	-	15	V_{GS} = 0	5	15	30	5	15	30	mA
Small-Signal, Short-Circuit Forward Transconductance	g_{fs}	1 kHz	15	5	-	7500	-	-	-	-	μmho
Small-Signal, Short-Circuit Reverse-Transfer Capacitance (Drain-to-Gate)	C_{rss}	1	15	5	-	0.25	0.35	-	0.25	0.38	pF
Input Capacitance	C_{iss}	1	15	5	-	5.5	-	-	5.5	-	pF
Admittance	-	RF | Mixer		RF | Mixer	-			-			-
Input Admittance	Y_{is}	100 MHz	15	5 | 3	0.155 + j 3.45			0.14 + j 3.38			mmho
Forward Transfer Admittance	Y_{fs}	100 MHz	15	5 | 3	7.4 + j 0.9			-			mmho
Output Admittance	Y_{os}	100 MHz | 10.7 MHz	15	5 | 3	0.21 + j 0.9			0.076 + j 0.153			mmho
Forward Conversion Transconductance	$g_{fs(c)}$	1 kHz	15	3	-	-	-	-	2800*	-	μmho
Maximum Available Power Gain	MAG	100	15	5	-	26	-	-	-	-	dB
Maximum Usable Power Gain (Unneutralized)	MUG	100	15	5	-	14	-	-	-	-	dB
Maximum Usable Power Gain (Neutralized)	MUG	100	15	5	14	17	-	-	-	-	dB
Maximum Available Conversion Gain	MAG_C	f_{in} = 100 f_{out} = 10.7	15	3	-	-	-	-	22	-	dB
Noise Figure	NF	100	15	5	-	3.5	5	-	-	-	dB

* Bulk (Substrate)-to-Source Volts (V_{BS}) = -3.

OPERATING CONSIDERATIONS

The flexible leads of the 40468A and 40559A are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons should be grounded, and appropriate precautions should be taken to protect the devices against high electric fields.

These devices should not be connected into, or disconnected from, circuits with the power on because high transient voltages may cause permanent damage to the devices.

TYPICAL y-PARAMETER CHARACTERISTICS

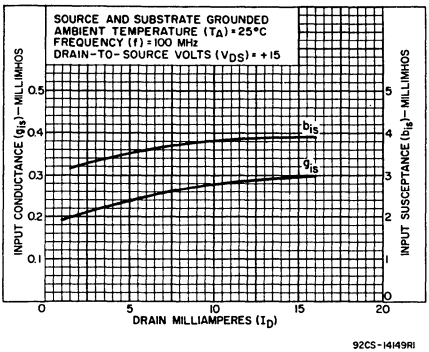


Fig.1 - Input admittance (y_{is}) vs drain current (I_D).

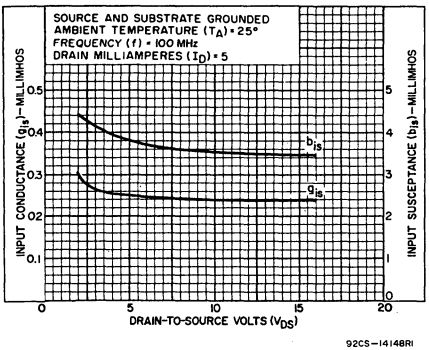


Fig.2 - Input admittance (y_{is}) vs drain-to-source voltage (V_{DS}).

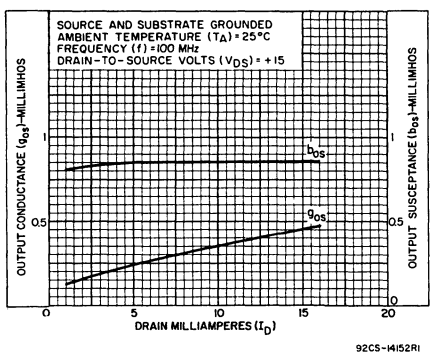


Fig.3 - Output admittance (y_{os}) vs drain current (I_D).

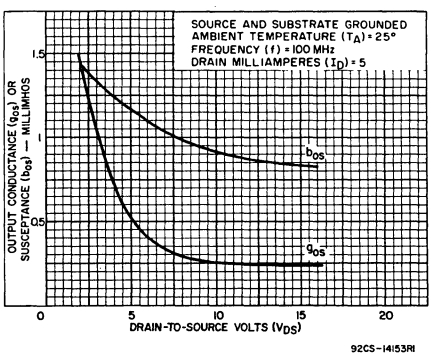


Fig.4 - Output admittance (y_{os}) vs drain-to-source voltage (V_{DS}).

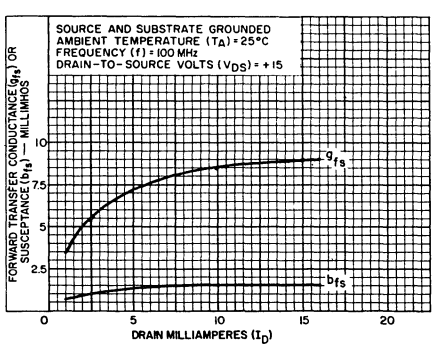


Fig.5 - Forward transadmittance (y_{fs}) vs drain current (I_D).

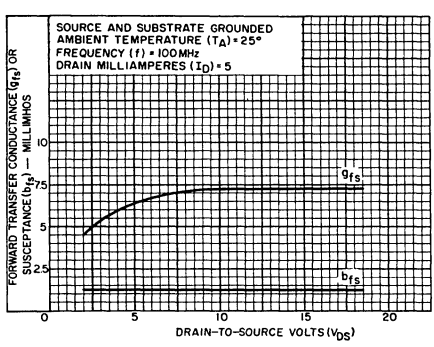


Fig.6 - Forward transadmittance (y_{fs}) vs drain-to-source voltage (V_{DS}).

TYPICAL y-PARAMETER CHARACTERISTICS

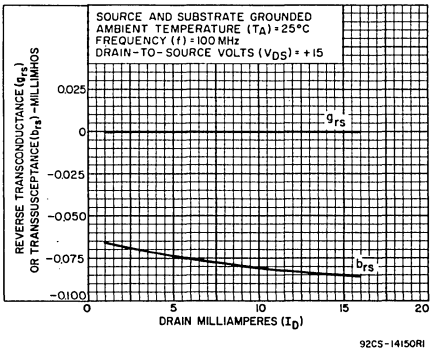


Fig.7 - Reverse transadmittance (y_{rs}) vs drain current (I_D).

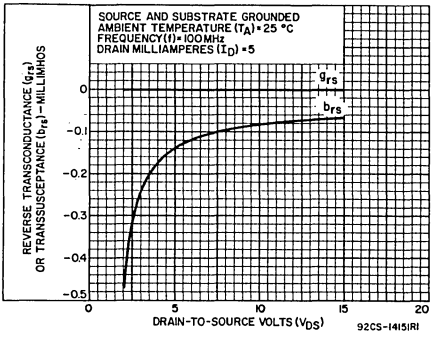


Fig.8 - Reverse transadmittance (y_{rs}) vs drain-to-source voltage (V_{DS}).

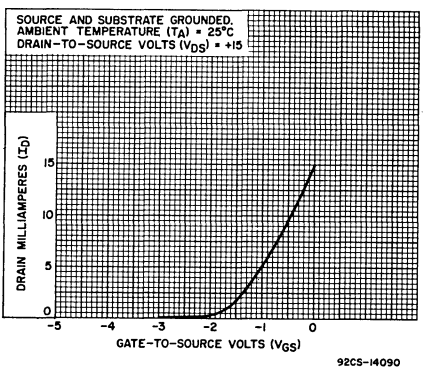
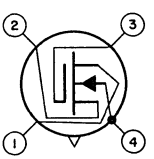


Fig.9 - Typical characteristic of drain current (I_D) vs gate-to-source voltage (V_{GS}).

TERMINAL DIAGRAM



LEAD 1 - DRAIN
LEAD 2 - SOURCE
LEAD 3 - INSULATED GATE
LEAD 4 - BULK (SUBSTRATE) AND CASE

RCA Solid State Division

MOS Field-Effect Transistors

40600
40601
40602

SILICON DUAL INSULATED-GATE FIELD-EFFECT TRANSISTORS



TO-72

N-Channel Depletion Types For VHF TV Receiver Applications

RCA 40600, 40601, and 40602 are n-channel depletion type, dual-insulated-gate, field-effect transistors utilizing the MOS construction. These devices have characteristics which make them highly desirable for rf-amplifier applications (40600), mixer applications (40601), and first-if-amplifier applications (40602) in vhf TV receivers and other types of commercial equipment operating at frequencies up to approximately 250 MHz.

These transistors feature a series arrangement of two separate channels, each channel having an independent control gate. In amplifier applications the 40600 and 40602 with their wide dynamic range provide substantially better cross-modulation performance than is obtainable with bipolar or single-gate field-effect transistors. In mixer applications the 40601 provides excellent isolation between the oscillator and rf signals because each of the two signal frequencies being mixed has its own control element. The wide dynamic range of the 40601 minimizes cross-modulation which is generally encountered in mixer stages.

Provision of two insulated gates also results in extremely low feedback capacitances (0.02 pF typ.), a feature which enables the 40600 and 40602 to provide high maximum useable power gains in unneutralized circuits – for example, 20 dB at 200 MHz typ. for the 40600, and 35 dB typ. at 44 MHz for the 40602. The gain of the rf and if stages can be controlled by applying agc voltage to gate No.2 and agc delay is easily obtained. Virtually no agc power is required for full gain reduction.

Types 40600, 40601, and 40602 are hermetically sealed in metal JEDEC TO-72 packages.

APPLICATIONS

- **VHF TV Receiver**
  - **40600 for rf amplifier applications**
  - **40601 for mixer applications**
  - **40602 for first-if-amplifier applications**

PERFORMANCE FEATURES

- **superior cross-modulation performance and greater dynamic range than bipolar and single-gate field-effect transistors**
- **permits use of vacuum-tube biasing techniques**
- **excellent thermal stability**

DEVICE FEATURES

- **extremely low feedback capacitance**
  C_{rss} = 0.02 pF typ.
- **high power gain**
  MUG_u = 20 dB typ. for 40600
  MAG = 35 dB typ. for 40602
  MAG_c = 14 dB typ. for 40601

Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	0 to +20	V
GATE No.1-TO-SOURCE VOLTAGE, V_{G1S}:		
Continuous (dc)	+1 to -8	V
Peak ac	+20 to -8	V
GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:		
Continuous (dc)	-8 to 40% of V_{DS}	V
Peak ac	-8 to +20	V
DRAIN-TO-GATE VOLTAGE, V_{DG1} or V_{DG2}	+20	V
DRAIN CURRENT, I_D (Pulsed):		
Pulse duration ≤ 20 ms, duty factor ≤ 0.15	50	mA
TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	400	mW
At ambient temperatures above 25°C	derate linearly at 2.67 mW/°C	
AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
LEAD TEMPERATURE (During soldering):		
At distances ≥ 1/32" from seating surface for 10 seconds max.	265	°C

ELECTRICAL CHARACTERISTICS, at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS 40600, 40601, 40602			UNITS
			Min.	Typ.	Max.	
Gate No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +15V, I_D = 200 μA V_{G2S} = +4V	-	-2	-	V
Gate No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +15V, I_D = 200 μA V_{G1S} = 0	-	-2	-	V
Gate No.1 Leakage Current	I_{G1SS}	V_{G1S} = -20V, V_{G2S} = 0, V_{DS} = 0	-	-	1	nA
Gate No.2 Leakage Current	I_{G2SS}	V_{G2S} = -20V, V_{G1S} = 0, V_{DS} = 0	-	-	1	nA
Drain Current	I_{DSS}	V_{DS} = +13V, V_{G1S} = 0, V_{G2S} = +4V	-	18	-	mA
Forward Transconductance	g_{fs}	V_{DS} = +13V, I_D = 10 mA V_{G2S} = +4V, f = 1 kHz	-	10000	-	μmho

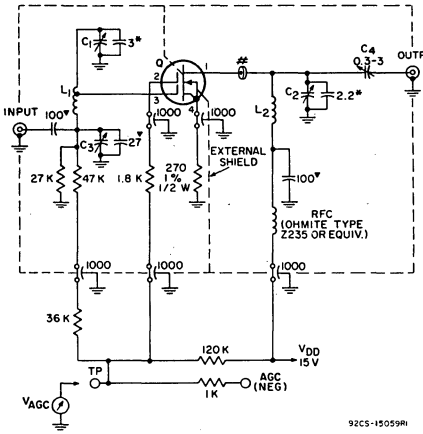
TYPICAL PERFORMANCE CHARACTERISTICS, at T_A = 25°C

CHARACTERISTICS	SYMBOLS	40600 RF AMPLIFIER f = 200 MHz	40602 IF AMPLIFIER f = 44 MHz	40601 MIXER f = 200 MHz	UNITS
		V_{G1S} is adjusted for I_D = 10 mA Gate No.2 at AC ground potential V_{DS} = 13V, V_{G2S} = +4V		Local-oscillator injection Voltage on Gate No.2 = 750 mV V_{DS} = 15V V_{G2S} = +0.6V V_{G1S} = 0.75V	
Small-Signal, Short Circuit Reverse-Transfer Capacitance (Drain-to-Gate No.1) at f = 1 MHz	C_{rss}	0.02 typ. 0.03 max.	0.02 typ. 0.03 max.	0.02 typ. 0.03 max.	pF
Output Capacitance	C_{oss}	2.2	2.2	2.2 at f = 44 MHz	pF
Input Capacitance	C_{iss}	5.5	5.5	5.5	pF
Input Resistance	r_{iss}	1.2	10	1.2	KΩ
Output Resistance	r_{oss}	2.8	12	12 at f = 44 MHz	KΩ
Magnitude of Forward Transadmittance	$\|Y_{fs}\|$	11000	11000	2700*	μmho
Phase Angle of Forward Transadmittance	$\angle\theta$	-46	-11	-	degrees
Maximum Available Power Gain	MAG	20	35	14**	dB
Maximum Usable Power Gain (Unneutralized)	MUG_u	20▲	1 Stage 28 2 Stages 26 3 Stages 24	- - -	dB dB dB
Power Gain See Fig.1 for measurement circuit	G_{PS}	17.5	-	-	dB
Noise Figure	NF	5 max.	-	-	dB

* Magnitude of forward conversion transadmittance

** Maximum available conversion gain

▲ Limited by practical design considerations



* Tubular ceramic.

▼ Disk ceramic.

\# Ferrite bead (½ used); Indiana General No. H1742C-(A-147) or F1157-1-H, or equivalent.

C_1, C_2: 1.5-5 pF variable air capacitor: E. F. Johnson Type 160-102, or equivalent.

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent.

C_4: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13, or equivalent.

L_1: 5 turns silver-plated 0.02" thick, 0.07"- 0.08"wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding.

L_2: Same as L_1 except winding length approx. 0.7"; no tap.

Fig.1 - 200 MHz Power Gain and Noise Figure Test Circuit for 40600 and 40602

TYPICAL SMALL-SIGNAL Y-PARAMETER CHARACTERISTICS at 200 MHz

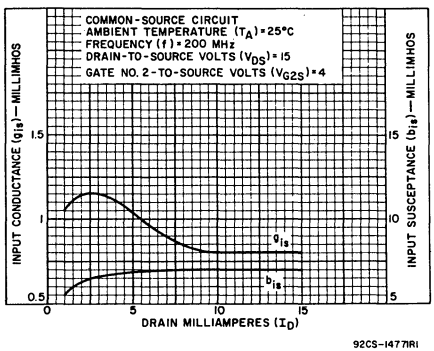


Fig.2 - Y_{is} vs. I_D

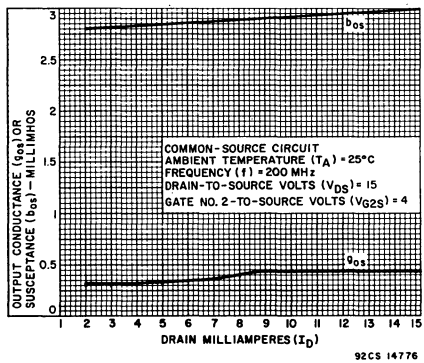


Fig.3 - Y_{os} vs. I_D

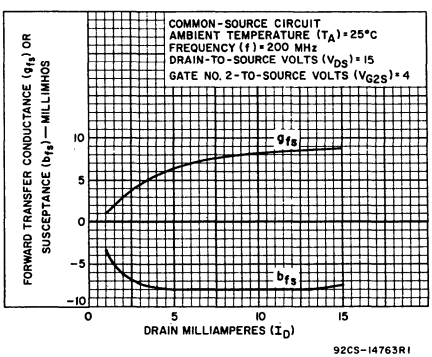


Fig.4 - Y_{fs} vs. I_D

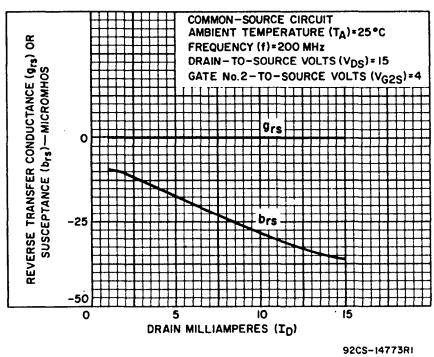


Fig.5 - Y_{rs} vs. I_D

TYPICAL SMALL-SIGNAL Y-PARAMETER CHARACTERISTICS at 200 MHz

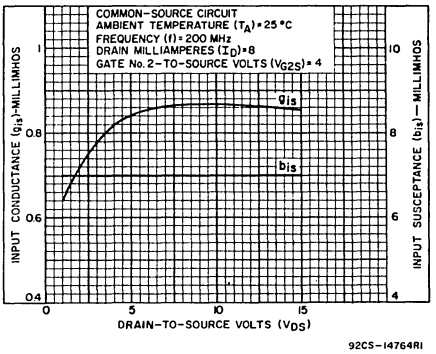


Fig.6 - Y_{is} vs. V_{DS}

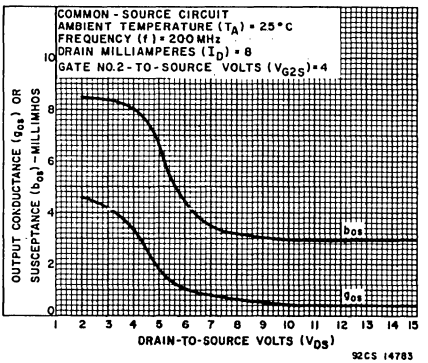


Fig.7 - Y_{os} vs. V_{DS}

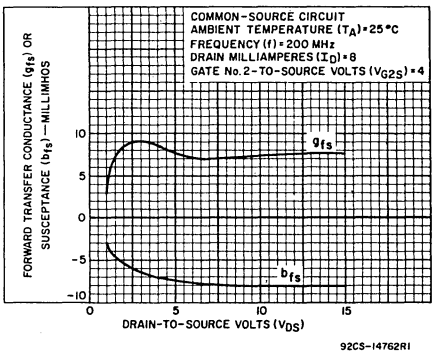


Fig.8 - Y_{fs} vs. V_{DS}

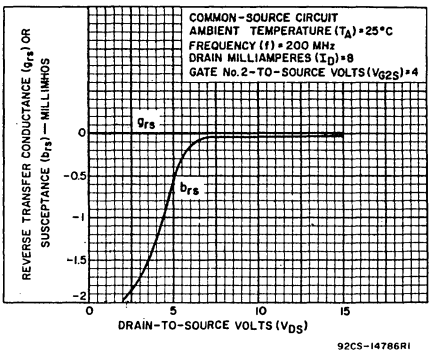


Fig.9 - Y_{rs} vs. V_{DS}

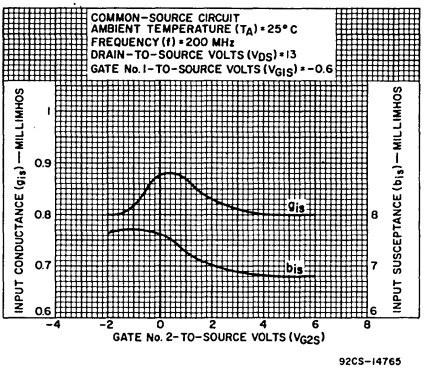


Fig.10 - Y_{is} vs. V_{G2S}

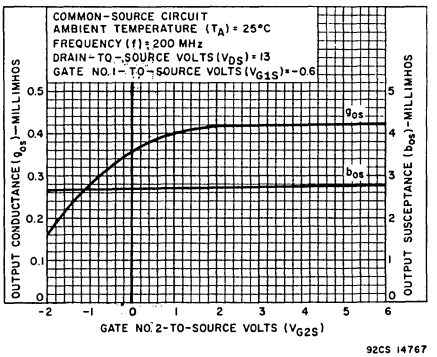


Fig.11 - Y_{os} vs. V_{G2S}

TYPICAL Y-PARAMETER CHARACTERISTICS at 200 MHz

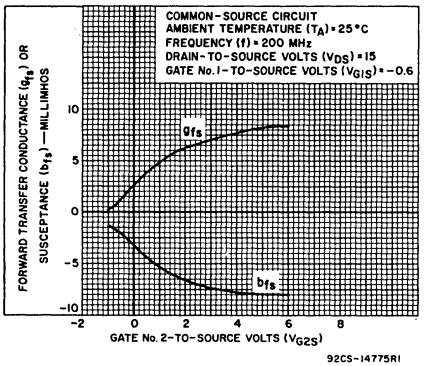


Fig.12 - Y_{fs} vs. V_{G2S}

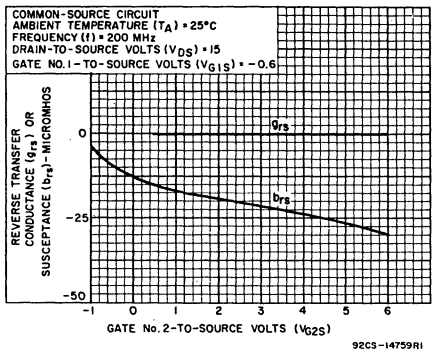


Fig.13 - Y_{rs} vs. V_{G2S}

TYPICAL SMALL-SIGNAL Y-PARAMETER CHARACTERISTICS at 44 MHz

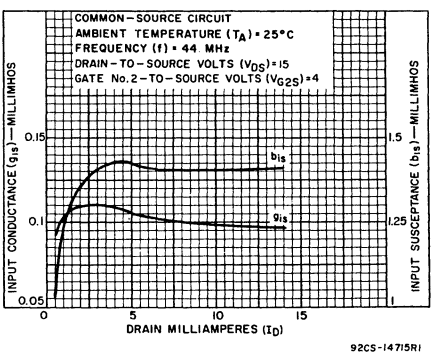


Fig.14 - Y_{is} vs. I_D

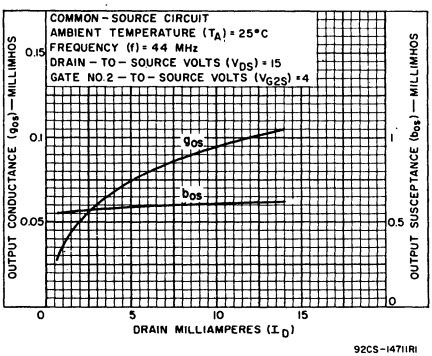


Fig.15 - Y_{os} vs. I_D

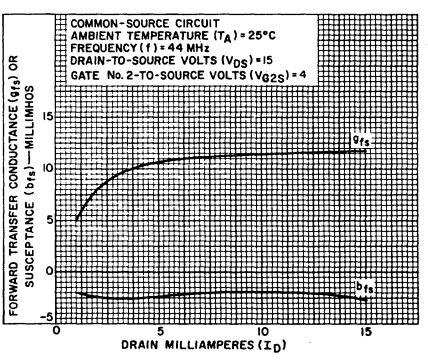


Fig.16 - Y_{fs} vs. I_D

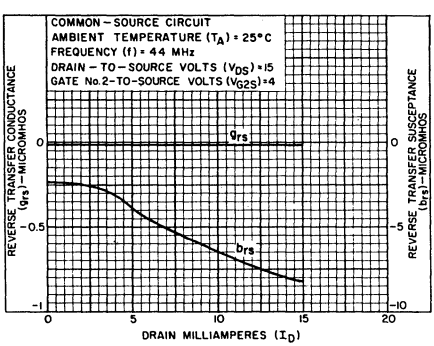


Fig.17 - Y_{rs} vs. I_D

TYPICAL SMALL-SIGNAL Y-PARAMETER CHARACTERISTICS at 44 MHz

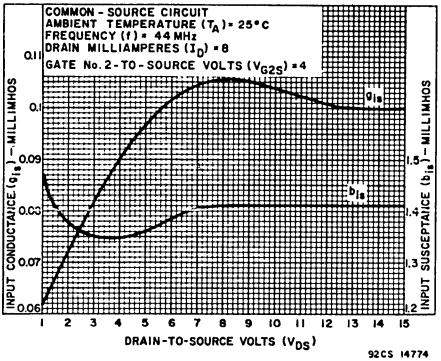


Fig.18 - Y_{is} vs. V_{DS}

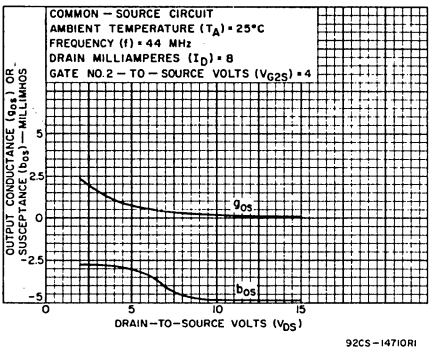


Fig.19 - Y_{os} vs. V_{DS}

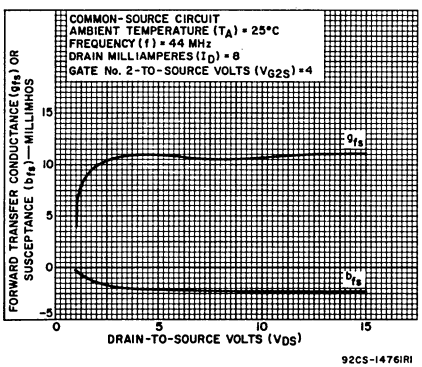


Fig.20 - Y_{fs} vs. V_{DS}

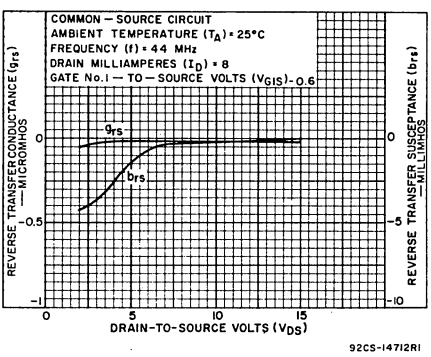


Fig.21 - Y_{rs} vs. V_{DS}

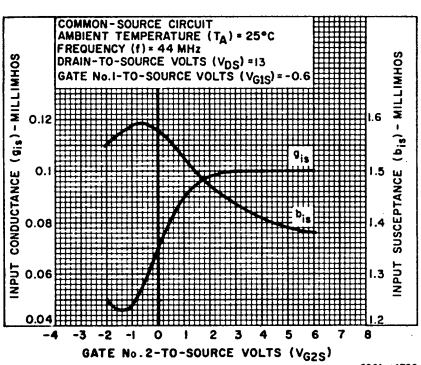


Fig.22 - Y_{is} vs. V_{G2S}

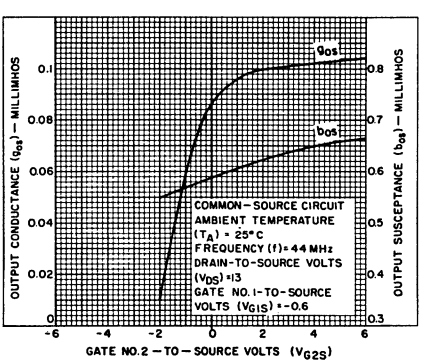


Fig.23 - Y_{os} vs. V_{G2S}

TYPICAL SMALL-SIGNAL Y-PARAMETER CHARACTERISTICS at 44 MHz

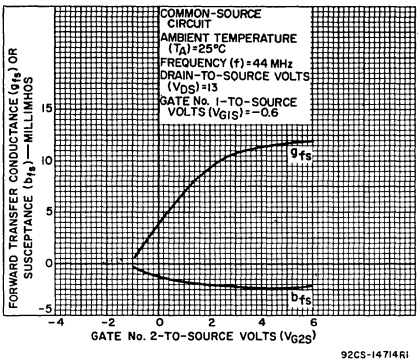


Fig.24 - Y_{fs} vs. V_{G2S}

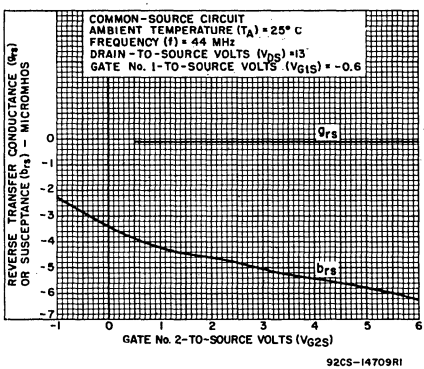


Fig.25 - Y_{rs} vs. V_{G2S}

TYPICAL SMALL-SIGNAL CHARACTERISTICS vs. FREQUENCY

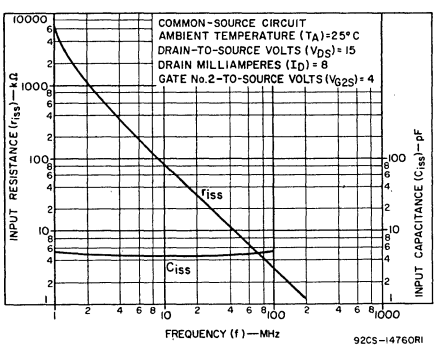


Fig.26 - C_{iss} and R_{iss} vs. f

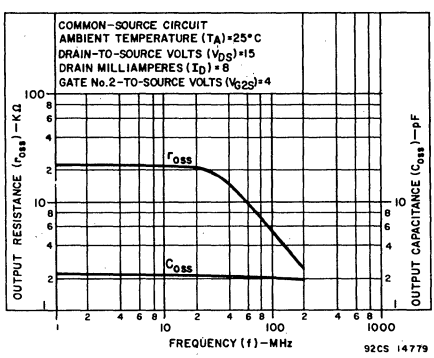


Fig.27 - C_{oss} and R_{oss} vs. f

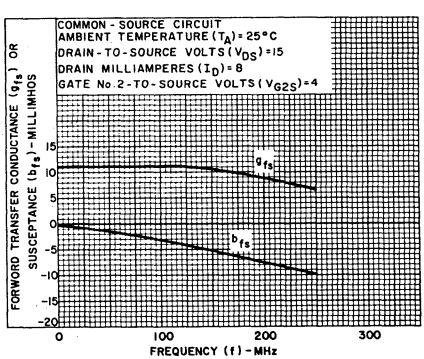


Fig.28 - Y_{fs} vs. f

TYPICAL TRANSFER CHARACTERISTICS

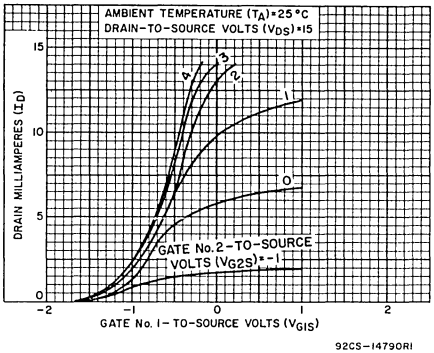


Fig.29 - I_D vs. V_{G1S}

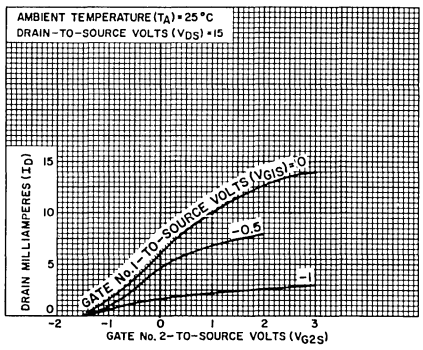


Fig.30 - I_D vs. V_{G2S}

TYPICAL OPERATING CHARACTERISTICS

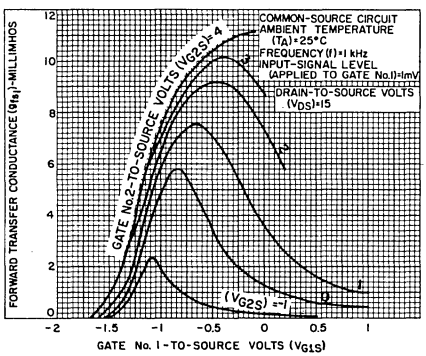


Fig.31 - g_{fs1} vs. V_{G1S}

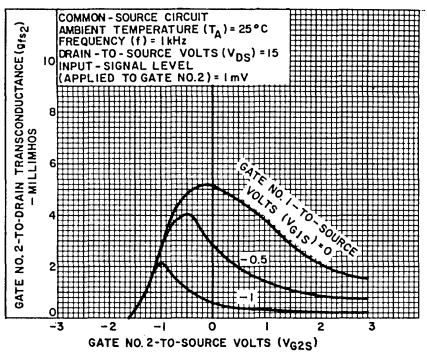
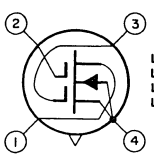


Fig.32 - g_{fs2} vs. V_{G2S}

TERMINAL DIAGRAM



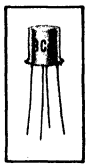
Lead 1 — Drain
Lead 2 — Gate No. 2
Lead 3 — Gate No. 1
Lead 4 — Source, Substrate and Case

RCA Solid State Division

MOS Field-Effect Transistors

40603
40604

SILICON DUAL INSULATED-GATE FIELD-EFFECT TRANSISTORS



TO-72

N-Channel Depletion Types For FM Tuner Applications

RCA 40603 and 40604 are n-channel silicon, depletion type, dual insulated-gate, field-effect transistors utilizing the MOS construction.

These devices have exceptional characteristics for rf-amplifier (40603) and mixer applications (40604) in FM tuners and other commercial equipment operating at frequencies up to approximately 150 MHz. These transistors feature a series arrangement of two separate channels, each channel having an independent control gate. For amplifier applications the 40603 with its wide dynamic range provides substantially better cross-modulation performance and relative freedom from spurious responses than is obtainable with bipolar or single-gate field-effect transistors. The mixing function performed by the 40604 is unique in that the signal applied to gate No.2 is used to modulate the input-gate (gate No.1) transfer characteristic. This technique is superior to conventional "square law" mixing, which can only be accomplished in the non-linear region of the device transfer characteristic.

Because of the low feedback capacitance (0.02 typ. pF) the 40603 can provide a power gain of 25 dB (typ.) at 100 MHz in an unneutralized amplifier circuit.

The gain of the rf stage can be controlled by applying agc voltage to gate No.2. Virtually no agc power is required for full gain reduction.

The 40603 and 40604 are hermetically sealed in JEDEC TO-72 packages.

Maximum Ratings, *Absolute-Maximum Values at* $T_A = 25^{\circ}C$:

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	0 to +20	V
GATE No.1-TO-SOURCE VOLTAGE, V_{G1S}:		
Continuous (dc)	-8 to +1	V
Peak ac	-8 to +20	V
GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:		
Continuous (dc)	-8 to 40% of V_{DS}	V
Peak ac	-8 to +20	V
DRAIN-TO-GATE VOLTAGE, V_{DG1} or V_{DG2}	+20	V
DRAIN CURRENT, I_D (Pulsed):		
Pulse duration $\leq$ 20 ms, duty factor $\leq$ 0.15	50	mA
TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	400	mW
At ambient temperatures above 25°C	derate linearly at 2.67 mW/°C	
AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
LEAD TEMPERATURE (During soldering):		
At distances $\geq$ 1/32" from seating surface for 10 seconds max.	265	°C

PERFORMANCE FEATURES

- large dynamic range permits large-signal handling before overload
- dual gates allow product mixing with extremely low harmonic generation
- greatly reduces spurious responses in FM receivers
- permits use of vacuum-tube biasing techniques
- excellent thermal stability
- superior cross-modulation performance and greater dynamic range than bipolar and single-gate field-effect transistors

DEVICE FEATURES

- extremely low feedback capacitance
  C_{rss} = 0.02 pF typ.
- high unneutralized RF power gain
  MUG = 25 dB (typ.) for 40603
- low noise figure
  NF = 2.5 dB typ. for 40603

ELECTRICAL CHARACTERISTICS, at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS				UNITS
			40603 RF AMPLIFIER		40604 MIXER		
			Typ.	Max.	Typ.	Max.	
Gate No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +15 V, I_D = 200 μA V_{G2S} = +4 V	-2	--	-2	--	V
Gate No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +15 V, I_D = 200 μA V_{G1S} = 0	-2	--	-2	--	V
Gate No.1 Leakage Current	I_{G1SS}	V_{G1S} = -20 V, V_{G2S} = 0, V_{DS} = 0	--	1	--	1	nA
Gate No.2 Leakage Current	I_{G2SS}	V_{G2S} = -20 V, V_{G1S} = 0, V_{DS} = 0	--	1	--	1	nA
Zero-Bias-Voltage Drain Current	I_{DSS}	V_{G2S} = +4 V, V_{G1S} = 0, V_{DS} = +13 V	18	--	18	--	mA
Small-Signal, Short-Circuit Reverse-Transfer Capacitance (Drain-to-Gate-No.1)	C_{rss}	V_{DS} = +13 V, I_D = 10 mA, f = 1 MHz V_{G2S} = +4 V	0.02	0.03	0.02	0.03	pF
Input Capacitance	C_{iss}	V_{DS} = +13 V, I_D = 10 mA V_{G2S} = +4 V, f = 1 MHz	5.5	--	5.5	--	pF
Output Capacitance	C_{oss}	V_{DS} = +13 V, I_D = 10 mA V_{G2S} = +4 V, f = 100 MHz	2.1	--	2.3	--	pF
Input Resistance	r_{is}	V_{DS} = +13 V, I_D = 10 mA V_{G2S} = +4 V, f = 100 MHz	3.5	--	3.5	--	kΩ
Output Resistance	r_{os}	V_{DS} = +13 V, I_D = 10 mA, V_{G2S} = +4 V; f = 100 MHz	4	--	--	--	kΩ
		V_{DS} = +13 V, I_D = 10 mA, V_{G2S} = +4 V; f = 10.7 MHz	--	--	20	--	kΩ
Forward Transconductance	g_{fs}	V_{DS} = +13 V, I_D = 10 mA V_{G2S} = +4 V, f = 1 kHz	10,000	--	2800*	--	μmho
Maximum Available Power Gain	MAG	V_{DS} = +13 V, I_D = 10 mA V_{G2S} = +4 V f = 100 MHz, f_{out} for 40604 (mixer) = 10.7 MHz	26	--	21	--	dB
Maximum Usable Power Gain (Unneutralized)	MUG		25▲	-	--	--	dB
Noise Figure	NF		2.5	--	--	--	dB

* conversion transconductance

▲ or limited by practical design considerations

TYPICAL Y-PARAMETER CHARACTERISTICS at 100 MHz

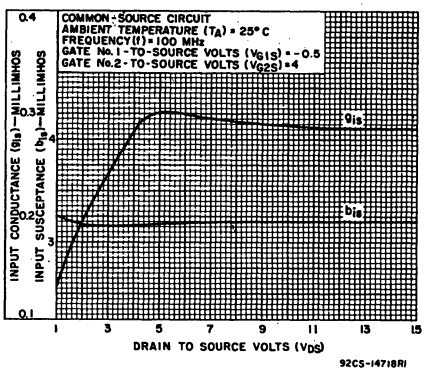


Fig.1 - Y_{is} vs. V_{DS}

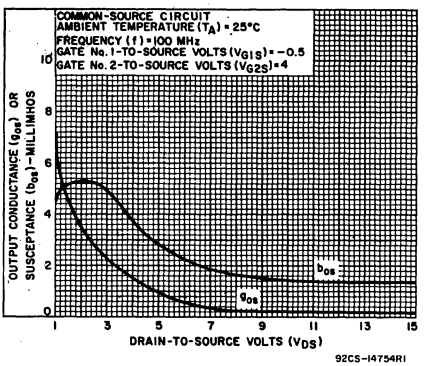


Fig.2 - Y_{os} vs. V_{DS}

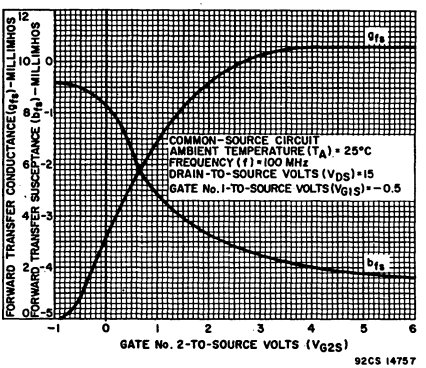


Fig.3 - Y_{fs} vs. V_{G2S}

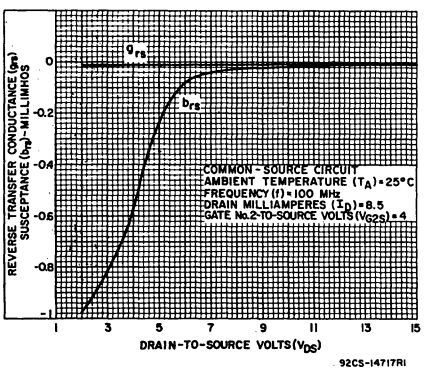


Fig.4 - Y_{rs} vs. V_{DS}

TYPICAL TRANSCONDUCTANCE CHARACTERISTIC

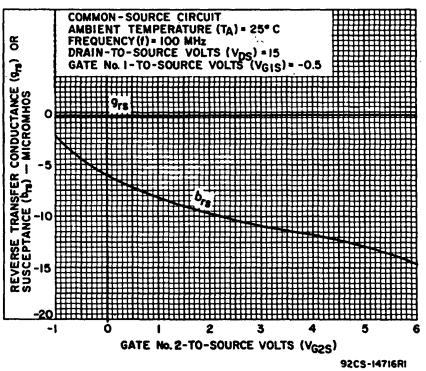


Fig.5 - Y_{rs} vs. V_{G2S}

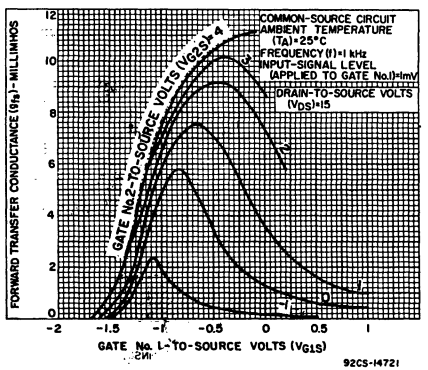


Fig.6 - Y_{fs} vs. V_{G1S}

TYPICAL Y-PARAMETER CHARACTERISTICS at 100 MHz

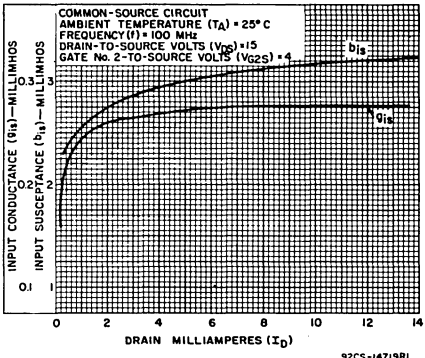


Fig.7 - Y_{is} vs. I_D

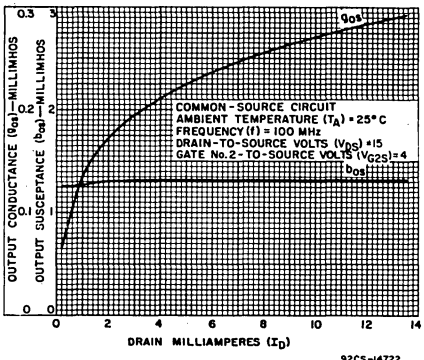


Fig.8 - Y_{os} vs. I_D

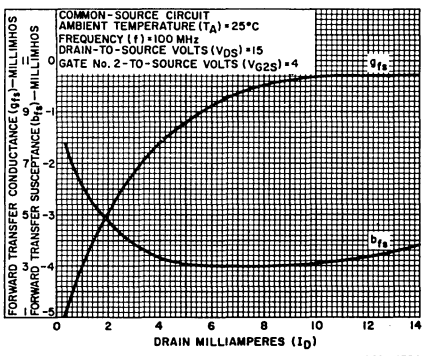


Fig.9 - Y_{fs} vs. I_D

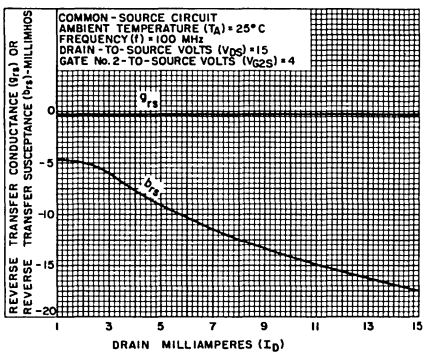
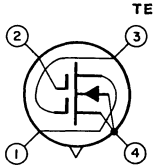


Fig.10 - Y_{rs} vs. I_D

TERMINAL DIAGRAM



Lead 1 — Drain
Lead 2 — Gate No. 2
Lead 3 — Gate No. 1
Lead 4 — Source, Substrate and Case

RCA Solid State Division

MOS Field-Effect Transistors

40673

RCA-40673 is an n-channel silicon, depletion type, dual insulated-gate field-effect transistor.

Special back-to-back diodes are diffused directly into the MOS* pellet and are electrically connected between each insulated gate and the FET's source. The diodes effectively bypass any voltage transients which exceed approximately ±10 volts. This protects the gates against damage in all normal handling and usage.

A feature of the back-to-back diode configuration is that it allows the 40673 to retain the wide input signal dynamic range inherent in the MOSFET. In addition, the low junction capacitance of these diodes adds little to the total capacitance shunting the signal gate.

The excellent overall performance characteristics of the RCA-40673 make it useful for a wide variety of rf-amplifier applications at frequencies up to 400 MHz. The two serially-connected channels with independent control gates make possible a greater dynamic range and lower cross-modulation than is normally achieved using devices having only a single control element.

The two gate arrangement of the 40673 also makes possible a desirable reduction in feedback capacitance by operating in the common-source configuration and ac-grounding Gate No. 2. The reduced capacitance allows operation at maximum gain *without neutralization;* and, of special importance in rf-amplifiers, it reduces local oscillator feedthrough to the antenna.

The 40673 is hermetically sealed in the metal JEDEC TO-72 package.

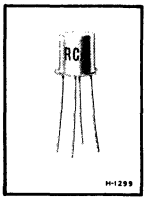
*Metal-Oxide-Semiconductor.

Maximum Ratings, *Absolute-Maximum Values, at T_A = 25°C*

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	-0.2 to +20	V
GATE No.1-TO-SOURCE VOLTAGE, V_{G1S}:		
Continuous (dc)	-6 to +1	V
Peak ac	-6 to +6	V
GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:		
Continuous (dc)	-6 to 30% of V_{DS}	V
Peak ac	-6 to +6	V
DRAIN-TO-GATE VOLTAGE, V_{DG1} OR V_{DG2}	+20	V
DRAIN CURRENT, I_D	50	mA
TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	330	mW
above 25°C	derate linearly at 2.2 mW/°C	
AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
LEAD TEMPERATURE (During soldering):		
At distances ≥1/32 inch from seating surface for 10 seconds max.	265	°C

SILICON DUAL INSULATED-GATE FIELD-EFFECT TRANSISTOR

N-Channel Depletion Type With Integrated Gate-Protection Circuits For RF Amplifier Applications up to 400 MHz



JEDEC TO-72

APPLICATIONS

- RF amplifier, mixer, and IF amplifier in military, industrial, and consumer communications equipment
- aircraft and marine vehicular receivers
- CATV and MATV equipment
- telemetry and multiplex equipment

PERFORMANCE FEATURES

- superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET s
- wide dynamic range permits large-signal handling before overload
- dual-gate permits simplified agc circuitry
- virtually no agc power required
- greatly reduces spurious responses in fm receivers
- permits use of vacuum-tube biasing techniques
- excellent thermal stability

DEVICE FEATURES

- back-to-back diodes protect each gate against handling and in-circuit transients
- low gate leakage currents —— I_{G1SS} & I_{G2SS} = 20 nA(max.) at T_A = 25°C
- high forward transconductance —— g_{fs} = 12,000 μmho (typ.)
- high unneutralized RF power gain —— G_{ps} = 18 dB(typ.) at 200 MHz
- low VHF noise figure —— 3.5 dB(typ.) at 200 MHz

ELECTRICAL CHARACTERISTICS, at T_A = 25°C unless otherwise specified

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
Gate-No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +15V, I_D = 200μA V_{G2S} = +4V	–	–2	–4	V
Gate-No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +15V, I_D = 200μA V_{G1S} = 0	–	–2	–4	V
Gate-No.1-Leakage Current	I_{G1SS}	V_{G1S} = +1 or–6 V V_{DS} = 0, V_{G2S} = 0	–	–	50	nA
Gate-No.2-Leakage Current	I_{G2SS}	V_{G2S} = ±6V V_{DS} = 0, V_{G1S} = 0	–	–	50	nA
Zero-Bias Drain Current	I_{DSS}	V_{DS} = +15V V_{G2S} = +4V V_{G1S} = 0	5	15	35	mA
Forward Transconductance (Gate-No.1-to-Drain)	g_{fs}	V_{DS} = +15V, I_D = 10mA V_{G2S} = +4V, f = 1kHz	–	12,000	–	μmho
Small-Signal, Short-Circuit Input Capacitance †	C_{iss}	V_{DS} = +15V, I_D = 10mA V_{G2S} = +4V, f=1MHz	–	6	–	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate No.1) ♦	C_{rss}		0.005	0.02	0.03	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}		–	2.0	–	pF
Power Gain (see Fig. 1)	G_{PS}	V_{DS} = +15V, I_D = 10mA V_{G2S} = +4V, f = 200 MHz	14	18	–	dB
Maximum Available Power Gain	MAG		–	20	–	dB
Maximum Usable Power Gain (unneutralized)	MUG		–	20*	–	dB
Noise Figure (see Fig. 1)	NF		–	3.5	6.0	dB
Magnitude of Forward Transadmittance	$\|Y_{fs}\|$		–	12,000	–	μmho
Phase Angle of Forward Trans-admittance	θ		–	–35	–	degrees
Input Resistance	r_{iss}		–	1.0	–	kΩ
Output Resistance	r_{oss}		–	2.8	–	kΩ
Protective Diode Knee Voltage	V_{knee}	$I_{DIODE(REVERSE)}$=±100μA	–	±10	–	V

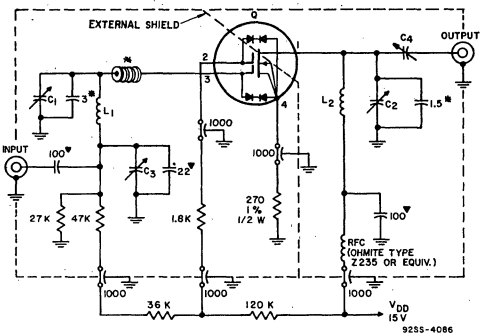
*Limited only by practical design considerations.

†Capacitance between Gate No. 1 and all other terminals

♦Three-terminal measurement with Gate No. 2 and Source returned to guard terminal.

OPERATING CONSIDERATIONS

The flexible leads of the 40673 are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons MUST be grounded.



#Ferrite bead (4); Pyroferric Co. "Carbonyl J" 0.09 in. OD; 0.03 in. ID; 0.063 in. thickness.

All resistors in ohms

All capacitors in pF

Q = 40673

▼ Disc ceramic.

*Tubular ceramic.

C_1: 1.8 – 8.7 pF variable air capacitor: E.F. Johnson Type 160-104, or equivalent.

C_2: 1.5 – 5 pF variable air capacitor: E.F. Johnson Type 160-102, or equivalent.

C_3: 1 – 10 pF piston-type variable air capacitor: JFD Type VAM-010; Johanson Type 4335, or equivalent.

C_4: 0.8 – 4.5 pF piston type variable air capacitor: Erie 560-013 or equivalent.

L_1: 4 turns silver-plated 0.02-in. thick, 0.075-0.085-in. wide, copper ribbon. Internal diameter of winding = 0.25 in, winding length approx. 0.80 in.

L_2: 4½ turns silver-plated 0.02-in. thick, 0.085-0.095-in. wide, 5/16-in. ID. Coil ≈ .90 in. long.

Fig. 1. 200-MHz Power gain and noise-figure test circuit

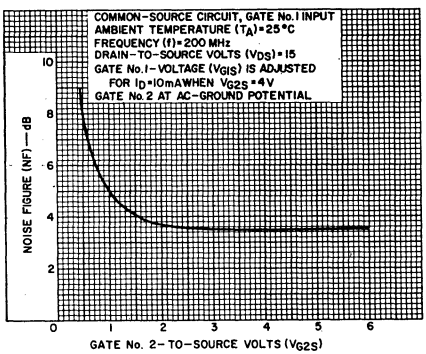


Fig. 2. NF vs. V_{G2S}

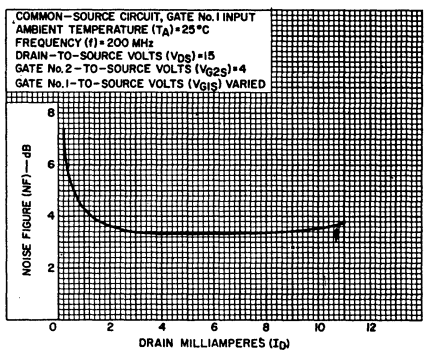


Fig. 3. NF vs. I_D

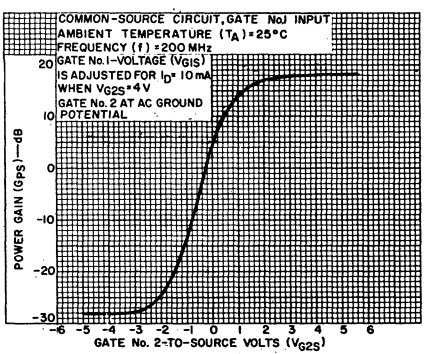


Fig. 4. G_{PS} vs. V_{G2S}

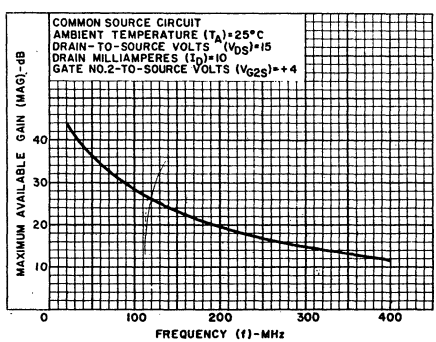


Fig. 5. MAG. vs. f

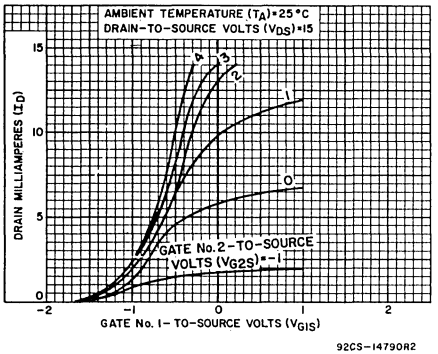


Fig. 6. I_D vs. V_{G1S}

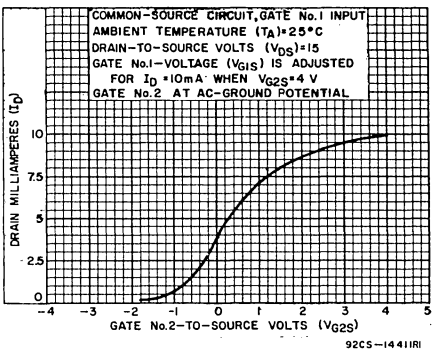


Fig. 7. I_D vs. V_{G2S}

Typical y Parameters vs. V_{DS}

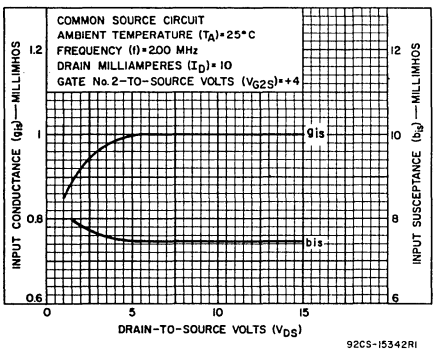


Fig. 8. y_{is} vs. V_{DS}

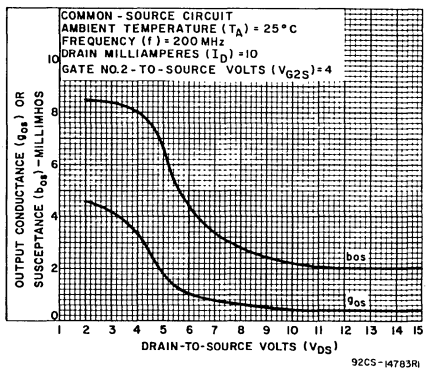


Fig. 9. y_{os} vs. V_{DS}

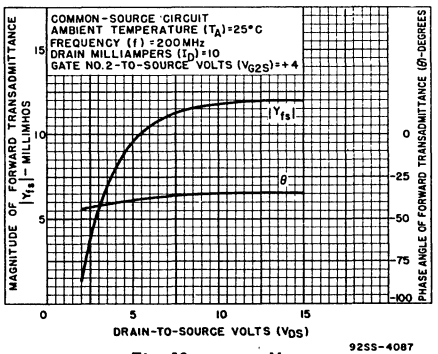


Fig. 10. y_{fs} vs. V_{DS}

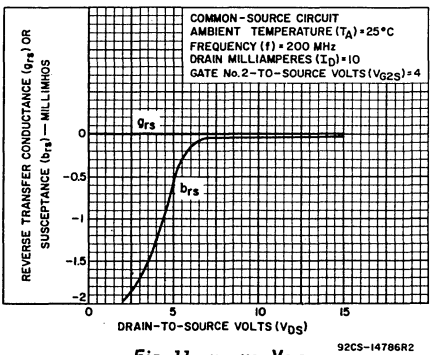


Fig. 11. y_{rs} vs. V_{DS}

Typical y Parameters vs. I_D

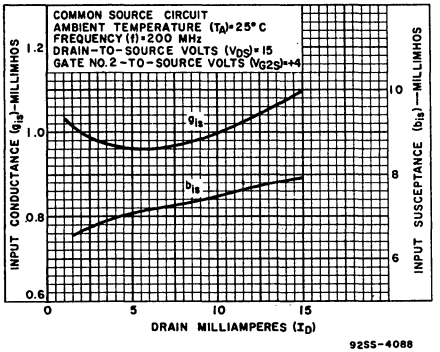


Fig. 12. y_{is} vs. I_D

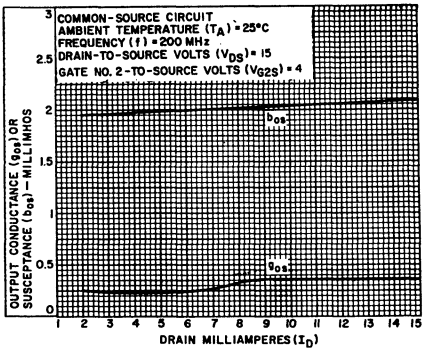


Fig. 13. y_{os} vs. I_D

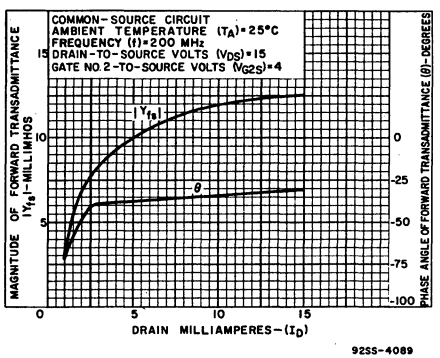


Fig. 14. y_{fs} vs. I_D

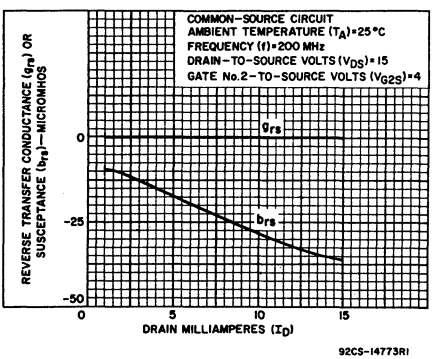


Fig. 15. y_{rs} vs. I_D

Typical y Parameters vs. V_{G2S}

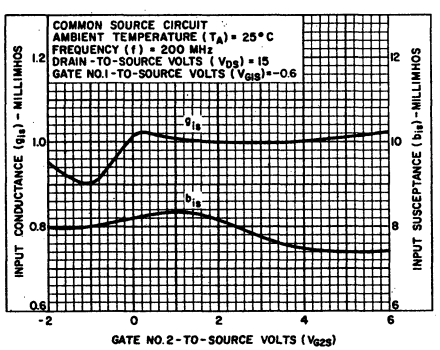


Fig. 16. y_{is} vs. V_{G2S}

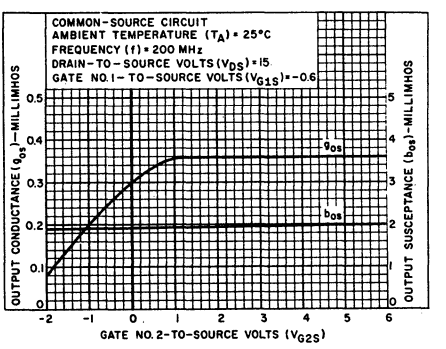


Fig. 17. y_{os} vs. V_{G2S}

Typical y Parameters vs. V_{G2S} (Cont'd)

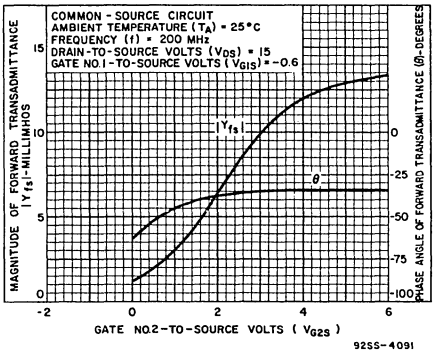


Fig. 18. y_{fs} vs. V_{G2S}

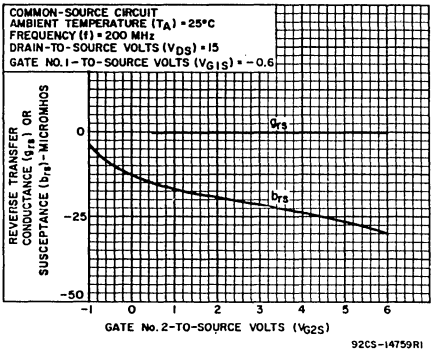


Fig. 19. y_{rs} vs. V_{G2S}

Typical y Parameters vs. Frequency

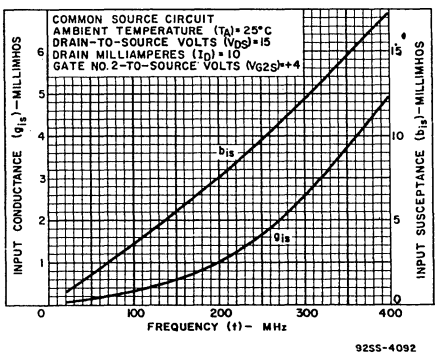


Fig. 20. y_{is} vs. frequency

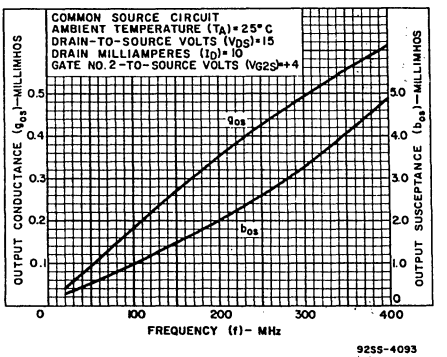


Fig. 21. y_{os} vs. frequency

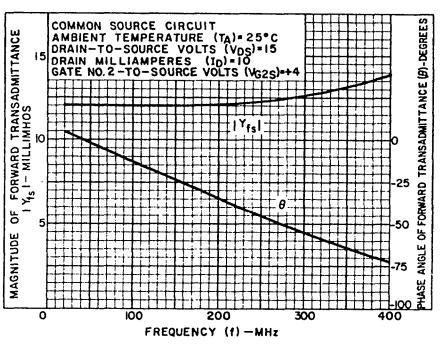


Fig. 22. y_{fs} vs. frequency

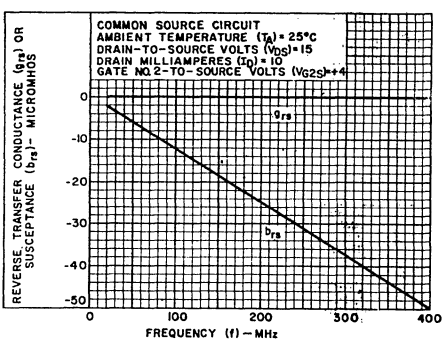


Fig. 23. y_{rs} vs. frequency

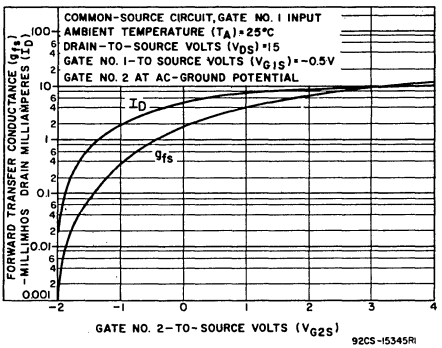


Fig. 24. g_{fs} and I_D vs. V_{G2S}

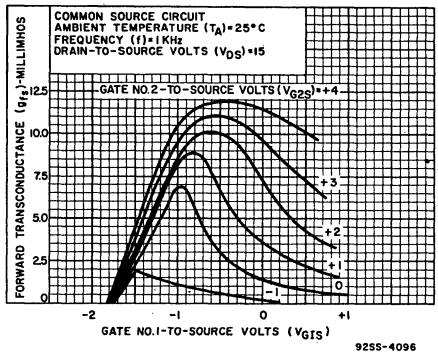


Fig. 25. g_{fs} vs. V_{G1S}

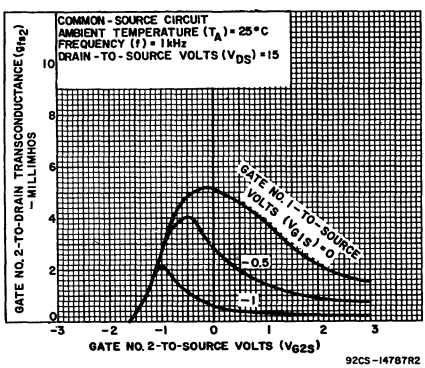
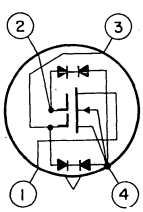


Fig. 26. g_{fs_2} vs. V_{G2S}

TERMINAL DIAGRAM



LEAD 1-DRAIN
LEAD 2-GATE No. 2
LEAD 3-GATE No. 1
LEAD 4-SOURCE, SUBSTRATE AND CASE

RCA
Solid State
Division

MOS Field-Effect Transistors

N-Channel Depletion Type

40819



H-1299

JEDEC TO-72

Silicon Dual-Insulated-Gate Field-Effect Transistor

With Integrated Gate-Protection Circuits

For RF Amplifier Applications up to 250 MHz

Device Features

- back-to-back diodes protect each gate against handling and in-circuit transients
- high forward transconductance: g_{fs} = 12,000 μmho (typ.)
- high unneutralized RF power gain: G_{ps} = 18 dB (typ.) at 200 MHz
- low VHF noise figure: 3.5 dB (typ.) at 200 MHz
- low gate leakage currents: I_{G1SS} & I_{G2SS} = 50 nA at T_A = 25° C
- increased drain-to-source voltage rating: V_{DS} = −0.2 to +25 V

RCA-40819 is an n-channel silicon, depletion type, dual insulated-gate field-effect transistor (FET).

The excellent overall performance characteristics of the RCA-40819 make it useful for a wide variety of rf-amplifier applications at frequencies up to 250 MHz. The two serially-connected channels with independent control gates make possible a greater dynamic range and lower cross-modulation than is normally achieved using devices having only a single control element.

The two-gate arrangement of the 40819 also makes possible a desirable reduction in feedback capacitance by operating in the common-source configuration and ac grounding Gate No.2. The reduced capacitance allows operation at maximum gain *without neutralization* and reduces local oscillator feedthrough to the antenna – features of special importance in rf and if amplifiers.

Special back-to-back diodes are diffused directly into the MOS* pellet and are electrically connected between each insulated gate and the FET's source. The diodes effectively bypass any voltage transients which exceed approximately ±10 volts and protect the gates against damage in all normal handling and usage.

The back-to-back diode configuration permits the 40819 to retain the wide input signal dynamic range inherent in the MOSFET. In addition, the low junction capacitance of these diodes adds little to the total capacitance shunting the signal gate.

Applications

- RF amplifier, mixer, and IF amplifier in military, industrial, and consumer communications equipment
- aircraft and marine vehicular receivers
- CATV and MATV equipment
- telemetry and multiplex equipment

Performance Features

- superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET s
- wide dynamic range permits large-signal handling before overload
- virtually no agc power required
- greatly reduces spurious responses in FM receivers
- dual gate permits simplified AGC circuitry

The 25-volt drain-to-source rating permits the use of higher voltage power supplies.

The 40819 is hermtically sealed in the metal JEDEC TO-72 package.

*Metal-Oxide-Semiconductor

Maximum Ratings

Continuous Working Voltage#, at T_A = 25°C:

Gate No.1-to-Source Voltage, V_{G1S}	−6 to +3	V
Gate No.2-to-Source Voltage, V_{G2S}	−6 to +6 or 40% of V_{DS} (whichever value is less)	V
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+25	V

#Continuous Working Voltage Ratings must be observed to maintain device characteristics. These ratings are based on long-term continuous voltage operation but may be exceeded for short durations (e.g. testing of device characteristics), provided the absolute Maximum Ratings are not exceeded.

Absolute Maximum Values, at T_A = 25°C:

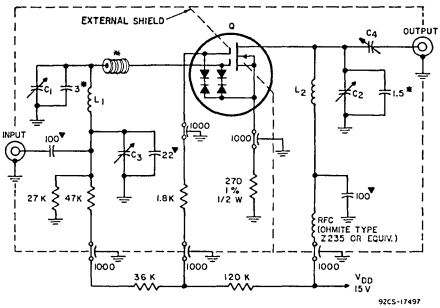
Drain-to-Source Voltage, V_{DS}	−0.2 to +25	V
Gate Terminal Current, I_{G1S} or I_{G2S}	±100	μA
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+31	V
Drain Current, I_D	50	mA
Transistor Dissipation, P_T:		
At T_A up to 25°C	330	mW
At T_A above 25°C	derate linearly 2.2 mW/°C	
Ambient Temperature Range:		
Operating and Storage	−65 to +175	°C
Lead Temperature (During Soldering):		
At distances 1/32 in from seating surface for 10 s max.	265	°C

ELECTRICAL CHARACTERISTICS, at T_A = 25° C unless otherwise specified

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
Gate-No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +15 V, I_D = 200 μA, V_{G2S} = +4 V	–	−2	−4	V
Gate-No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +15 V, I_D = 200 μA, V_{G1S} = 0	–	−2	−4	V
Gate-No.1-Leakage Current	I_{G1SS}	V_{G1S} = ± 6 V, V_{DS} = 0, V_{G2S} = 0	–	–	50	nA
Gate-No.2-Leakage Current	I_{G2SS}	V_{G2S} = ± 6 V, V_{DS} = 0, V_{G1S} = 0	–	–	50	nA
Zero-Bias Drain Current	I_{DSS}	V_{DS} = + 15 V, V_{G2S} = +4 V, V_{G1S} = 0	5	15	35	mA
Forward Transconductance (Gate-No.1-to-Drain)	g_{fs}	V_{DS} = +15 V, I_D = 10 mA, V_{G2S} = +4 V, f = 1 kHz	–	12,000	–	μmho
Small-Signal, Short-Circuit Input Capacitance†	C_{iss}	V_{DS} = +15 V, I_D = 10 mA, V_{G2S} = +4 V, f = 1 MHz	–	6	–	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate No.1)♦	C_{rss}		0.005	0.02	0.03	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}		–	2	–	pF
Power Gain (see Fig. 1)	G_{PS}	V_{DS} = +15 V, I_D = 10 mA, V_{G2S} = +4 V, f = 200 MHz	14	18	–	dB
Maximum Available Power Gain	MAG		–	20	–	dB
Maximum Usable Power Gain (unneutralized)	MUG		–	20*	–	dB
Noise Figure (see Fig. 1)	NF		–	3.5	6.0	dB
Magnitude of Forward Transadmittance	$\|Y_{fs}\|$		–	12,000	–	μmho
Phase Angle of Forward Transadmittance	θ		–	−35	–	degrees
Input Resistance	r_{iss}		–	1	–	kΩ
Output Resistance	r_{oss}		–	2.8	–	kΩ
Protective Diode Knee Voltage	V_{knee}	I_{diode} (reverse) = ±100 μA	–	±10	–	V

* Limited only by practical design considerations. † Capacitance between Gate No.1 and all other terminals.

♦ Three-terminal measurement with Gate No.2 and Source returned to guard terminal.



#Ferrite bead (4); Pyroferric Co. "Carbonyl J" 0.09 in OD; 0.03 in ID; 0.063 in thickness.

Q = 40673
▼Disc ceramic.
* Tubular ceramic.

All resistors in ohms

All capacitors in pF

C_1: 1.8 – 8.7 pF variable air capacitor: E. F. Johnson Type 160-104, or equivalent.

C_2: 1.5 – 5 pF variable air capacitor: E. F. Johnson Type 160-102, or equivalent.

C_3: 1 – 10 pF piston-type variable air capacitor: JFD Type VAM-010; Johanson Type 4335, or equivalent.

C_4: 0.8 – 4.5 pF piston type variable air capacitor: Erie 560-013 or equivalent.

L_1: 4 turns silver-plated 0.02-in thick, 0.075-0.085 in wide, copper ribbon. Internal diameter of winding = 0.25 in, winding length approx. 0.80 in.

L_2: 4-1/2 turns silver-plated 0.02 in thick, 0.085-0.095-in wide, 5/16-in; ID Coil = .90 in long.

Fig. 1. 200 MHz power gain and noise figure test circuit

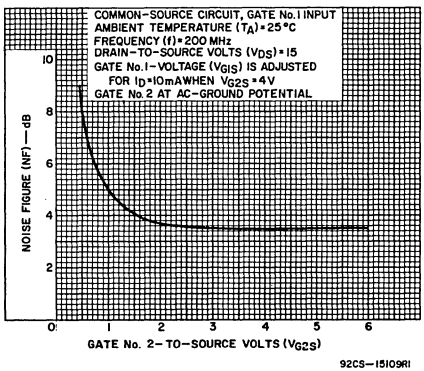


Fig. 2. NF vs. V_{G2S}

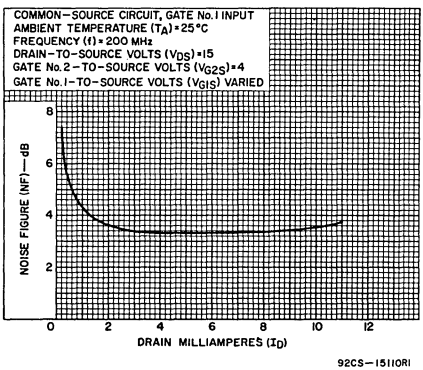


Fig. 3. NF vs. I_D

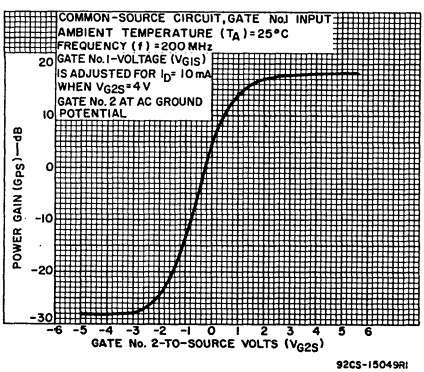


Fig. 4. G_{PS} vs. V_{G2S}

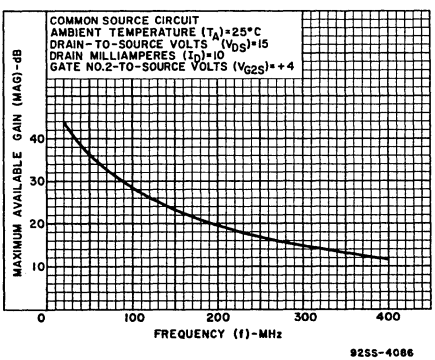


Fig. 5. MAG vs. f

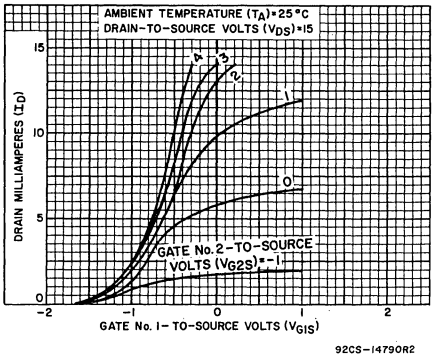


Fig. 6. I_D vs. V_{G1S}

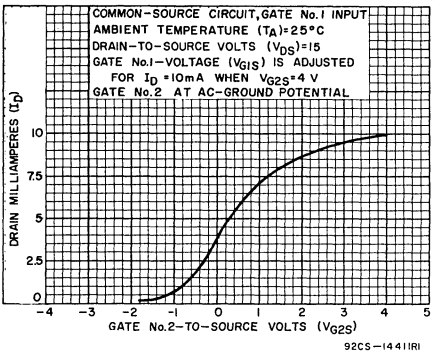


Fig. 7. I_D vs. V_{G2S}

Typical y Parameters vs. V_{DS}

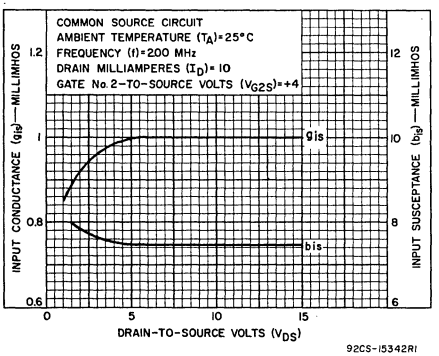


Fig. 8. y_{is} vs. V_{DS}

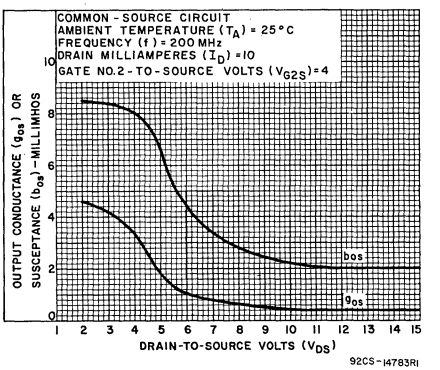


Fig. 9. y_{os} vs. V_{DS}

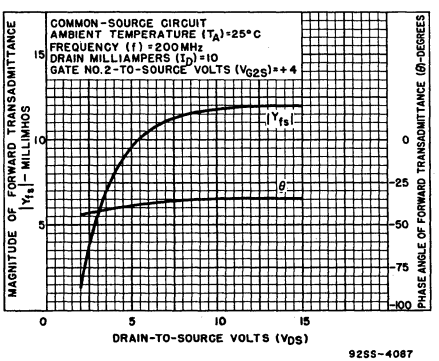


Fig. 10. y_{fs} vs. V_{DS}

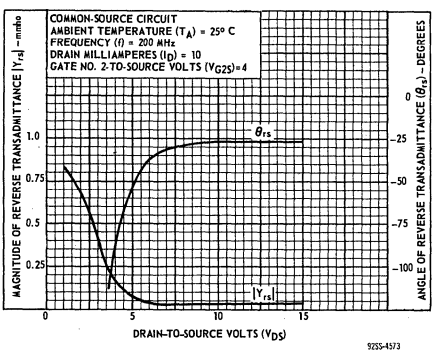


Fig. 11. y_{rs} vs. V_{DS}

Typical y Parameters vs. I_D

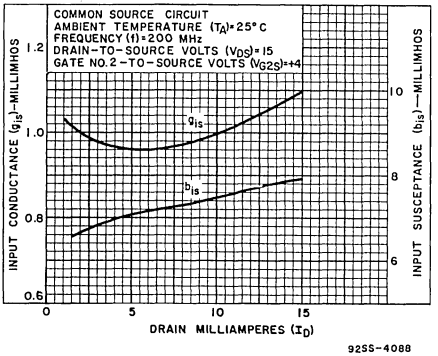


Fig. 12. y_{is} vs. I_D

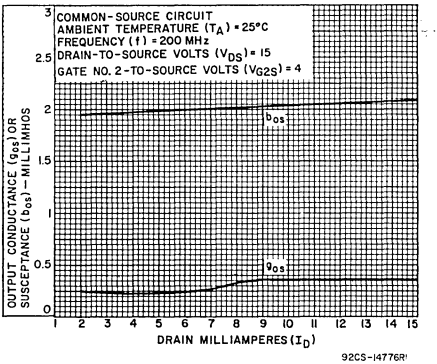


Fig. 13. y_{os} vs. I_D

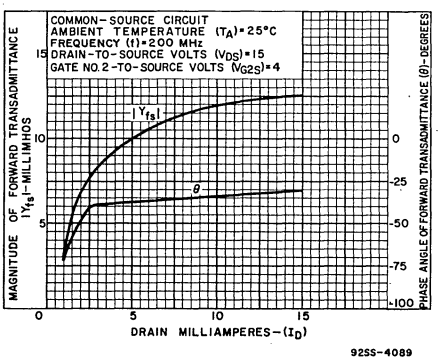


Fig. 14. y_{fs} vs. I_D

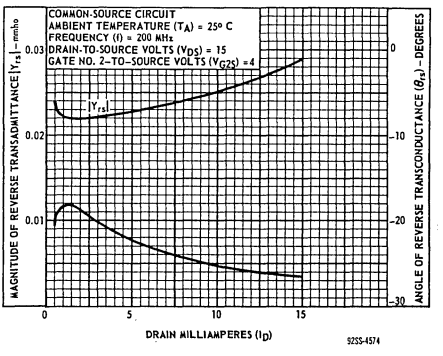


Fig. 15. y_{rs} vs. I_D

Typical y Parameters vs. V_{G2S}

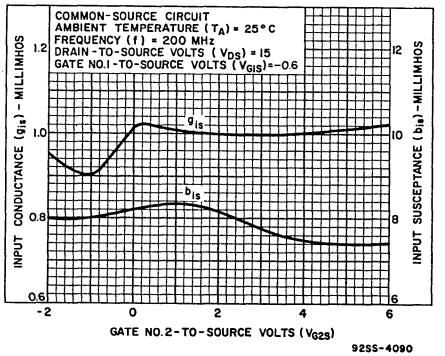


Fig. 16. y_{is} vs. V_{G2S}

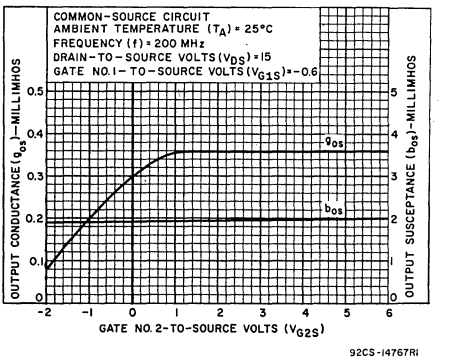


Fig. 17. y_{os} vs. V_{G2S}

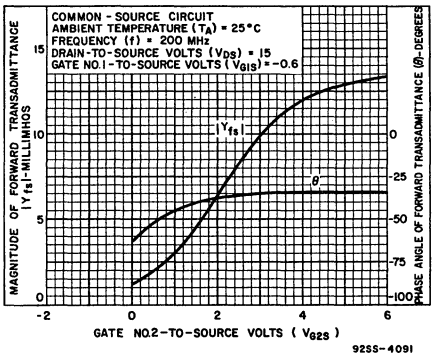


Fig. 18. y_{fs} vs. V_{G2S}

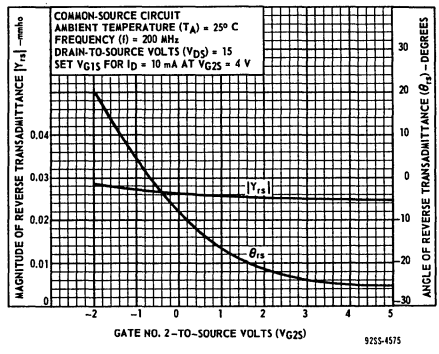


Fig. 19. y_{rs} vs. V_{G2S}

Typical y Parameters vs. Frequency

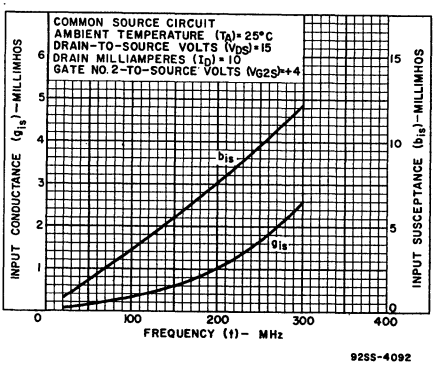


Fig. 20. y_{is} vs. frequency

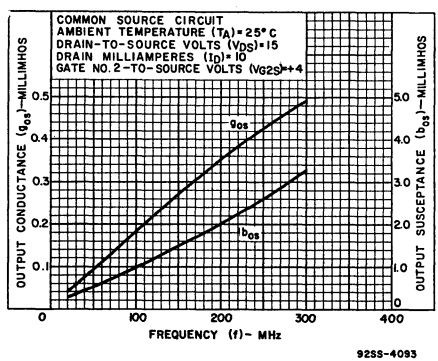


Fig. 21. y_{os} vs. frequency

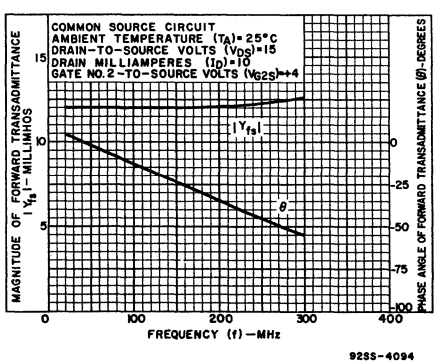


Fig. 22. y_{fs} vs. frequency

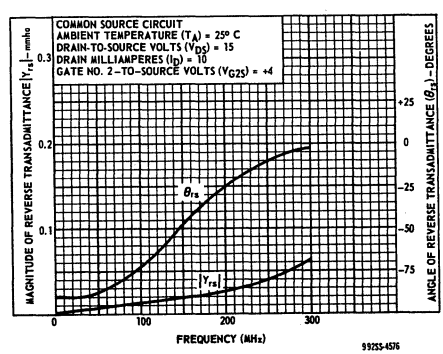


Fig. 23. y_{rs} vs. frequency

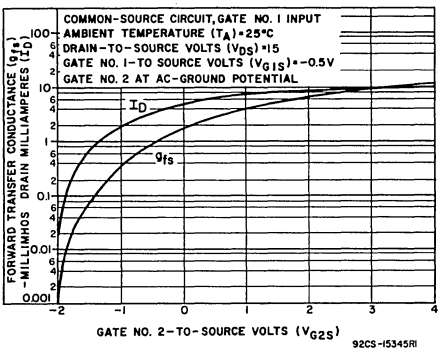


Fig. 24 g_{fs} and I_D vs. V_{G2S}

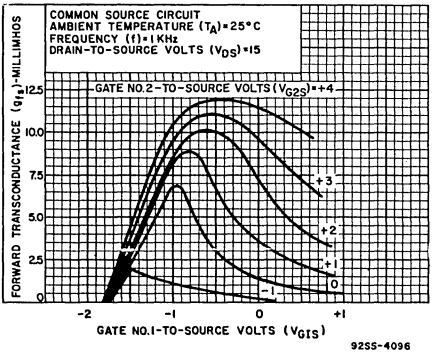


Fig. 25. g_{fs} vs. V_{G1S}

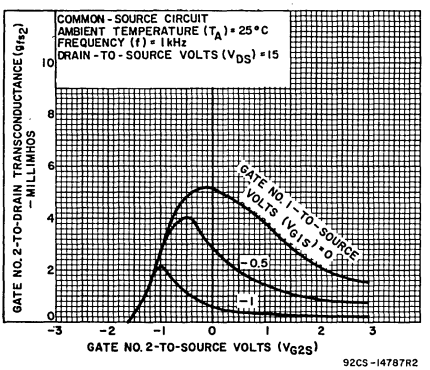
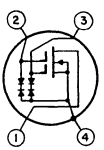


Fig. 26. g_{fs2} vs. V_{G2S}

OPERATING CONSIDERATIONS

The flexible leads of the 40819 are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons MUST be grounded.

TERMINAL DIAGRAM



LEAD 1 - DRAIN
LEAD 2 - GATE No.2
LEAD 3 - GATE No.1
LEAD 4 - SOURCE, SUBSTRATE, AND CASE

RCA
Solid State
Division

MOS Field-Effect Transistors

N-Channel Depletion Types

40820—40821



JEDEC TO-72 H-1299

Silicon Dual-Insulated-Gate Field-Effect Transistors

With Integrated Gate-Protection Circuits
For VHF-TV Tuner Applications

40820 — RF Amplifier **40821** — Mixer

Device Features

- back-to-back diodes protect each gate against handling and in-circuit transients
- high forward transconductance: g_{fs} = 12,000 μmho (typ.)
- high unneutralized RF power gain: G_{ps} = 17 dB (typ.) at 200 MHz (40820)
- low VHF noise figure: 3.5 dB (typ.) at 200 MHz (40820)
- low gate leakage currents: I_{G1SS} & I_{G2SS} = 50 nA

RCA-40820 and 40821 are n-channel silicon, depletion type, dual-insulated-gate, MOS▲ field-effect transistors for RF amplifier (40820) and mixer (40821) applications in VHF-TV receivers and other commercial equipment operating at frequencies up to 250 MHz.

These devices designed for VHF performance, provide excellent power gain, low-noise figures and have wide dynamic range. The dual-gate feature offers good cross-modulation performance over the AGC range and reduces feedback capacitance by shielding Gate No. 1 from the drain. The very low feedback capacitance also eliminates the need for circuit neutralization and reduces local oscillator feed-through to the antenna.

Virtually no AGC power is required because of the high gate input resistance of the MOS FET types. Automatic AGC delay can be achieved with a very slight change in the input impedance by the application of AGC voltage to Gate No. 2.

▲ Metal-Oxide-Semiconductor.

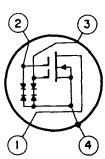
Performance Features

- superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET s
- wide dynamic range permits large-signal handling before overload
- virtually no agc power required
- dual gate permits simplified AGC circuitry

The dual-gate arrangement also makes it possible to isolate the local oscillator signal from the incoming signal by applying each signal to a separate gate.

Integral back-to-back diodes protect the gates against damage in normal handling and usage by limiting transient voltages that exceed ±10 volts. The 40820 and 40821 are hermetically sealed in metal JEDEC TO-72 packages.

TERMINAL DIAGRAM



LEAD 1 — DRAIN
LEAD 2 — GATE No.2
LEAD 3 — GATE No.1
LEAD 4 — SOURCE, SUBSTRATE, AND CASE

Maximum Ratings

Continuous Working Voltage#, at $T_A = 25°C$:	40820	40821	
Gate No. 1-to-Source Voltage, V_{G1S}	−6 to +3	−4.5 to +3	V
Gate No. 2-to-Source Voltage, V_{G2S}	-6 to +6 or 40% of V_{DS} (whichever value is less)	-4.5 to +4.5 or -4.5 to 40% of V_{DS} (whichever value is less)	V
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+20	+20	V
Absolute Maximum Values, at $T_A = 25°C$:			
Drain-to-Source Voltage, V_{DS}	−0.2 to +20	−0.2 to +20	V
Gate Terminal Current, I_{G1S} or I_{G2S}	±100	±100	μA
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+26	+24.5	V
Drain Current, I_D	50	50	mA
Transistor Dissipation:			
At T_A up to 25°C	330	330	mW
At T_A above 25°C	derate linearly 2.2 mW/°C		
Ambient Temperature Range:			
Operating and Storage	−65 to +175	−65 to +175	°C
Lead Temperature (During Soldering):			
At distances 1/32 in from seating surface for 10 s max.	265	265	°C

Continuous Working Voltage Ratings must be observed to maintain device characteristics. These ratings are based on long-term continuous voltage operation but may be exceeded for short durations (e.g. testing of device characteristics), provided the Absolute Maximum Ratings are not exceeded.

TYPICAL CHARACTERISTICS

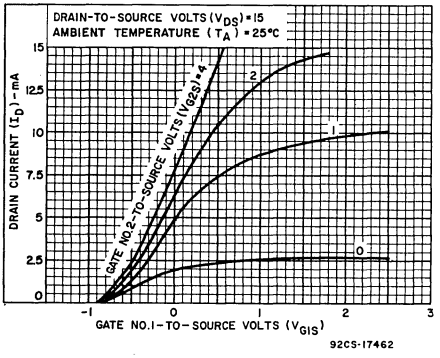


Fig.1 – I_D vs. V_{G1S} for types 40820 and 40821.

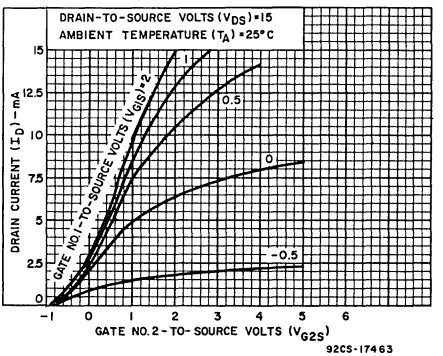


Fig.2 – I_D vs. V_{G2S} for types 40820 and 40821.

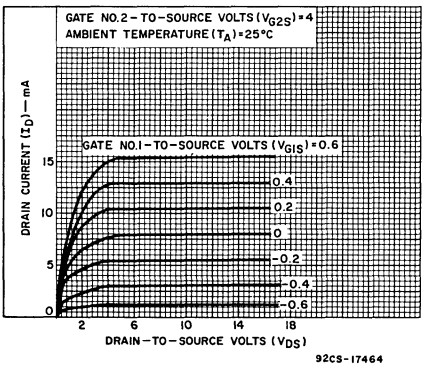


Fig.3 – I_D vs. V_{DS} for types 40820 and 40821.

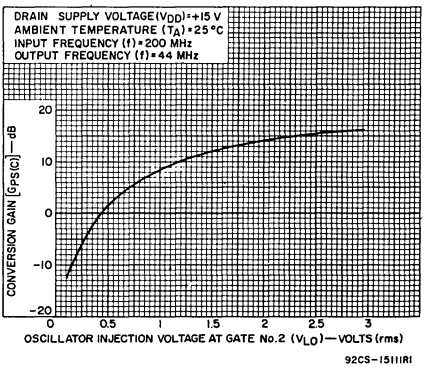


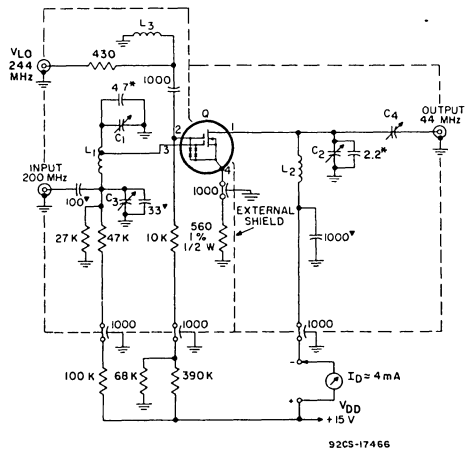
Fig.4 – $G_{PS(C)}$ vs. V_{LO} for type 40821.

ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		LIMITS 40820 Min.	40820 Typ.	40820 Max.	40821 Min.	40821 Typ.	40821 Max.	UNITS
Gate No. 1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS}= +15V, I_D=200μA, V_{G2S}= +4V		–	–1	–3	–	–1	–3	V
Gate No. 2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS}= +15V, I_D=200μA, V_{G1S}=0		–	–1	–3	–	–1	–3	V
Gate-to-Source Forward Breakdown Voltage: Gate No. 1	$V_{(BR)G1SSF}$	I_{G1SSF} I_{G2SSF} 100 μA	V_{G2S} V_{DS} 0	–	9	–	–	11	–	V
Gate No. 2	$V_{(BR)G2SSF}$		V_{G1S} V_{DS} 0	–	9	–	–	11	–	V
Gate-to-Source Reverse Breakdown Voltage: Gate No. 1	$V_{(BR)G1SSR}$	I_{G1SSR} I_{G2SSR} 100 μA	V_{G2S} V_{DS} 0	–	9	–	–	11	–	V
Gate No. 2	$V_{(BR)G2SSR}$		V_{G1S} V_{DS} 0	–	9	–	–	11	–	V
Gate No. 1-Terminal Forward Current	I_{G1SSF}	V_{DS} V_{G2S} 0	V_{G1S} 6 V	–	–	50	–	–	–	nA
			V_{G1S} 4.5 V	–	–	–	–	–	50	nA
Gate No. 1-Terminal Reverse Current	I_{G1SSR}	V_{DS} V_{G2S} 0	V_{G1S} –6 V	–	–	50	–	–	–	nA
			V_{G1S} –4.5 V	–	–	–	–	–	50	nA
Gate No. 2-Terminal Forward Current	I_{G2SSF}	V_{DS} V_{G1S} 0	V_{G2S} 6 V	–	–	50	–	–	–	nA
			V_{G2S} 4.5 V	–	–	–	–	–	50	nA
Gate No. 2-Terminal Reverse Current	I_{G2SSR}	V_{DS} V_{G1S} 0	V_{G2S} –6 V	–	–	50	–	–	–	nA
			V_{G2S} –4.5 V	–	–	–	–	–	50	nA
Zero-Bias Drain Current	I_{DS}	V_{DS} +15 V, V_{G1S} 0, V_{G2S} +4 V		0.5	8	15	0.5	8	20	mA
Forward Transconductance (Gate No. 1-to-Drain)	g_{fs}	V_{DS} +15 V I_D 10 mA V_{G2S} +4 V	f 1 kHz	–	12000	–	–	12000	–	μmho
Small-Signal, Short-Circuit Input Capacitance♦	C_{iss}		f 1 MHz	–	6	8.5	–	6	9	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate-No. 1)♦	C_{rss}			0.005	0.02	0.03	0.005	0.02	0.04	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}			–	2	–	–	2	–	pF
Power Gain (see Fig. 6)	G_{PS}			14	17	–	–	–	–	dB
Noise Figure (see Fig. 6)	NF		f 200 MHz	–	4.5	6	–	–	–	dB
Conversion Gain	$G_{PS(C)}$		f 200/44 MHz	–	–	–	11	–	–	dB

♦ Capacitance between Gate No. 1 and all other terminals.

♦ Three-terminal measurement with Gate No. 2 and Source returned to guard terminal.



Q = 40821
▼Disc. ceramic.
* Tubular ceramic.
All resistors in ohms
All capacitors in pF

C_1, C_2: 1.5–5 pF variable air capacitor: E.F. Johnson Type 160-102 or equivalent.

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent.

C_4: 0.9–7 pF compression-type capacitor: ARCO 400 or equivalent.

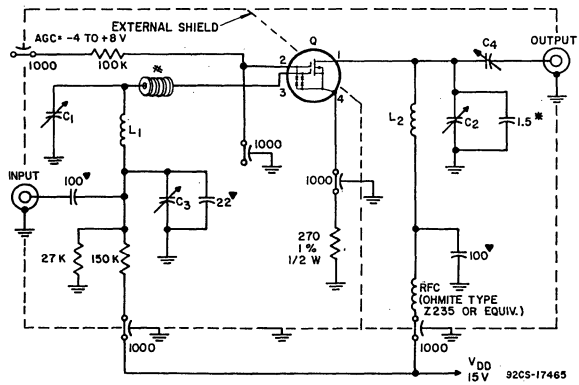
L_1: 5 turns silver-plated 0.02" thick, 0.07"–0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding.

L_2: Ohmite Z-235 RF choke or equivalent

L_3: J. W. Miller Co. #4580 0.1 μH RF choke or equivalent.

Note: If 50Ω meter is used in place of sweep detector, a low pass filter must be provided to eliminate local oscillator voltage from load.

Fig. 5 – Conversion power gain test circuit for type 40821.



#Ferrite bead (4); Pyroferric Co. "Carbonyl J" 0.09 in OD; 0.03 in ID; 0.063 in thickness.

Q = 40820
▼ Disc ceramic.
* Tubular ceramic.

All resistors in ohms

All capacitors in pF

C_1: 1.8 – 8.7 pF variable air capacitor: E. F. Johnson Type 160-104, or equivalent.

C_2: 1.5 – 5 pF variable air capacitor: E. F. Johnson Type 160-102, or equivalent.

C_3: 1 – 10 pF piston-type variable air capacitor: JFD Type VAM-010; Johanson Type 4335, or equivalent.

C_4: 0.8 – 4.5 pF piston type variable air capacitor: Erie 560-013 or equivalent.

L_1: 4 turns silver-plated 0.02-in thick, 0.075-0.085 in wide, copper ribbon. Internal diameter of winding = 0.25 in, winding length approx. 0.80 in.

L_2: 4-1/2 turns silver-plated 0.02 in thick, 0.085-0.095-in wide, 5/16-in; ID Coil ≈0.90 in. long.

Fig. 6 – 200 MHz power gain and noise figure test circuit for type 40820.

Table 1 – y parameters vs. frequency

CHARACTERISTICS	SYMBOL	FREQUENCY (MHz)				UNITS
		50	100	200	250	
Y Parameters						
Input Conductance	g_{is}	0.08	0.33	1.0	1.6	mmho
Input Susceptance	b_{is}	1.8	3.6	7.5	9.8	mmho
Magnitude Forward Transadmittance	$\|y_{fs}\|$	12	12	12	12.3	mmho
Angle of Forward Transadmittance	$<y_{fs}$	–2	–13	–35	–45	degrees
Output Conductance	g_{os}	0.10	0.18	0.36	0.42	mmho
Output Susceptance	b_{os}	0.5	1.0	2.0	2.6	mmho
Magnitude of Reverse Transadmittance	$\|y_{rs}\|$	8	12	25	40	μmho
Angle of Reverse Transadmittance	$<y_{rs}$	–88	–73	–25	–10	degrees

TEST CONDITIONS: Drain-to-Source Volts (V_{DS}) = 15, Drain Milliamperes (I_D) = 10, Gate No.2-to-Source Volts (V_{G2S}) = 4

TYPICAL CHARACTERISTICS

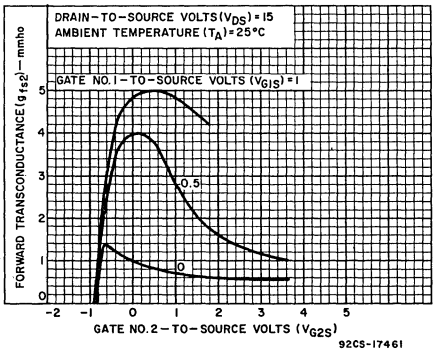


Fig. 7 – g_{fs} vs. V_{G2S} for types 40820 and 40821.

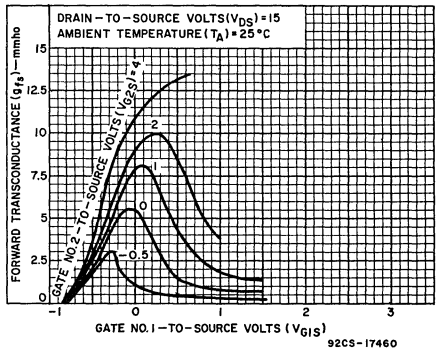


Fig. 8 – g_{fs} vs. V_{G1S} for types 40820 and 40821.

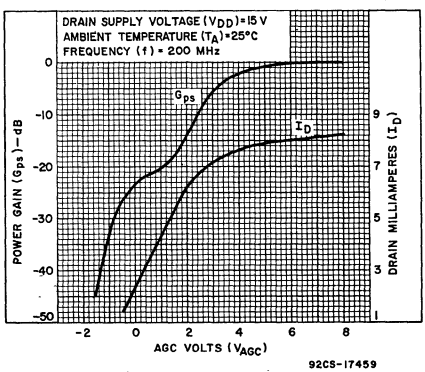


Fig. 9 – G_{PS} vs. V_{AGC} for type 40820.

OPERATING CONSIDERATIONS

The flexible leads of the 40820 and 40821 are usually soldered to the circuit elements. As is the case with any high-frequency semiconductor device, the tips of soldering irons MUST be grounded.

TYPICAL y PARAMETERS

y parameters vs. V_{DS}

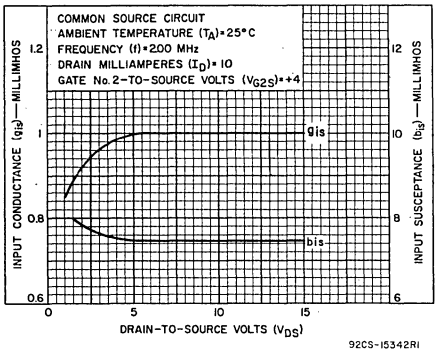


Fig. 10 – y_{is} vs. V_{DS}

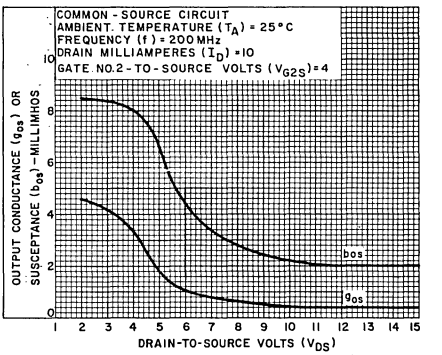


Fig. 11 – y_{os} vs. V_{DS}

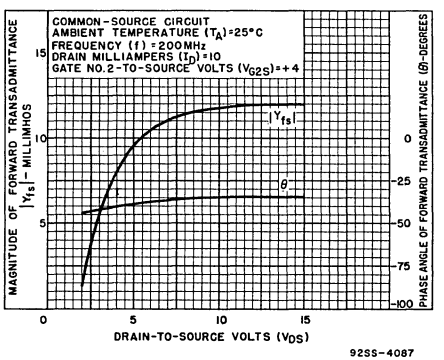


Fig. 12 – y_{fs} vs. V_{DS}

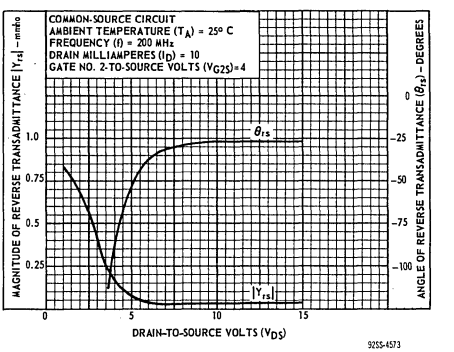


Fig. 13 – y_{rs} vs. V_{DS}

y parameters vs. I_D

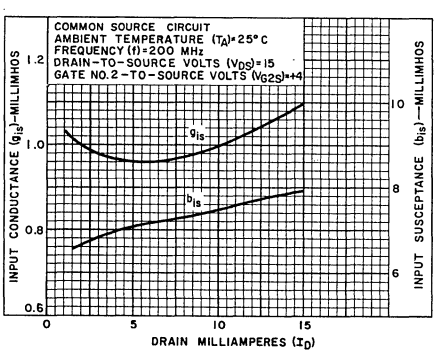


Fig. 14 – y_{is} vs. I_D

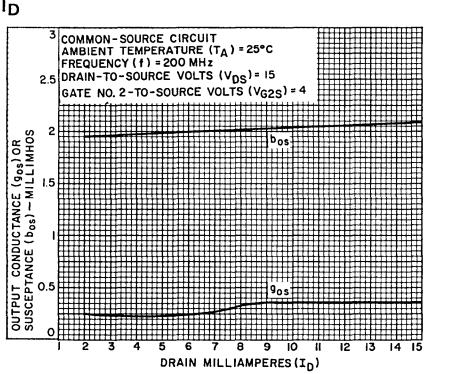


Fig. 15 – y_{os} vs. I_D

TYPICAL y PARAMETERS

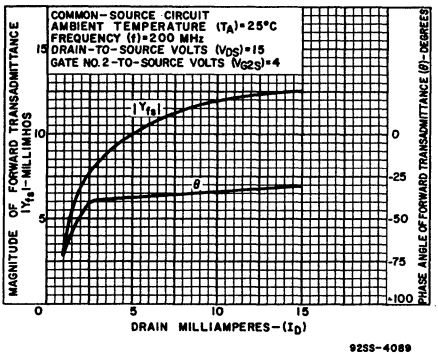


Fig. 16 – y_{fs} vs. I_D

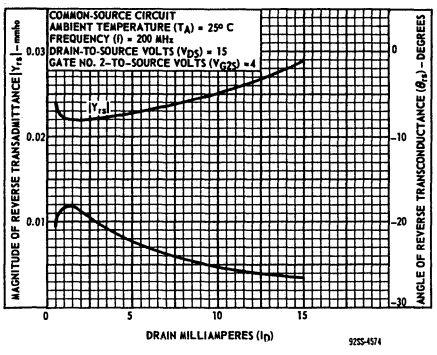


Fig. 17 – y_{rs} vs. I_D

y parameters vs. V_{G2S}

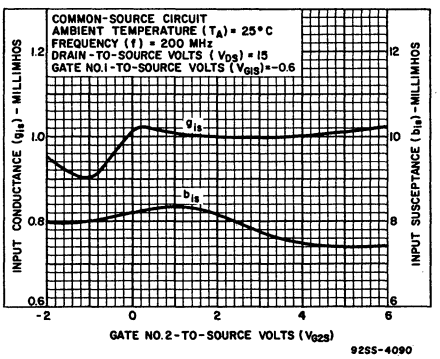


Fig. 18 – y_{is} vs. V_{G2S}

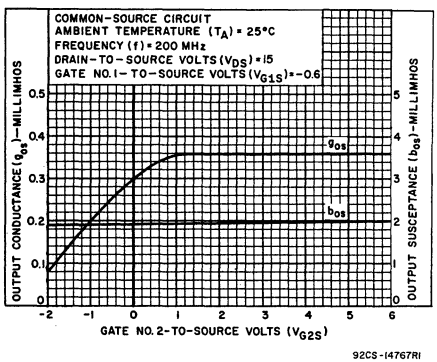


Fig. 19 – y_{os} vs. V_{G2S}

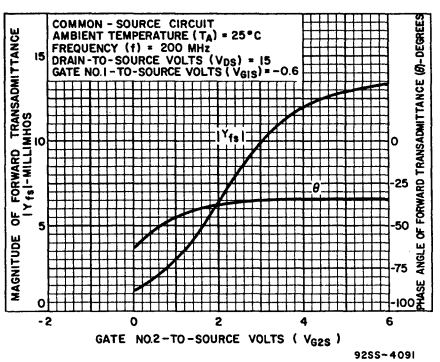


Fig. 20 – y_{fs} vs. V_{G2S}

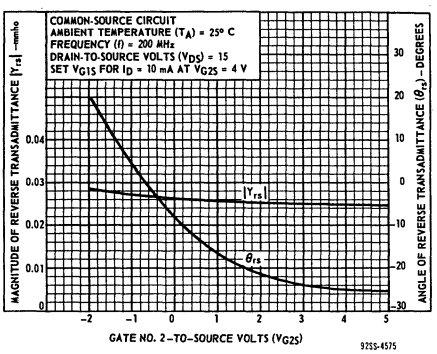


Fig. 21 – y_{rs} vs. V_{G2S}

RCA
Solid State
Division

MOS Field-Effect Transistors

N-Channel Depletion Types

40822-40823



JEDEC TO-72 H-1299

Silicon Dual-Insulated-Gate Field-Effect Transistors

With Integrated Gate-Protection Circuits

For FM Tuner Applications

40822 – RF Amplifier 40823 – Mixer

Device Features

- back-to-back diodes protect each gate against handling and in-circuit transients
- high forward transconductance: g_{fs} = 12,000 μmho (typ.)
- high unneutralized RF power gain: G_{ps} = 24 dB (typ.) at 100 MHz (40822)
- low VHF noise figure: 2 dB (typ.) at 100 MHz (40822)
- low gate leakage currents: I_{G1SS} & I_{G2SS} = 50 nA at T_A = 25°C

RCA-40822 and 40823 are n-channel silicon, depletion type, dual-insulated-gate, field-effect transistors for RF amplifier (40822) and mixer (40823) applications in FM receivers and other commercial equipment operating at frequencies up to 150 MHz.

These devices designed for VHF performance, provide excellent power gain, low-noise figures and have wide dynamic range. The dual-gate feature offers good cross-modulation performance over the AGC range and reduces feedback capacitance by shielding Gate No. 1 from the drain. The very low feedback capacitance also eliminates the need for circuit neutralization and reduces local oscillator feed-through to the antenna.

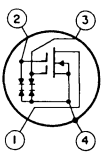
Virtually no power is required in AGC utilizing the 40822 and 40823. In addition, these devices minimize input impedance variations and automatically achieve AGC delay when AGC is applied to Gate No. 2. The dual-gate arrangement also makes it possible to isolate the local oscillator signal from the incoming signal by applying each signal to a specific gate.

Performance Features

- superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET's
- wide dynamic range permits large-signal handling before overload
- virtually no agc power required
- greatly reduces spurious responses in FM receivers
- dual gate permits simplified AGC circuitry

Back-to-back diodes, diffused directly into the MOS pellet, protect the gates against damage in normal handling and usage by limiting transient voltages that exceed +10 volts. The 40822 and 40823 are hermetically sealed in metal JEDEC TO-72 packages.

TERMINAL DIAGRAM



LEAD 1 – DRAIN
LEAD 2 – GATE No.2
LEAD 3 – GATE No.1
LEAD 4 – SOURCE, SUBSTRATE AND CASE

Maximum Ratings

Continuous Working Voltage[#], *at* $T_A = 25°C$:	40822	40823	
Gate No. 1-to-Source Voltage, V_{G1S}	−6 to +3	−4.5 to +3	V
Gate No. 2-to-Source Voltage, V_{G2S}	-6 to +6 or 40% of V_{DS} (whichever value is less)	-4.5 to +4.5 or 40% of V_{DS} (whichever value is less)	V
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+20	+20	V
Absolute Maximum Values, at $T_A = 25°C$:			
Drain-to-Source Voltage, V_{DS}	−0.2 to +18	−0.2 to +18	V
Gate Terminal Current, I_{G1S} or I_{G2S}	±100	±100	μA
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+24	+22.5	V
Drain Current, I_D	50	50	mA
Transistor Dissipation:			
At T_A up to 25°C	330	330	mW
At T_A above 25°C	derate linearly 2.2 mW/°C		
Ambient Temperature Range:			
Operating and Storage	−65 to +175	−65 to +175	°C
Lead Temperature (During Soldering):			
At distances 1/32 in from seating surface for 10 s max.	265	265	°C

[#] Continuous Working Voltage Ratings must be observed to maintain device characteristics. These ratings are based on long-term continuous voltage operation but may be exceeded for short durations (e.g. testing of device characteristics), provided the Absolute Maximum Ratings are not exceeded.

TYPICAL CHARACTERISTICS FOR TYPES 40822 AND 40823

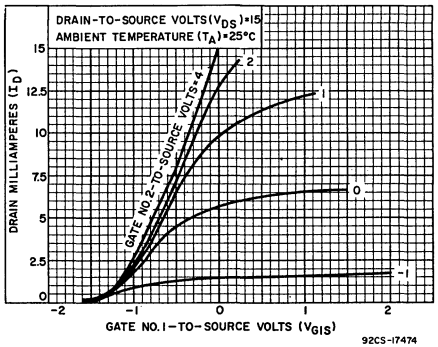


Fig. 1 - I_D vs. V_{G1S}

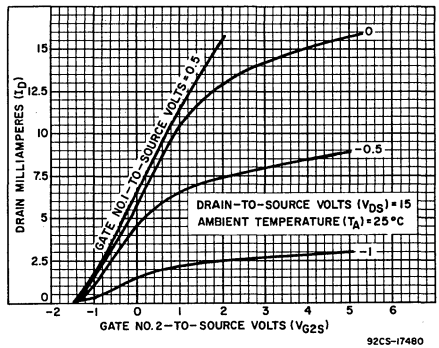


Fig. 2 - I_D vs. V_{G2S}

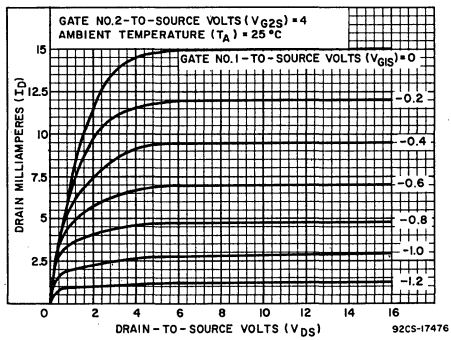


Fig. 3 - I_D vs. V_{DS}

ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$

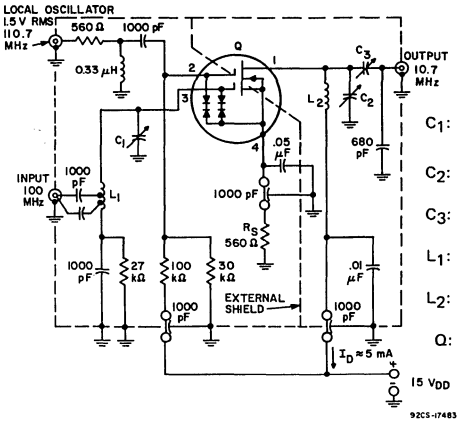
CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		LIMITS 40822 Min.	40822 Typ.	40822 Max.	40823 Min.	40823 Typ.	40823 Max.	UNITS
Gate No. 1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS}= +15V, I_D=200μA, V_{G2S}= +4V		–	–2	–4	–	–2	–4	V
Gate No. 2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS}= +15V, I_D=200μA, V_{G1S}=0		–	– 2	–4	–	– 2	– 4	V
Gate-to-Source Forward Breakdown Voltage: Gate No. 1	$V_{(BR)G1SSF}$	I_{G1SSF} = I_{G2SSF} = 100 μA	$V_{G2S} = V_{DS} = 0$	–	9	–	–	11	–	V
Gate No. 2	$V_{(BR)G2SSF}$		$V_{G1S} = V_{DS} = 0$	–	9	–	–	11	–	V
Gate-to-Source Reverse Breakdown Voltage: Gate No. 1	$V_{(BR)G1SSR}$	I_{G1SSR} = I_{G2SSR} = 100 μA	$V_{G2S} = V_{DS} = 0$	–	9	–	–	11	–	V
Gate No. 2	$V_{(BR)G2SSR}$		$V_{G1S} = V_{DS} = 0$	–	9	–	–	11	–	V
Gate No. 1-Terminal Forward Current	I_{G1SSF}	$V_{DS} = V_{G2S} = 0$	V_{G1S} = 6 V	–	–	50	–	–	–	nA
			V_{G1S} = 4.5 V	–	–	–	–	–	50	nA
Gate No. 1-Terminal Reverse Current	I_{G1SSR}	$V_{DS} = V_{G2S} = 0$	V_{G1S} = –6 V	–	–	50	–	–	–	nA
			V_{G1S} = –4.5 V	–	–	–	–	–	50	nA
Gate No. 2-Terminal Forward Current	I_{G2SSF}	$V_{DS} = V_{G1S} = 0$	V_{G2S} = 6 V	–	–	50	–	–	–	nA
			V_{G2S} = 4.5 V	–	–	–	–	–	50	nA
Gate No. 2-Terminal Reverse Current	I_{G2SSR}	$V_{DS} = V_{G1S} = 0$	V_{G2S} = -6 V	–	–	50	–	–	–	nA
			V_{G2S} = –4.5 V	–	–	–	–	–	50	nA
Zero-Bias Drain Current	I_{DS}	V_{DS} = +15 V, V_{G1S} = 0, V_{G2S} = +4 V		5	15	30	5	15	35	mA
Forward Transconductance (Gate No. 1-to-Drain)	g_{fs}	V_{DS} = +15 V I_D = 10 mA V_{G2S} = +4 V	f = 1 kHz	–	12000	–	–	12000	–	μmho
Small-Signal, Short-Circuit Input Capacitance †	C_{iss}		f = 1 MHz	–	6.5	9.5	–	6.5	10	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate-No. 1)♦	C_{rss}			0.005	0.020	0.030	0.005	0.025	0.045	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}			–	2	–	–	2	–	pF
Power Gain (see Fig. 5)	G_{PS}		f = 100 MHz	19	24	–	–	–	–	dB
Noise Figure (see Fig. 5)	NF			–	2	3.5	–	–	–	dB
Conversion Gain	$G_{PS(C)}$		f = 100 to 10.7 MHz	–	–	–	14	18	–	dB

† Capacitance between Gate No. 1 and all other terminals.

♦ Three-terminal measurement with Gate No. 2 and Source returned to guard terminal.

OPERATING CONSIDERATIONS

The flexible leads of the 40820 and 40821 are usually soldered to the circuit elements. As is the case with any high-frequency semiconductor device, the tips of soldering irons MUST be grounded.



C_1: 1.3-5 pF variable air capacitor: E.F. Johnson Type 160-102 or equivalent.

C_2: 2.7-19.6 pF variable air capacitor: E.F. Johnson Type 160-110 or equivalent.

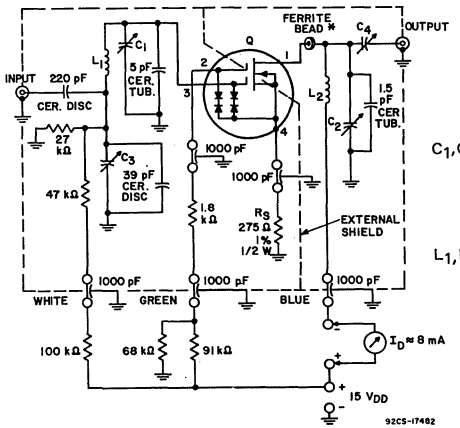
C_3: 80 pF max. compression-type capacitor: Arco 405 or equivalent

L_1: 8 turns No. 22 wire on 1/4" diameter air core. One turn spacing between windings. Tapped at one turn from low end.

L_2: 37 turns No. 34 wire on 3/16" diameter air core. Unloaded Q = 63

Q: 40823.

Fig. 4 - 100/10.7-MHz conversion power gain test circuit for type 40823.



C_1, C_2: 1.3-5.4 pF variable air capacitor

C_3: 1-10 pF variable air capacitor, piston type: Johanson Co., No. 4335

C_4: 1-15 pF variable air capacitor, precision piston type: Roanwell Corp. SG11129/AG

L_1, L_2: 0.22 μH RF choke (7T): Miller, No. 4584

*Ferramic toroid (1/2 used): Indiana General, No. CF101-(0-6)

Fig. 5 - 100-MHz power gain and noise figure test circuit for type 40822.

TYPICAL CHARACTERISTICS FOR TYPES 40822 AND 40823

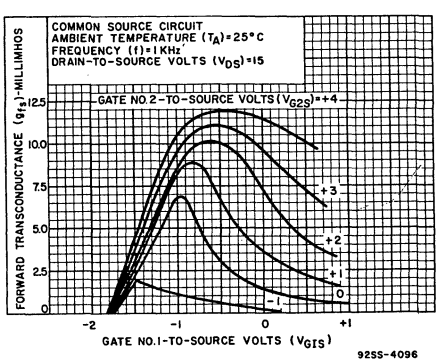


Fig. 6 - g_{fs} vs. V_{G1S}

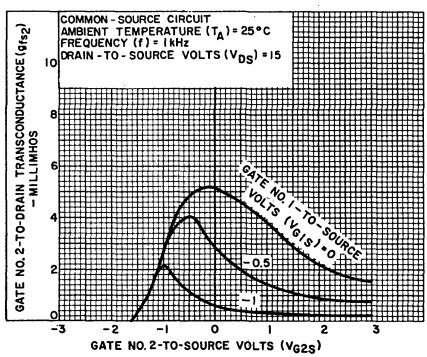


Fig. 7 - g_{fs2} vs. V_{G2S}